SIEMENS

Revision Record

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Open Source Software

Commercial Software

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SCADA - Applications

v2.20

User Guide

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NOTE

For your own safety, observe the warnings and safety instructions contained in this document, if available.

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Revision Record

Version	Date	Author/Department	Approver/Department	Modifications
2.2.0.0	12.2017			Released Version

Preface

Notes on Safety

This manual is not a complete index of all safety measures required for operation of the equipment (module or device). However, it comprises important information that must be followed for personal safety, as well as to avoid material damage. Information is highlighted and illustrated as follows according to the degree of danger:



WARNING

WARNING means that death or severe injury **may** result if the measures specified are not taken.

✧ Comply with all instructions, in order to avoid death or severe injuries.



CAUTION

CAUTION means that medium-severe or slight injuries **can** occur if the specified measures are not taken.

✧ Comply with all instructions, in order to avoid moderate or minor injuries.



NOTE

Important information about the product, product handling or a certain section of the documentation which must be given particular attention.

Qualified Electrical Engineering Personnel

Only qualified electrical engineering personnel may commission and operate the equipment (module, device) described in this document. Qualified electrical engineering personnel in the sense of this manual are people who can demonstrate technical qualifications as electrical technicians. These persons may commission, isolate, ground and label devices, systems and circuits according to the standards of safety engineering.

Only competent and authorized personnel should work with this product after becoming thoroughly familiar with all warnings, safety notices, operating instructions and maintenance procedures.

Proper Use

The product must not be used for any other purposes than that described in the technical documentation. If it is used together with third-party devices and components, these must be recommended or approved by Siemens AG. The successful and safe operation of this product is dependent on adequate transportation and proper handling, storage, installation, operation and maintenance.

Scope

This document provides specific and detailed information on how to use a particular product or product component.



NOTE

Note that the screenshots used in this document contain sample data which may not be available in some systems.

Typical Users

This document is designed for users that are already familiar with operational and technical aspects of power generation and power transmission and distribution as well as product concepts:

- **System Engineers**

  The System Engineer is able to install and to customize the system. He needs deep knowledge about the internal structures and processes of the network control system.

- **Operators**

  The Operator is controlling the network or the plants with the system therefore he needs to know how the network control system has to be operated.

- **Data Engineers**

  The Data Engineer translates the real network of a utility or the generation plants to the data model of the network control system. He also maintains the data model.

- **Technical Specialists**

  The Technical Specialist is responsible for a special technical area which is related to the network control system. This can be the planning department, which plans changes in the network and uses network analyses applications of the network control system or the RTU department, which has to connect the RTU to the network control system. The Technical Specialist has to know how specific functions of the network control system work in detail.

Structure of the Manual

- **Introduction**:

  Basic information about the component

- **Workflow/Operations**:

Main use cases, workflows and operational steps for optimal use of the component

- References:

  Reference and description of user interfaces and dialogs

This document is part of a set of manuals that describes the complete product.

Various other product modules may be mentioned or discussed in this document. For more detailed information or if you have any questions about these products, please contact your Siemens representative.

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The product contains, among other things, Open Source Software developed by third parties. The Open Source Software used in the product and the license agreements concerning this software can be found in the Readme_OSS. These Open Source Software files are protected by copyright. Your compliance with those license conditions will entitle you to use the Open Source Software as foreseen in the relevant license. In the event of conflicts between Siemens license conditions and the Open Source Software license conditions, the Open Source Software conditions shall prevail with respect to the Open Source Software portions of the software. The Open Source Software is licensed royalty-free. Insofar as the applicable Open Source Software License Conditions provide for it you can order the source code of the Open Source Software from your Siemens sales contact - against payment of the shipping and handling charges - for a period of at least 3 years since purchase of the Product. We are liable for the Product including the Open Source Software contained in it pursuant to the license conditions applicable to the Product. Any liability for the Open Source Software beyond the program flow intended for the Product is explicitly excluded. Furthermore any liability for defects resulting from modifications to the Open Source Software by you or third parties is excluded. We do not provide any technical support for the Product if it has been modified.

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Among other things, this product contains Commercial Software that was developed by third parties and requires the placement of a logo or trademark legends, or other acknowledgements:



[javalogo, 1, --_--]

Java is a trademark of Oracle.

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1 Introduction

The Spectrum Power™ User Interface allows you to monitor and control the power system.

This user guide describes:

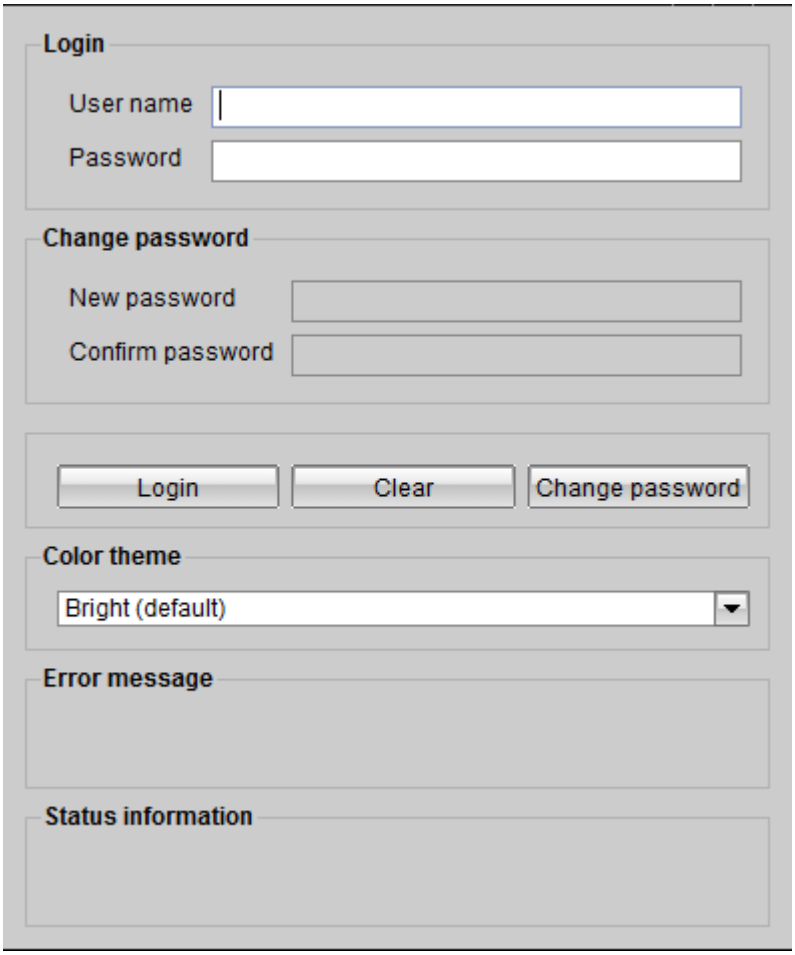
- Basic Signaling Display
- Heads-Up Display
- Console and User Management Display
- Online Diagram Viewer
- Station Explorer
- Summaries
- List of Displays
- Calculated Points Display
- System Network Status Display
- Technological Area (TA)-Related Information Displays
- Reminder Summary
- Curve Tool
- Operational Dashboard
- Corporate Viewer

2 Login

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2.1 Overview

The operator can login to Spectrum Power 7 using the login window by entering the 'User name' and 'Password' fields as defined in Console and User Management (CUMAN). For additional information, refer to 6 Console and User Management.

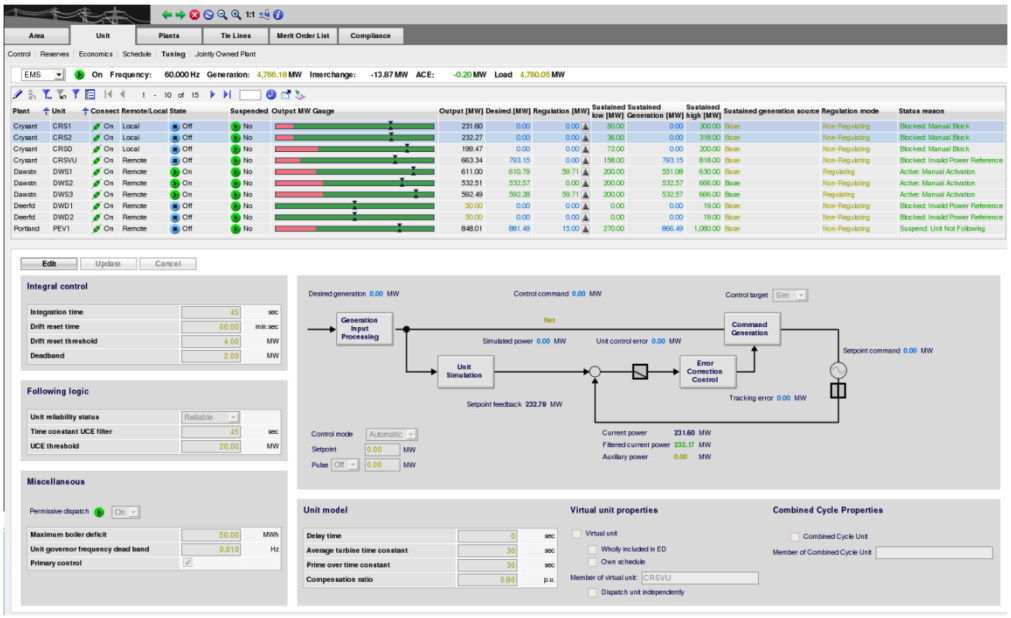


[Sc_Loginwindow, 2, en_US]

Figure 2-1 Login Window

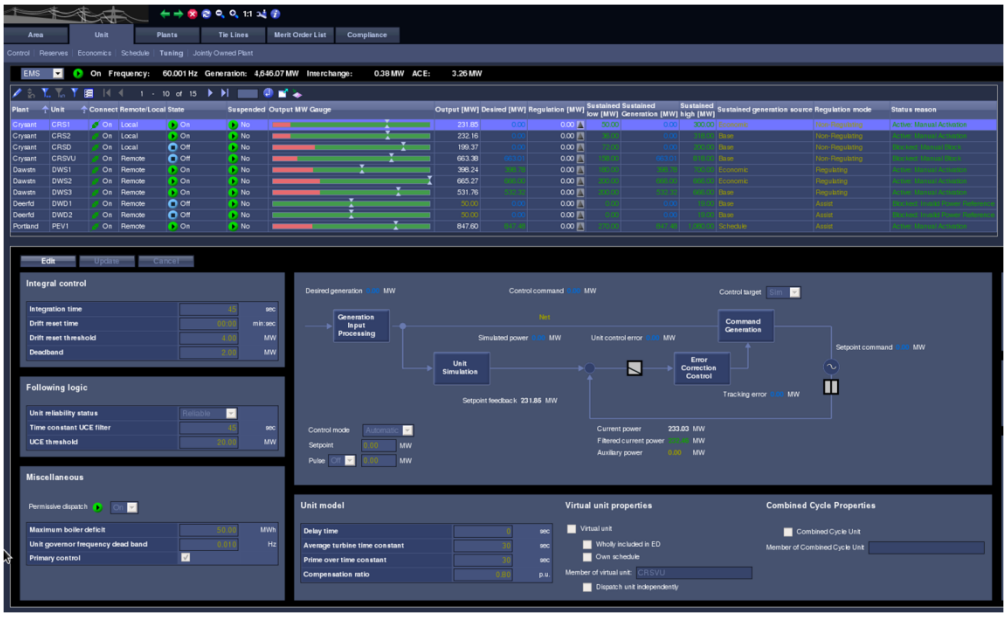
Spectrum Power supports two color themes for application displays, Basic Signaling Display, Online Diagram Viewer, Curve Tool, and Operational Dashboard. The operator can select the preferred color theme upon log-in.

The selected color theme is applied after Login



[sc_lighttheme, 1, en_US]

Figure 2-2 Bright Theme



[sc_darktheme, 1, en_US]

Figure 2-3 Dark Theme

3 Basic Signaling Display

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3.1 Introduction

A key element for operator interaction with Spectrum Power is the Basic Signalling Display (BaSiDi)[1] which is automatically displayed after logging in to the real-time operating system.

The BaSiDi contains the following buttons:

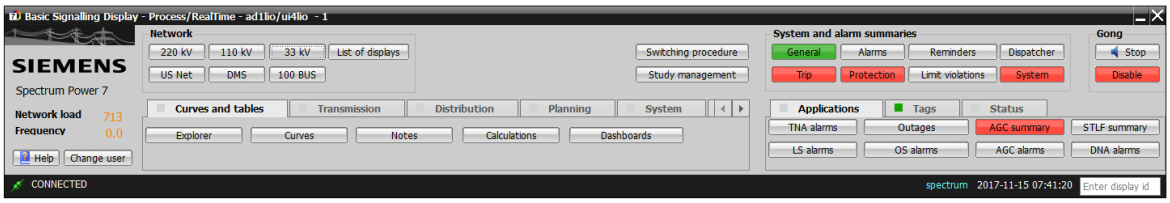
- Network diagrams
- Summaries
- Applications

Some buttons also indicate that one or more alarms have not been acknowledged on the corresponding network diagram or summary.

[1] This is also called the Basic Signaling Window

3.2 User Interface

3.2.1 Overview



[sc_basidi, 2, en_US]

Figure 3-1 Basic Signalling Display

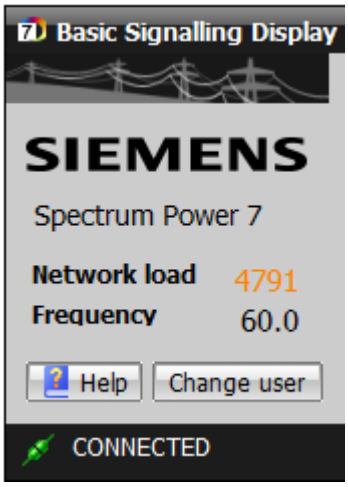
The Basic Signalling Display consists of buttons for various functions and an area where the date, time, and total load are displayed.

Flashing buttons indicate that one or more alarms have not been acknowledged on the corresponding network diagram or summary. The Basic Signalling Display cannot be removed from the screen.

NOTE

The BaSiDi might change based on customer needs.

3.2.2 System Information Area



[sc_NetworkLoadandFrequency_12032014, 4, en_US]

Figure 3-2 Network Load and Frequency

Click **Help** to invoke the online help.

Clicking the button **Change User** (for application level login) opens a display for entering the login information (user name, password) for a different user. This allows to log in as a different user without log out of the current user.

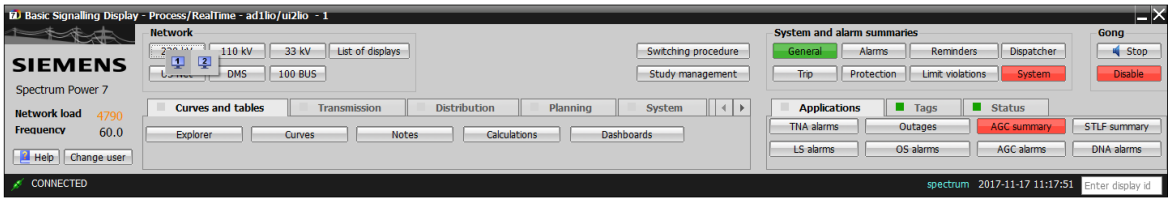
If the real-time communications (RTC) systems are operational and all systems work as expected, the **CONNECTED** sign shows. If all RTC servers are stopped, display and operation capabilities can temporarily be lost on the UI servers and the UI server changes to **DISCONNECTED**.



[sc_mmi-offline, 1, en_US]

Figure 3-3 UI Offline

The operator can specify on which a monitor a display should be opened by right-clicking on the button. The display opens in that monitor by default in the future.



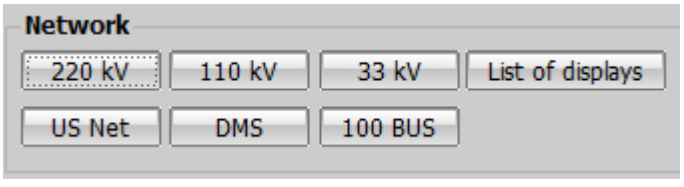
[sc_selectmonitor, 2, en_US]

Figure 3-4 Select Monitor

3.2.3 Buttons

Network Diagrams

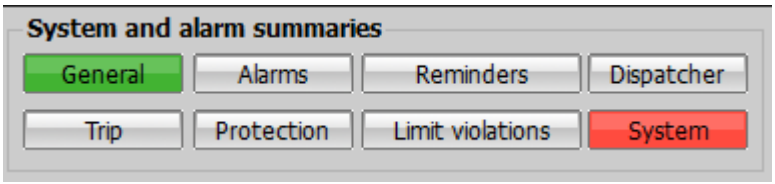
The Basic Signalling Display contains a number of display call-up buttons. The display call-up buttons allow you to invoke network diagrams and Spectrum Power applications.



[sc_ButtonsToInvokeSingleLineDiagrams, 3, en_US]

Figure 3-5 Buttons to Invoke Network Diagrams

System and Alarm Summaries

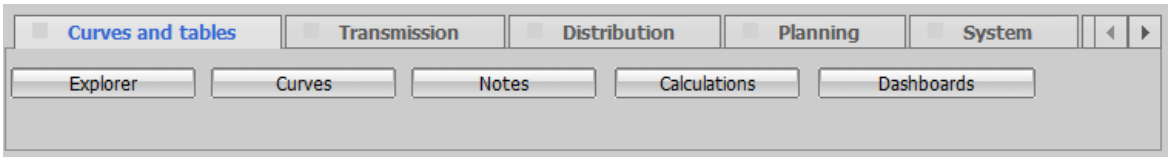


[sc_ButtonsToInvokeSystemAndAlarmSummaries_12032014, 3, en_US]

Figure 3-6 Buttons to Invoke System and Alarm Summaries

A blinking red button indicates that the associated summary consists of one or more alarm entries that are not acknowledged.

Tools and Application Displays

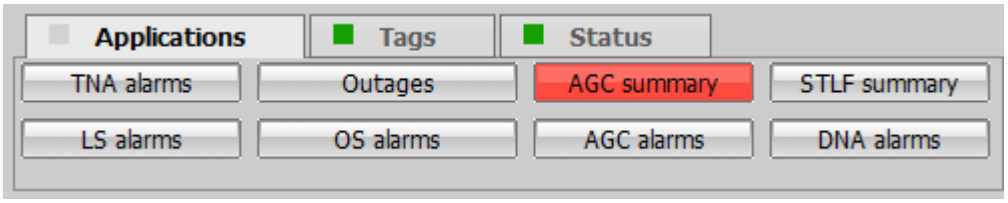


[sc_ButtonsToCallUpToolsAndDisplayApplications_12032014, 3, en_US]

Figure 3-7 Buttons to Call-up Tools and Display Applications

Tabs are available in the Basic Signalling Display that show buttons for calling up tools and application displays.

Application Summary, Tag Summary, and Status Summary



[sc_ButtonsToViewApplicationSummaryTagSummaryAndStatusSummary_12042014, 3, en_US]

Figure 3-8 Buttons to View the Application Summary, Tag Summary, and Status Summary

3.3 Operational Steps

3.3.1 Opening a Summary

When a new alarm message arrives, the button of the corresponding alarm summary in the BaSiDi starts flashing.

To open an alarm summary:

- Click the button for this summary.

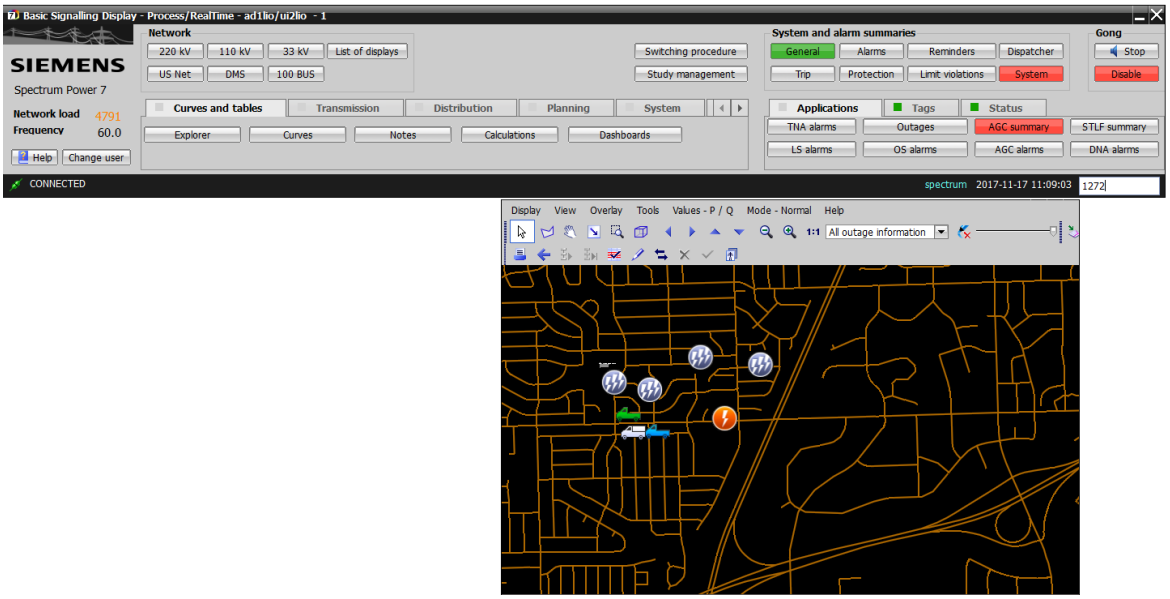
  The summary opens, displaying the latest alarm messages.

3.3.2 Calling up a Display using View Number

Each display has a view number associated with it. To call up a display, click the corresponding button or enter the view number:

- In the field for Direct Display Call-Up (DDCU), which is at the right bottom section of the Basic Signalling display, enter the view number and press **Enter**.

  The corresponding display opens.



[sc_basidi-displaycallup, 2, en_US]

Figure 3-9 Direct Display Callup

4 Heads-Up Display (HUD)

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4.1 Introduction

An alternate to BaSiDi is the Heads-Up Display (HUD) which provides a central system overview and common access point for starting user workflows and Spectrum Power application.

When the HUD is launched, it exhibits smart resizing behavior and consumes full monitor width. Position of HUD can be changed to top or bottom of the screen.

HUD layout is assigned to a user role and cannot be changed while logged in.

4.2 User Interface

4.2.1 Overview

NOTE

With v2.20, HUD is released only for EMS customers.



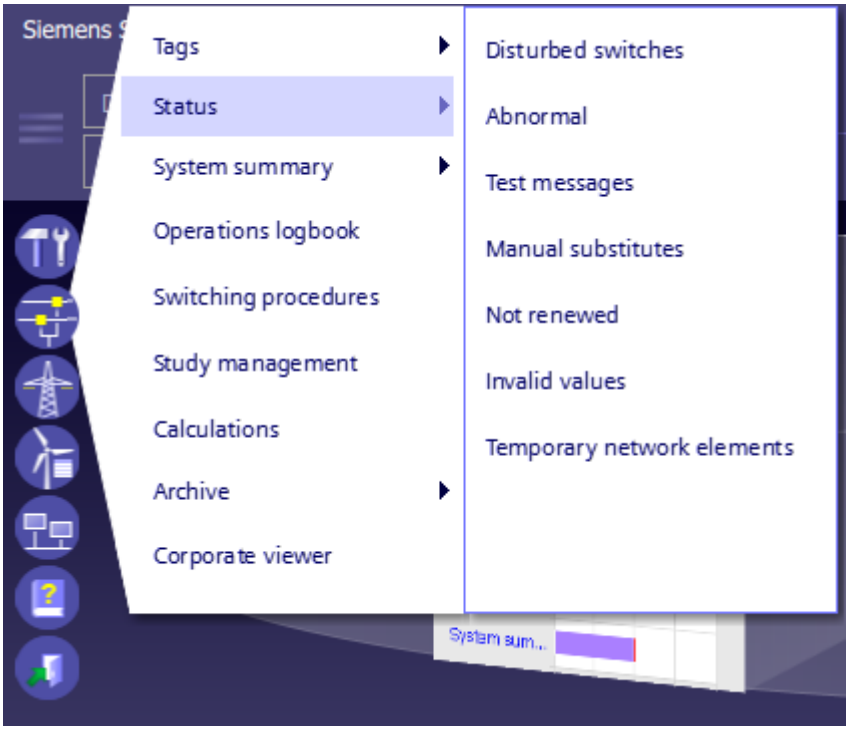
[dw_Heads-Up Display Marked 2, 1, en_US]

Figure 4-1 Heads-Up Display

1) Navigation menu
2) Display callup and Dashboards
3) Cards
4) Alarm bar
5) Worldmaps
6) Mute audio alarms
7) Gauges

4.2.2 Navigation Menu

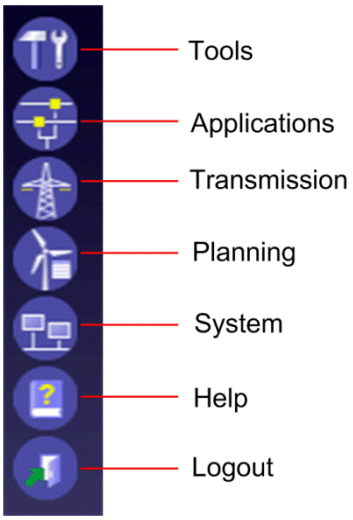
When the HUD is maximized, the operator can hover over and select the Navigation menu options to invoke displays from HUD.



[sc_Navigation Menu, 1, en_US]

Figure 4-2 Navigation Menu

Navigation menu supports multiple levels. An arrow ▶ icon indicates that the menu item has a sub menu. When the operator hovers over the item, a clapped-out menu opens. Click on a menu item to open the display. The display uses the remaining height of screen in an optimized way depending on position of HUD.



[dw_NavigationMenu, 1, en_US]

Figure 4-3 Navigation Menu Entries

NOTE

The menu items shown in the actual customer system depends on the scope of the delivery and customer needs. Usually, menu entries are customized for each project.

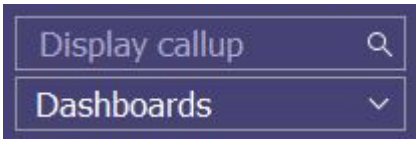
Menu items in Navigation menu:

- Tools
  - Explorer
  - Curves
  - Notes
  - List of displays
  - Desktop layout
  - Camera
  - New dashboard
  - User preference
- Applications
  - Tags
    - ◊ Removed from operation
    - ◊ Control inhibit
    - ◊ Permit to work
    - ◊ Permit to test
    - ◊ Object in test
    - ◊ Switching in progress
    - ◊ Used for disconnection
  - Status
    - ◊ Disturbed switches
    - ◊ Abnormal
    - ◊ Test messages
    - ◊ Manual substitutes
    - ◊ Not renewed
    - ◊ Invalid values
    - ◊ Temporary network elements
  - System summary
    - ◊ General
    - ◊ System
  - Operations logbook
  - Switching procedures
  - Study management
  - Calculations
  - Archive
    - ◊ Disturbance data
    - ◊ Values
    - ◊ Messages
    - ◊ Switching procedures
  - Corporate viewer
- Transmission

  - Network analysis
    - ◊ Real-time monitor
    - ◊ Study
    - ◊ Joint studies
    - ◊ Day-Ahead Congestion Forecast
    - ◊ Two Days-Ahead Congestion Forecast
    - ◊ Intra-Day Congestion Forecast
    - ◊ Maintenance study
    - ◊ Outage Scheduler
  - Power applications
  - Short-term load forecast
  - Current operation plan
  - Training
  - Training - Transfer
- System
  - Users
  - Authorities
  - System status
  - ICCP
  - Multi site
  - Communication status
- Help
- Logout
  - Change user
  - Log out

4.2.3 Display Callup and Dashboards

Display callup and Dashboards fields are present on top left corner of the HUD.



[sc_Display Callup and Dashboards, 1, en_US]

Figure 4-4 Display Callup and Dashboards

Display callup provides an input field for entering the view number of a display to call it up directly.

The drop-down Dashboards box displays recently opened dashboards for the current user session. The operator can open a previously saved dashboard and can save new dashboards.

4.2.4 Cards

HUD supports cards which display real-time information and provide access to details regarding Automatic Generation Control (AGC), Load Shed, Transmission Network Analysis (TNA), Alarms, and Trends on Curve tool.



[sc_Cardsgen, 1, en_US]

Figure 4-5 Example of Cards

NOTE

Cards shown in the actual customer system depend on the scope of the delivery and customer needs.

Various types of cards in HUD are:

- Generation Control card

  The Generation Control card shows the essential information regarding frequency, Area Control Error (ACE), and interchange. All data is synchronized with the Power Apps display.

  Clicking on this card opens the main display of Power Apps in a new window on the same monitor.

  Font colors for analog values reflect quality codes. The trend displays invalid values as gaps.

- Load Shed card

  The Load Shed card shows real-time status information that is synchronized with Load Shed display.

  Clicking on Load Shed card opens the Load Shed application.

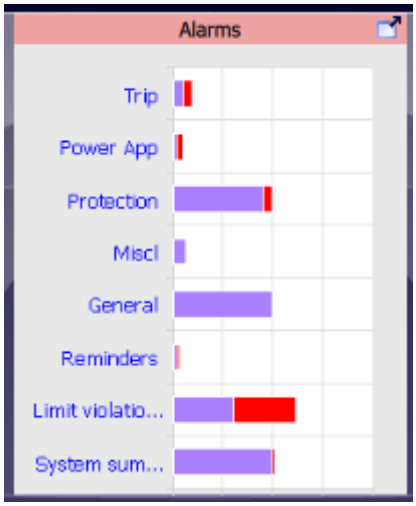
- Trending card

  The Trending card shows correct historical/trending information of three data points that are preconfigured. Such information is synchronized with curve tool data for the three TAs. Invalid values are shown as gaps.

  Clicking on Trending card opens curve tool in a new window.

- Alarms card

  Alarm card displays alarm categories which are linked to corresponding alarm summaries. Alarm card is updated real-time. If a new alarm occurs, the header of the card blinks to indicate the new alarm. Header of alarm card and alarm icons are in synchronization regarding blinking and blinking frequency.



[sc_AlarmsCard, 1, en_US]

Figure 4-6 Blinking Header of Alarm card

Alarm card shows eight alarm categories which are linked to summaries. Clicking on each alarm category opens the summary of the corresponding category.

The bars on the Alarm card show the total number of alarms. Unacknowledged alarms are displayed in red.

- Transmission Network Analysis card

  The operator can view a summary of the currently active base case and contingency violations to get a quick impression on the network status. Additionally, the card provides an overview of planned outages that are currently active, about to start or about to end. Click on Transmission Network Analysis card to open TNA Operator RT Monitor display.

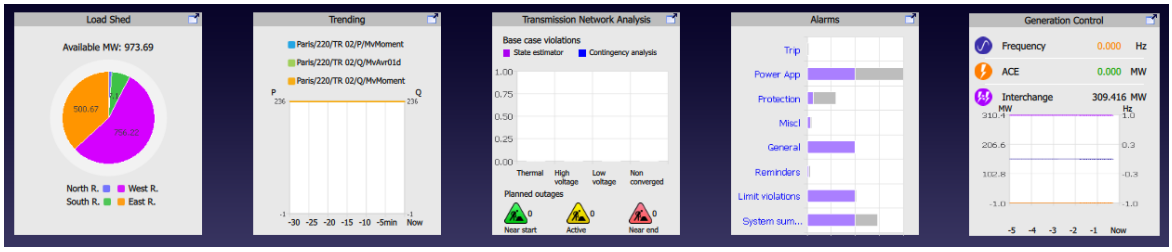
Cylinder vs Flat View of Cards

When an operator login from Login window, the cylinder view of cards in the HUD is loaded by default.

The operator can change view of cards, and can switch from cylinder view to flat view (and vice versa). This change does not affect any display information.

In flat view:

- Cards are placed next to each other.
- Width of the cards remains the same. Distance between cards is also same.
- Clicking on a card directly opens the corresponding application display.

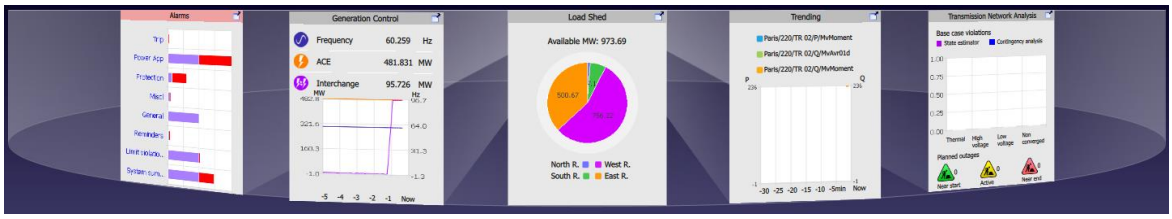


[sc_FLat view full, 1, en_US]

Figure 4-7 Flat View of Cards

In cylinder view:

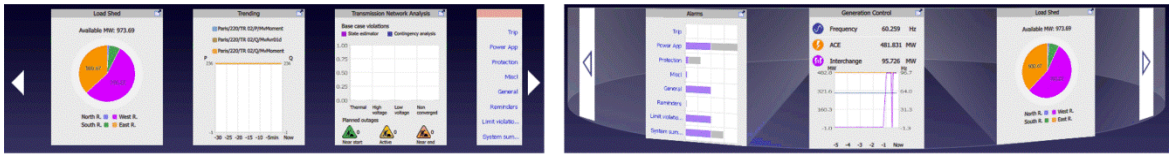
- Cards are placed on a cylinder. The width of the middle card is maximum. A single click on the middle card opens the corresponding application display.
- Single clicking on the side cards brings the card to middle.
- Double clicking on the side card brings the card to middle and opens the corresponding application display.



[sc_Cylinde view full, 1, en_US]

Figure 4-8 Cylinder View of Cards

When there is less area available than number of cards, arrows appear on left and right side of both cylinder and flat views.

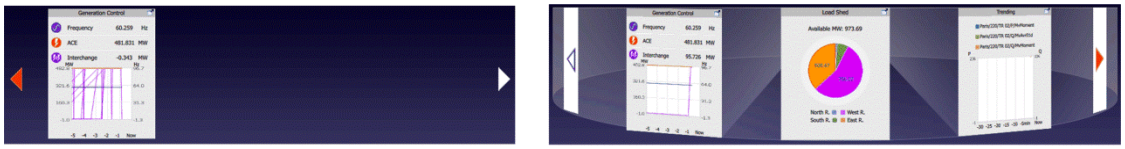


Flat view Cylinder view

[dw_Arrow in Flat vs Cylinder, 1, en_US]

Figure 4-9 Arrows in Flat vs Cylinder View

- Clicking left arrow shows previous card.
- Clicking right arrow shows next card.
- If a new alarm occurs related to a hidden card:
  - Arrow on left or right starts to blink in red.
  - Clicking on a blinking arrow shows the blinking card within the middle of the HUD.



Flat view Cylinder view

[dw_Blinking Arrow in Flat vs Cylinder view, 1, en_US]

Figure 4-10 Blinking Arrows in Flat vs Cylinder View

4.2.5 Alarm Bar

Top part of the HUD contains the alarm bar. Icons on alarm bar starts blinking when new alarms occur. The blinking color represents the alarm severity: alarms with high severity blink red; alarms with medium severity blink yellow. Blinking stops when the operator acknowledges all alarms of that alarm category. For additional information, refer to 4.3.2 Opening an Alarm Summary Using Alarm Bar.



[sc_Alarm Bar, 1, en_US]

Figure 4-11 Alarm Bar

The number adjacent to the corresponding alarm icon in the alarm bar indicates the count of new alarms.

When there is less real estate available for all the alarm icons, arrows appear on left and right side of the alarm bar. If an alarm occurs related to a currently hidden alarm icon, the arrow on the left or right side of the alarm bar starts blinking in red. The operator can navigate to the hidden blinking alarm icon by clicking on the blinking arrow.



[sc_BlinkingArrowAlarm, 1, en_US]

Figure 4-12 Blinking Arrow on Alarm Bar

The operator can quickly navigate to the corresponding alarm summary for starting problem analysis by clicking on the alarm icons.

4.2.6 Mute Audio Alarms

HUD supports muting of the latest alarms or all alarms.

To mute the current alarm, click  .

To mute all alarms, click  .

4.2.7 Gauges

Gauges are updated real-time and display the value of configured measurements like Load, Area Control Error (ACE), and Reserves. The specific measurements are configured for each customer system.



[sc_Guages, 1, en_US]

Figure 4-13 Gauges

A gauge can display up to three lower limits and up to three upper limits. Depending on current value, the areas for nominal value and between the limits gets colored differently.

White color indicates that the current value is neutral. Yellow, orange, and red color indicate that the current value is in critical zones, for example, upper limit 1, upper limit 2, and upper limit 3 respectively.

The default colors for the three lower limit areas are the same as for the three upper limit area.

When there is an ‘invalid data’ quality on gauges:

- White colored text 'invalid' is shown instead of the value.
- The needle is not displayed.

4.2.8 Status Bar

Status bar of HUD displays:

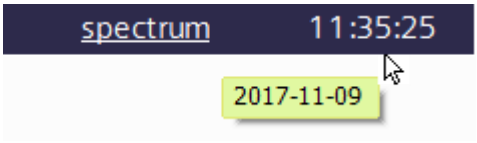
- Connection status and connection icon in status bar. In case of offline scenario, the status bar displays a notification to the operator.
- Name of currently logged in operator. By clicking on the user name, the CUMAN display is opened, showing the detailed access rights for this operator. For additional information, refer to 6 Console and User Management.
- Current time in hours:minute:seconds format.



[dw_Status Bar, 1, en_US]

Figure 4-14 HUD Status Bar

When the operator hovers with mouse pointer over time in the status bar, and stays on it, a tooltip appears showing current date. The tooltip disappears when the mouse pointer is moved away from time in status bar.



[sc_Tooltip for Time and Date, 1, en_US]

Figure 4-15 Tooltip for Time and Date

Specific time format and time zone are configurable and are in sync with other Spectrum Power 7 displays.

The operator can view details on current server connection. In case of broken connection, the operator can view feedback immediately.

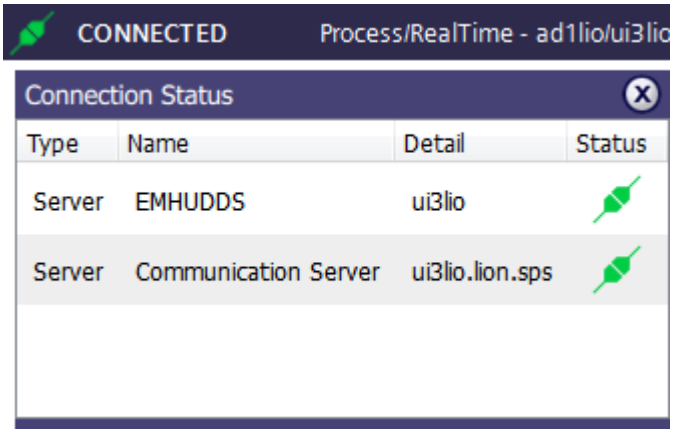
The connection status is shown using connection icon in the status bar.



[sc_Status bar, 1, en_US]

Figure 4-16 HUD Status Bar

Clicking on the connection icon opens the Connection Status popup and console names are visible.



[sc_Connection Status, 1, en_US]

Figure 4-17 HUD Connection Status

Janus connection status is shown as ‘Communication Server’ on Connection Status popup.

4.2.9 COM Offline Scenario

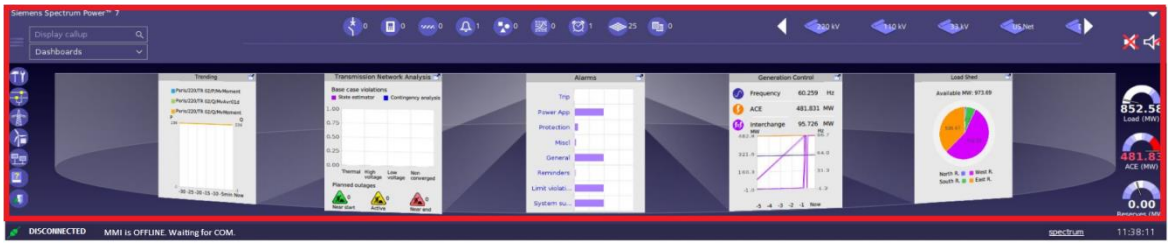
HUD handles COM offline scenario in a way similar to BaSiDi by showing a red border and a feedback message.

During COM offline scenario:

- The operator receives clear feedback if the information on the HUD is not updated anymore.
- The operator stays logged in.

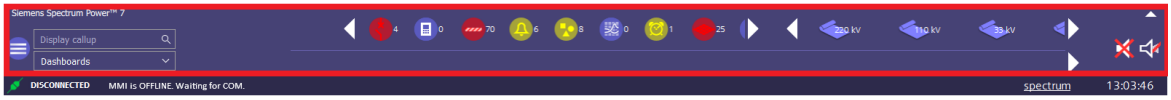
When both COM servers are offline:

- A red border appears around maximized and minimized mode of HUD window.
- An error message ‘MMI is offline, waiting for COM’ is displayed in the status bar of HUD which informs the operator that the user interface is offline.
- The HUD accepts inputs from operator and works in the same way as before. However, the displayed information is not updated and all follow-up dialogs and windows have the same red border. The user can close the HUD using menu ‘Logout’ in the Navigation menu.



[sc_COM Offline Scenario Max, 1, en_US]

Figure 4-18 Maximized HUD With Red Border During COM Offline Scenario



[SC_Offline Scenario Min, 1, en_US]

Figure 4-19 Minimized HUD With Red Border During COM Offline Scenario

When the COM server is online again within LifeCheck timeout, the red border disappears, the error message in status bar is removed, and all values/status in HUD are updated.

After 'LifeCheck timeout, a dialog' appears and font color of error message 'MMI is Offline, waiting for COM' changes to red.

When the operator confirms LifeCheck dialog, the HUD and all open Spectrum Power windows get closed.

4.2.10 Configuration of Different User Roles

A system operator or administrator can configure different layouts of HUD, for example, different alarm icons, cards, menu entries, and worldmap shortcuts for different authority roles.

The HUD layout is assigned to a authority role and loaded automatically upon login. The HUD layout cannot be changed while logged in. For additional information, refer to 6 Console and User Management.

4.2.11 Show HUD On All Monitors

The operators can choose whether HUD is to be displayed on one monitor or on all monitors through **Tools** > **User preference**s in Navigation menu. For additional information, refer to 4.3.8 Using HUD on All Monitors.

When HUD is displayed on one monitor, by default HUD is shown on top of the monitor, spanning the whole monitor width with expanded layout.

When operator chooses to display HUD on all monitors, position of HUD on other monitors is consistent to position of HUD on main monitor.

The settings are stored with desktop layout management and is available while login again.

When a user triggers a new window using HUD, the new window is shown on the same monitor as the HUD window from where the action was triggered.

4.2.12 HUD Layout

Maximum and Minimum Layout of HUD

The operator can switch between HUD layout without readjusting all other currently open windows on that monitor. HUD remains clearly visible and all other windows are rearranged accordingly. For additional information, refer to 4.3.6 Minimizing and Maximizing HUD.

When HUD is maximized, the size of other Spectrum Power 7 windows located on the same screen gets adapted, so that none of these windows are overlapped by HUD.

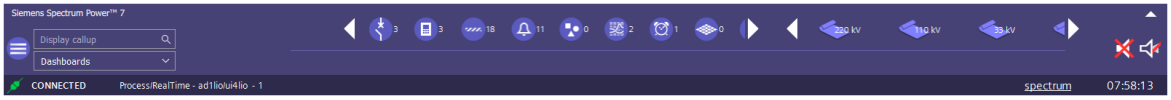


[sc_HUD, 1, en_US]

Figure 4-20 HUD With Maximized Layout

When HUD is maximized, Spectrum Power 7 windows on other monitors remains unchanged in size or position.

When HUD is minimized, the size of open Spectrum Power 7 windows automatically increases and consumes the space released by HUD.



[sc_HUD min, 1, en_US]

Figure 4-21 HUD With Minimized Layout

When an open Spectrum Power 7 window is maximized which is located on the same monitor as HUD, the Spectrum Power 7 window uses only the remaining available space below or above HUD.

Position of HUD Window

The operator can position the HUD, so that it does not overlay any other displays. The HUD exhibits smart resize behavior when placing it at top or bottom of a monitor. The HUD consumes full monitor width by default.

When operator drops the HUD, and the position where the operator dropped the HUD is above half of the screen size, HUD will be placed on top. If it is dropped below the half of the screen size, then it will be placed at the bottom of the screen. Spectrum Power windows that are on the same screen as the HUD are rearranged and do not get overlapped.

4.3 Operational Steps

4.3.1 Opening a Display Using Navigation Menu

To open a summary using Navigation menu:

- Hover over the Navigation menu to open the list of menu items.
- If a menu item has sub-items, click on ▶ to show the sub-menu item.
- Clicking on menu item or sub-menu item opens the corresponding display as new window.

4.3.2 Opening an Alarm Summary Using Alarm Bar

To open the alarm summary:

- Click on the corresponding alarm icon on the alarm bar.
- The corresponding summary opens as new window displaying the latest alarm messages.

4.3.3 Viewing New Alarms

When a new alarm occurs, the alarm icon in alarm bar, the header of the alarm card and chart corresponding to related alarm category in alarm card start blinking to give the operator clear indication of a new alarm. The number adjacent to the corresponding alarm icon in alarm bar indicates the count of current alarms.

The operator can view details of new alarm by clicking the link of related alarm category on alarm card. This opens the corresponding alarm summary as new window.

The operator can open the corresponding alarm summary by clicking on the blinking alarm icon as well.

All blinking stop after the alarm is acknowledged.

4.3.4 Viewing Transmission Network Analysis Result

To open further details:

- Click on Transmission Network Analysis card to open TNA operator RT Monitor display.

4.3.5 Calling Up a Display Using View Number

To call up a display, click on **Display callup** 

- In Display callup field, enter view number and press **Enter**.
- The corresponding display opens as a new window.

4.3.6 Minimizing and Maximizing HUD

HUD layout can be minimized on demand to hide cards.

- To minimize, click on placed on the upper right corner of HUD.
- To maximize, click on again.



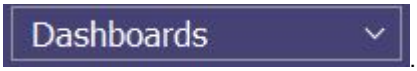
[sc_MaxMin HUD, 1, en_US]

Figure 4-22 Minimizing Icon in HUD

When a HUD is minimized or maximized, the size of other open Spectrum Power 7 windows gets adjusted accordingly, so that, none of these windows are overlapped by HUD. HUD remains always on top.

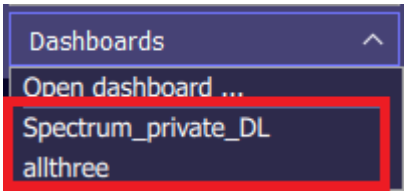
4.3.7 Opening a Dashboard

To open a dashboard:

- Click on **Dashboards** 

. A drop down list appears.
- Click on **Open dashboard ...**. The Open dashboard window is displayed, and any available dashboard can be opened. For additional information, refer to 18 Operational Dashboard.

If an already opened dashboard is closed using the close button, the entry in recent list is unchanged. It can be opened again using related entry in recent list.



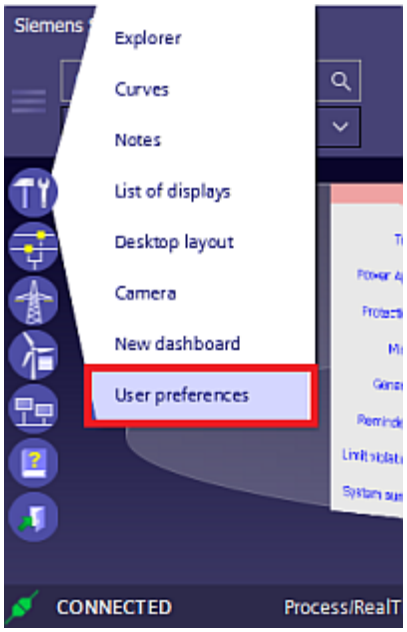
[sc_RecentListDashboards, 1, en_US]

Figure 4-23 Example of List of Recent Dashboards

4.3.8 Using HUD on All Monitors

When HUD is shown on one monitor and operator wants to view HUD on all monitors:

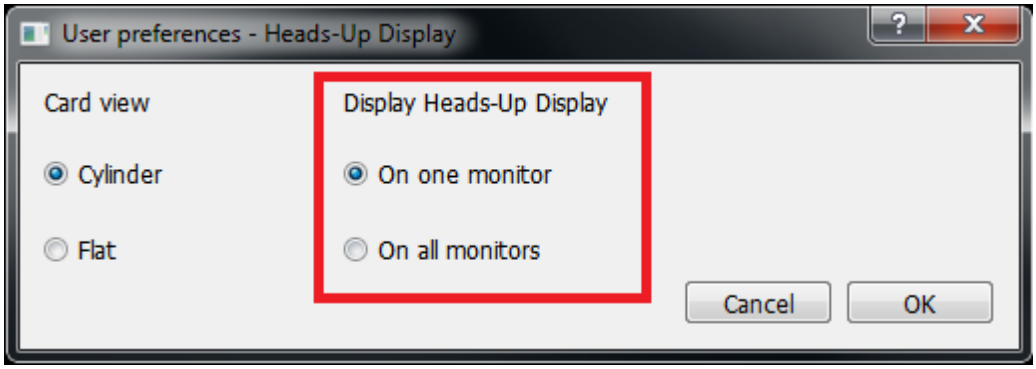
- Click on **Tools** in Navigation menu.
- Navigate and select **User preferences** in Tools menu.



[sc_HUD-Tools menu, 1, en_US]

Figure 4-24 Navigation Menu

- User preferences window provides option to open the HUD on one monitor or all monitors.
- Select **On all monitors** and click **OK**.



[sc_User Preference Monitor, 1, en_US, 1, en_US]

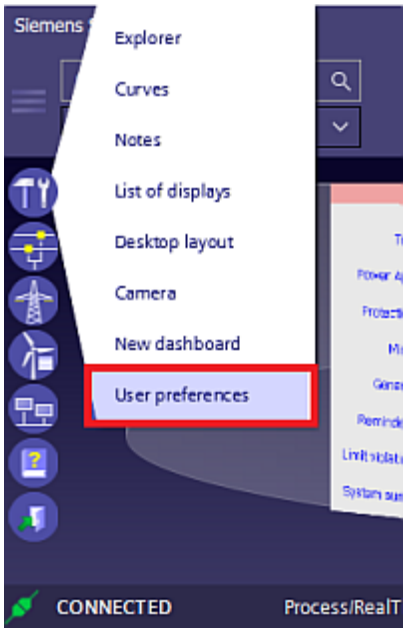
Figure 4-25 User Preferences Window

HUD is displayed on all monitors and all system wide information (for example, alarms, Gong off/on, gong temporary off, card contents) are synchronized.

4.3.9 Switching Between Cylinder View and Flat View of Cards

If an operator wants to switch between cylinder view and flat view of cards:

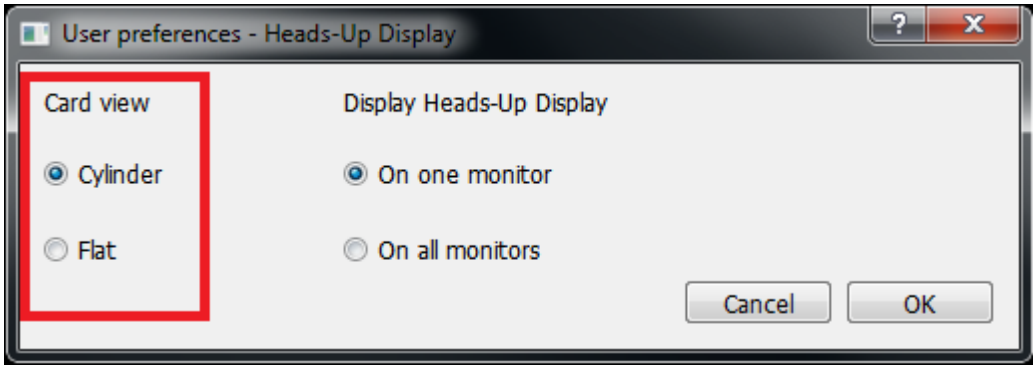
- Click on **Tools** in Navigation menu.
- Navigate and select **User preferences** in Tools menu.



[sc_HUD-Tools menu, 1, en_US]

Figure 4-26 Navigation Menu

- In User preferences window, select **Cylinder** or **Flat** based on preference.
- Click **OK** to load cards in cylinder or flat view.



[sc_User Preference View, 1, en_US, 1, en_US]

Figure 4-27 User Preferences Window

5 Direct Display Call-Up and Server Operating Modes

Connection Status: Shows if the system is connected or disconnected. The system automatically re-connects to another server.

Table 5-1 Server Operating Modes

	Training Mode
	Back-up Mode
	Test Mode
No Icon	Real-time Mode.

Server Operation Modes:

- Real-time (not indicated as this is the normal use scenario for operators)
- Training
- Back-up
- Test

Direct Display Callup

The Direct Display Call-Up (DDCU) entry box is located in the lower left section of the ODV that provides an input field for entering the view number of a display to call it up directly.



[sc_Display Callup ODV, 1, en_US]

Figure 5-1 Direct Display Callup

6 Console and User Management

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6.1 Introduction

The Console and User Management (CUMAN) display allows an authorized user to manage all console and user authorities in the system.

Click the corresponding button in the Basic Signalling Display to open CUMAN display. The Consoles tab of CUMAN is displayed.

Table 6-1 Console Function Authorities

Function Label	Description
Realtime	The user may start the BaSiDi.
Control	The user may control network elements using Supervisory Control functions.
Tag/Flag	The user may use tags on network elements.
ManUpdat	The user may update a network element's status using the Manual Updating function.
ValUpdat	The user may manually substitute an analog value.
Studies	The user may enter Study mode and use the Study Management display.
Training	The user may enter Training mode and perform OTS functions.
Admin	The user may change access rights.
LoadShed	The user may perform load shed functions.
OmsAdmin	The user has the rights of Outage Management administrator.
SPM App	The user may start the SPM (Switching procedure management) application.
SPM Auth	The user may authorize switching procedures.
SPM Crea	The user may create and modify switching procedures.
DMS DPF	The user may run DSPF and view the output data.
DMS FISR	The user may run FISR and view the output data.
DMS VVC	The user may run VVC and view the output data.
DMS FLOC	The user may run FLOC and view the output data.
DMS SCC	The user may run SCC and view the output data.
DMS OFR	The user may run OFR and view the output data.
DMS Opt	The user may change certain solution options in real-time mode. The user may change all solution options in study mode.
DMS Adva	The user may change advanced DNA solution options.
PW Auth	The user may set a Permit to Work tag.
Acknowl	The user may acknowledge pending alarms in summaries and network diagrams.
TNE	The user may set and remove temporary network elements.
INITSOS	The user may perform INITSOS in the system status display.
FailOver	The user may use the failover functionality of the system status display.
OMS AppR	The user is allowed to use Outage Management.
OMS AppW	The user is allowed to change Outage Management data.

Function Label	Description
CMS AppR	The user is allowed to use Crew Management.
CMS AppW	The user is allowed to change Crew Management data.
TCS AppR	The user is allowed to use Trouble Call Management.
TCS AppW	The user is allowed to change Trouble Call Management data.
ChgLimit	The user is allowed to change limits online.
ChgDeadB	The user is allowed to change dead bands.
ChgRedDS	The user is allowed to change redundant data sources.
UserTech	The user is allowed to change the assignment of technological areas to users.
UnderFrq	The user is allowed to use underfrequency load shed.
StormMod	The user is allowed to use Storm Mode.
SysCnfOM	The user is allowed to change Outage Management system configuration.
RulMgrOM	The user is allowed to use the Outage Management rule manager.
HIS Admi	The user has the rights of HIS administrator.
DelMsg	The user is allowed to delete messages.
SVRAuth	Authority for projection mapboard remote control.
CdlAuth	Common Data Link Authority
TimeAuth	The user may change the time source.
RtesdAut	The user may use the RTU equipment status display.
CurveAut	The user may use the curve tool.



NOTE

Authorities that are not mentioned in the table are not used.

6.2 User Interface

6.2.1 Title bar

Title bar icons are used to navigate to the previous and next display, refresh the display, print, and so on.

Table 6-2 Title Bar Icons

Icons	Description
	Opens the previous window.
	Opens the next window.
	Stops loading of data.
	Updates the table in the display.
	Export to Excel
	Zoom out
	Zoom in
	Reset zoom
	Online Help
	Displays general information about the display.

6.2.2 Selection Tabs

The following selection tabs are displayed:

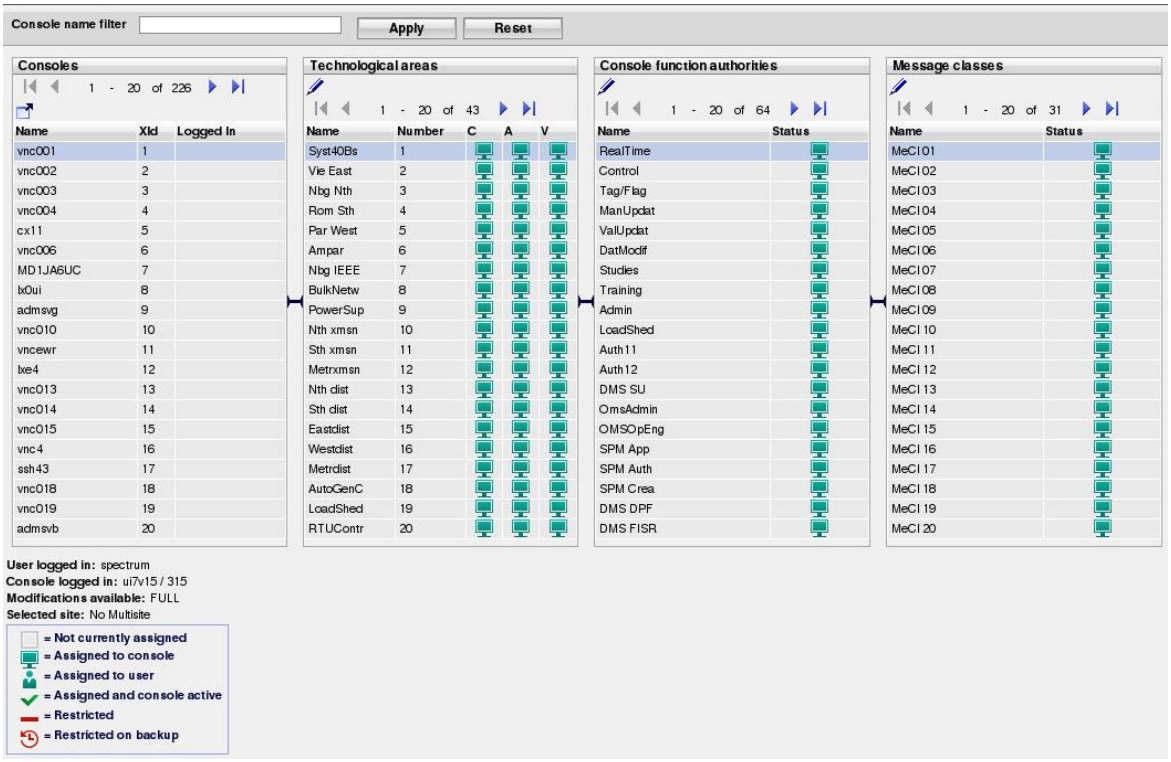
- Consoles
- Users
  - User Properties
  - Areas Of Control
  - Technological areas
  - Function Authorities
  - Message Classes
- Manage Areas of Control
- Assignment Control
- Blocked list
- Unassigned authorities
- Alarm Forwarding

6.2.3 Toolbar

Each tab contains one or more tables. Each table provides a toolbar, which can contain the following icons depending on the actions allowed for the table:

- 
  Switches the table to edit mode.
- 
  Buttons to scroll to the first, previous, next, and last page.
- 
  Opens the TA table in a new window.

6.2.4 Consoles



[sc_cuman_consoles, 4, en_US]

Figure 6-1 Console and User Management Consoles

This display has the following components:

- A filter line to select a particular console.
- Four tables:
  - **Consoles**

    Showing console name, XId, and (if the console is logged in) the user name under which the console is logged in.

Selecting an item in the Consoles table displays the data for the selected console in the other tables.

– **Technological areas**

This table has five columns. The first two columns identify Technological Areas by Name and Number. The next three columns indicate the selected console authorities for Control (C), Alarming (A), and Viewing (V).

The content of these three columns varies, depending on the authorities assigned to the console and if a user is logged in at the selected console, the authorities assigned to the logged-in user are as follows:

◊ indicates that the selected console does not have authority for the selected Technological Area. (And, if a user is currently logged-in at the console, the user does not have authority for the Technological Area.)
◊ indicates that the selected console has authority for the selected Technological Area.
◊ If a user is logged in at the selected console, indicates that the logged-in user has authority for the Technological Area but the Console does not.
◊ If a user is logged in at the selected console, indicates that the logged-in user and the selected console have the authority for the Technological Area.
◊ indicates that this authority is restricted for the selected site.
◊ indicates that this authority is restricted to backup for the selected site.
◊ indicates preferred authority or competence.
◊ indicates alternative authority or competence.

An authorized user can make the changes by clicking , followed by clicking the desired cell to toggle between the available values.

– **Console function authorities**

This table has two columns:

◊ A list of authority names.
◊ Indicates whether each authority is assigned to the selected console.

An authorized user can make the changes as described under **Technological areas**.

– **Message classes**

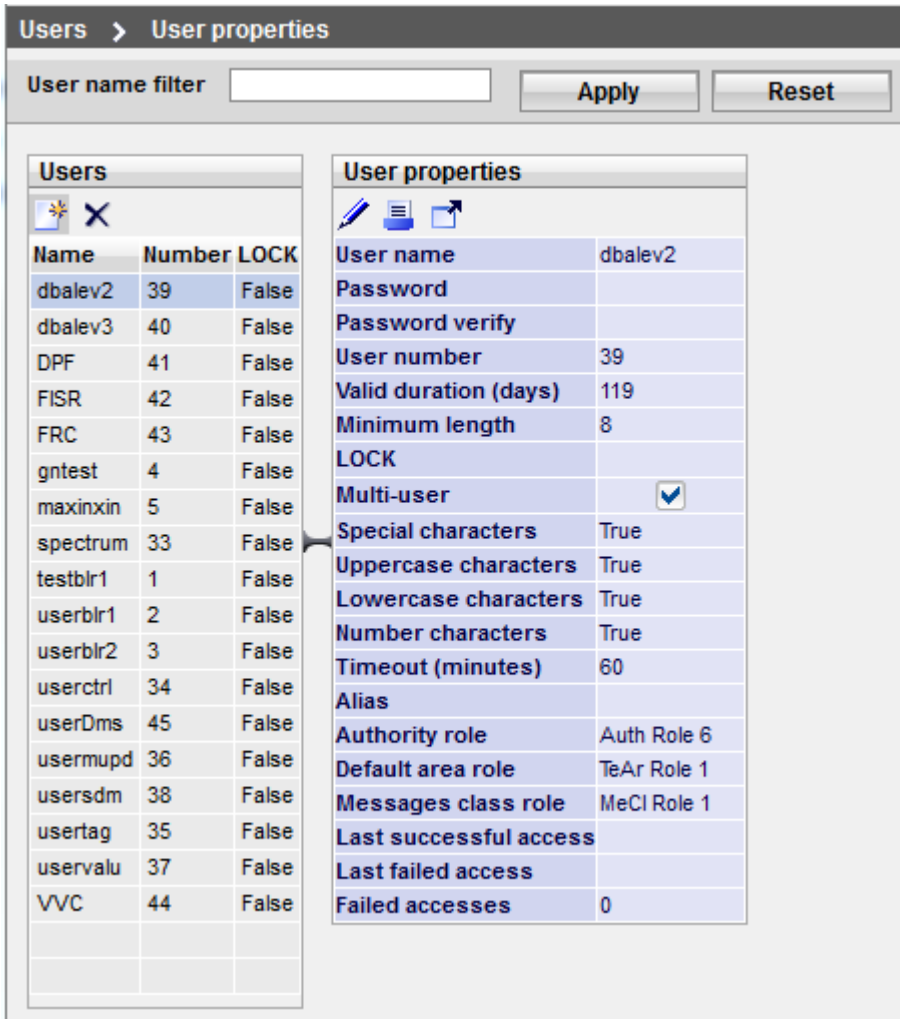
This table has two columns:

◊ A list of message classes.
◊ Indicates whether the selected console has authority for the message class.

An authorized user can make the changes as described under **Technological areas**.

6.2.5 Users

6.2.5.1 User Properties



[sc_CumanUserProperties_12032014, 4, en_US]

Figure 6-2 Cuman User Properties

This display has the following components:

- A filter line to select a certain user.
- A table showing the list of users with name, number, and information whether the user is locked. Selecting a user in this table causes detailed information for that user to be displayed in the user properties table.
- Icon 

  Click this icon to create a new Spectrum Power user. When the button is clicked, a pop-up window is displayed where the following information has to be entered:

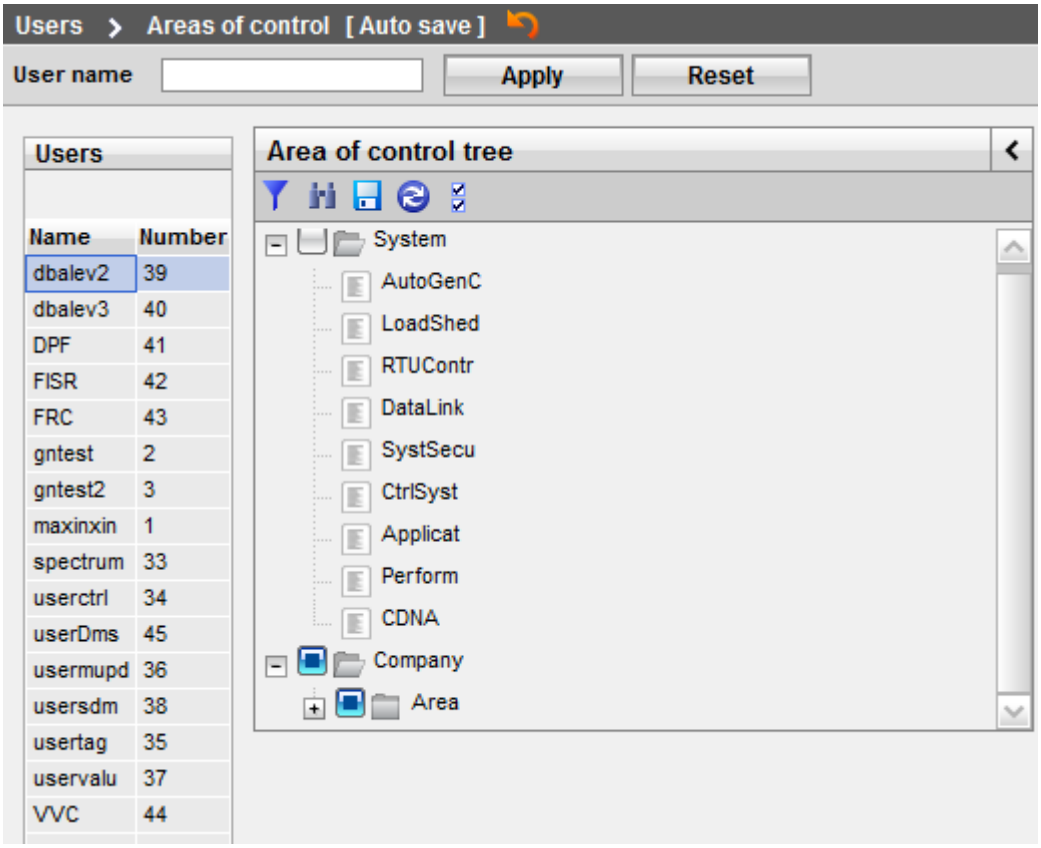
  – **User name**: This has to be unique.
  – **Password**: Initial Password.

- **User number**: Numeric user identifier. This has to be unique. If none is entered, a number is assigned automatically.
- **Valid Duration (days)**: Days before the user is required to change the password.
- **Minimum length**: Minimum number of characters in the password.
- **Lock**: This user is locked. (A user can also be locked due to successive failed login attempts.)
- **Multi-User**: If TRUE, this user name can be logged in simultaneously on multiple consoles.
- **Special characters**: If TRUE, any password has to contain at least one non-alphanumeric character.
- **Uppercase, lowercase, number character**: If TRUE, the password for this user name has to have both alphabetic and numeric characters.
- **Timeout**: Timeout for this user
- **Alias**: Alias name
- **Authority role**
- **Default area group**
- **Messages class group**

- A table showing the user properties.

6.2.5.2 Areas of Control

The Areas of Control display allows the user to transfer authority of all or parts of his Areas of Control.



[sc_AreaOfControl_12032014, 4, en_US]

Figure 6-3 Areas of Control Display

This display has the following components:

- A filter line that allows you to filter for a certain user.
- A table that lists all users.
- An Areas of Control tree that shows the assignment of areas of control to the user selected in the users table

The toolbar provides the following icons:

	Filter Tree
	Search
	Save Changes
	Refresh Tree
	Select All Points

Assigning/de-assigning Areas of Controls is possible by clicking the corresponding checkboxes.

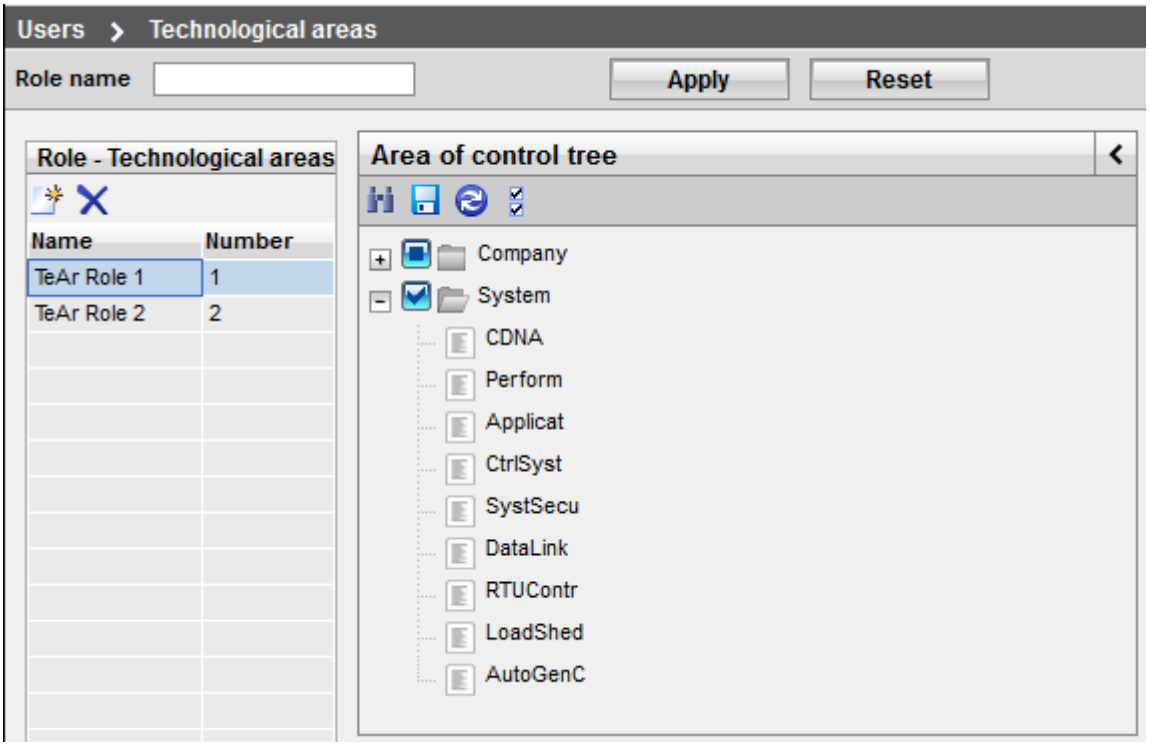


NOTE

A ⚠ beside a node in the tree indicates that the currently selected user is the only user with responsibility for the respective node, who is currently logged in.

6.2.5.3 Technological Areas

This display allows the definition and storage of default Area of control (AOC) assignments, which will be assigned to a user in the user properties display.



[sc_DefineAreaGroups_12042014, 6, en_US]

Figure 6-4 Technological Areas Display

This display has the following components:

- A filter line to select a Technological Area group.
- **New role** creates a new technological area role.
- **Role - Technological areas** displays the name and number of the defined Technological Area role.
- The Areas of Control tree on the right side to select the Areas of Controls that shall be assigned to the technological area group selected in the table on the left.

The toolbar provides the following icons:

	Filter Tree
	Search
	Save Changes
	Refresh Tree
	Select All Points

Assigning/de-assigning Areas of Controls is possible by clicking the corresponding checkboxes.

6.2.5.4 Function Authorities

This tab can be used to define, modify, or delete authority roles. These roles can then be assigned to a user or to a console.



[sc_CumanUserFunctionauthorites, 1, en_US]

Figure 6-5 Cuman Authorities

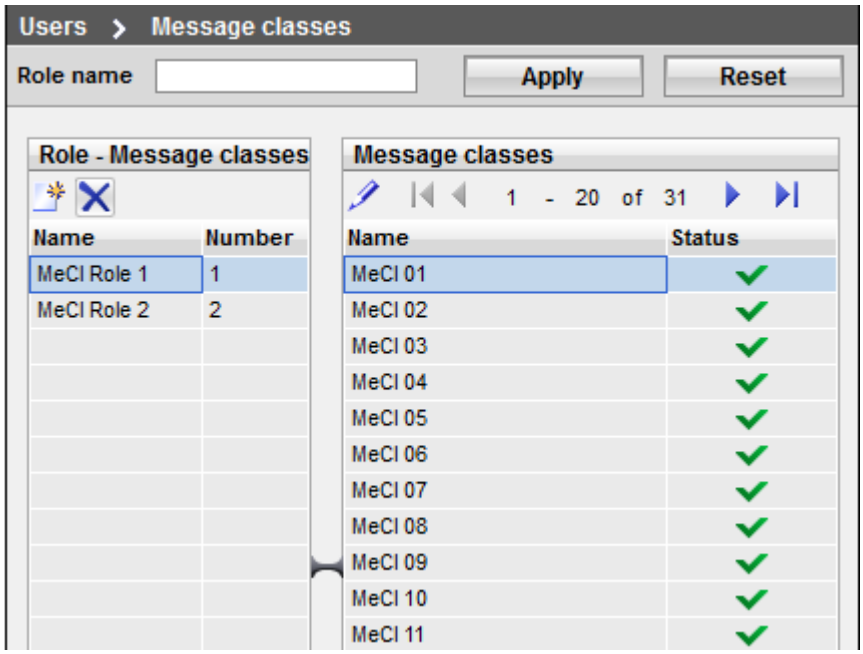
This display has the following components:

- A filter line to select an authority role.
- **Role - Authorities** table shows the name and number of the defined authority roles as well as the Heads-Up Display that will be shown when the user logs in.
  - The icon creates a new authority role.
  - The icon allows the user to edit the selected Heads-Up Display that will be shown when the user logs in.
  - The icon deletes the selected authority role.

- **Console function authorities** table shows name and status of the Console Function Authorities assigned to the authority role selected in the **Role - Authorities** table.

6.2.5.5 Message Classes

This tab can be used to define, modify, or delete message class roles. These roles can then be assigned to a user or to a console.



[sc_CumanMessageClasses_12032014, 5, en_US]

Figure 6-6 Cuman - Message Classes

This display has the following components:

- A filter line to select a message class group.
- **Group - Message classes** table shows the name and number of the defined message class group.
  - The Icon creates a new message class group.
- **Messages classes** table shows name and status of the Messages Classes assigned to the message class group selected in the **Group - Message classes** table.

6.2.6 Crew Members

This tab is used to enter crew member names for use by manual update.

NOTE

If Outage Management is part of the system, the Crew members tab is read-only. In this case crew members can only be added and their data be modified using the Crew Management of Outage Management. The Outage Management data is synchronized with SCADA and the crew members entered using Outage Management are displayed in this tab. For detailed information on Crew Management, refer to the Crew Management User Guide.



[sc_cuman-crewmembers, 2, en_US]

Figure 6-7 Crew Members

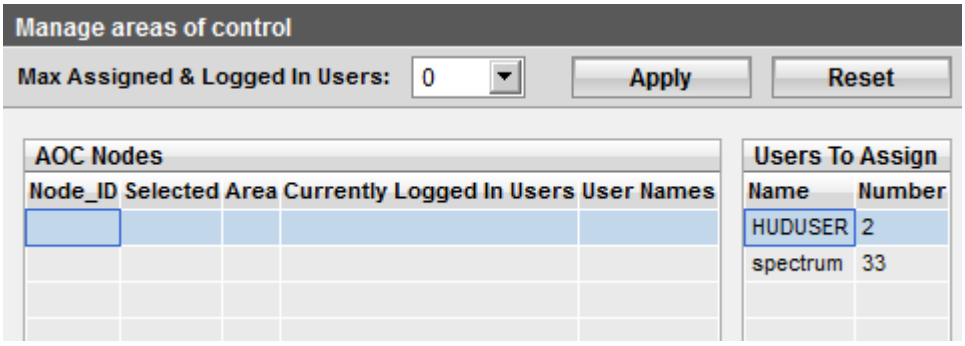
This display shows a table with all currently defined crew members.

- Last name, first name: Name of the crew member.
- Area group: area group (as defined in the Define Area Groups tab) to which the crew member is assigned.

  The user can add, edit of delete crew members.

6.2.7 Manage Areas of Control

This display provides an overview of Areas of Control assignments. It refers only to the currently logged in users. It displays only the last Areas of Control nodes (that is, nodes whose immediate children are technological areas). Additionally the operator can assign areas of control to a user.



[sc_ManageAreasOfControl_12032014, 5, en_US]

Figure 6-8 Manage Areas of Control Display

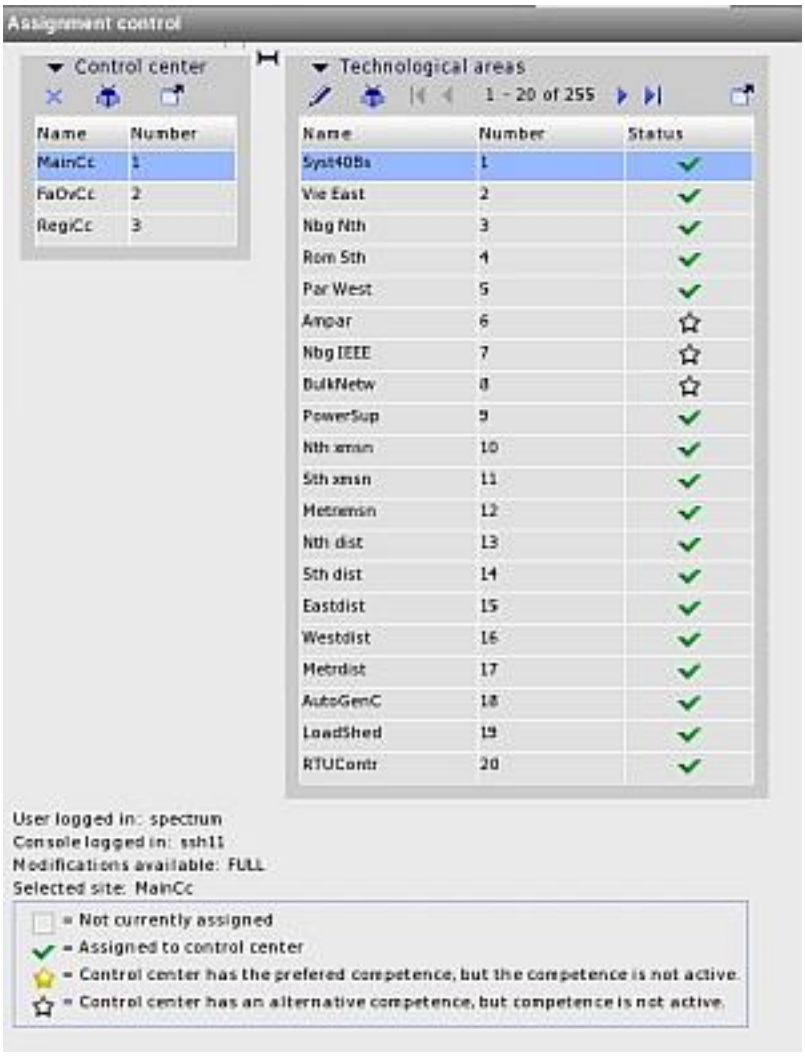
This display has the following components:

- A filter line where you can select which Areas of Controls will be displayed in the table, based on the number of assignments: setting the number of assignments of the Areas of Controls to be displayed to 0

means to display those nodes that are currently not assigned, <=1 means all Areas of Controls that are currently not assigned or assigned to one user, and so on.

- A table with all Areas of Control nodes conforming to the filter criteria; the right-most column “User Names” shows all users to whom the Areas of Control is currently assigned, separated by a comma.
- A table with all currently logged in users.

6.2.8 Assignment Control



[sc_CumanAssignmentControl_12032014, 3, en_US]

Figure 6-9 Cuman Assignment Control

This tab shows a list of all control centers in the Multisite network. For each control center the Technological Area assignments are displayed:

- **Control center** table
- **Technological areas** table shows a list of all Technological Areas with name and number as well as whether they are assigned to the selected control center.

6.2.9 Blocked List



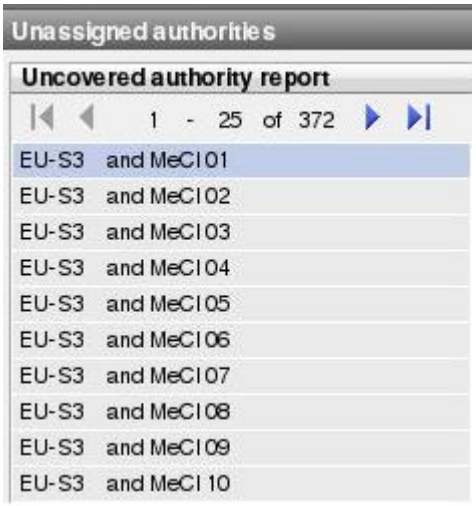
[sc_CumanBlockedList_12032014, 2, en_US]

Figure 6-10 Cuman Blocked List

This tab manages the list of words that the system does not permit to be used as passwords.

The display contains a single table with a list of blocked passwords.

6.2.10 Unassigned Authorities



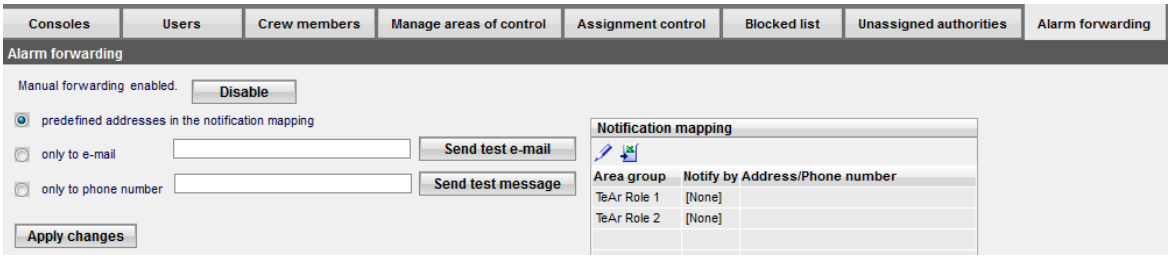
[sc_CumanUnassignedAuthorities_12032014, 3, en_US]

Figure 6-11 Cuman Unassigned Authorities

This tab provides a list of all technological areas and messages classes that are not assigned to one of the currently logged in consoles.

6.2.11 Alarm Forwarding

This display is used to enable or disable alarms and configure Alarm Forwarding.



[sc_cuman-alarmforwarding, 5, en_US]

Figure 6-12 Alarm Forwarding

- **Disable/Enable:**

  Disables or enables alarm forwarding.

  Depending on the current status, either **Disable** (when alarm forwarding is currently enabled) or **Enable** (when alarm forwarding is currently disabled) is shown.

  When alarm forwarding is enabled, the following options can be defined:

- **Predefined addresses in the notification mapping:**

  The messages are sent to the addresses/phone numbers as defined in the **Notification mapping** table.

- **Only to e-mail:**.

  When selecting this option, the messages are only sent to the address entered into the field beside the radio button. Clicking button **Send test e-mail** sends a test e-mail to the entered address to check whether it is working correctly.

- **Only to phone number:**

  When selecting this option, the messages are only sent to the telephone number entered into the field beside the radio button. Clicking button **Send test message** sends a test message to the entered address to check whether it is working correctly.

- **Apply changes:**

  Clicking this button saves the changes

Notification mapping

This table shows which messages (depending on the area group) shall be sent to which mail address/telephone number.

Click to change the table to edit mode. Now the fields **Notify by** and **Address/phone number** can be modified.

Click to export the table to an Excel file.

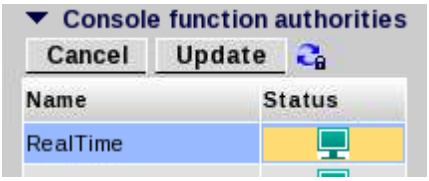
6.3 Operational Steps

6.3.1 Modifying Authorities

To modify authorities, perform the following:

- In the **Authorities** tab, select an authority role in the table Role – Authorities.
- Click in the Authorities table.
- To assign an authority, click the corresponding field in the Status column.

  The field shows , indicating that the authority is assigned to the user.

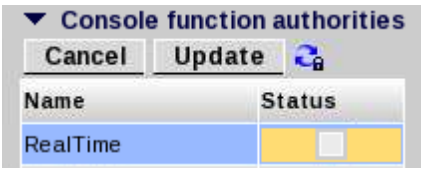


[sc_AssigningAuthorities_12032014, 1, en_US]

Figure 6-13 Assigning Authorities

- To remove the authority assignment, click the corresponding field in the column Status again.

  is removed from the field.



[sc_RemoveAuthority_12032014, 1, en_US]

Figure 6-14 Remove Authority

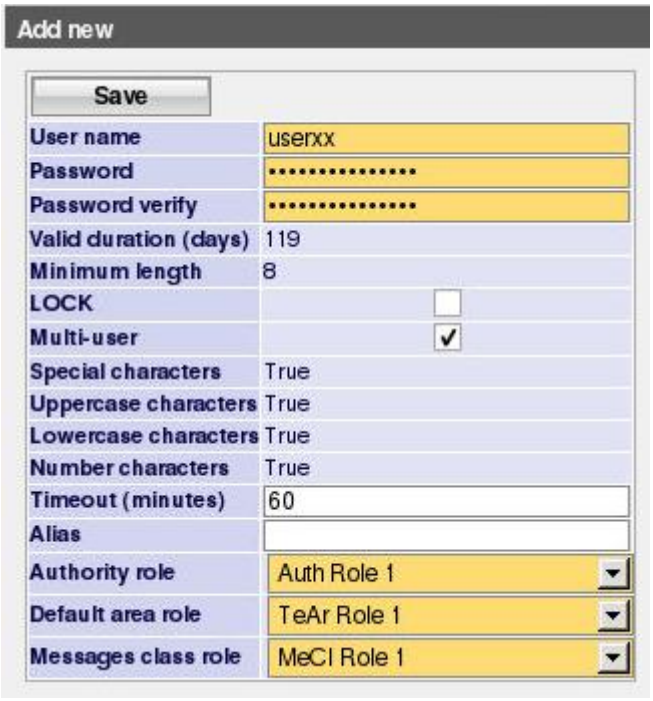
- Click **Update** to save the changes.

6.3.2 Adding a User

To add a user, perform the following:

- In the Users tab, click .

  Enter the user parameters in the display that opens.



[sc_CumanNewUser_12032014, 4, en_US]

Figure 6-15 Cuman - New User

- Click **Save**.

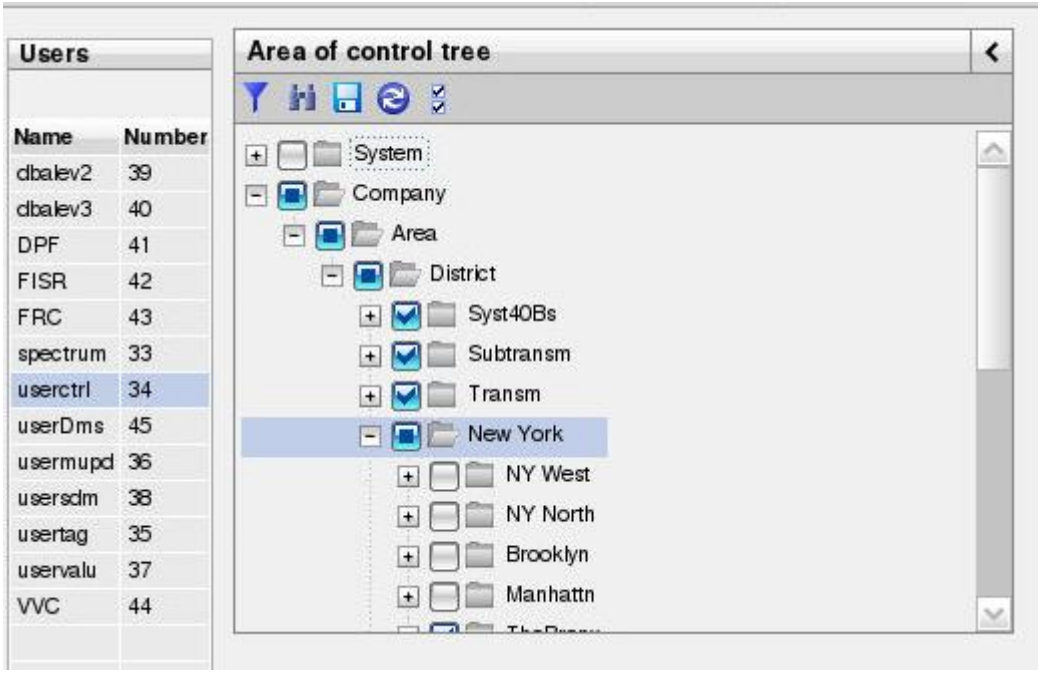
  The new user is displayed in the Users table.

6.3.3 Assign Areas of Control to a User

To assign an area of control to a user, perform the following steps:

- In the **Users > Areas of control** tab, select the user to whom you want to assign areas of control in the table at the left side.

  The tree at the right side shows the areas of control currently assigned to the user.



[sc_Area of control tree, 1, en_US]

Figure 6-16 Area of Control Tree

indicates that some (but not all) of the child areas of the area are already assigned to the user.

indicates that this area of control (and all its child areas) are assigned to the user.

indicates that this area is currently not assigned to the user.

- Navigate to the area you want to assign to the user.
- Click a icon to assign the area to the user.

  A hook is set left of the area name

  NY North
- Click to save the changes.

  The change is saved and the area is assigned to the user.

6.3.4 Defining a Technological Area Role

For managing areas in an efficient way, it is possible to define technological area roles consisting of multiple areas. To define a technological area role, perform the following:

- In the **Users > Technological Areas** tab, click the icon .

  A display for entering a name for the new technological area role is displayed.

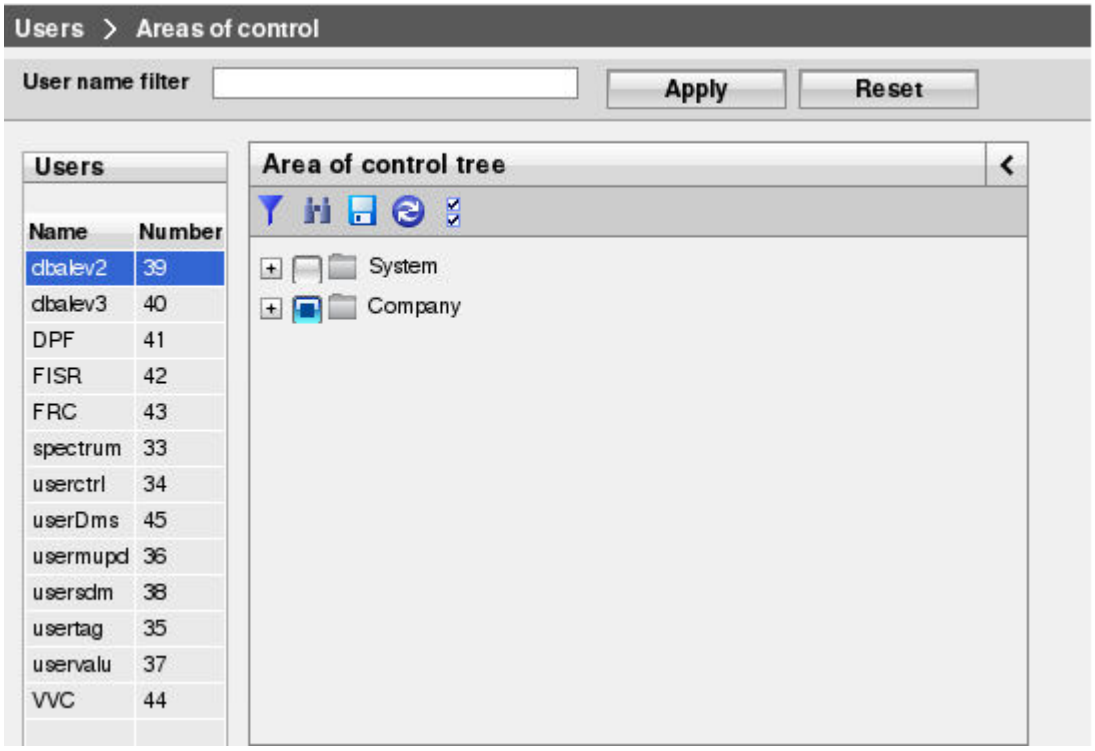


[sc_UserNewAreaGroup1_12032014, 4, en_US]

Figure 6-17 Add New Technological Area Role

- Enter a name and optionally, a comment, and click **Save**.

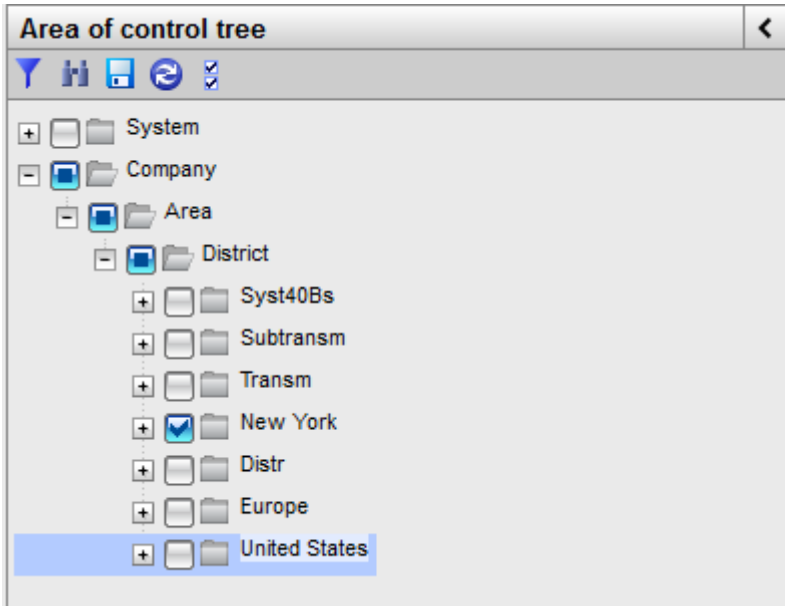
  The new technological area role is saved and shown in the table of technological area roles.



[sc_UserNewAreaGroup2_12032014, 4, en_US]

Figure 6-18 Technology Area Roles

- Technological area roles are used to manage user rights based on defined areas. For example, regions or districts. Each area contains one or multiple equipments. To assign a technological area role to the area group, click the corresponding item in the detail tree.



[sc_UserNewAreaGroup3_12032014, 2, en_US]

Figure 6-19 Area of Control Tree

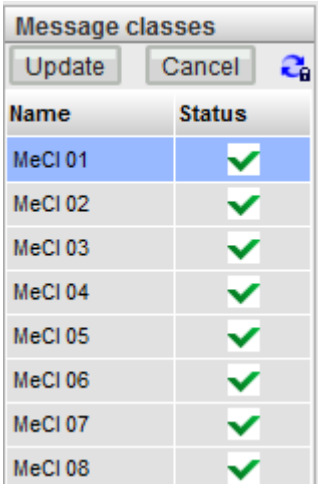
- Click the icon to save the changes.

6.3.5 Modifying Message Classes

Messages can be grouped into a message class. To modify the message class, perform the following:

- In the **Message classes** tab, select a message class group from the Role - Message classes table.
- Click in the **Message classes** table.
- To assign a message class, click the corresponding field in the Status column.

  indicates that the message class is assigned to the user.

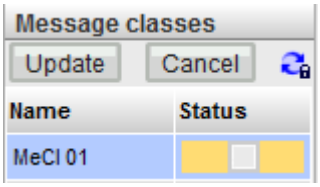


[sc_Messageclasses_Modify, 2, en_US]

Figure 6-20 Cuman - Assign Message Class

- To remove the message class assignment, click the corresponding field in the Status column again.

  is removed from the field.



[sc_RemoveMessageclass, 1, en_US]

Figure 6-21 Cuman - Remove Message Class

- Click **Update** to save the changes.

6.3.6 Defining a New Crew Member

When manually updating the status of a network element, the operator can select the name of the crew member who performed the operation. To define a new crew member, perform the following steps:

NOTE

If Outage Management control is part of the system, crew members can only be added and their data be modified using the Crew Management of Outage Management control. For detailed information on Crew Management, refer to the Crew Management User Guide

- Select the Crew members tab.

  The Crew members tab showing all currently defined crew members is displayed.

  Click the icon and then click **New Row**.

- Enter the last name and first name of the new crew member and select the area group to which the crew member shall be assigned.



[sc_cuman-define-crewmembers1, 2, en_US]

Figure 6-22 Add New Crew Member

- Click **Update**.

  The new crew member is displayed in the table.



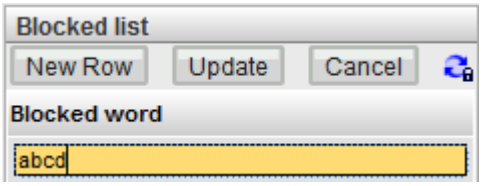
[sc_cuman-define-crewmembers2, 1, en_US]

Figure 6-23 New Crew Member

6.3.7 Adding or Deleting Blocked Passwords

This section applies to users setting their own password. To add or delete blocked passwords, perform the following:

- Select the **Blocked list** tab.
- To add a blocked password, click in the table.
- Click **New Row**.
- Enter the word to be blocked.



[sc_blockedlist, 1, en_US]

Figure 6-24 Blocked List

- Click **Update**.

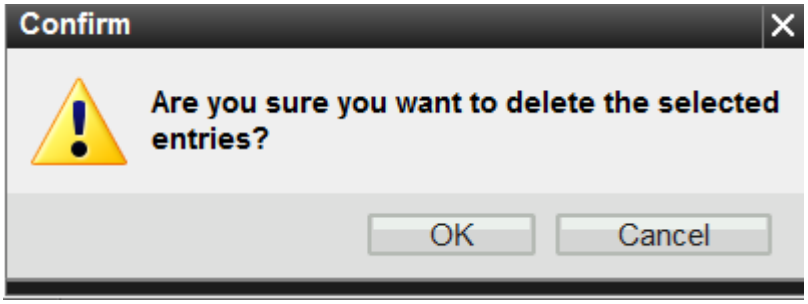
  The new word is entered into the table.



[sc_blockedword, 1, en_US]

Figure 6-25 Blocked List - Blocked Word

- To delete a word from the blocked list, select the word and click in the toolbar.
- Click **Ok** to confirm deletion.



[Sc_deleteconfirmation, 1, en_US]

The corresponding blocked passwords are deleted.

6.3.8 Changing Alarm Forwarding Parameters

To enable or disable alarm forwarding, perform the following steps:

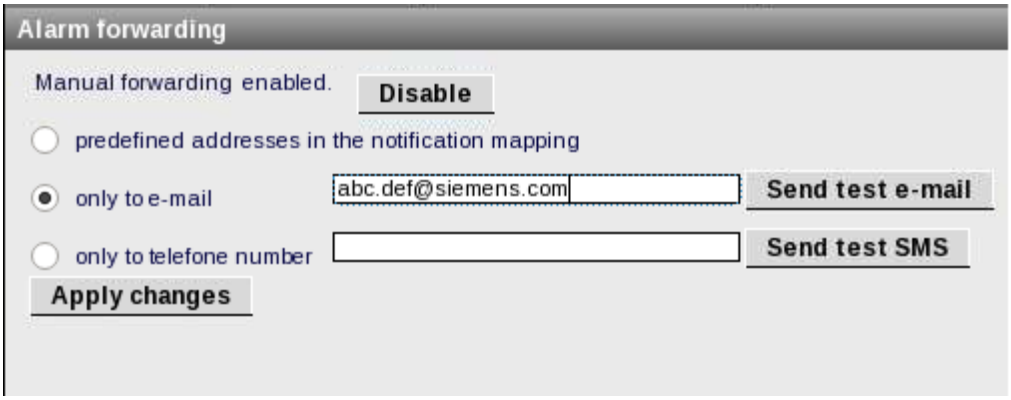
- Select the **Alarm Forwarding** tab.
- If the button **Disable** is shown, alarm forwarding is currently enabled. Click this button

  If alarm forwarding is disabled, the button is now labelled **Enable**.
- Click **Enable** to enable Alarm forwarding again.

  If alarm forwarding is enabled, the button is labelled **Disable** again

To send all alarms to a certain e-mail address:

- Select the **only to e-mail** radio button and enter the desired e-mail address



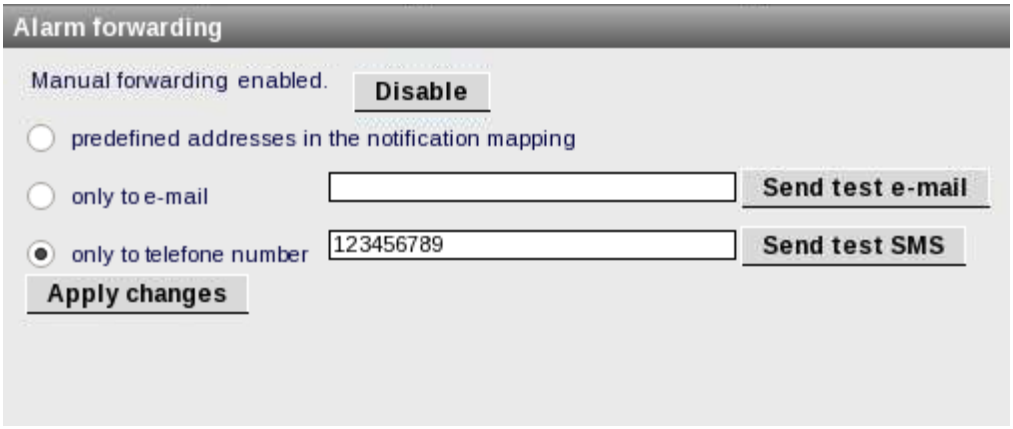
[sc_cuman-onlytoemail, 2, en_US]

Figure 6-26 Alarm Forwarding to E-mail

- Click the button **Send test e-mail** to check whether the address is correct.
- Click **Apply changes** to save the changes.

  All alarms are forwarded to the entered e-mail address.

Sending all alarms to a telephone number is achieved by selecting **only to telephone number** radiobutton, and then proceed as described above for forwarding to e-mail address.

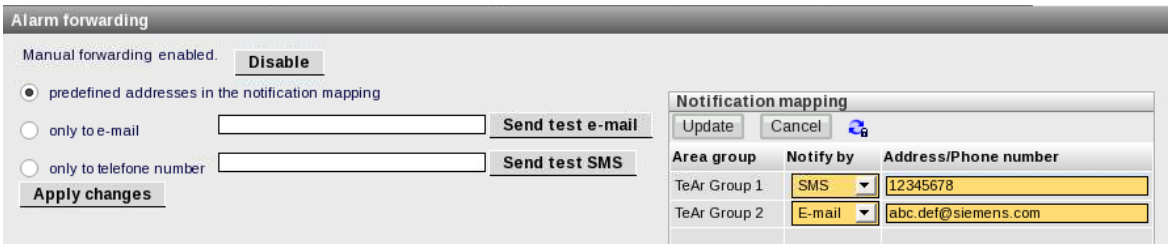


[sc_cuman-onlytophone, 2, en_US]

Figure 6-27 Alarm Forwarding to Phone

To send alarms to different e-mail addresses, depending on the concerned technological area:

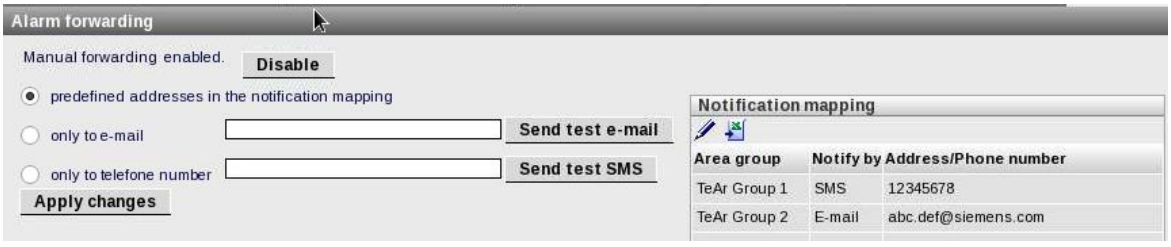
- Select the radiobutton predefined addresses in the notification mapping, if not already selected.
- Click on the top of the **Notification mapping** table.
- The table switches to edit mode.
- For each technological area, select SMS or E-mail in the field **Notify by** and enter an address or phone number.



[sc_cuman-editnotificationmapping, 3, en_US]

Figure 6-28 Alarm Forwarding to a Particular Technological Area

- Click **update** to save the changes



[sc_cuman-notificationmapping-edited, 3, en_US]

Figure 6-29 Alarm Forwarding - Saving Changes

6.3.9 Configuring HUD Layout for Different Authority Roles

The system administrator can define different HUD layouts for different user/authority roles

Create new HUD Layout

To create new HUD layout:

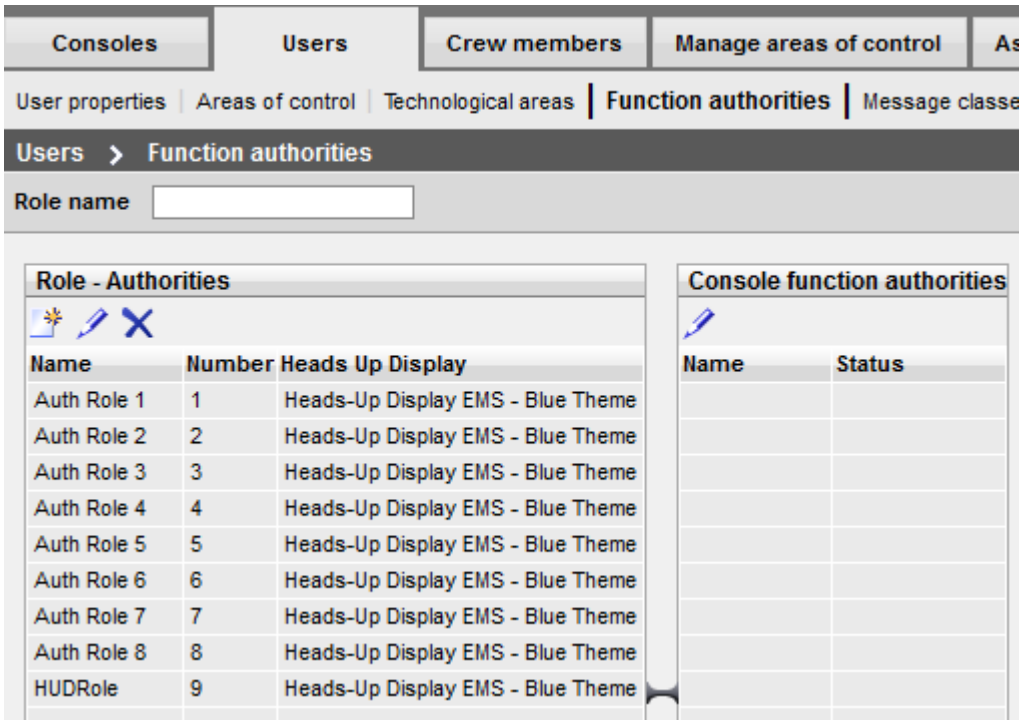
- Assign unique view number for the layout.
- Create a set of QML files.
- Set up configuration for the layout.

Setting up a new HUD layout makes it available in CUMAN.

Configuring HUD for an Authority Role

To assign already created HUD layout to an authority role:

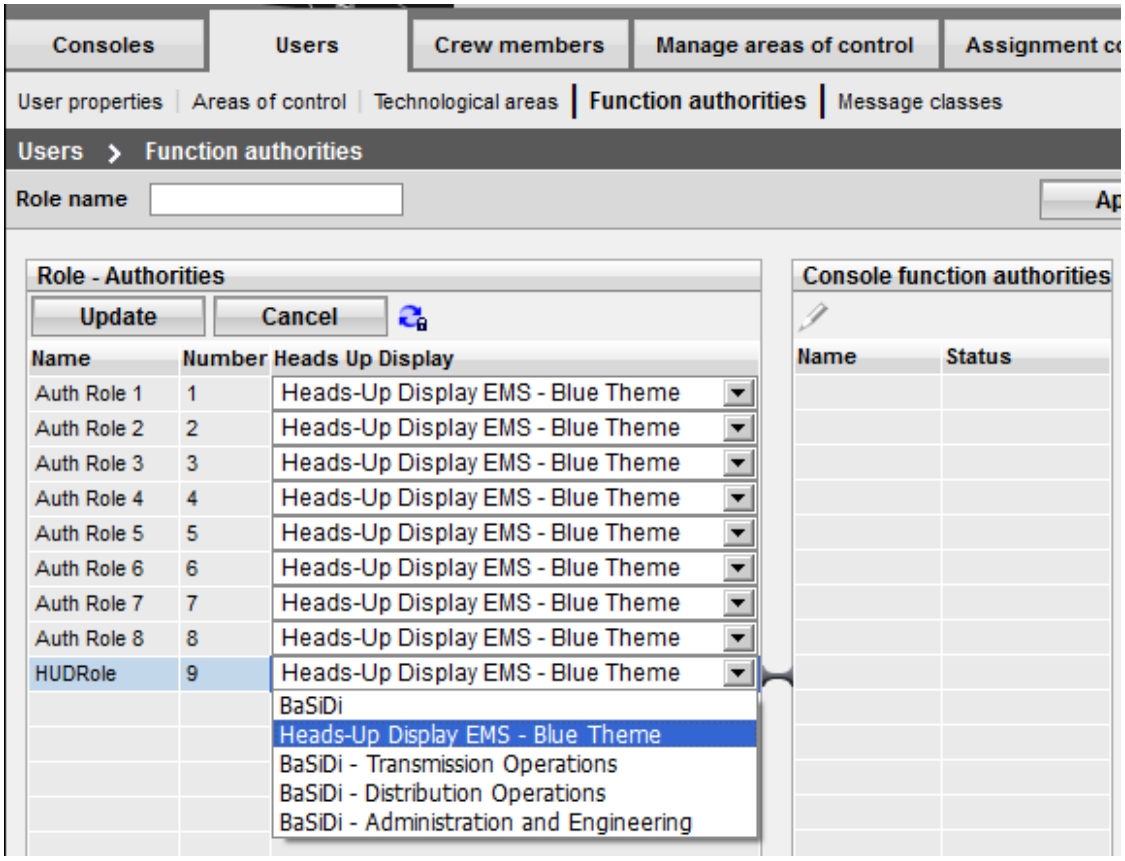
- Open CUMAN and select **Users** > **Function authorities** tab.



[sc_CUMAN Display, 1, en_US]

Figure 6-30 CUMAN Display

- Select Authority Role. This will show currently assigned HUD layout and list of available HUD.
- Choose one already configured layout.
- Click **Update**. System saves the chosen layout.



[sc_CUMAN Display 2, 1, en_US]

Figure 6-31 CUMAN Display

When a user login into the system, the configured HUD layout for current role shows up without additional selection steps, and HUD reflects current user rights.

7 Online Diagram Viewer

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7.1 Introduction

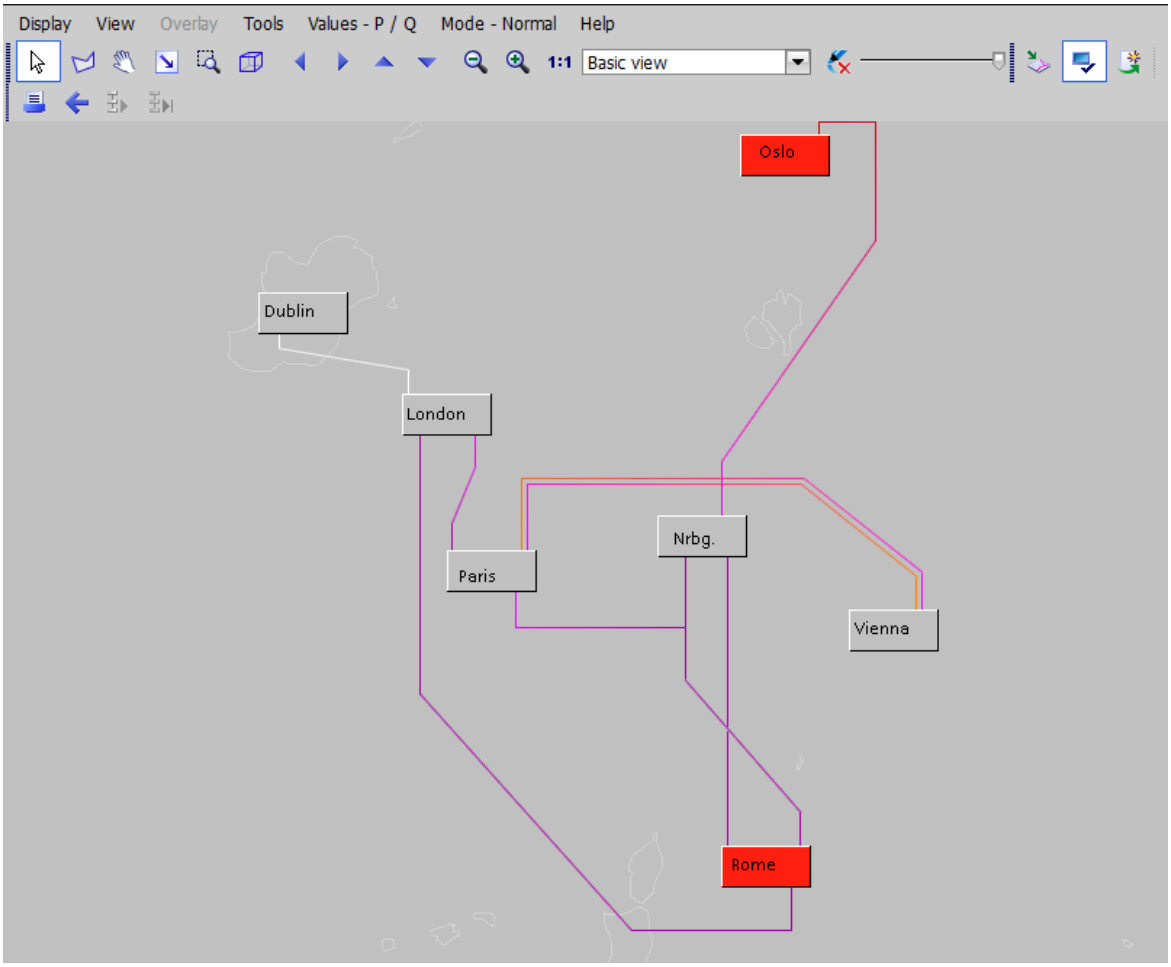
The Online Diagram Viewer (ODV) displays schematic network diagrams and geo-spatial network diagrams. The interaction and visualization capabilities are consistent for both types of diagrams.

Geo-spatial network diagram: The coordinates of graphical elements have a relation to the real world. The user can view the geographic position of equipments or outages and so on. Furthermore, the user can view streets, customers, land usage or satellite images.

Schematic network diagram: Is a simplified notation for visualizing an electrical power system. Elements on the diagram do not represent the physical size or location of the electrical equipment. The display is optimized to provide the user a good overview of the network topology.

7.2 User Interface

7.2.1 Overview



[sc_SingleLineDiagram_12032014, 5, en_US]

Figure 7-1 Network Diagram

7.2.2 Menu Bar

Menu bar has the following options:

- **Display**

  A menu to activate display related functions, for example, printing the display, and so on.

- **View**

  The following options can be selected from the view window:

  – Always on Top: Display remains on top
  – Navigator: Opens Navigator window that gives a minimized version of the window

- Animation: Animation can be turned on or off. For more information refer 7.3.16 Using Display Profiles for Enhanced Situation Awareness
- Display flow direction: Defines how values for flow directions are represented in the network diagram. The values can be represented by arrow directions, - or + signs, or colors.
- Display TA: Defines how TA names are represented in the network diagram and curve tool. The operator can choose to display names, text (long name) or ID (numeric value).

  Note: This setting applies to all currently open ODV and curve tool windows and is not saved with desktop layout management to prevent confusion.

- **Overlay**

  A menu that lists the valid overlay planes and their visibility states of the network diagram for the current magnification level. The Overlay Control Menu allows you to toggle the visibility status of an overlay plane. If an overlay plane is selected (made visible), a highlighted check box is displayed at the right side of the plane name in the Overlay menu.

- **Tools**

  A menu that provides various tools.

  For example, showing display information, making a snapshot of the display, and so on.

- **Values**

  A menu that allows you to select various analog value pairs for representation.

  For example, P/Q, P/Q estimated, I/I%, I/I% estimated, and so on.

- **Modes**

  A menu that allows you to preselect an application that processes subsequent network element selections.

  At the right side of the menu name, the currently selected mode is indicated. The following menu items are provided:

  - **Mode: Normal**

    This is the default mode.

  - **Mode: Trace**

    Switches the network diagram to trace mode.

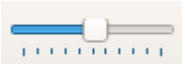
  - **Mode: TNE**

    Allows you to insert temporary network elements.

7.2.3 Upper Toolbar

The toolbar is shown by default on top. You can change the position of the toolbar by dragging and dropping the toolbar on the handle shown at the left side. You can also arrange multiple toolbars within one row.

Table 7-1 Toolbar Icons

Icon	Description
	Pointer mode
	Polygon selection mode
	Middle button pan mode
	Middle button rubberband pan
	Middle button rubberband zoom
	3D navigation
	Step zoom out
	Step zoom in
	Default view
	Allows the user to select which advanced visualization is displayed. Refer to the corresponding chapters display profile for additional details.
	Clears the highlighting of elements (for example, due to tracing)
	Slider to change the background opacity of the network diagram.
	View Displays and highlights the currently selected TAs in the ODV set as the Master ODV (refer to next icon)
	Master Sets the ODV window as the master ODV window for view actions. There can only be one master window.



NOTE

The user can drag and drop a TA or element (like a crew or outage) onto an ODV display for viewing the location.

7.2.4 Lower Toolbar

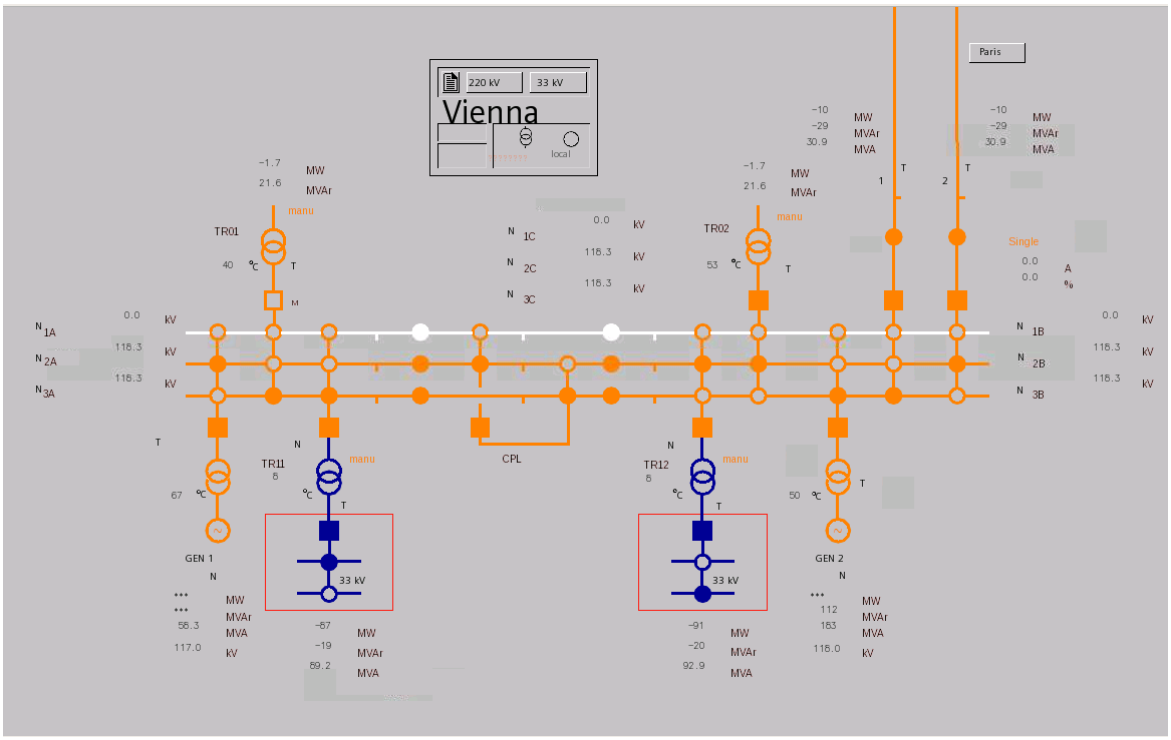
One or more rows of buttons that allow you to activate frequently used facilities of Spectrum Power network diagrams.

The toolbar content and the accessibility of the provided toolbar buttons depend on the following:

- User authorities
- Selected network diagram plane

- Current situation

7.2.5 Display Area



[sc_Single-lineDiagramDisplayArea_12032014, 2, en_US]

Figure 7-2 Network Diagram - Display Area

The display area usually contains a network diagram, which can be zoomed-in.

For example, network overview map, substation diagram, a network diagram of the computer system, and so on.

Network diagrams (for example, network overview map and substation diagrams) support the visualization of processes in your power network by dynamically changing the representation of the shown network equipment. Various equipment-specific, dynamic characteristics which are of interest for the operation of your network are dynamically changed to reflect the status of the power system.

General Information



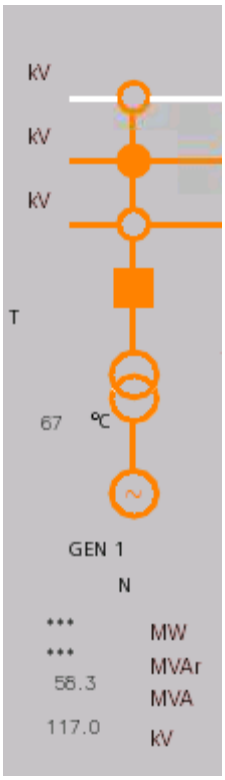
[sc_SingleLineDiagramGeneralInformation_12032014, 1, en_US]

Figure 7-3 Network Diagram - General Information

This area exists on a substation diagram. General information consists of the following:

- Button to invoke the summary of the alarms and messages of this substation.
- Buttons to invoke the other substation diagrams.
- Name of the substation.
- Status of the RTU.
- Status of the connection.

Equipment Information



[sc_SingleLineDiagramEquipmentInformation_12032014, 2, en_US]

Figure 7-4 Network Diagram - Equipment Information

The status information of following equipments is displayed:

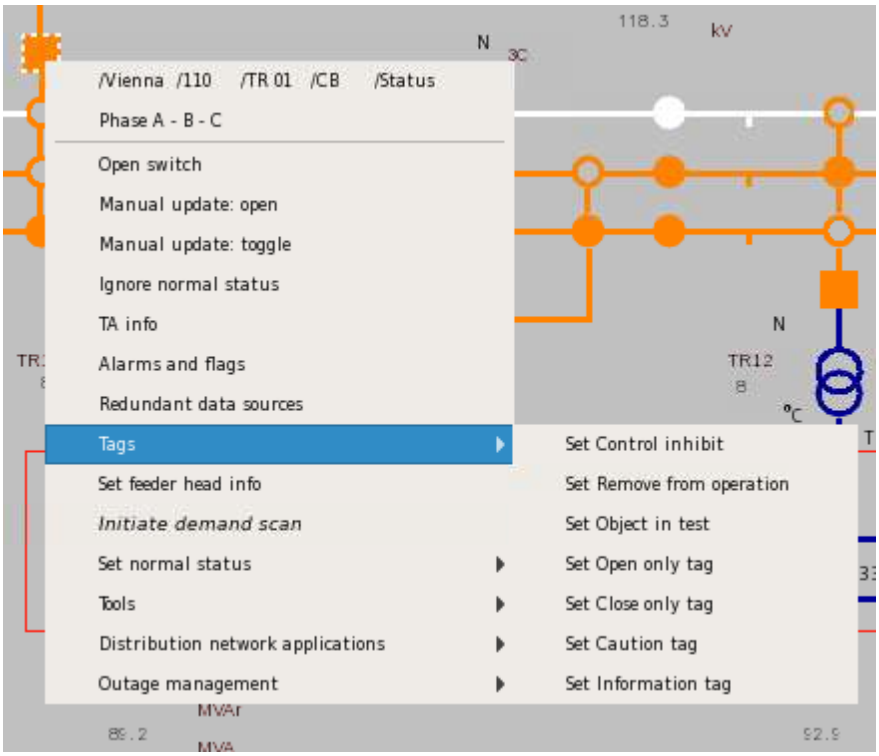
- Switches
- Switchbays
- Network Components (for example, line, busbar, and so on.)
- Transformer Tap Changers
- Transformer Parallel Operation Devices
- Parallel Lines
- Connections

Context Menu

The Context Menu allows the operator to initiate operations on the selected network element.

To invoke the Context Menu on a network diagram, right-click the selected network element.

On the header of a Context Menu the Technological Address (TA) of the selected network element are displayed.



[sc_odv-contextmenu, 2, en_US]

7.2.6 Status Bar

The status bar provides the following information:

Connection status

Server Operation Modes

Table 7-2 Server Operating Modes

	Training Mode
	Back-up Mode
	Test Mode
No Icon	Real-time Mode.

Username: The currently logged in user. Username is linked to the CUMAN user Refer to 6.2.5 Users for more details.

Direct Display Call-up: This field allows you to enter a view name or view number, and the corresponding display is opened.

7.3 Operational Steps

7.3.1 Panning

To pan, perform the following:

- Click in the toolbar of the network diagram to activate panning.
- Move the cursor over the display area of the network diagram and press the left button of the mouse.
- Without releasing the left button of the mouse, drag the cursor in the required direction.

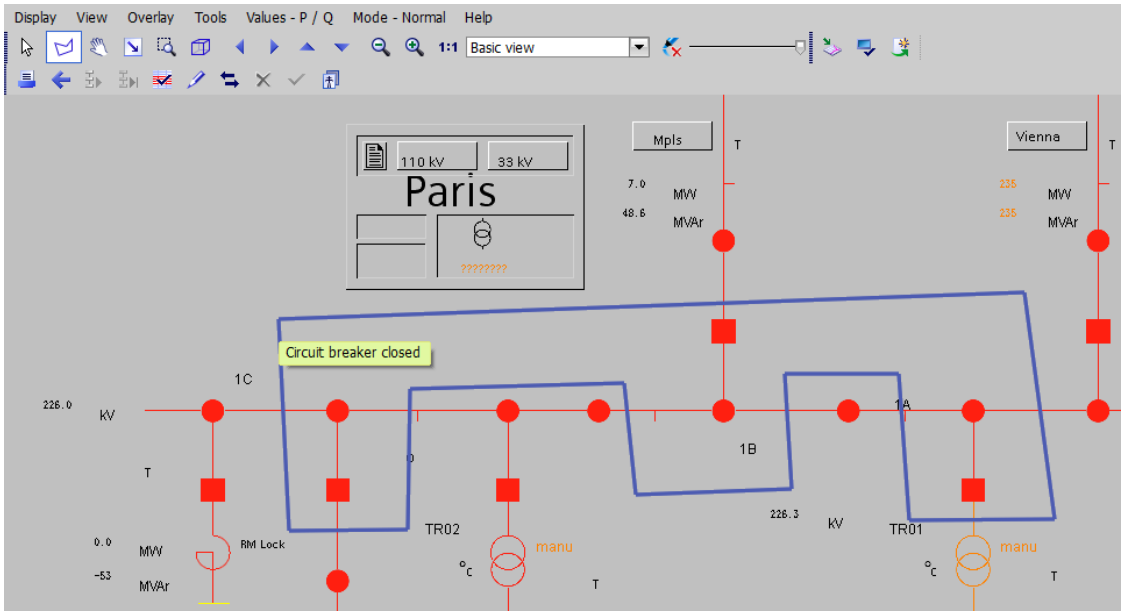
  The network diagram moves through the display area of the network diagram accordingly.
- After the required network diagram portion is visible in the display area of the network diagram, release the wheel button of the mouse.

7.3.2 Polygon Selection Mode

To select multiple TAs using polygon selection:

- Open ODV network diagram.
- Select polygon selection mode icon from the toolbar. The polygon selection mode is activated. Once the polygon selection mode is active, the operator can select vertices of the polygon by clicking over the worldmap.
- Define successive polygon edges by clicking on each additional vertex positions.

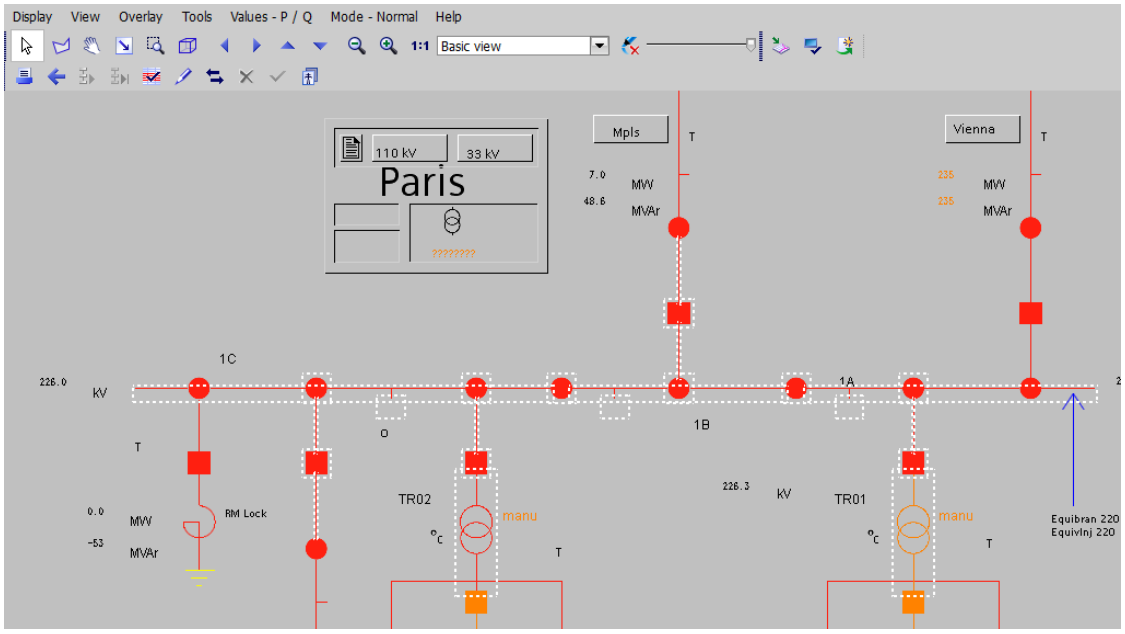
  The operator can view sample line from previous clicked vertex to current cursor position. Polygon shape supports upto a max of 100 vertices.



[sc_Defining Polygon Edges, 1, en_US]

Figure 7-5 Defining Polygon Edges

- To complete the polygon, click on the first vertex of the selection or double click anywhere on the network diagram. Once the polygon creation is completed, the operator can view the selected TAs. When operator has completed a polygon selection, the system switches back to pointer mode automatically, keeping the selected items.



[sc_Selected TAs, 1, en_US]

Figure 7-6 Selected TAs

The operator can keep previous selection alive along with the current selection by pressing shift before starting polygon construction, and during construction of the current polygon.

In case of any correction, current selection has to be cancelled and the user has to again start drawing polygon.

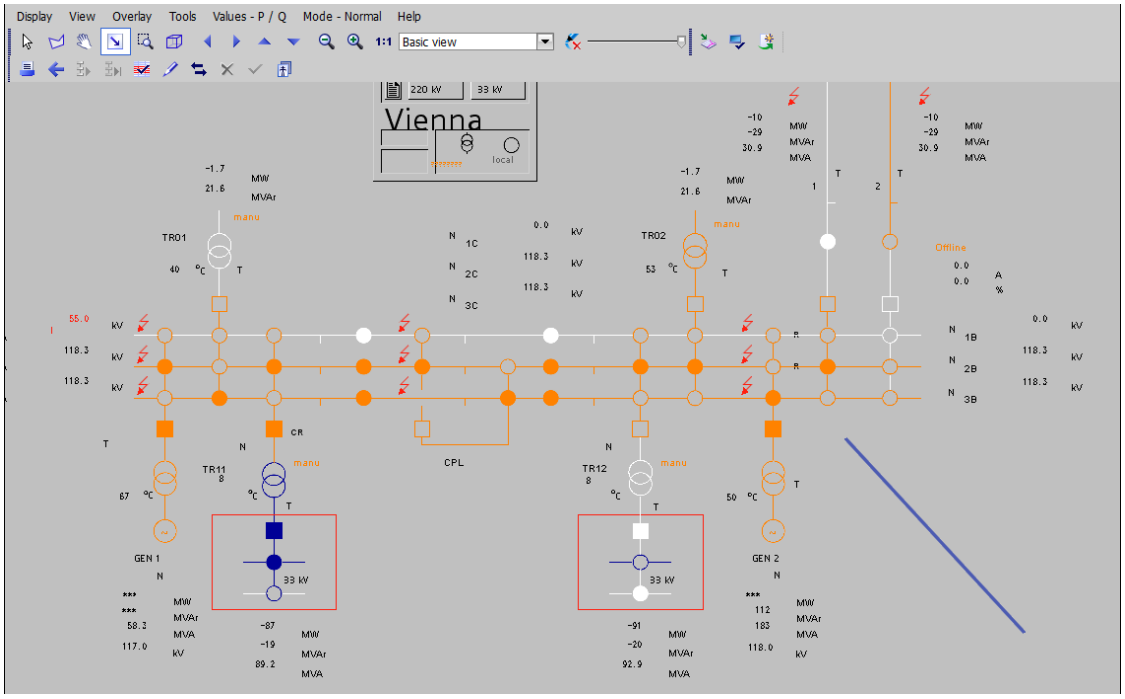
Panning and zooming are only supported while in polygon selection mode only before the start of the construction of the polygon. Panning and zooming are not supported while the polygon shape construction is under progress.

7.3.3 Rubberband Panning

To do rubberband panning, perform the following:

- Click  in the toolbar of the network diagram to activate the rubberband panning.
- Move the cursor over the display area of the network diagram and press the left button of the mouse.
- Without releasing the left button of the mouse, drag the cursor into the required direction.

  A line (rubberband) is displayed within the display area of the network diagram. It is redrawn every time according to the mouse movement. The rubberband is a vector whose direction determines the shift direction of the subsequent network diagram shift. Length of the rubberband determines the shift distance.



[sc_RubberbandPanning_12032014, 3, en_US]

Figure 7-7 Rubberband Panning

- Release the left button of the mouse.

  The network diagram is shifted according to the specified rubberband and the other portion of the network diagram is displayed in the display area of the network diagram.

7.3.4 Rubberband Zooming

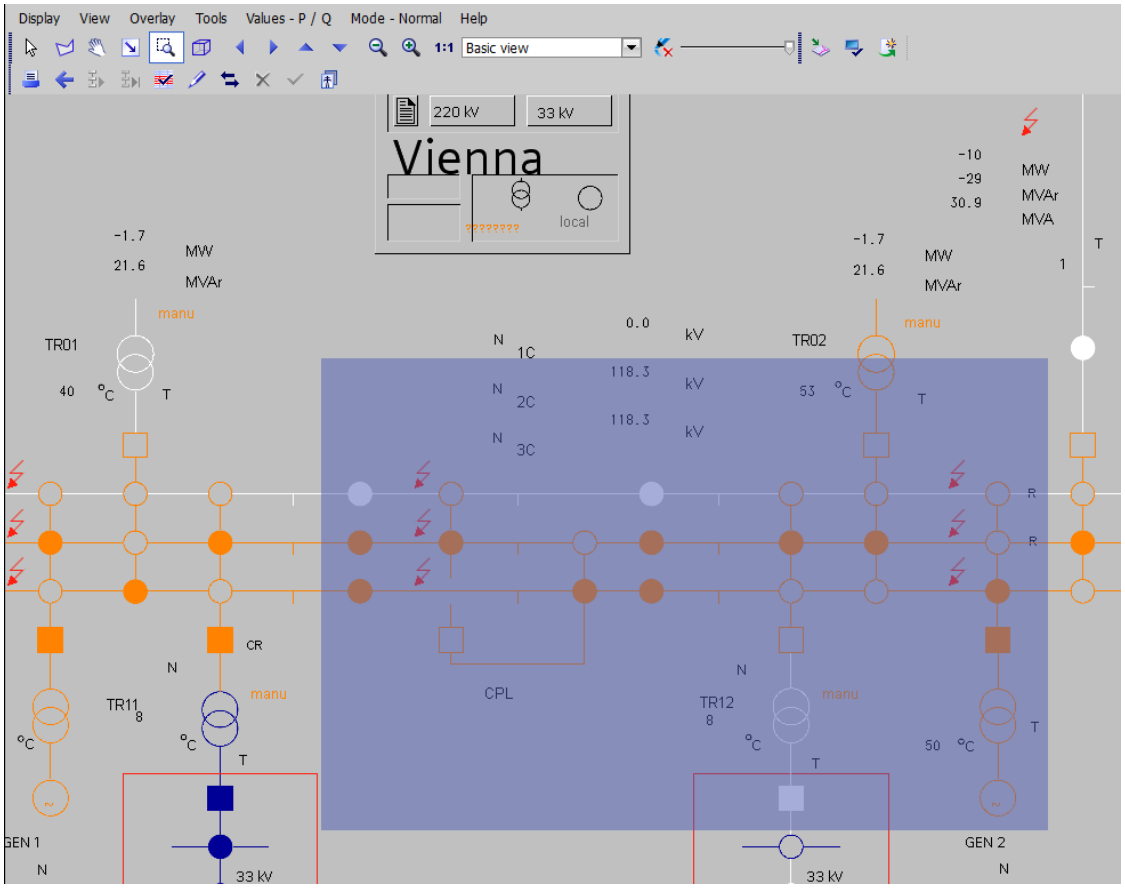
To do rubberband zooming, perform the following:

- Click in the toolbar of the single-line diagram to activate rubberband zooming.
- Move the cursor to the network diagram location that represents the upper left corner of the required network diagram portion and press the left button of the mouse.
- Drag the cursor.

  A rectangle (rubberband rectangle) is displayed within the display area. The rubberband rectangle indicates the selected network diagram portion. It is continuously redrawn according to your mouse movement.

- Release the left button of the mouse.

  This completes the selection of the required network diagram portion. The rubberband rectangle becomes static.



[sc_RubberbandZooming_12032014, 3, en_US]

Figure 7-8 Zoom

The selected network diagram portion is enlarged to fit within the display area of the single-line diagram.

Zooming can also be performed using the icons and in the title bar of the ODV, or using the wheel button of the mouse.

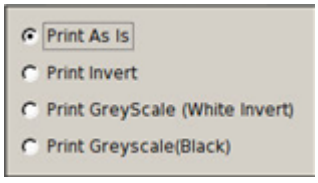
Clicking the icon 1:1 restores the normal size of the display.

7.3.5 Printing a Network Diagram

To print a network diagram, perform the following:

- Click in the lower toolbar.

  A display for selecting a print option is opened.

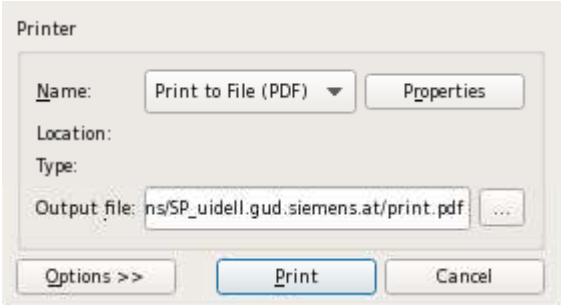


[sc_Printer Options, 1, en_US]

Figure 7-9 Printing Options

- Select the radio button.

  Print Options display is closed and a display to select the printer is opened.



[sc_odv-print, 3, en_US]

Figure 7-10 Printer Selection

- Select a printer and click **Print** to print the network diagram on the selected printer.

7.3.6 Displaying an Element in the Master ODV

To display an element selected in one display in a different network, do the following:

- Open 2 ODVs.
- In one of the ODVs, click the icon .
- In the second ODV, select an element that shall be displayed in the ODV defined as master.
- Click the icon

  The Master ODV is exposed to front and shows the selected TA (highlighted).

7.3.7 Displaying the TAs Selected in a Tabular Display in an ODV

To display TAs selected in a tabular display (for example, the Station Explorer) in a network diagram:

- Open an ODV and the Station Explorer.
- In the Station Explorer, select one or several TAs.
- In the second ODV, select an element that shall be displayed in the ODV defined as master.
- Go to the ODV and click .

  The selected TAs are displayed in the ODV and are highlighted

NOTE

Displaying TAs selected in a tabular display in an ODV can also be performed using the icon .

When the operator requests a view on map for a given list of TA elements, the worldmap opens displaying and highlighting each of them, or the existing child TA elements in the closest upper hierarchy level for those TA elements from the original TA list in the request, for which a graphical representation was not found.

If the worldmap is not highlighting the selected TA element, but child elements of any of the parent hierarchy levels, a notification in the status bar is displayed for the user.

7.3.8 Clear the Highlighting of Elements

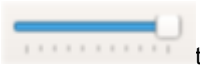
To clear the highlighting of elements, do the following:

- If elements are highlighted in the ODV, for example due to alarms, click the icon  to clear the highlighting.

  All elements are displayed in their normal state.

7.3.9 Change the Background Opacity

To change the background opacity, do the following:

- In the ODV, drag the slider  to the left or right side, until you achieve the desired opacity.

  The ODV changes accordingly.

7.3.10 Acknowledging Alarms

Acknowledging a Single Alarm

To acknowledge the alarm of a single TA, perform the following:

- Right-click the TA.

  A context menu opens.

- Select **Acknowledge** from the context menu.

  The alarms referring to the selected TA are acknowledged.

Acknowledging All Visible Alarms

To acknowledge all the alarms that are visible, perform the following:

- Click  in the toolbar.

  All alarms referring to TAs in the currently visible display section are acknowledged.

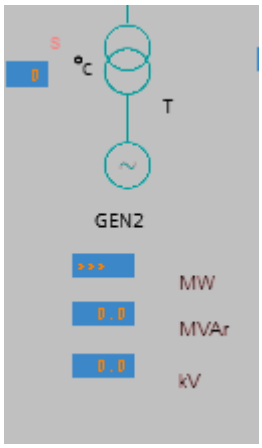
7.3.11 Updating Analog Values Manually

Updating using Toolbar Button

To manually update the analog value using toolbar, perform the following:

- Click in the toolbar.

  The network diagram changes to manual update mode. The background color of values that can be modified changes color.



[sc_ManualUpdate_12032014, 3, en_US]

Figure 7-11 Manual Update

- Update the value and press Enter.
- Click in the lower toolbar.

  The new values are displayed.

The modified values are tagged with Removed from Operation.

Updating using Context Menu

To manually update the analog value using context menu, perform the following:

- Right-click the respective value and select Substitute Value from the context menu.

  The background color of the analog value changes indicating that these values can be modified.
- Update the value and press Enter.
- Click in the lower toolbar.

  The new value is displayed. The value is tagged with Removed from Operation.

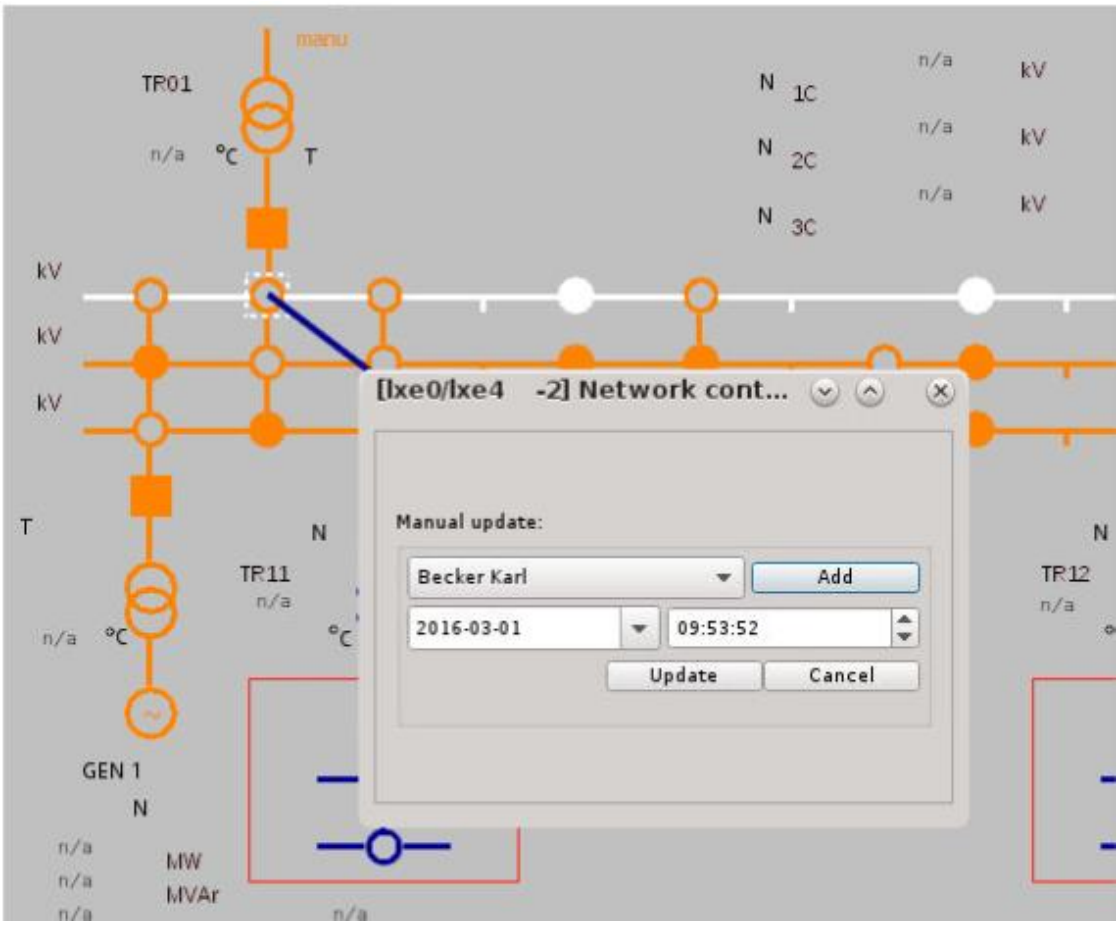
7.3.12 Updating the Status of a Digital Element

To manually update the status of a digital element, perform the following:

- Right-click the desired network element.

- In the context menu, select the item for manual update.

  A display for selecting the name of the crew member who operated the element and the time when the update occurred is opened.



[sc_manualupdatedialog, 1, en_US]

Figure 7-12 Manual Update Dialog

- Select the name of the crew member from the dropdown list or click button **Add** to define a new crew member.

  After clicking Add, the Crew members tab of the Cuman display is opened where you can enter the name and technological area of the new crew member (for information on the Cuman Crew members tab, refer to 6.2.6 Crew Members).

- Select date and time when the element was operated by the crew member and click **Update**. The new status of the network element is displayed.

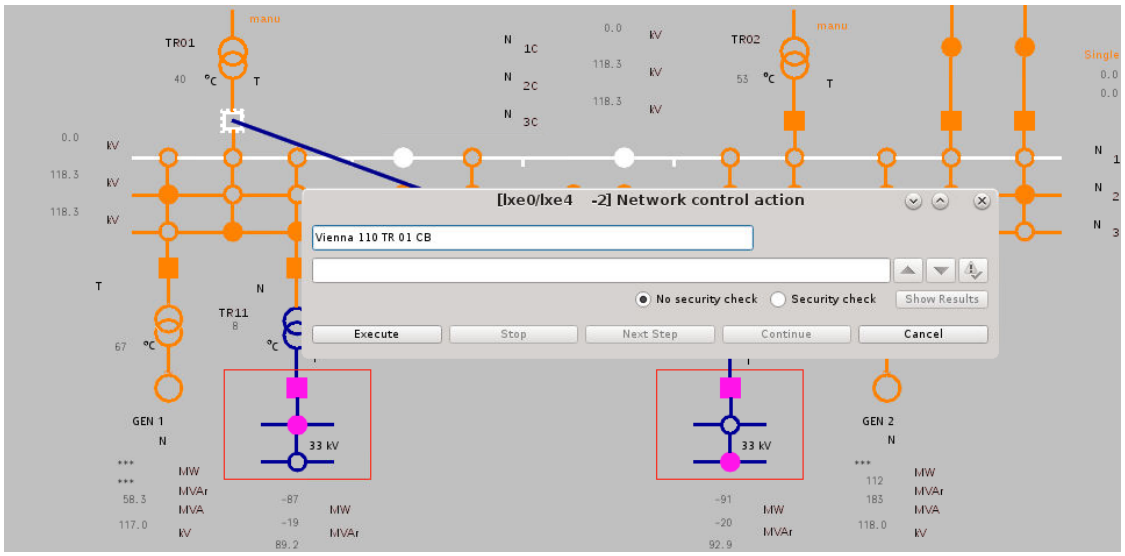
7.3.13 Performing a Supervisory Control Action

To perform a supervisory control action, perform the following:

- Right-click the graphic symbol of the element.

- Select the required operation in the context menu.

  The Network Control Action window opens. If interlocking conditions are violated, a diagnostic report is displayed.



[sc_supervisory-control, 3, en_US]

Figure 7-13 Network Control Action

- Click to unlock the currently displayed diagnostic report.

  Repeat this step for each diagnostic report.

NOTE

If is disabled, the currently displayed diagnostic reports are configured as **cannot be unlocked**.

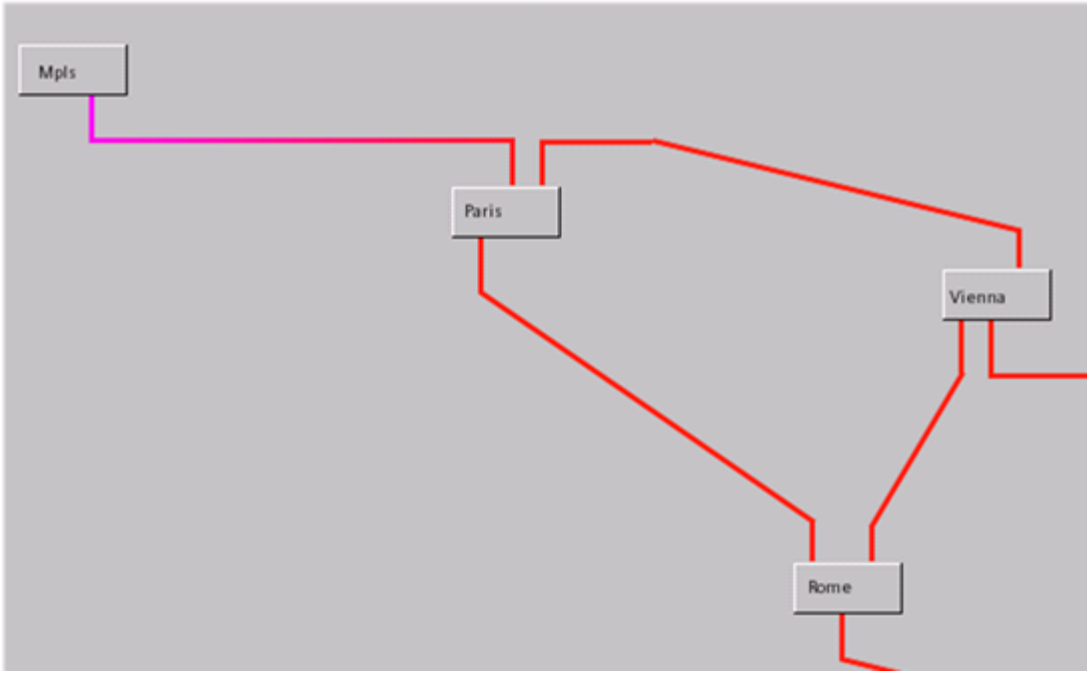
- Select the appropriate radio buttons to decide whether a security check should be performed by network analysis.
- When all diagnostic reports are unlocked, click **Execute**.

  The supervisory control action is performed. The graphic symbol changes to reflect the new status. Additionally, the general summary shows what action has been performed.

If you do not want to perform the operation, click **Cancel**

One Side Open Lines

If you have opened a line at one end, the system uses a color gradient to indicate the new energization status of the line. The gradient is shown from the defined one-side open color (magenta by default) to the currently used color for that line.



[sc_energization status, 1, en_US]

Figure 7-14 Energization Status for Lines

- If you do not want to perform the operation, click **Cancel**.

7.3.14 Inserting Notes in Network Diagrams and Setting Tags on Elements

For information on the inserting of notes and setting of tags, refer to the Chapters 11 List of Displays and 14 TA-Related Information Displays.

7.3.15 Opening and Navigation using the Navigator Window

Opening the Navigator Window

The Navigator window provides a minimized version of the network diagram for quick reference and easy user orientation. To open the navigator window:

- Click on **view** in the Menu bar

  The view menu opens

- Select **Navigator**

  The Navigator window opens.

  The operator can move, resize, and close the navigator window.

Navigating Using the Navigator Window

- Click on a part in the navigator window.

  The main window reflects the same position as the navigator window.

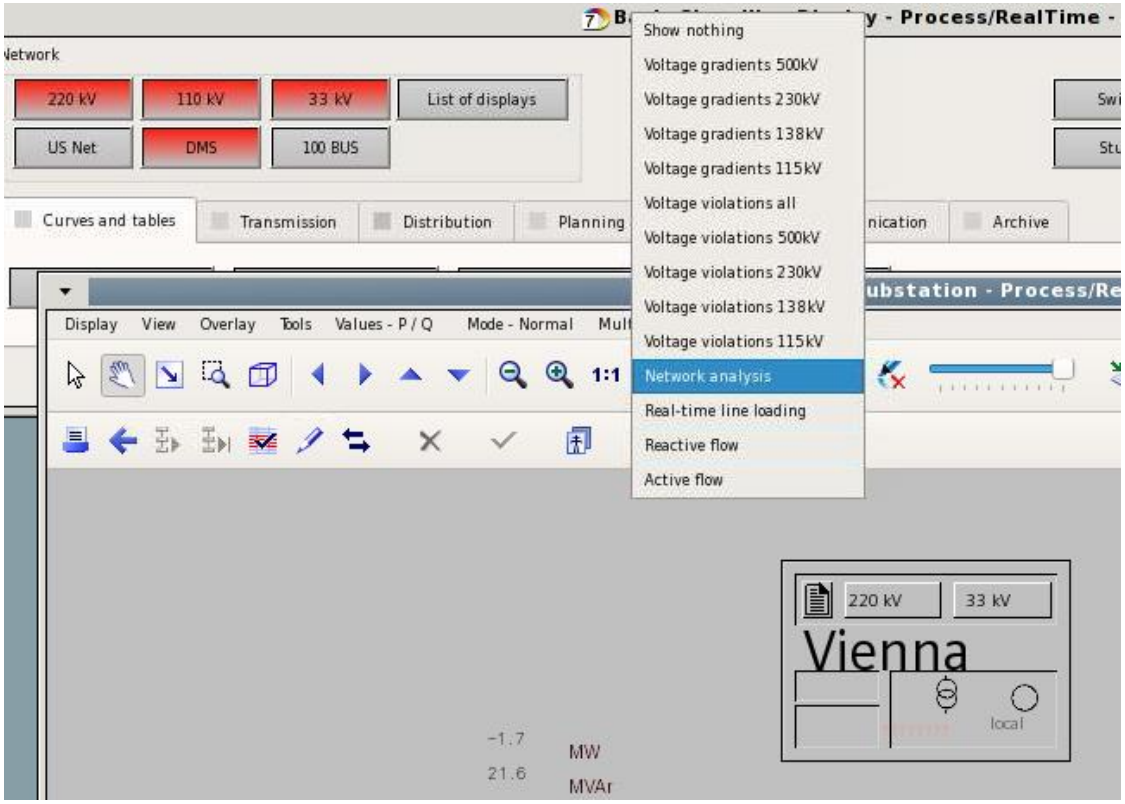
- Drag the outline in the navigator window.

  The view point in the main window shifts according to the navigator window.

7.3.16 Using Display Profiles for Enhanced Situation Awareness

To select which advanced visualization option is displayed in the network diagram, perform the following:

- Click on the display profile drop-down list in the ODV.
- A list of the of pre-configured display profiles is displayed.



[sc_show-na, 1, en_US]

Figure 7-15 Display Profile for Advanced Visualization

- Select an entry

  The display is updated according to the new setting, displaying the selected information.

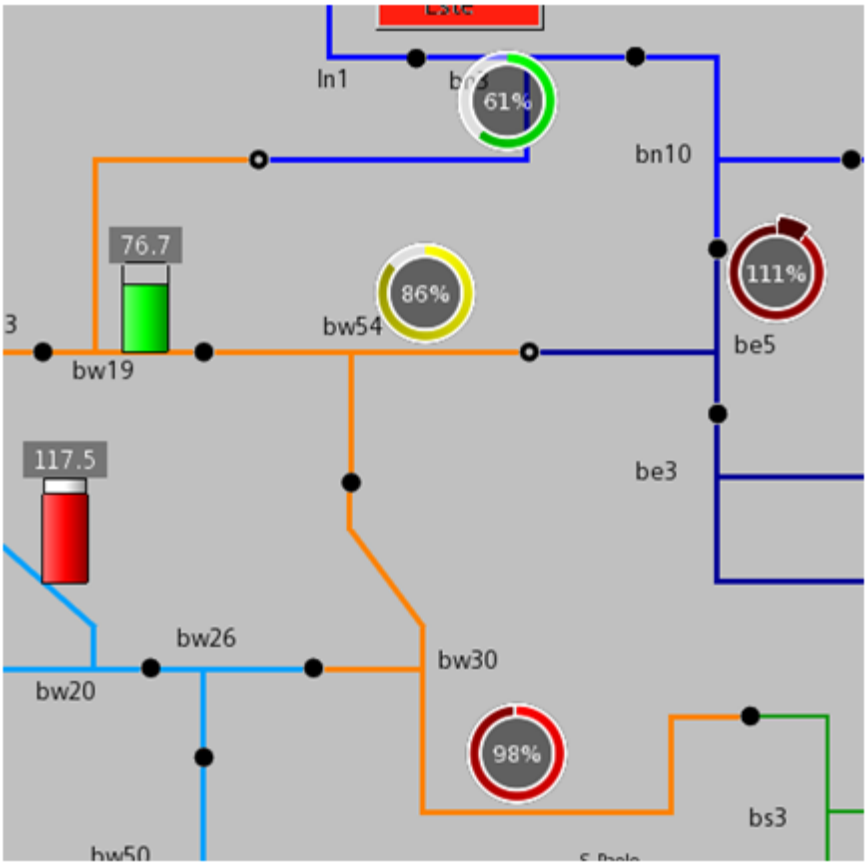
To visualize data on top of a network diagram, you can select pre-defined display profiles from the ODV toolbar. There are different profiles for transmission and distribution networks. The detailed list of profiles depends on the delivery scope.

- Gauges

  Gauges or bar charts indicate value relation to their limits. Center-filled gauges are used to represent measurements that have both, upper and lower limits, for example, voltages. Center filled gauges are supported in 3D

- Pie Charts

  Pie charts indicate the percentage load on the power lines. The pie chart symbol changes to indicate an open line.



[sc_pie_charts, 1, en_US]

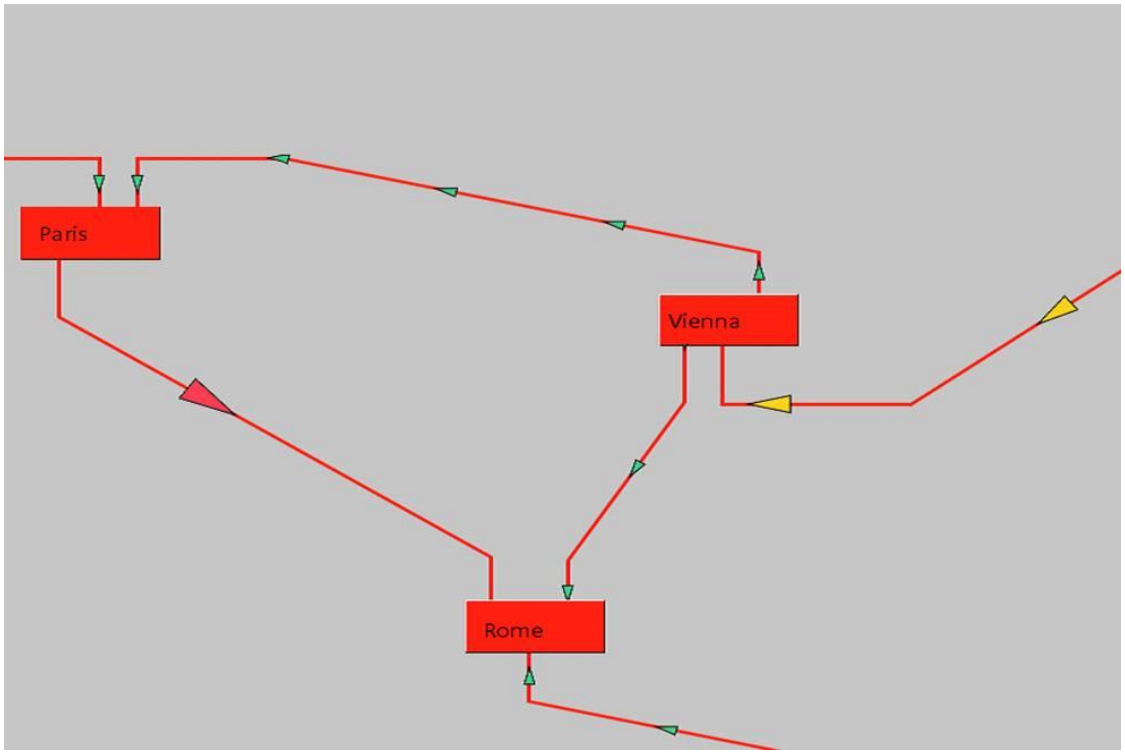
Figure 7-16 Pie Charts and Gauges

- Animated power flow arrows

  Animation can be shown for active and reactive power flows.

  By default, the animation is on. You can turn it off by using the ODV main menu to disable flow arrows.

  Flow arrow size and color can change dynamically based on load. The arrows can also be configured to blink.



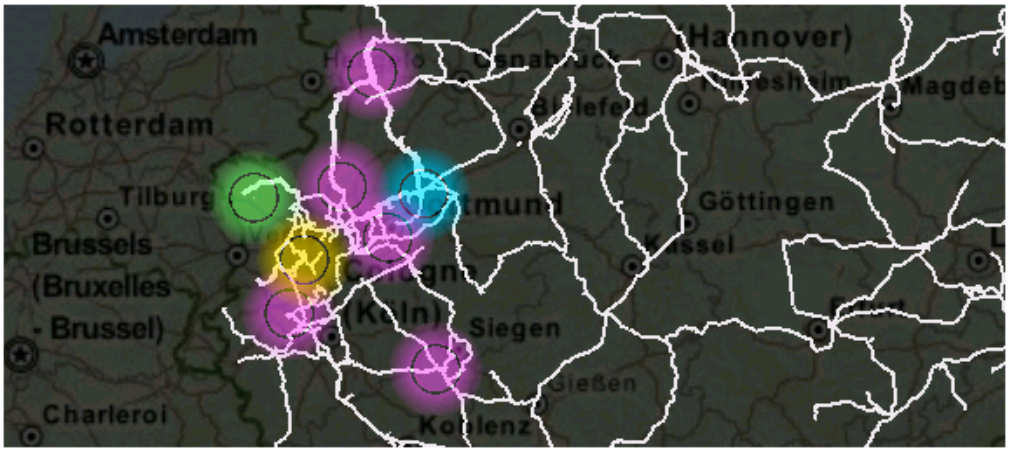
[sc_flowarrows, 2, en_US]

Figure 7-17 Power Flow Arrows Animated

- 2D bubbles and 3D cones

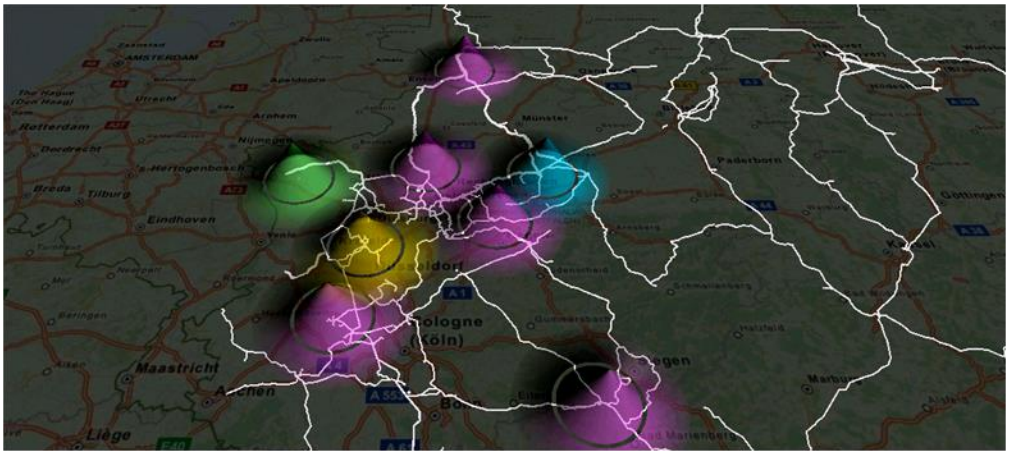
  2D bubbles and 3D cones are used for visualizing significant deviations from normal state by single-colored bubbles with a fixed radius around a bus. Two colors indicate the type of deviation (high voltage = yellow, low voltage = orange). The severity is shown by the color density with low density indicating upcoming problems and high density representing severe problems. Violations for multiple voltage levels can be shown at one glance.

  Violated limits are coded additionally by showing a small black ring. With increasing excess of the limit value, the bubble grows beyond the initial radius, creating a halo effect around the black ring. By clicking the 3D icon in the ODV toolbar, critical violations are shown as cones. Low voltage violations are cones pointing downwards, high voltage violations as cones pointing upwards. The height of the cone indicates the severity of the violation. Each cone matches a bubble with a circle in the 2D view, supporting the user's orientation in the system. Non-critical deviations (limit not yet violated) stay flat for indicating potentially upcoming but not yet critical problems. For accessing further details on the violation, you can click the cone or bubble to call-up an info box.



[sc_Bubbles, 1, en_US]

Figure 7-18 2D Bubbles

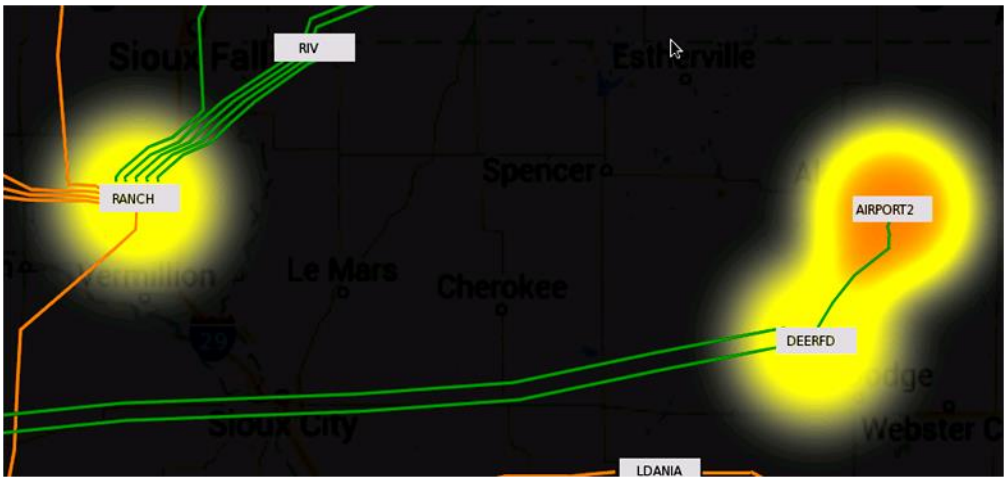


[sc_Cones, 1, en_US]

Figure 7-19 3D Cones

- Contour Gradients

  For visualizing continuous values like voltages, you can use the contour profiles.



[sc_Contour, 1, en_US]

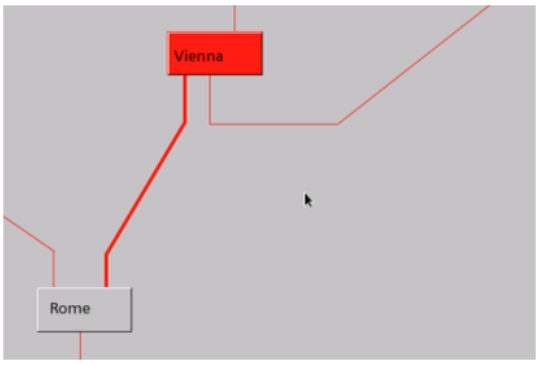
Figure 7-20 Contour Gradients

- Line Loading

  Provides an overview of the network load on a single line diagram. The thickness of the line depends on the current load on that particular line.

  When the load on a line increases, the thickness of the line increases accordingly. Unknown states are shown with a question mark icon.

  Optionally, flashing can be shown for the most severe states. You can switch off flashing by turning animation off in the ODV main menu

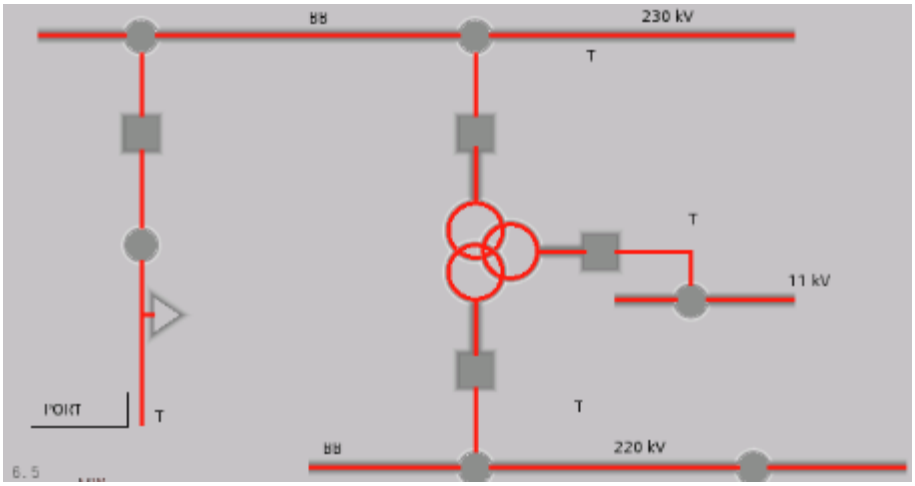


[sc_ThickLines, 1, en_US]

Figure 7-21 Line Loading

- Unassigned Areas

  Colored outline indicate areas not assigned to the operator. The operator can get a quick overview of the assigned and unassigned areas to get better situational awareness.



[sc_unowned, 2, en_US]

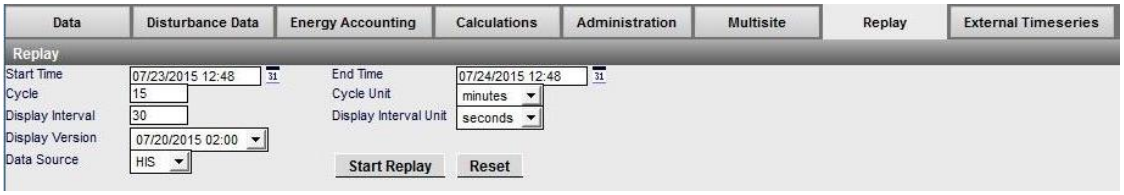
Figure 7-22 Unassigned Areas

7.3.17 Using Replay Mode

In replay mode, you can view historical network states by selecting a start and end date from continuous data recording.

For starting the replay mode, perform the following:

- Select the **Archive** tab in the BaSiDi and click on any button to open the Historical Information System (HIS).
- Click the **Replay** tab.
- Define a **Start Time** and **End Time**.
- Select a **Display Version**.
- Click the **Start Replay** button.



[sc_HISReplay_HIS, 2, en_US]

Figure 7-23 HIS Replay Mode

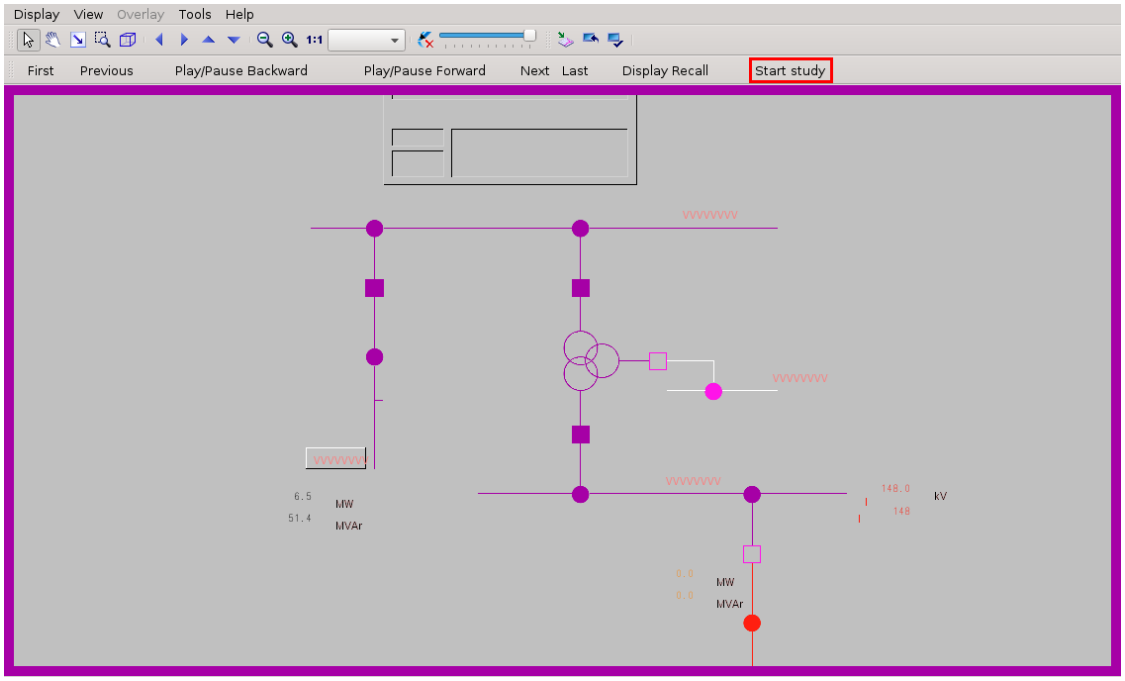
- An ODV window is displayed with a frame indicating that Replay mode is active and a toolbar for the Replay.

Creating TNA Study Using HIS Replay Mode

To create a TNA study:

- Create a replay using the preceding procedure.
- Pause the replay.
- Click **Start Study**.

A TNA study session starts.



[sc_TNAstudyreplay, 2, en_US]

Figure 7-24 Replay Mode

8 Station Explorer

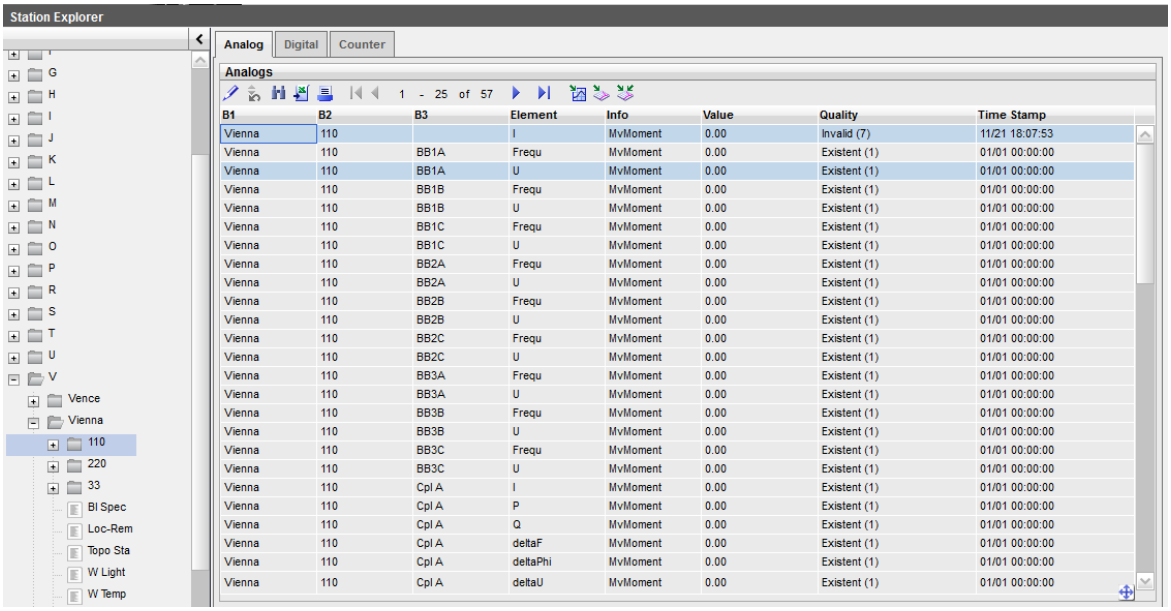
8.1	Introduction	116
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8.1 Introduction

Station Explorer is used to display and handle the current data of analogs, digitals and counters.

8.2 User Interface

8.2.1 Overview



[sc_stationexplorer, 3, en_US]

Figure 8-1 Station Explorer - Analog Values

8.2.2 Title Bar

Title bar icons are used to navigate to the previous and next display, refresh the display, print, and so on.

Table 8-1 Title Bar Icons

Icon	Description
	Opens the previous window.
	Opens the next window.
	Stops loading of data.
	Updates the table in the display.
	Favorites (Revert to original, add favorite, organize favorites)
	Export
	Zoom out
	Zoom in
	Reset zoom
	Online Help
	Displays general information about the display.

8.2.3 Tree View

The TA tree is used to select the B1, B2, and B3 levels of the information to be displayed.

To view TAs in the data section, perform the following:

- Scroll to the substation of your interest and then click the desired substation.
- The corresponding TAs are displayed in the Data Section.

8.2.4 Data Section

8.2.4.1 Selection Tabs

Following tabs are provided for the different values types:

- Analog
- Digital
- Counter

8.2.4.2 Toolbar

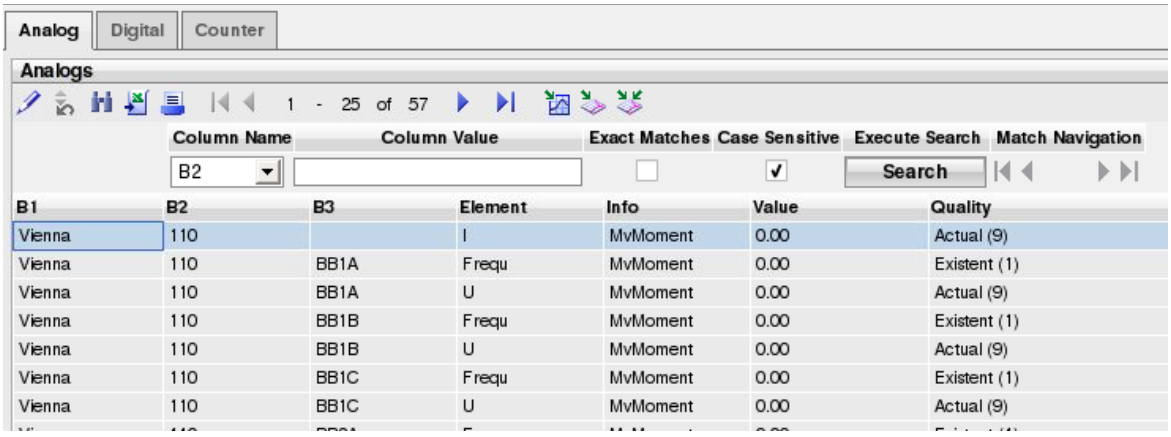
Following are the toolbar buttons:

- 
  Edit button is available only for analog and counter values. It allows to edit certain parameters of a TA.

- 
  Restore Sort is used to sort the order of the list of TAs to the default sorting order.
- 
  Search opens a line where you can enter a filter to reduce the amount of displayed TAs. The filter line provides input fields for all columns that can be used as filter. To close the filter line, click this icon again.
- 
  Exports the display content to a file.
- 
  Prints the display content
- 

  Scroll icons are used to scroll forward or backward in a multi-page display, or to jump to the beginning, end, or to a particular page.
- 
  Opens the curve tool showing a curve for the selected TA.
- 
  View all on Map opens the network diagram for all displayed TAs.
- 
  View all on Map opens the network diagram for all displayed TAs.

8.2.4.3 Search



[sc_DataSectionSearch, 1, en_US]

Figure 8-2 Data Section - Search

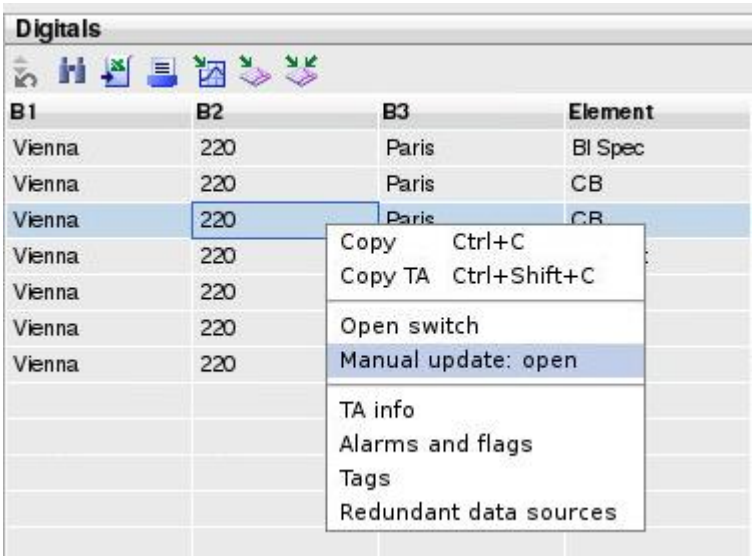
The search line is displayed on top of the List of TAs. It allows to define search criteria to reduce the amount of data displayed.

It can be searched for B1, B2, B3, Element, or Info column selecting whether only exact matches are taken and whether the search is performed considering the case of the letter entered in the Column Value field.

8.2.4.4 List of TAs

Each data row consists of columns B1, B2, B3, Element, Info, Time, Value/State, Quality (for Analogs, Counters), and 'Tags, Limits, Quality' (TLQ for Digitals) for a TA. The data rows can be sorted according to each column by clicking the header of the desired column.

Right-clicking a line opens a pop-up menu for performing supervisory control operations on the respective TA or for opening a tab of the TAInfo display for this TA.



[sc_stationexplorer-contextmenu, 2, en_US]

Figure 8-3 Data Section - Context Menu

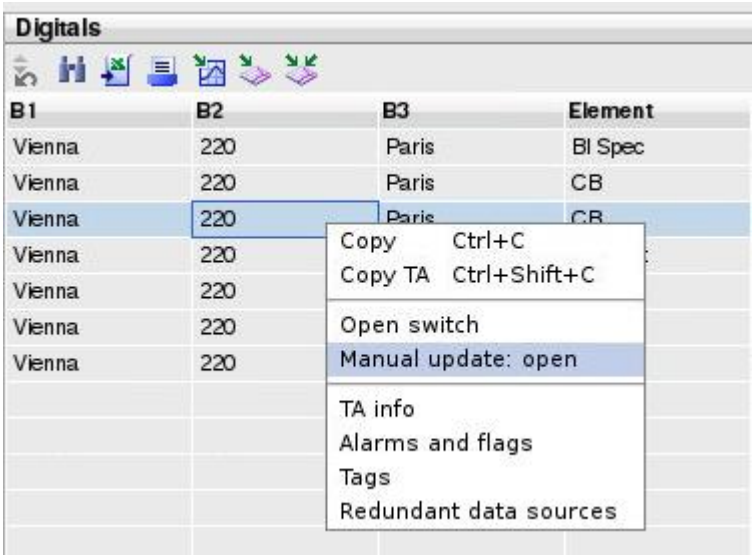
8.3 Operational Steps

8.3.1 Updating the Status of a Digital Element Manually

To update the status of a digital element manually, perform the following:

- In the Station Explorer, select the Digital tab.
- Select the desired element in the TA tree.
- Right- click the line of the element whose status you want to update.

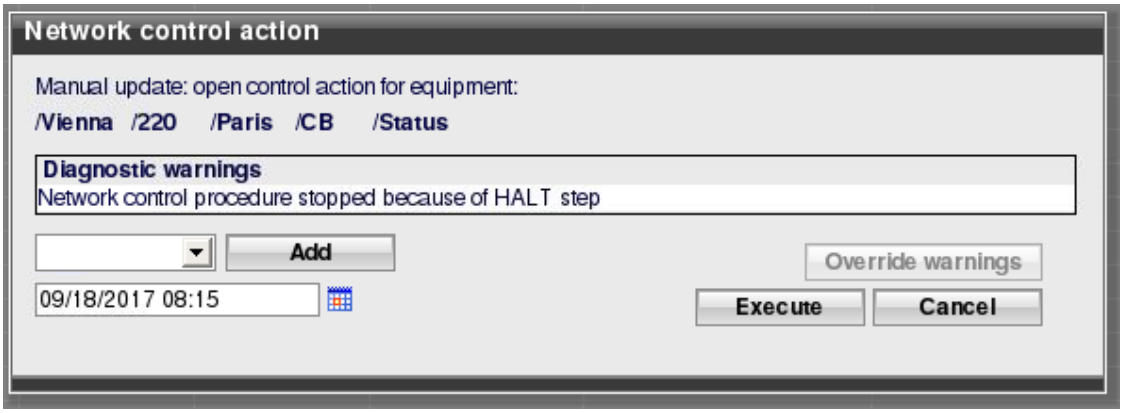
  A context menu opens.



[sc_stationexplorer-contextmenu, 2, en_US]

- Select **Manual update: open**.

  A display opens, showing diagnostic information, which allows you to confirm the operation.



[sc_NetworkControlAction, 1, en_US]

- Click **Execute**.

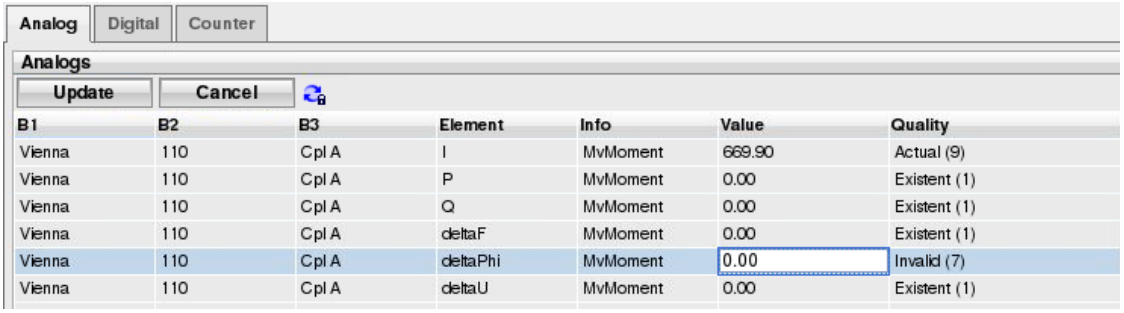
  The status of the digital element is updated.

8.3.2 Updating an Analog Value Manually

To update the analog value manually, perform the following:

- In the Station Explorer, select the Analog tab.
- Select the desired element in the TA tree.
- Click in the toolbar.

  The table changes to edit mode and you can edit the fields in the Value column.



B1	B2	B3	Element	Info	Value	Quality
Vienna	110	Cpl A	I	MvMoment	669.90	Actual (9)
Vienna	110	Cpl A	P	MvMoment	0.00	Existent (1)
Vienna	110	Cpl A	Q	MvMoment	0.00	Existent (1)
Vienna	110	Cpl A	deltaF	MvMoment	0.00	Existent (1)
Vienna	110	Cpl A	deltaPhi	MvMoment	0.00	Invalid (7)
Vienna	110	Cpl A	deltaU	MvMoment	0.00	Existent (1)

[sc_AnalogEdit, 1, en_US]

Figure 8-4 Analog Edit

- Update the desired value.
- Click **Update**.

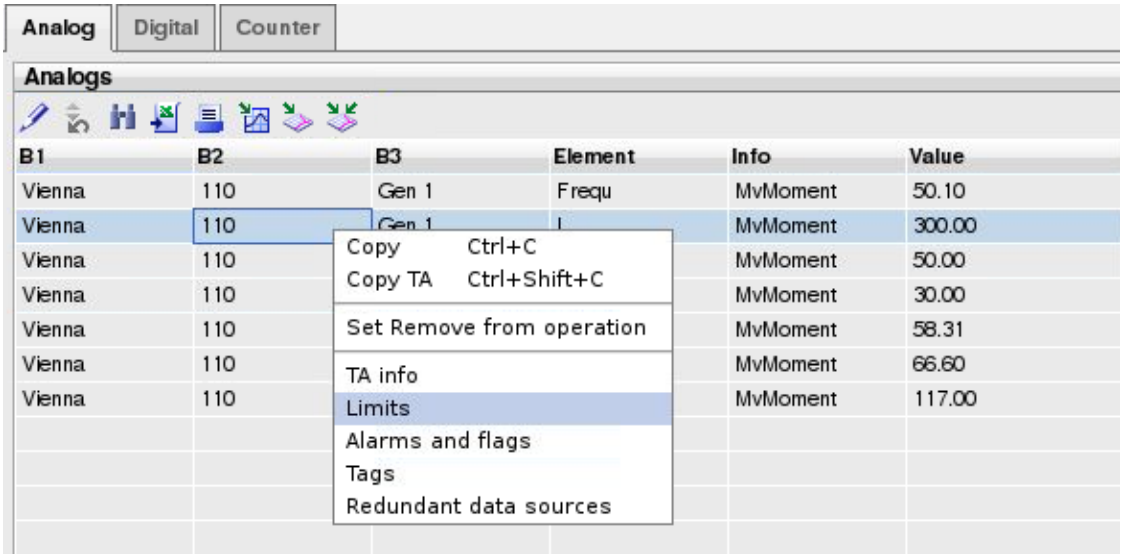
  The updated value is saved.

8.3.3 Calling Up a TA Info Display or a Network Diagram for an Element

To call-up a TA Info display for an element, perform the following:

- In the Station Explorer, select the Analog tab.
- Select the desired element in the TA tree.
- Right-click the line of the TA for which you want to call up the TA Info display.

  A context menu opens.



[sc_CallUpContextMenu, 1, en_US]

Figure 8-5 Callup: Context Menu

- Select the desired entry (for example, **Limits**).

  The Limits tab of the TA Info display is opened.

To call-up a network diagram for an element, perform the following steps:

- In the Station Explorer, select the Analog tab.
- Select the desired element in the TA tree.
- Click a line in the Analog table.
- Click in the toolbar.

  A network diagram showing the selected TA is opened (the selected TA is highlighted).

NOTE

Alarms and flags, Limits, Tags, Redundant data sources, or TAInfo calls up the respective tab of the TA Info display.

9 Summaries

9.1	Introduction	126
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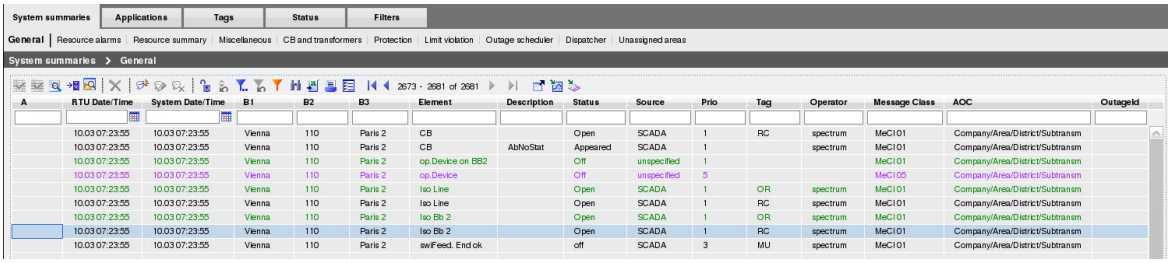
9.1 Introduction

Summaries provide the messages of power system events in tabular form. Different types of power system events are provided in different (event-specific) summaries.

Click the corresponding button in the BaSiDi to open Summaries display. Refer to 3 Basic Signaling Display for additional details.

9.2 User Interface

9.2.1 Overview



[sc_GeneralSummary, 1, en_US]

Figure 9-1 General Structure of the Alarm Summary Display

9.2.2 Title Bar

The title bar provides icons used to navigate to the previous and next display, refresh the display, print, and so on.

Table 9-1 Title Bar Icons

Icon	Description
	Opens the previous window.
	Opens the next window.
	Stops loading of data.
	Updates the table in the display.
	Favorites (Revert to original, add favorite, organize favorites)
	Export to Excel
	Zoom out
	Zoom in
	Reset zoom
	Online help
	Displays general information about the display.

9.2.3 Menus

System and Alarm Summaries

Following is the list of System and Alarm summaries:

- General Summary
- Resource Alarms
- Resource Summary
- Miscellaneous
- CB Trips and Transformer Alarms
- Protection Alarms
- Limit Violation Alarms
- Outage Scheduler alarms
- Dispatcher
- Unassigned areas

Apps

Following is the list of supported applications:

- DNA Limit Violation Alarms
- Power Application Alarms
- Power Application Dispatcher Messages

- Load Shedding
- OM Alarms
- Underfrequency Load Shed

Tags

Following is the list of supported tags:

- Removed from Operation
- Control Inhibit
- Permit to Work
- Permit to Test
- Alarm Inhibit
- Object in Test
- Open only
- Close only
- Caution tag
- Information tag
- Switch in Progress
- Used for Disconnection

Status

Following is the list of supported status types:

- Disturbed Switching Elements
- Abnormal Status
- Test Messages
- Manual substitutions
- Not Renewed Values
- Invalid Values
- Temporary Network Elements
- Manual limit substitutions

Filtered Alarm Lists

Following is the list of filtered alarms:

- Unacknowledged
- End Date
- B1 System
- TA

9.2.4 Summary

9.2.4.1 Toolbar

- 

  Acknowledge Selection

  Clicking this button acknowledges the selected alarms.

- 

  Acknowledge Current Page

  Click to acknowledge all alarms on the page.

- 

  TAInfo

  Opens the TAInfo display for the selected alarm.

- 

  Historical Messages

  Opens the HIS display to select TAs in the archive to display the messages.

- 

  Station Explorer

  Opens the Station Explorer for the selected alarm

- 

  Delete Selected Messages

  Deletes the selected alarms

- 

  New Comment

  Inserts a comment for an alarm. After clicking this button, a pop-up window where the comment can be inserted is shown.

- 

  Edit Comment

  Opens a pop-up window where the comment can be edited.

- 

  Delete Selected Comment

  Deletes the selected comment.

- **Tracking Control**

  The tracking icon determines how the alarm summary is updated:

  – 

By default, the alarm lists have tracking mode enabled when it is first opened. While in tracking mode, the operator is looking at the latest alarms for the given list. Clicking the tracking button in the toolbar will exit tracking mode, leaving the visible list in place on the display. New alarms can still be seen coming into the visible page until it goes past the page boundaries (last visible row).

– 

When tracking is off, the alarm summary table is not updated automatically when a new alarm occurs. In addition, when tracking is off, a red border is displayed around the summary. The icon and the border starts flashing when a new alarm occurs. To view the new alarms, the icon has to be clicked to turn on tracking mode again and update the summary.

When new alarms are generated while the user is not in tracking mode the red list border blinks on and off. The tracking button in the toolbar also blinks to indicate to the user where they can click to enable tracking mode again. Clicking the tracking button in the toolbar will bring the user to the latest alarms in the list.

- 

  Restore Sort: Restores the sort order of the table to the default sorting.
- 

  Advanced Filter

  Opens the Filter display for defining filter criteria.
- 

  Reset Advanced Filter

  Resets the current filter settings to the default filter.
- 

  Filter opens the In-line Filter line where you can enter a filter to reduce the amount of displayed TAs. The filter line provides input fields for all columns that can be used as filter. Click this icon again to close the filter line.

  By default, the Inline Filter is turned on, when the display is opened.
- 

  Opens a line for entering search criteria.
- 

  Export

  Exports the summary to a file
- 

  Print

  Prints the summary
- 

  Adjust Columns

  Hides/shows columns

- 

  Paging Icons

  These icons are used to page forward or backward in a multi-page summary, or to jump to the beginning, end, or to a particular page in the summary.

- 

  Open in new window

- 

  Show curve

  Opens the curve tool for the TA of the selected alarm. Supports single and multi select.

- 

  View on Map

  Opens the network diagram for the selected alarm.

9.2.4.2 Alarm Summary Table

This table contains the alarm messages. An **A** to the left of an alarm indicates that the alarm has not been acknowledged.

The table consists of the following columns:

- A

  Acknowledge indication or General check indication

- RTU Date/Time

  Date/time when the message occurred in the RTU

- System Date/Time

  Date/time when the message arrived in the Spectrum system

- TA (B1, B2, B3, Element)

  Technological address of the message - one column for each TA part

- Description

  Description text

- Status

  Status

- Source

  Source of the message

- Prio

  Priority of the message

- Tag

  Tag of the message

- Operator

  Operator who cause the message

- Message class

  Message class to which the message belongs

- AOC

  Areas of control to which the message refers

- Outage ID

  If the message is related to an outage, the outage record ID is shown in this column. Clicking the outage ID opens the Outage Details display for this outage. (For additional information on the Outage Details display, refer to the Outage Management User Guide).

9.3 Operational Steps

9.3.1 Acknowledging Alarms

It is not required to acknowledge all the alarms. Whether an acknowledgement is required is determined when the alarm is generated and is indicated by an **A** to the left of the alarm in the alarm summary.

You can perform any of the following:

- Acknowledge all the alarms on the current page by clicking the icon .
- Acknowledge one or more individual alarms by selecting the corresponding alarms and clicking the icon .

The alarms are acknowledged and removed from the summary.

9.3.2 Deleting Alarms

To delete alarms, proceed as follows:

- Select the alarms that you want to delete and click .

The alarms are removed from the list.

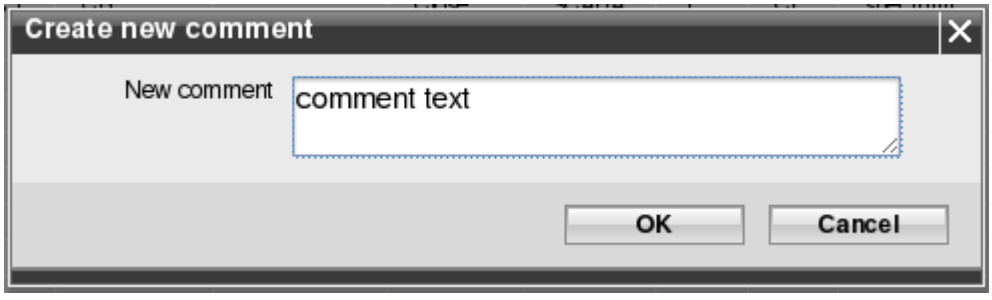
9.3.3 Inserting or Deleting Comments

Inserting Comments

To insert comments, perform the following:

- Select the line for which you want to insert a comment and click .

  The display shows an input field for entering a comment.



[sc_AlarmListInsertCommentField_12032014, 2, en_US]

Figure 9-2 Summary - Insert Comment Field

- Enter a comment and click **OK**.

  The comment is displayed below the selected line.



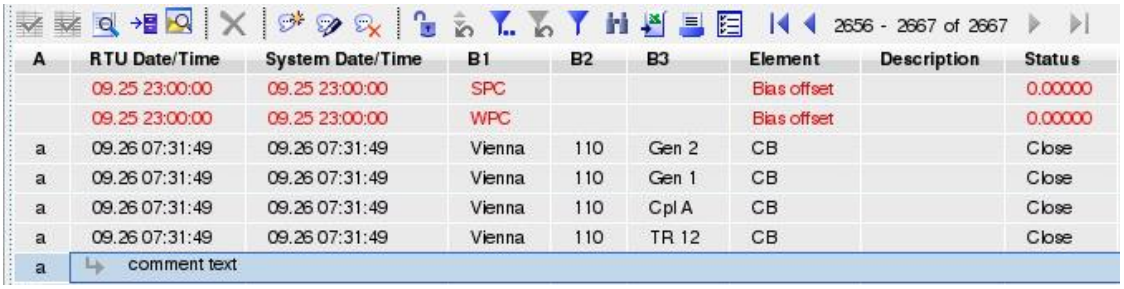
[sc_SummaryCommentinserted, 1, en_US]

Figure 9-3 Summary - Comment Inserted

Deleting Comments

To delete comments, perform the following:

- Select the comment to be deleted and click the icon  .



[sc_SummaryCommentDelete, 1, en_US]

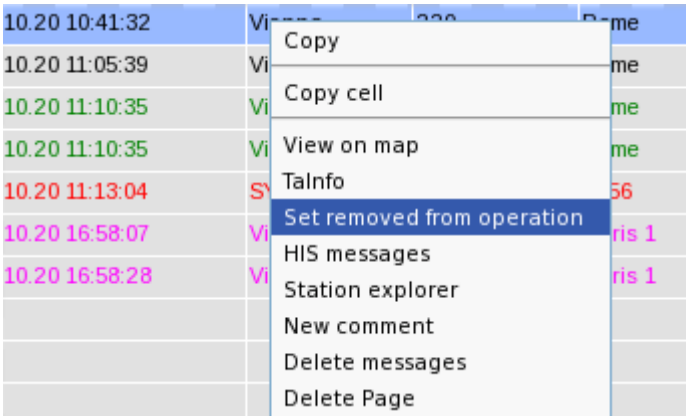
Figure 9-4 Summary - Delete Comment

The selected comment is removed.

9.3.4 Blocking an Alarm

Blocking an alarm means to set the tag Removed from operation on the network element. For this purpose, perform the following steps:

- Open the summary with the desired alarm.
- Right-click the alarm and select **Set removed from operation** from the context menu.



[sc_summary_block, 1, en_US]

Figure 9-5 Summary - Block an Alarm

The tag Removed from operation is set on the network element.

9.3.5 Viewing the TA Referenced by a Summary Entry in the Network Diagram

To view a TA referenced by a summary entry in the corresponding network diagram, perform the following steps:

- Right-click the desired summary entry.
- From the context menu, select the item View on map.
- Alternatively, you can also select the summary entry with the left mouse button and click the icon in the toolbar.

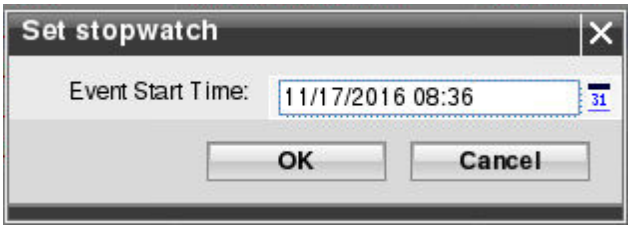
  The network diagram containing the TA is opened, the TA which the summary entry references is high-lighted.

9.3.6 Setting a Stopwatch for an Alarm

For setting a stopwatch for an alarm in a summary, perform the following steps:

- Open the summary containing the alarm.
- Right-click the alarm and select **Set stopwatch**.

  A display is opened that shows the time when the alarm occurred.

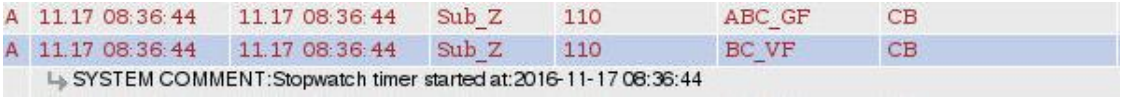


[sc_alarmtimer-setstopwach-dialog, 1, en_US]

Figure 9-6 Set Stop Watch

- Change the start time, if required, and click **OK**.

  A comment is added to the alarm in the summary when the stopwatch has started.



[sc_alarmtimer-setstopwachfinished, 1, en_US]

Figure 9-7 Stopwatch Timer Started

- When the alarm disappears, another comment is added showing the time when the stopwatch has stopped as well as the duration.



[sc_timerstopped, 1, en_US]

Figure 9-8 Stopwatch Timer Completed

10 Operations Logbook

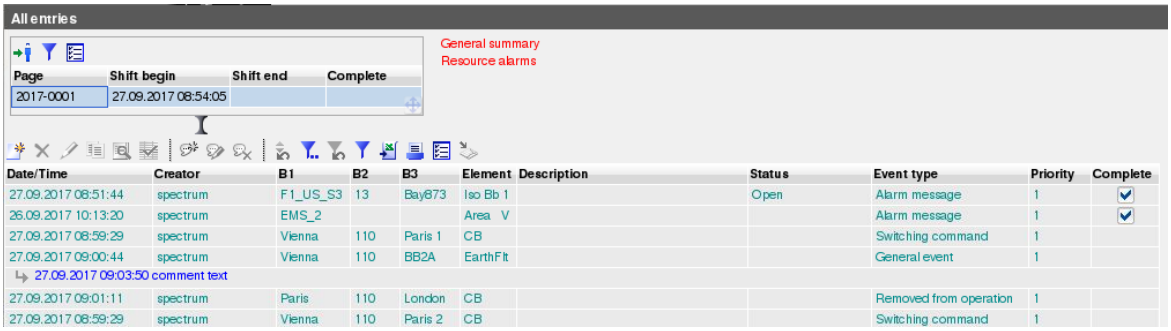
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10.1 Introduction

The Operations Logbook allows the operator to store activities and incidents at a dedicated place. There is a separate logbook page for each shift. Entries may be added manually, or also some alarm messages may be parameterized to be added to the logbook automatically when they occur.

10.2 User Interface

10.2.1 Overview



[sc_OperationsLogbook, 1, en_US]

Figure 10-1 General Structure of the Operations Logbook

10.2.2 Title Bar

The title bar provides icons used to navigate to the previous and next display, refresh the display, print, and so on.

Table 10-1 Title Bar Icons

Icons	Description
	Opens the previous window.
	Opens the next window.
	Stops loading of data.
	Updates the table in the display.
	User Displays (Revert to original, create user displays, organize user displays)
	Export to Excel
	Zoom out
	Zoom in
	Reset zoom
	Online Help
	Displays general information about the display.

10.2.3 Table of Shift Pages

A separate page exists for each shift. At the end of a shift, the operator on duty hands over to the next operator and creates a new shift page.

10.2.3.1 Toolbar

Table 10-2 Title Bar Icons

Icons	Description
	Shift change Opens a display for creating a new shift page.
	Inline filter Inserts a line at the top of the table for defining a filter, which is directly applied to the table content. Clicking this icon a second time switches off the filter mode again.
	Adjust Columns Hides or shows columns

10.2.3.2 Table

- Page

  ID of the selected shift page

- Shift begin

  Date and time when the shift has started

- Shift end

  Date and time when the shift has ended (empty when the shift is still ongoing)

- Complete

  A hook indicates whether the selected shift is still ongoing or whether it has already been finished.

10.2.3.3 Alarm Summary Links

Clicking a link at the right side of this table, opens the corresponding alarm summary in a new window. The color of the links indicates whether the summaries contain entries or not: when the link is displayed in blue color, the summary contains no entries, when it is displayed in red, it does contain entries.

10.2.4 Logbook Table

10.2.4.1 Toolbar

Table 10-3 Title Bar Icons

Icon	Description
	New message
	Edit
	Delete
	Duplicate message
	TAInfo Opens the TAInfo display for the selected message
	Acknowledge Clicking this button acknowledges the selected messages.
	New Comment Inserts a comment for a message. After clicking this button, a pop-up window where the comment can be inserted is shown.
	Edit comment After clicking this button, a pop-up window where the comment can be edited is shown.
	Delete Selected Comment Deletes the selected comment.
	Restore sort Restores the current sorting order to the default sorting.
	Filter Opens the Filter display for defining filter criteria.
	Reset filter Resets the current filter settings to the default filter.
	Inline filter Inserts a line at the top of the table for defining a filter, which is directly applied to the table content. Clicking this icon a second time switches off the filter mode again.
	Export Opens a submenu for selecting the file type and scope (all rows or only displayed rows) for the export.
	Print Opens a display with options for what to print and where to print it.
	Adjust Columns Hides or shows columns
	View on map Opens a single-line diagram showing the selected records.

10.2.4.2 Table

- Date/time

  Date and time of the message

- Creator

  Name of the operator who created the message

- B1, B2, B3, Element

  TA to which the message refers

- Description

  Description of the message

- Status

  Status information

- Event type

  Type of the message

  - General event

    Free message

  - Error location

    Short description of a fault and its location

  - Commissioning

    A device is set into operation

  - Removed from operation

    A device is removed from operation

  - Switching command

    A device is switched

- Priority

  Priority of the message

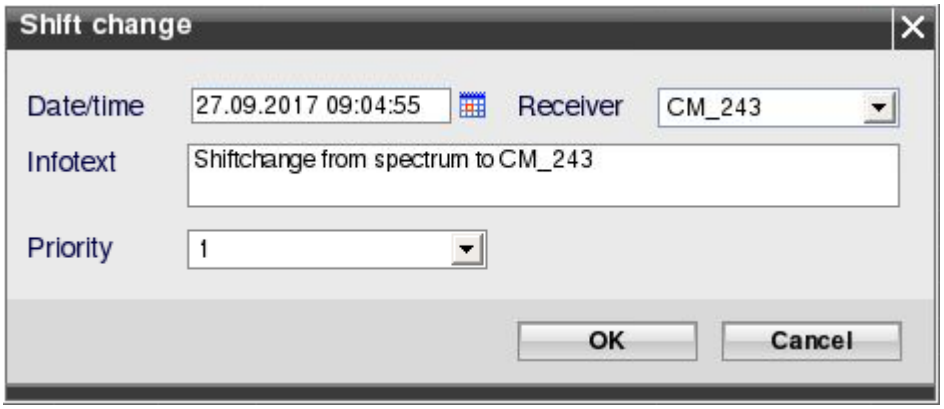
- Complete

  A hook in this column indicates that work on this message has been completed. Messages that have been completed cannot be modified anymore.

10.3 Operational Steps

10.3.1 Inserting a New Shift Page

- Click the icon on top of the table of shift pages
- Fill in the fields in the shift change display



[sc_LogbookShiftchange, 1, en_US]

Figure 10-2 Shift Change

- Click **OK**

  A new entry is displayed in the table of shift pages.

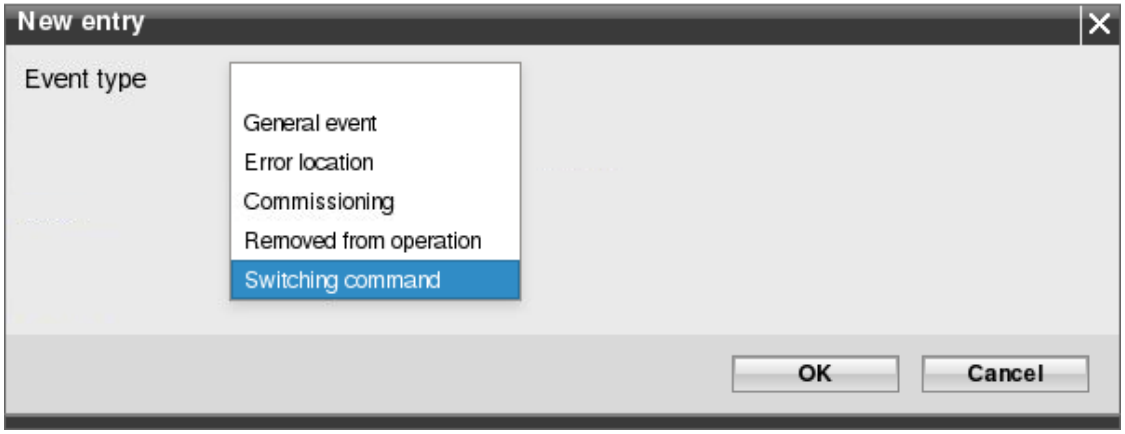
  The previous shift is set to complete.

  The logbook table shows all entries that were not set to complete in the previous shift.

10.3.2 Adding a New Logbook Entry

- Click the icon above the logbook table.

  A display for selecting the type of the new logbook entry opens.

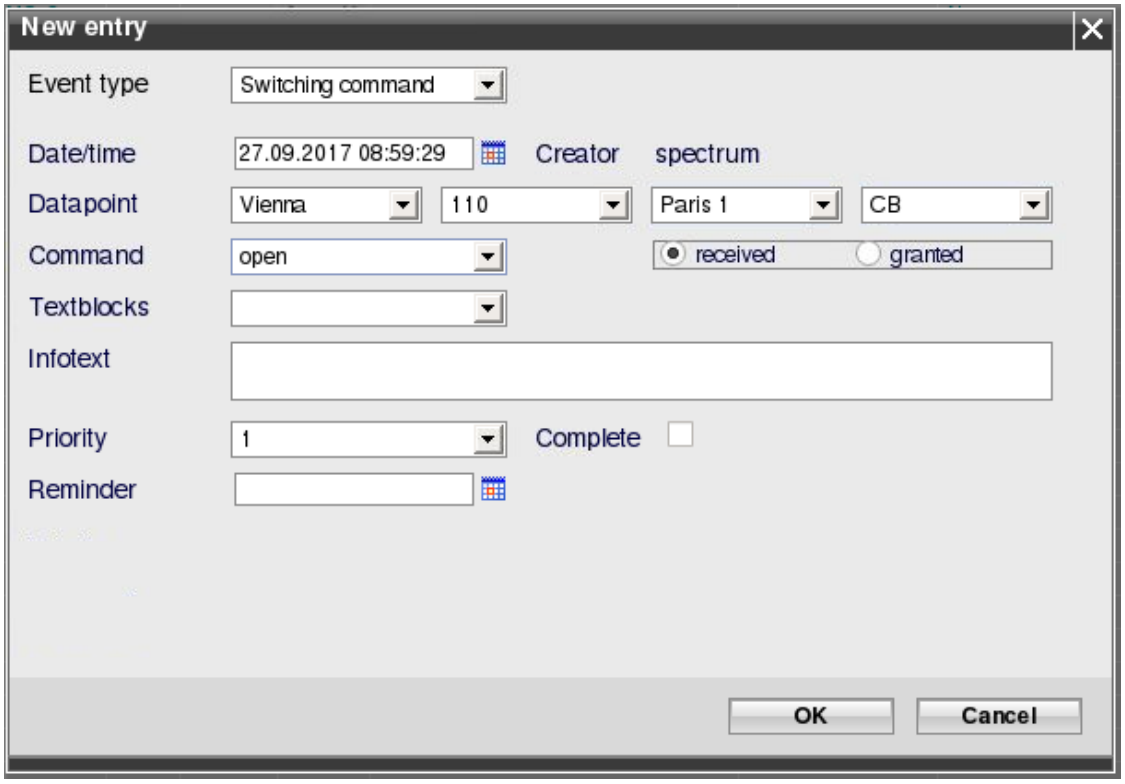


[sc_LogbookNewentry, 1, en_US]

Figure 10-3 Adding a New Logbook Entry

- Select a type

  The display shows additional input fields for inserting all information required for this type.



[sc_NewLogbookEntry, 1, en_US]

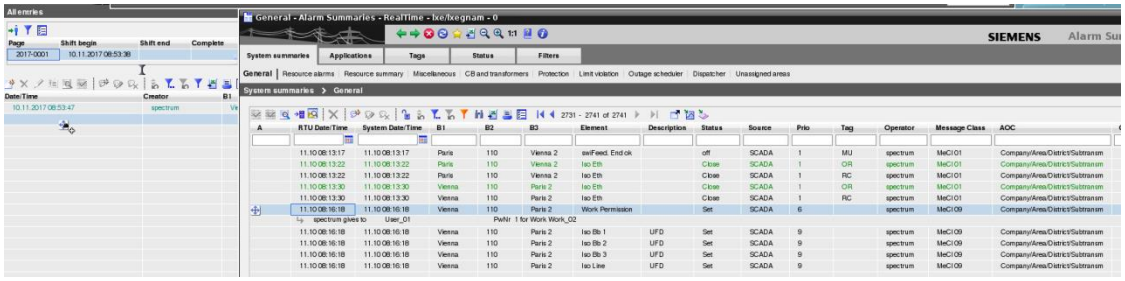
Figure 10-4 New Logbook Entry

- Fill in the fields and click **OK**.

  The display is closed and the logbook table shows the new entry.

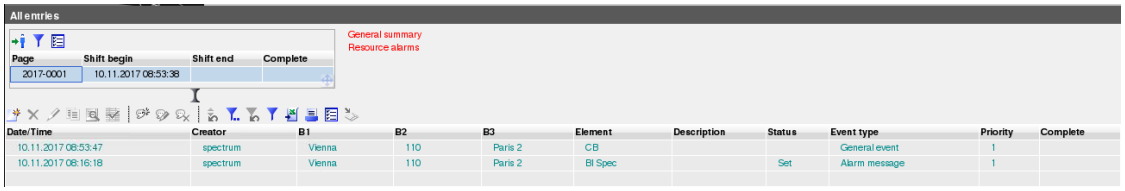
- You can also add an entry from an alarm summary to the logbook via drag and drop.

  For this purpose select the desired message in the alarm summary and drag and drop it to the logbook.



[logbook-add-draganddrop2, 1, en_US]

The message from the alarm summary is added as a new line to the logbook.

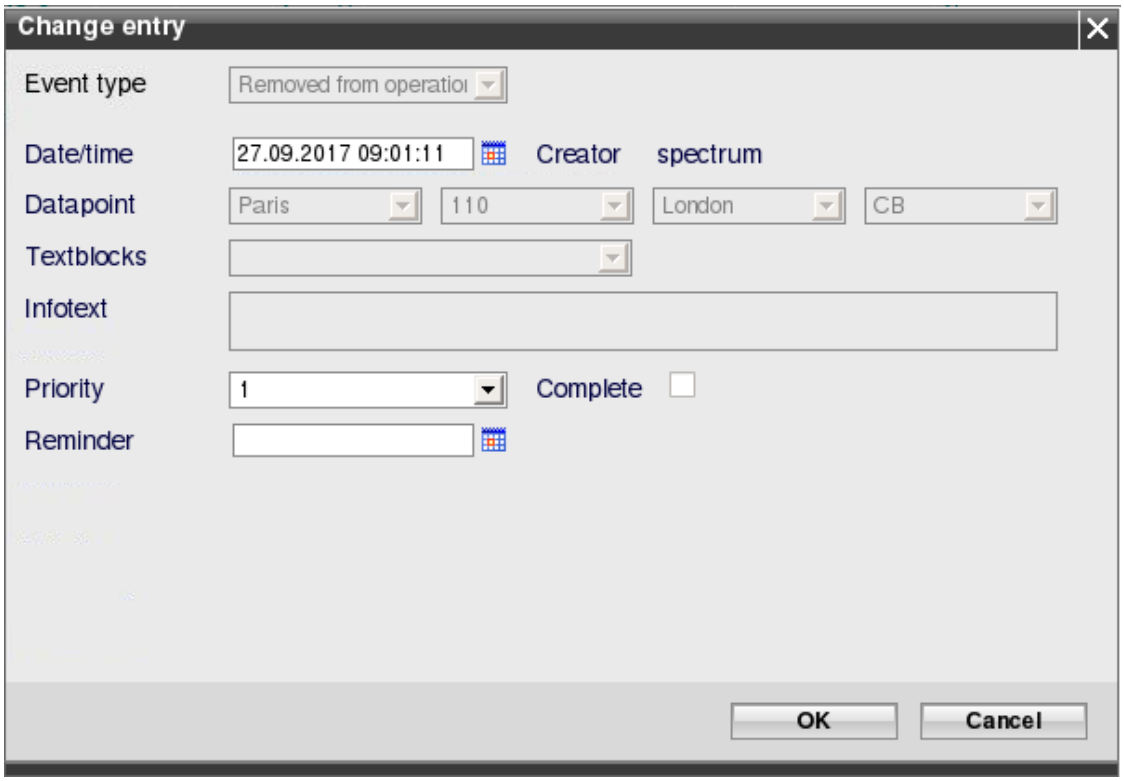


[logbook-add-draganddrop3, 1, en_US]

10.3.3 Editing an Existing Logbook Entry

- Select the entry you want to edit in the logbook table and click the icon .

  A display showing all information for the selected entry opens.



[sc_LogbookEdit, 1, en_US]

Figure 10-5 Edit an Entry

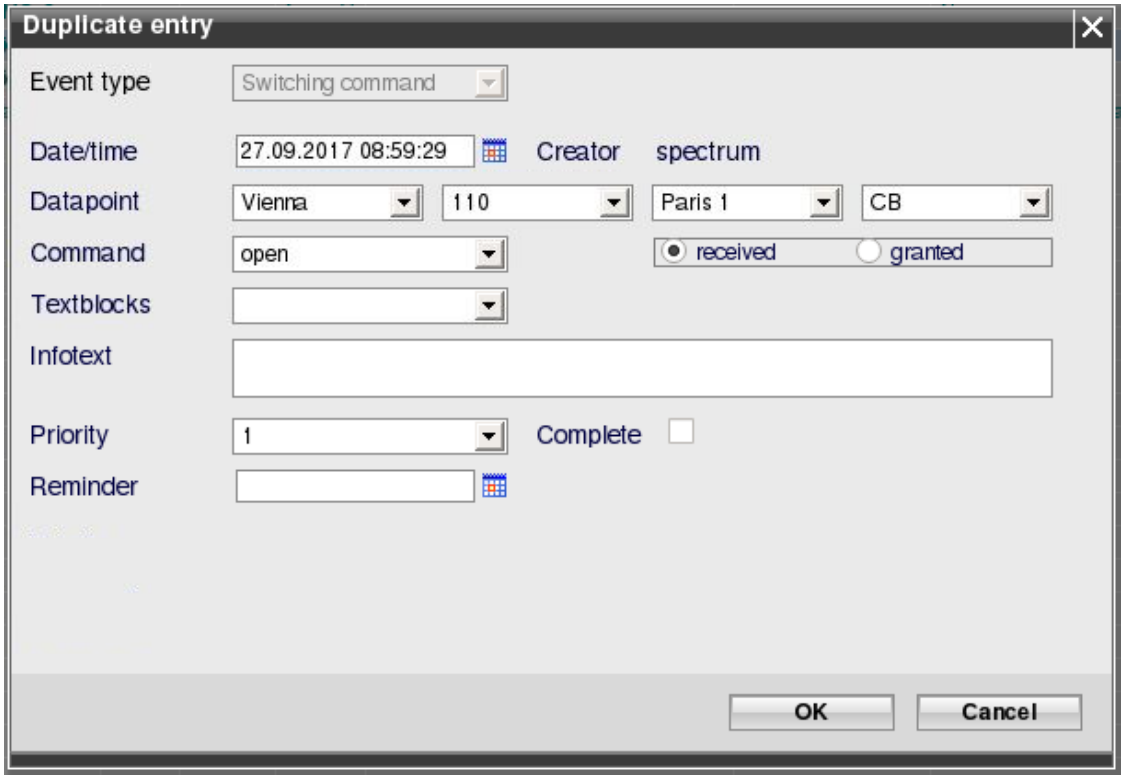
- Modify the information as required and click **OK**

  The display is closed and the logbook table shows the modified entry.

10.3.4 Duplicating a Logbook Entry

- Select the entry you want to duplicate in the logbook table and click the icon .

  A display showing all information for the selected entry opens.



[sc_LogbookDuplicate, 1, en_US]

Figure 10-6 Duplicate Entry

- Modify the information as required and click **OK**.

  The display is closed and the logbook table shows the new duplicated entry.

10.3.5 Setting a Logbook Entry to “Complete”

- Select the entry you want to set to **complete** in the logbook table and click the icon .

  A display showing all information for the selected entry opens, asking to confirm deletion.

- Click **OK**

  The display is closed and the entry is shown struck through and the hook in column **Complete** is set.

10.3.6 Acknowledging a Reminder

When a reminder has been defined for an entry and this reminder has expired, the entry is highlighted in red color.

- Select this entry and click the icon 

  The entry is displayed in normal color again, the hook in column **Complete** is set.

10.3.7 Inserting and Deleting a Comment for a Logbook Entry

- Select the entry for which you want to add a comment and click the icon  .

  A display for inserting a comment opens.

- Enter a comment and click **OK**.

  The display is closed and the comment is displayed below the logbook entry

- To delete a comment, select the respective comment and click the icon  .

  A display asking for confirmation opens.

- Click **OK**.

  The display closes and the comment is displayed struck through.

11 List of Displays

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11.1 List of Displays

The List of displays provides the following features:

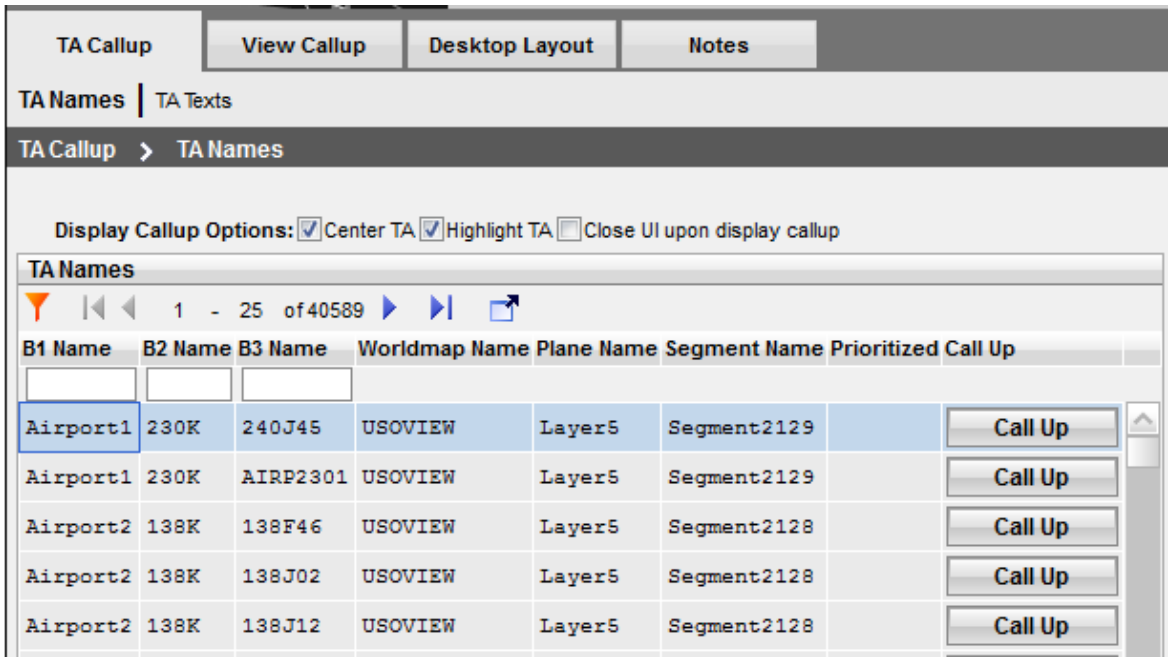
- Display selection using TA Names, TA Texts, or View numbers.
- Desktop Layout Management: Saving, Modifying, and Loading Desktop Layouts.
- Displaying free notes that are entered in a worldmap.

Click the corresponding button in the BaSiDi or menu item on Heads-Up Display to open the List of Displays.

11.2 User Interface

11.2.1 Header

11.2.1.1 Overview



[sc_TACallup, 3, en_US]

11.2.1.2 Title Bar

Title bar icons are used to navigate to the previous and next display, refresh the display, print, and so on.

Table 11-1 Title Bar Icons

Icon	Description
	Opens the previous window.
	Opens the next window.
	Stops loading of data.
	Updates the table in the display.
	Print
	Zoom out
	Zoom in
	Reset zoom
	Displays general information about the display.
	Closes the window.



NOTE

The export icon  is not available for the **TA Callup** tab.

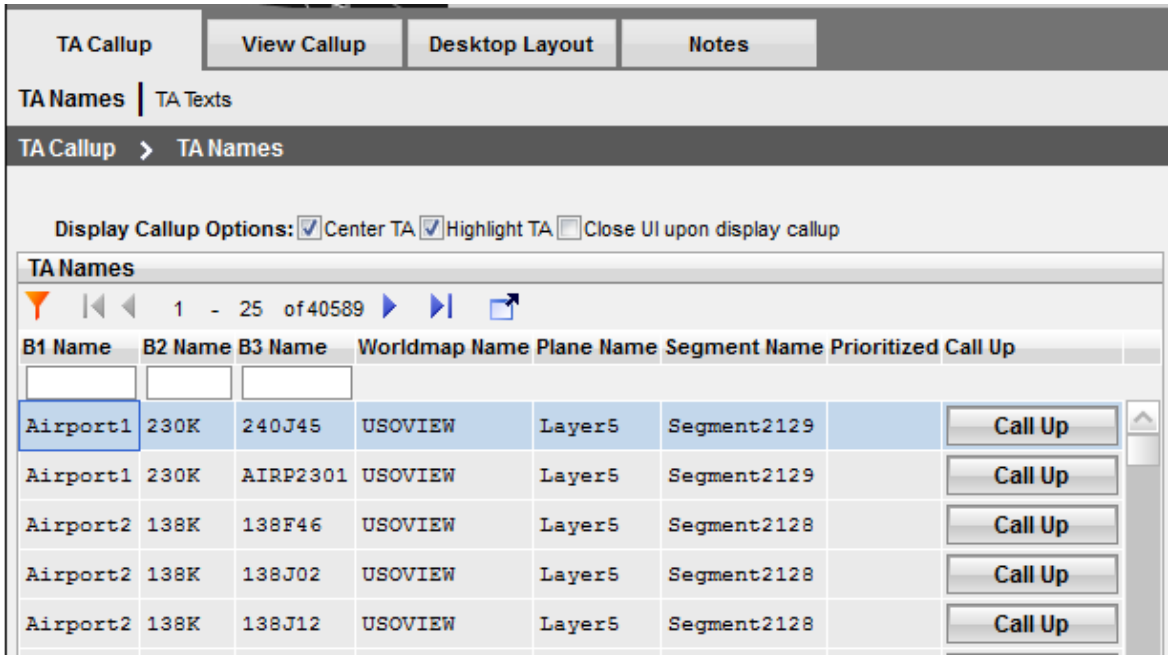
11.2.1.3 Selection Tabs

Following is the list of Selection tabs:

- TA Callup
- View Callup
- Desktop Layout
- Notes

11.2.2 TA Callup Tab

11.2.2.1 Overview



[sc_TACallup, 3, en_US]

Figure 11-1 List of Displays TA Names - TA Callup

11.2.2.2 Display Call-up Section

Selection Type

Following are the Selection types:

- TA Names
- TA Texts

Display Call-up Options

Following are the display call-up options:

- **Center TA**

  The selected TA is displayed in the center of the worldmap.

- **Highlight TA**

  The selected TA blinks in the worldmap.

- **Close UI upon display call-up**

  List of displays is closed after worldmap is opened.

Toolbar

Following are the toolbar options:

- 
  Restore Sort
- 
  Filter opens the in-line filter line where you can enter a filter to reduce the amount of displayed TAs. The filter line provides input fields for all columns that can be used as filter. Click this icon again to close the filter line. By default, the filter is enabled.
- 

  Buttons to page to the first, previous, next, and last page.
- 
  Opens the TA table in a new window.

List of TAs

TA Name Tab

Following is the list of column names in the TA Names tab:

- B1 Name
- B2 Name
- B3 Name

  Identification of the TA.
- Worldmap Name

  Name of the worldmap containing the TA.
- Plane Name

  Name of the plane containing the TA.
- Segment Name

  In case of several segments within a plane, this specifies the name of the segment containing the TA.
- Prioritized

  If this flag is set for a plane, in case of multiple results of TA Callup (for example, during drag and drop of a TA into a network diagram), this plane is used.
- Call Up

  Button to call up the corresponding display.

TA Text Tab

Following is the list of TA Text tab:

- B1 Text
- B2 Text
- B3 Text

Identification of the TA.

- Worldmap Name

  Name of the worldmap containing the TA.

- Plane Name

  Name of the plane containing the TA.

- Segment Name

  In case of several segments within a plane, this specifies the name of the segment containing the TA.

- Prioritized

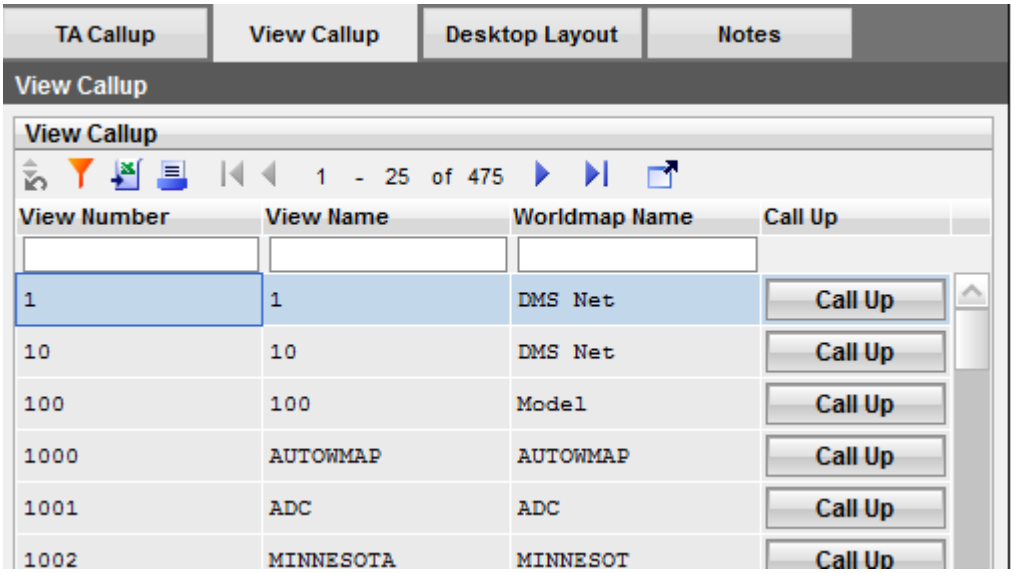
  If this flag is set for a plane, in case of multiple results of TA Callup (for example, during drag and drop of a TA into a network diagram) this plane is used.

- Call Up

  Button to call up the corresponding display.

11.2.3 View Callup Tab

11.2.3.1 Overview



[sc_ViewCallup, 3, en_US]

Figure 11-2 List of Displays TA - View Callup

11.2.3.2 Toolbar

Toolbar has the following options:

- 

  Restore Sort

- 

Filter Icon opens the Inline Filter line where you can enter a filter to reduce the amount of displayed TAs. The filter line provides input fields for all columns that can be used as filter. Clicking this icon again closes the filter line. Filter is enabled by default.

- 

  Buttons to page to the first, previous, next, and last page.

- 

  Opens the table in a new window.

11.2.3.3 List of Views

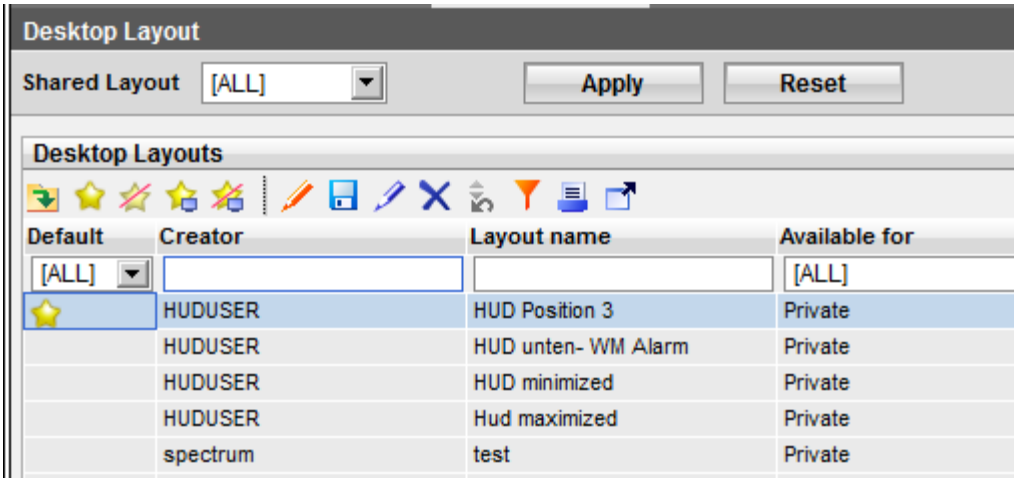
Following is the list of Views:

- View Number
- View Name
- Worldmap Name
- Call Up

  Button to call-up the corresponding display.

11.2.4 Desktop Layout Tab

11.2.4.1 Overview



[sc_DLM, 4, en_US]

Figure 11-3 List of Displays - Desktop Layout

11.2.4.2 Toolbar

Following are the toolbar options:

- 

  Open desktop layout

- 
  Set the desktop layout as default layout
- 
  Remove the default desktop layout
- 
  Set the desktop layout as system default desktop layout
- 
  Remove system default desktop layout.
- 
  Adds a new layout
- 
  Saves desktop layout
- 
  Edit desktop layout
- 
  Delete desktop layout
- 
  Restore sort order to default state
- 
  Filter opens the Inline Filter line where you can enter a filter to reduce the amount of displayed desktop layouts. The filter line provides input fields for all columns that can be used as filter. Clicking this icon again closes the filter line. Filter is enabled by default .
- 
  Print
- 
  Open in new window

11.2.4.3 Table of Desktop Layouts

Desktop layouts table has the following options:

- Shared Layout

  Filters the list of desktop layouts that are available for each user group.
- Default

  Specifies whether the corresponding layout is the default layout.
- Creator

Specifies the name of user who created the layout.

- Layout name

  Name of the layout (defined by the user when creating the layout).

- Available for

  Specifies if the Desktop layout is public (for all users), private (for the currently logged in user) or for a specific user group.

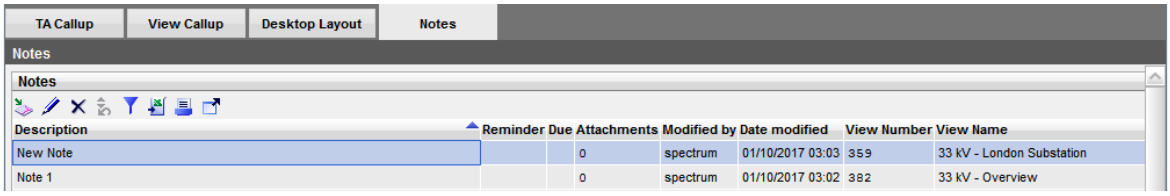
- System default

  Specifies whether the corresponding layout is the current system default layout.

11.2.5 Notes Tab

11.2.5.1 Overview

The notes display shows the notes that are created in a network diagram using context menu item **add note**.The notes can be opened in their corresponding network diagram.



[sc_notes_new, 3, en_US]

Figure 11-4 List of Displays TA Names - Free Notes

11.2.5.2 Toolbar

Toolbar has the following options:

- 

  View on Map

- 

  Edit note

- 

  Delete note

- 

  Restore Sort

- 

  Filter opens the Inline Filter line where you can enter a filter to reduce the amount of displayed TAs. The filter line provides input fields for all columns that can be used as filter. Click this icon again to close the filter line.

- 

  Open note in new window

11.2.5.3 List of Notes

Following is the list of Notes:

- Reminder

  If a reminder for the note is set

- Due

  If a reminder is still due

- Attachments

  Number of attachments added

- Modified by

  The operator who modified the note last

- Date Modified

  The date when the note was last modified

- View Number

  Number of the view where the note is located

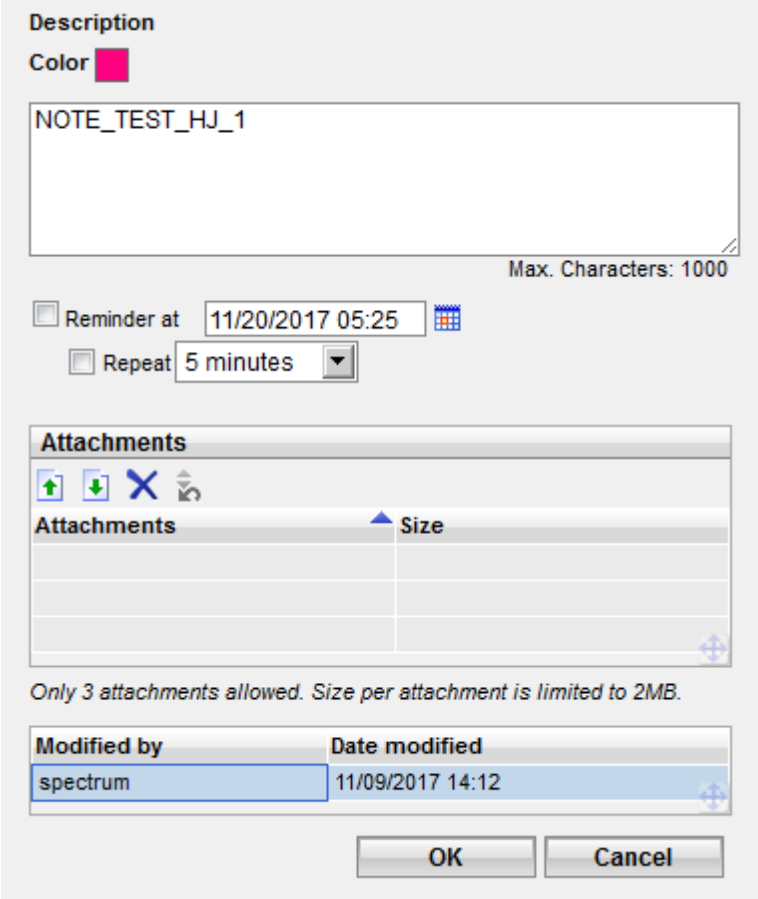
- View Name

  Name of the view where the note is located

11.2.5.4 Notes Dialog

For adding or editing a note, you can specify the following options:

- Font color
- Description
- Reminder
- Attachments:
  - Upload: opens a dialog where you can pick a file that shall be uploaded
  - Download
  - Delete: Removes attachment from the note
  - Reset sort order: Sets sort order of columns back to default



[sc_notesdialog, 1, en_US]

Figure 11-5 Example for Notes Dialog

NOTE

You can upload up to three attachments, each having a maximum size of 2MB .

11.3 Operational Steps

11.3.1 Calling Up a Display

To call-up a display, perform the following:

- Select the **TA Callup** tab of the List of display.
- In the Display Callup Options section, click Center TA and Highlight TA.
- Click the Call Up button of the respective line of the TA Names table.

  The network diagram containing the selected TA is opened with the TA displayed in the center of the display and is flashing.

  The operator can filter COLD displays by a copied TA. The copied TA can be pasted into filter area using keyboard, context menu, or drag and drop.

Pasting TA using keyboard

- Right click on an element (for example, circuit breaker) in a network diagram and select the TA name on context menu header to add TAinfo to clipboard.
- In a COLD display, click any inline filter (TA column) and press CTRL+SHIFT+V to paste TA.

Pasting TA using context menu

- Right click on an element (for example, circuit breaker) in a network diagram and select the TA name on context menu header to add TAinfo to clipboard.
- In a COLD display, right click on any inline filter and choose 'Paste TA'.

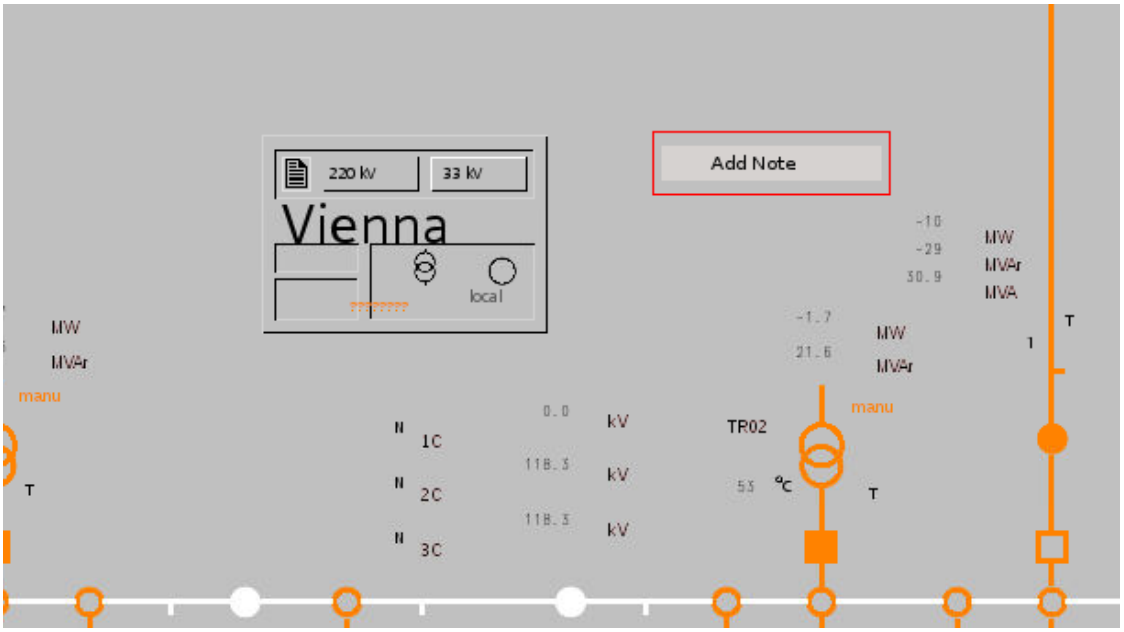
Pasting TA using drag and drop

- Click and drag an element (for example, circuit breaker) in a network diagram.
- Drop the element on COLD display. Inline filter gets populated with appropriate values for each column (B1, B2, B3, and so on).

11.3.2 Adding a Note to a Network Diagram

To add a note to a network diagram, perform the following steps:

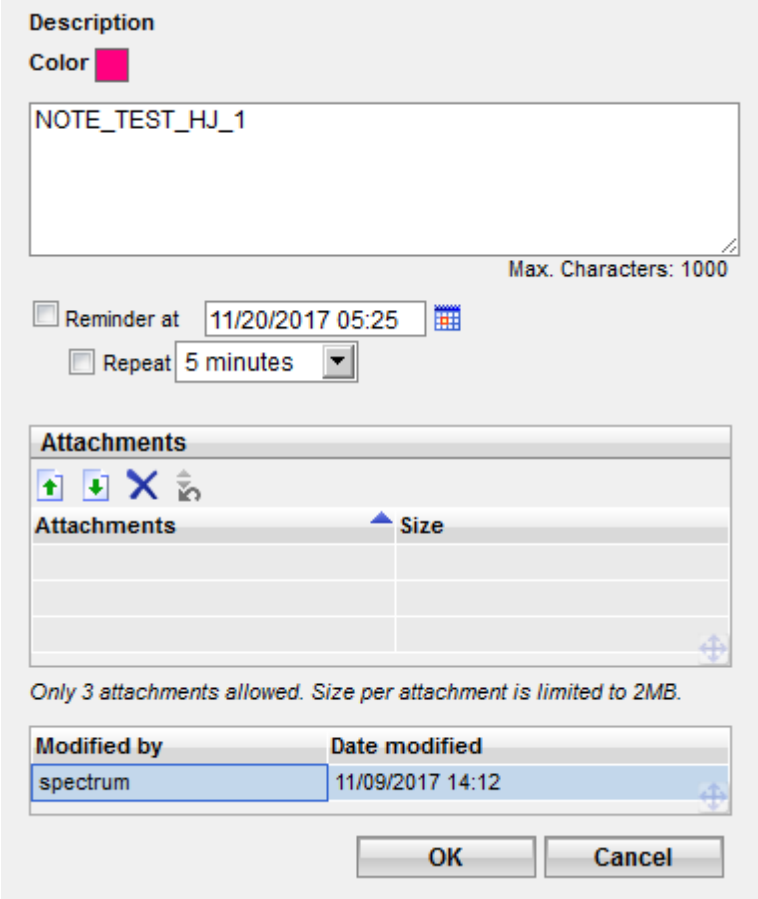
- Open the desired network diagram, right-click at the place in the network diagram where you want to add the note and select the item **Add Note** from the context menu.



[sc_addnote1, 1, en_US]

Figure 11-6 Add Note

The Note Editor for entering a text for the note, setting a reminder, and adding attachments opens.

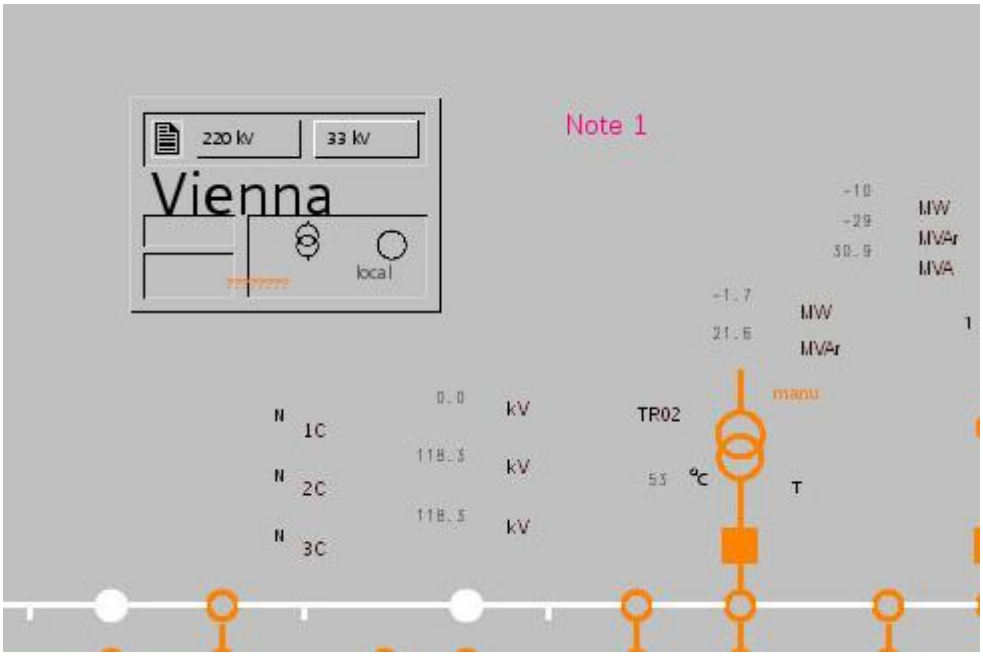


[sc_addnote, 2, en_US]

Figure 11-7 Note Editor

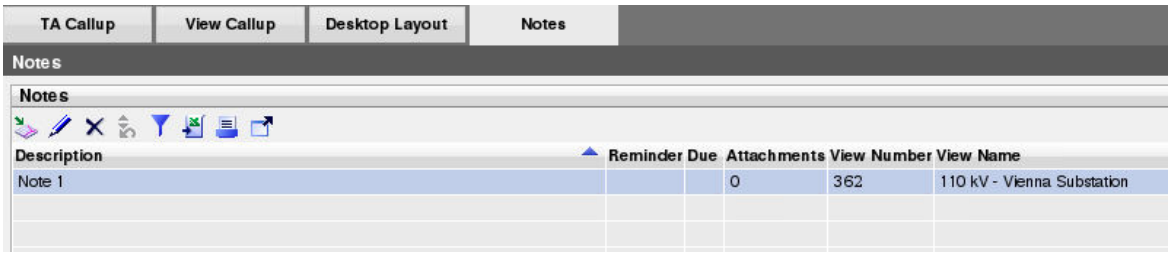
- Enter the desired data and click **OK**.

  The note is displayed in the network diagram and in the Notes tab of the List of Displays.



[sc_addnote3, 1, en_US]

Figure 11-8 Note Display in Network Diagram



[sc_addnote4, 2, en_US]

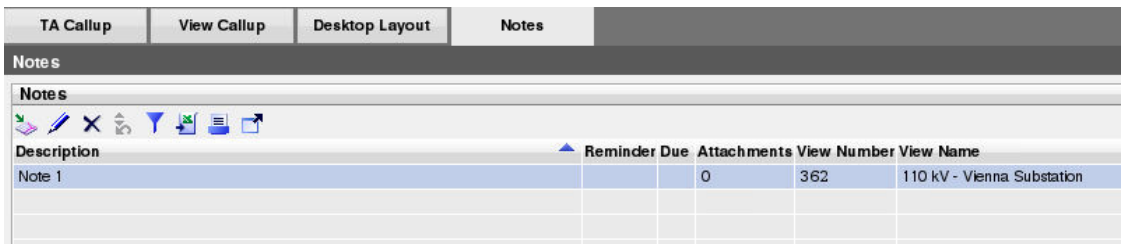
Figure 11-9 Note Display Tab

11.3.3 Displaying Notes and Opening the Corresponding Network Diagram

To display Notes and open the corresponding network diagram, perform the following:

- Open the List of displays and select the Notes tab.

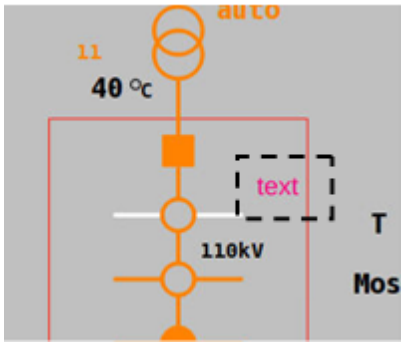
  The table shows one row for each note.



[sc_addnote4, 2, en_US]

- Select one row and click the view on map icon .

The corresponding network diagram displaying the note is shown. Upon mouse-over, the note is displayed in a tooltip.

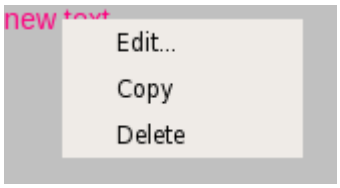


[ugsca022-310113-01.tif, 1, en_US]

Figure 11-10 List of Displays TA - Free Notes on Network Diagram

- For displaying the entire note text, right-click the note.

  A context menu is opened.



[sc_ContextMenufoNotes, 1, en_US]

Figure 11-11 Context Menu for Notes

- Select **Edit...** in the context menu.

  A display showing the complete note text is opened.

You can perform the following using Notes:

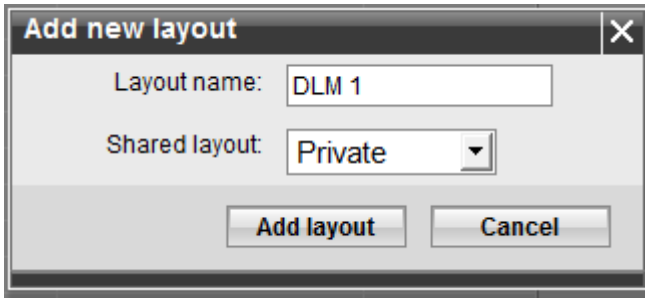
- Pick color for the note font.
- Add the note text. The note text is limited to 1000 characters.
- Set a reminder for the free note, the reminder can be recurring.
- Add upto 3 attachments, each attachment limited to 2 MB for a note.
- Check the operator and time that the note was last modified.

11.3.4 Defining a Desktop Layout

To define a desktop layout, perform the following:

- Click  icon in the toolbar.
- The **Add New Layout** dialog opens.

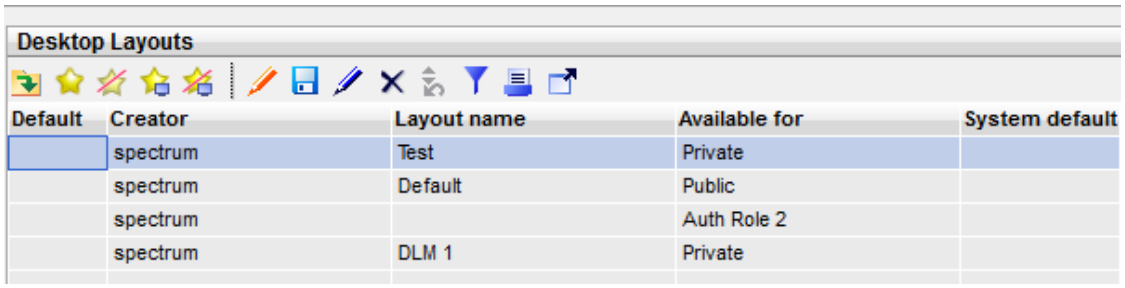
  Enter the Layout name and select the user group that the layout has to be shared with from the **Shared Layout** drop down list. Select **Private** to not share the layout.



[sc_newlayout, 2, en_US]

Figure 11-12 List of Displays TA - Add New Layout

- Click **Add Layout**.
- The layout is added as shown.



[sc_Desktoplayout, 2, en_US]

Figure 11-13 List of Displays TA - Create New Layout

11.3.5 Setting a Desktop Layout as the Default Layout

To set a desktop layout as a default layout, perform the following:

- Select the Layout that you want to set as the default Layout.
- Click the Icon.

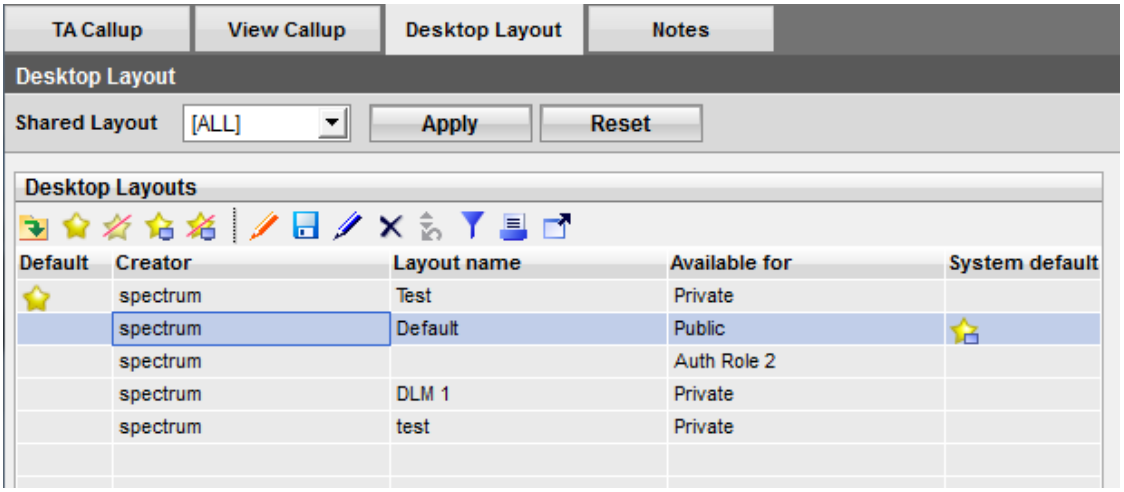
  A is displayed in the **Default** column. When you log on the next time, this layout is displayed.

11.3.6 Setting a Desktop Layout as the System Default Layout

The system administrator can set a system default desktop layout. To set a system default layout:

- Select the Layout that you want to set as the default Layout.
- Click the icon.

  The desktop layout is set as the system default layout.



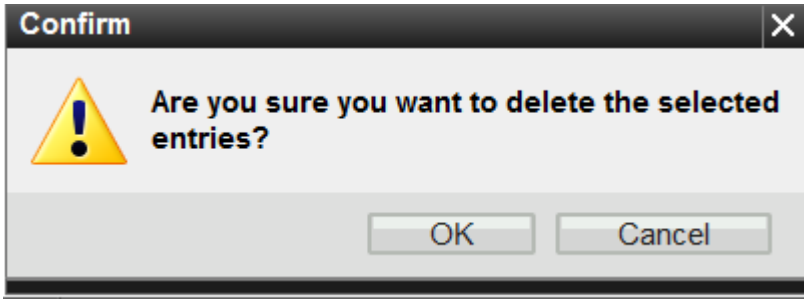
[ic_defaultDLM, 2, --_--]

Figure 11-14 Default Desktop Layout

11.3.7 Deleting a Desktop Layout

To delete a desktop layout, perform the following:

- Select the desktop layout that you want to delete.
- Click the ✕ icon in the toolbar.
- Click **Ok** to confirm deletion.



[Sc_deleteconfirmation, 1, en_US]

The corresponding desktop layouts are deleted.

12 Calculated Points Display

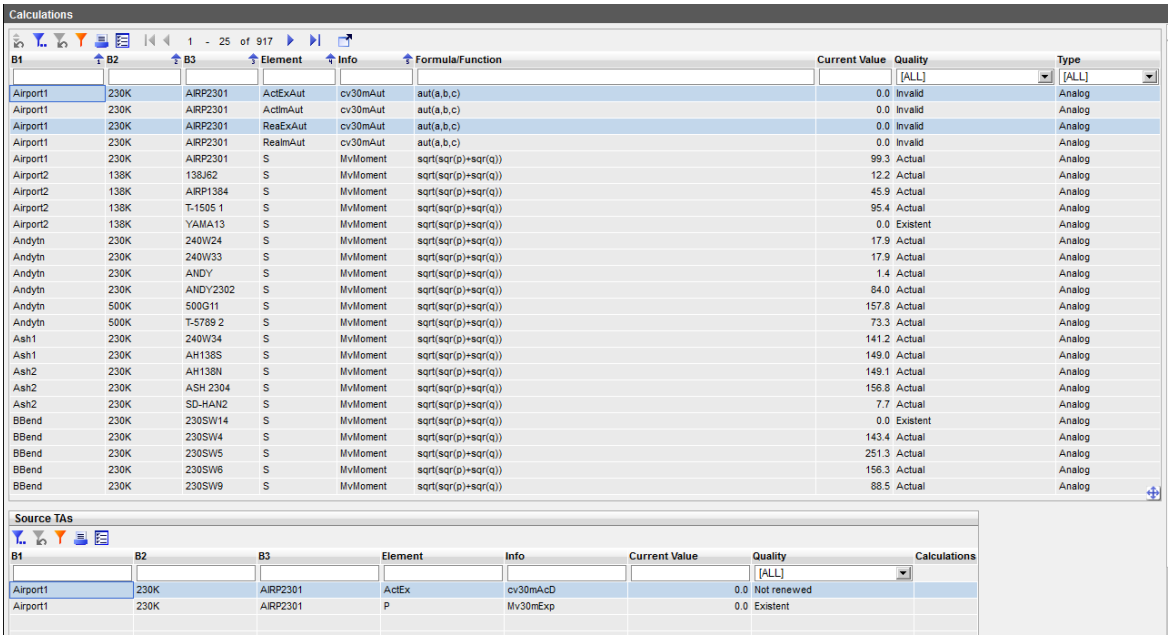
12.1	Calculated Points Display	172
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12.1 Calculated Points Display

The Calculated Points display lists all data points calculated by formulas with their source TAs. Click the **Calculations** button under the **Curves and tables** tab in the BaSiDi and HUD to open Calculated Points Display.

12.2 User Interface

12.2.1 Overview



[sc_Calculations, 1, en_US]

Figure 12-1 Calculated Points

12.2.2 Title Bar

Title bar icons are used to navigate to the previous and next display, refresh the display, print, and so on.

Table 12-1 Title Bar Icons

Icon	Description
	Opens the previous window.
	Opens the next window.
	Stops loading of data.
	Updates the table in the display.
	Zoom out
	Zoom in
	Reset zoom
	Displays general information about the display.

12.2.3 Calculated Values Table

12.2.3.1 Toolbar

The Toolbar provides the following icons:

Table 12-2 Title Bar Icons

Icon	Description
	Restore sort Restores the current sorting order to the default sorting.
	Filter Opens the Filter display for defining filter criteria.
	Reset filter Resets the current filter settings to the default filter.
	Inline filter Inserts a line at the top of the table for defining a filter, which is directly applied to the table content. Clicking this icon a second time switches off the filter mode again.
	Print Opens a display with options for what to print and where to print it.
	Adjust columns Hides/shows columns

12.2.3.2 Table

Table provides the following information:

- **B1**, **B2**, **B3**, **Elem**, **Info**: TA of the calculated value.
- **Formula**: Formula used to calculate this value.

- **Current value**: Current result of the calculation.
- **Quality**: Quality of the calculated value.
- **Type**: analog or digital

12.2.4 Source TA Table

Shows all TAs that are used as a source for the variables of the formula selected in the Calculated Values table.

- **B1**, **B2**, **B3**, **Elem**, **Info**

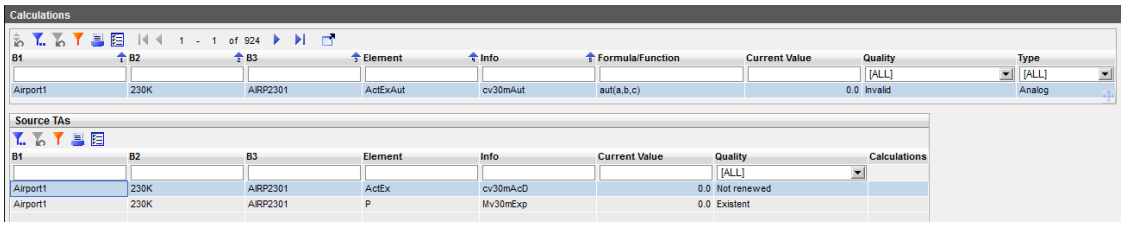
  TA of the source value.

- **Quality**

  Quality of the value

- **Calculations**

  If the Source TA is a calculated value, a click on the link Show in this column opens a new window showing the calculation and operands of this source TA:



[sc_calculations2, 3, en_US]

Figure 12-2 Calculations

12.3 Operational Steps

12.3.1 Displaying the Source TAs of a Calculated Value

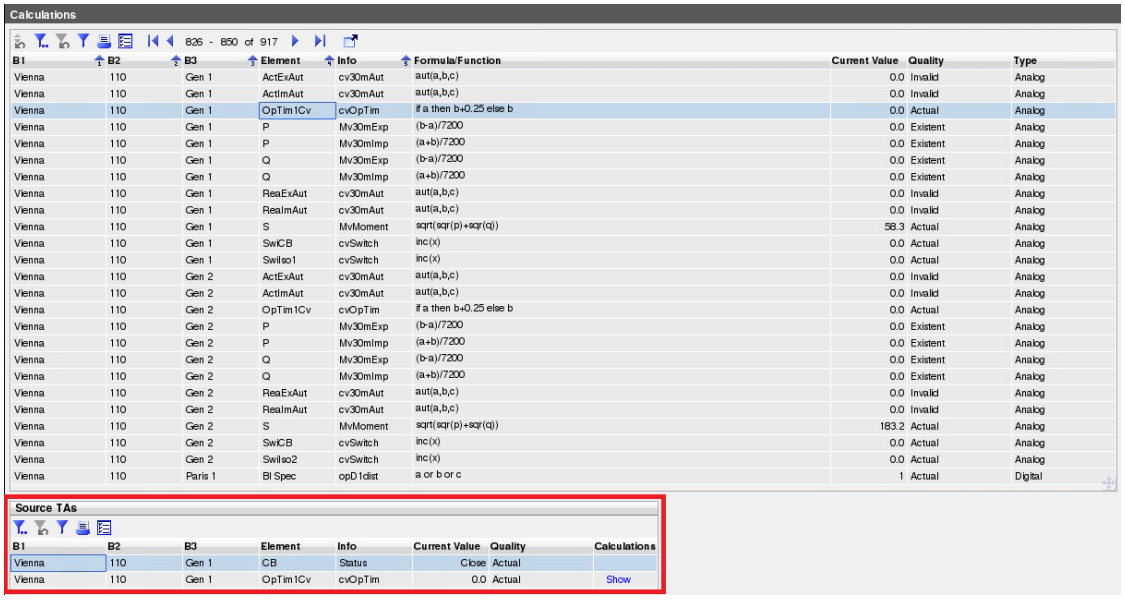
To display the source TAs of a calculated value, perform the following:

- Reduce the amount of data displayed in the Calculated Values table by setting filter criteria in the Filter section and click **Apply**.

  The calculated values corresponding to the filter criteria are displayed in the Calculated Values table.

- Click on a line in the Calculated Values table.

  The source values for the calculation of the selected value are displayed in the Source TAs table.



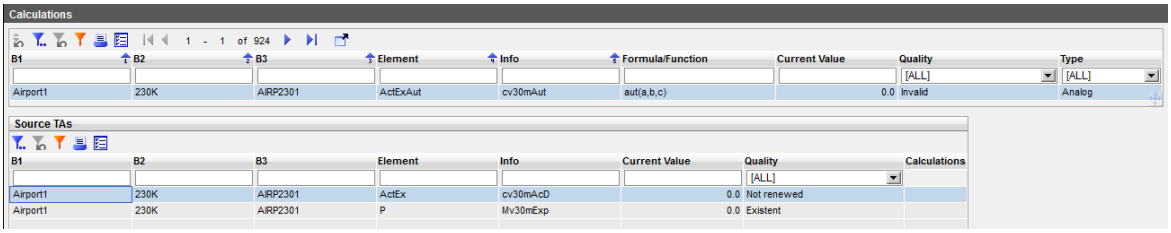
[sc_CalculationShow, 1, en_US]

Figure 12-3 Source TAs - Calculated Points

If one of the source TAs is a calculated value, perform the following to display the calculation of this Source TA:

- Click the link **Show** of the corresponding line in the Source TAs table.

The calculation and the source values of the selected value source TA are displayed in a new window.



[sc_calculations2, 3, en_US]

Figure 12-4 Calculated Points

13 System and Network Status Display

13.1	System Network Status Display	178
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13.1 System Network Status Display

The System Network Status Display provides information on the status of the system computers, control centers in the multisite system, and other related displays.

Click the corresponding button in the BaSiDi and HUD to open the System Network Status Display. The Computers tab in System Status is displayed.

13.2 User Interface

13.2.1 Title Bar

Title bar icons are used to navigate to the previous and next display, refresh the display, print, and so on.

Table 13-1 Title Bar Icons

Icons	Description
	Opens the previous window.
	Opens the next window.
	Stops loading of data.
	Updates the table in the display.
	Zoom out
	Zoom in
	Reset zoom
	Online Help
	Displays general information about the display.

13.2.2 Selection Tabs

Following is the list of Selection tabs:

- Computers
- Consoles
- SIDS
- Clocks
- NetIfs
- WebServers
- LDAP
- Multisite
- Multisite Data Responsibility
- SNMP Device Summary
- SNMP Point Summary

NOTE

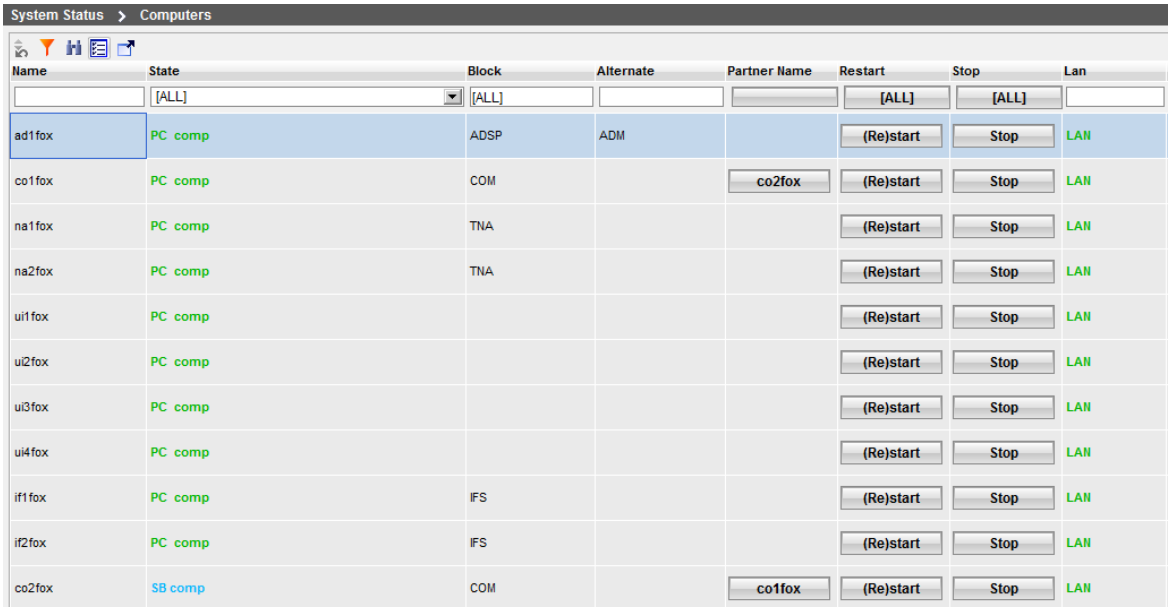
Depending on the availability of data, not all tables can be filled with data.

13.2.3 Toolbar

Depending on the selected tab and situation, the following icons may be available in the toolbar:

- Restore Sort: Restores the sort order of the table to the default sorting.
- Opens the Inline Filter line where you can enter a filter to reduce the amount of displayed TAs. The filter line provides input fields for all columns that can be used as filter. Clicking this icon again closes the filter line.
- Opens a line for entering search criteria.
- Hides/shows columns
- Opens note in new window

13.2.4 Computers



[sc_SystemStatusComputers_12032014, 6, en_US]

Figure 13-1 System Status - Computers

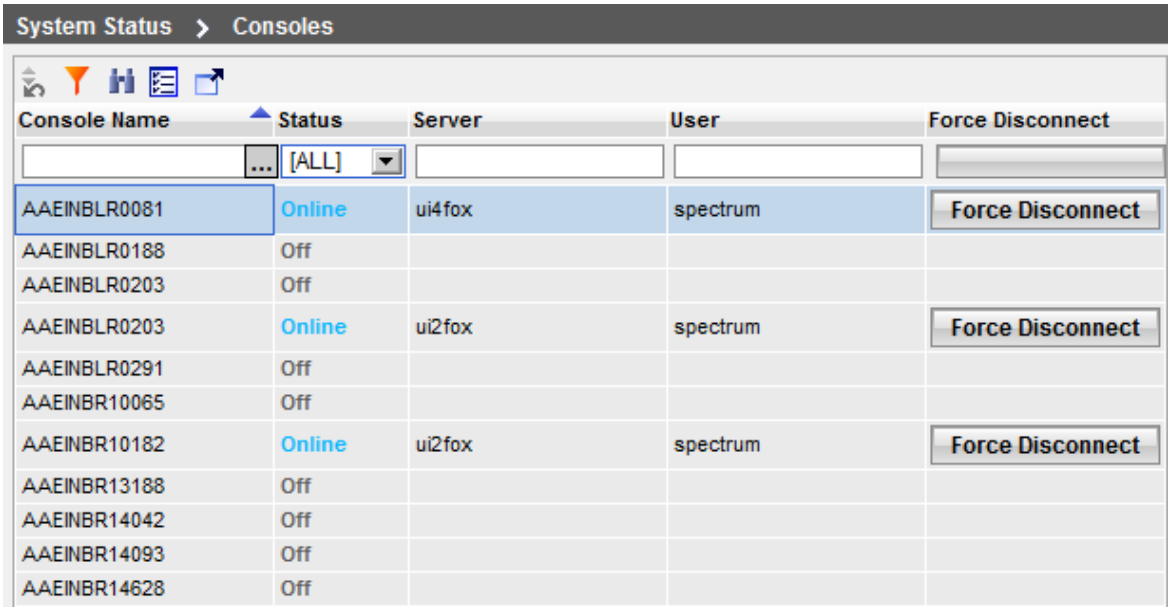
This tab provides a list of all the computers with the following information:

- The state of each computer.
- The blockfunction of each computer.
- The alternate blockfunction.
- Buttons for the user to **Restart** and **Stop** the individual computer.
- The Partner Name
- The LAN connections of each computer.

NOTE

The **(Re)start** and **Stop** buttons are disabled if the permission for INITSOS is not given to the operator.

13.2.5 Consoles



[sc_SystemStatusConsoles_12032014, 6, en_US]

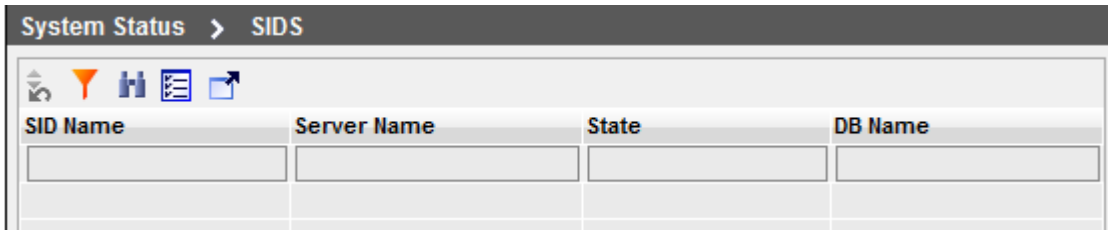
Figure 13-2 System Status - Consoles

This tab provides a list of all consoles logged in to the system with the following:

- The name of each console.
- The status (Off, Online, and so on) of each console.
- The name of the computer associated with the console.

The Force Disconnect button is deactivated.

13.2.6 SIDS



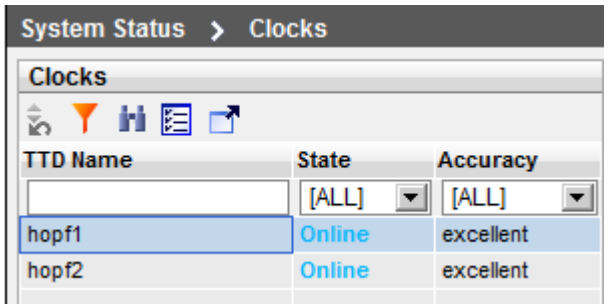
[sc_SystemStatusSIDS, 6, en_US]

Figure 13-3 System Status - SIDS

This tab provides a list of the following:

- The name of each System ID (SID).
- The server name.
- The state of each SID.
- The data base name of the SID.

13.2.7 Clocks



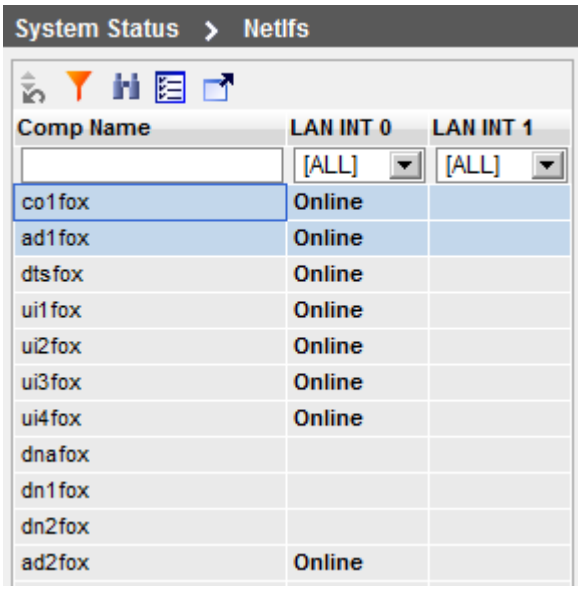
[sc_SystemStatusClocks, 6, en_US]

Figure 13-4 System Status - Clocks

This tab provides a list of all clocks with the following:

- On or Off status of the external clocks used to synchronize the system time.
- Accuracy.

13.2.8 NetIfs



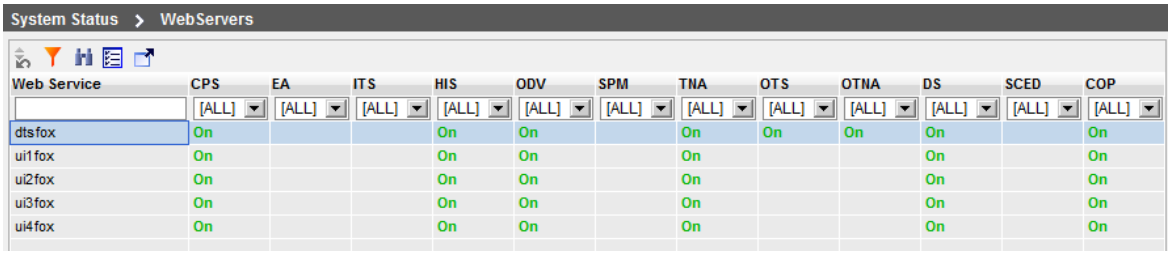
[sc_SystemStatusNetIfs_12032014, 5, en_US]

Figure 13-5 System Status - NetIfs

This tab provides a list of all network interfaces with the following:

- Computer name.
- Status of both interfaces to LAN A and both interfaces to LAN B.

13.2.9 WebServers



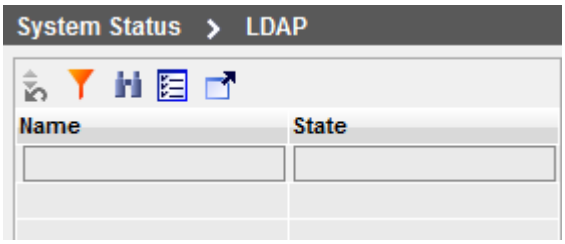
[sc_SystemStatusWebServers_12032014, 5, en_US]

Figure 13-6 System Status - WebServers

This tab provides a list of all JBOSS base WebUI webservers with the following:

- The Webserver name.
- Status of each Web application.

13.2.10 LDAP



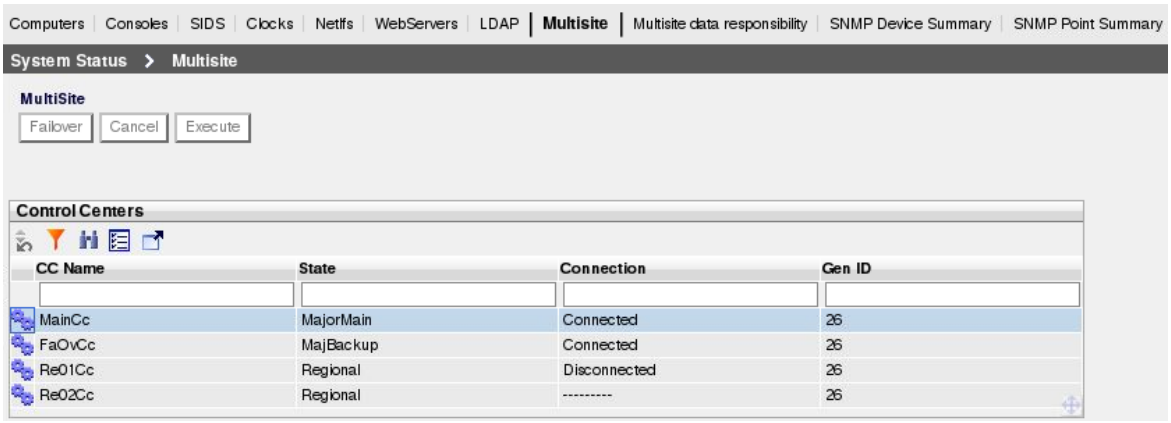
[sc_SystemStatusLDAP_12032014, 5, en_US]

Figure 13-7 System Status - LDAP

This tab provides a list of all LDAP servers with the following:

- Name
- Status

13.2.11 Multisite



[sc_SystemStatusMultisite_12032014, 5, en_US]

Figure 13-8 System Status - Multisite

This tab shows all control centers in the multisite system with information on state, connection status, and database generation number. When selecting a control center, the connection status fields of the other control centers indicate whether they have a connection to the selected control center.

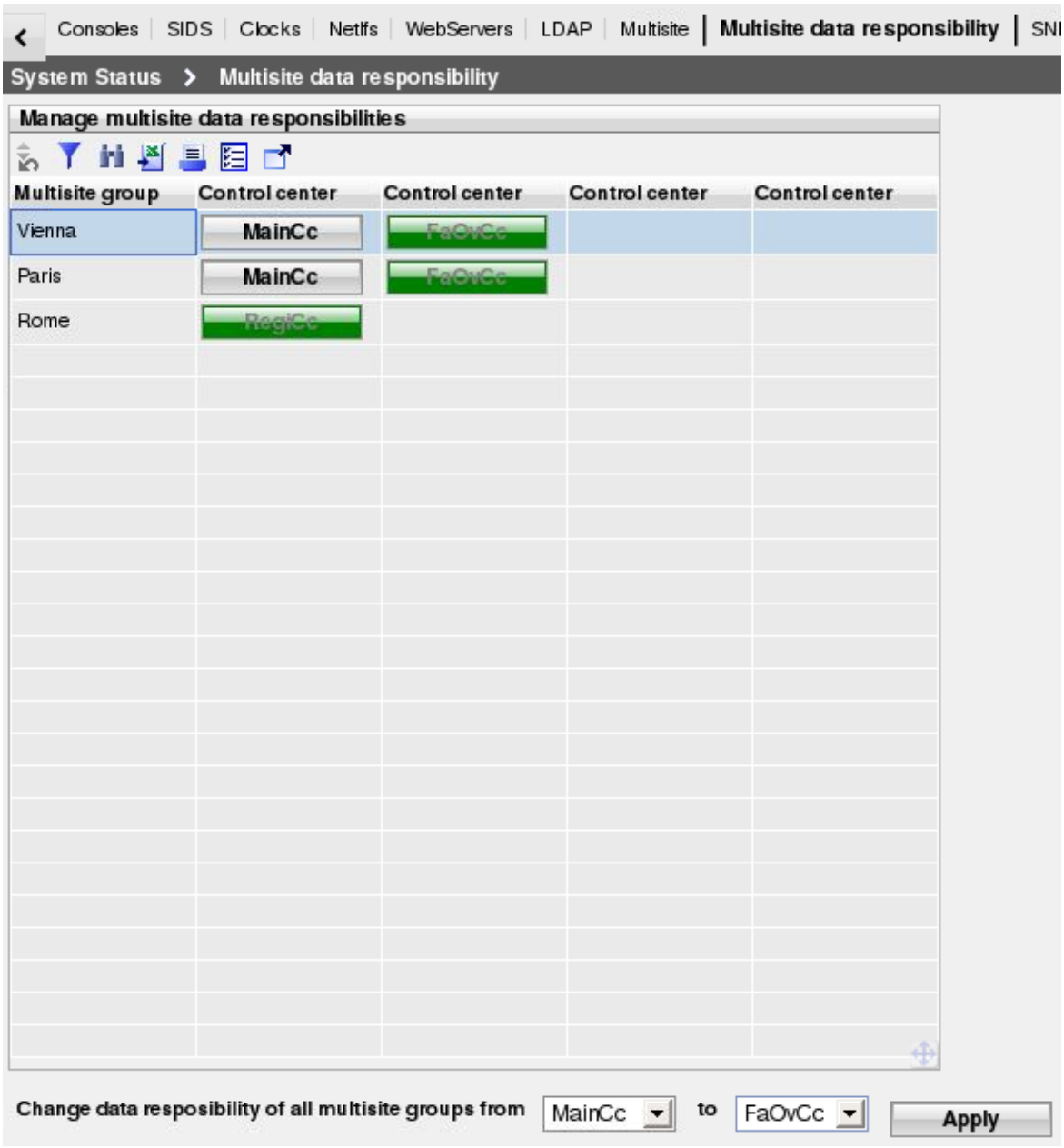
Additionally this tab provides a button failover that allows the current control center to take over data responsibility. This operation is only possible if the current control center is a backup control center.

After the button Failover has been clicked, buttons Cancel and Execute allow to start or cancel the taking over of data responsibility. If the database generation numbers are not identical, a message is displayed and it has to be confirmed whether to start failover though the database generation numbers are not identical, or whether to cancel the failover.

NOTE

The **Failover** button is disabled if the permission for Failover is not given to the operator. The system also checks if the multisite system allows failover.

13.2.12 Multisite Data Responsibility



[sc_MultisiteDataResponsibility, 1, en_US]

Figure 13-9 Multisite Data Responsibility

This table provides the following items:

- Multisite group

List of multisite groups as defined in IMM

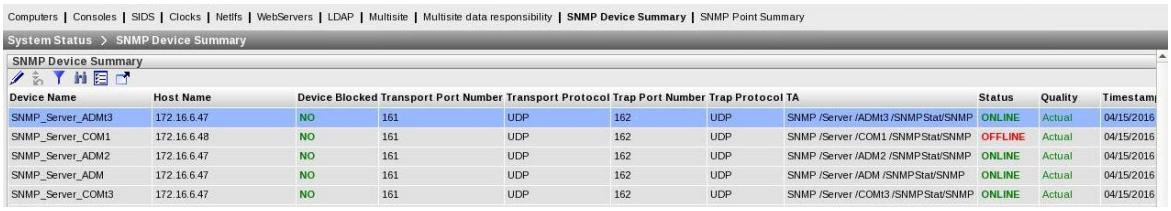
- Control Center

  Shows the control centers with data responsibility for the individual groups as defined in IMM; the current CC with data responsibility for this group is shown in green color; clicking on any other Control Center field in this row switches over data responsibility for this group to the respective control center

Using the line below the table allows you to change the data responsibility for all groups from one CC to another (selectable from the drop-down list.)

13.2.13 SNMP Device Summary

This tab provides a list of all SNMP devices along with their status:

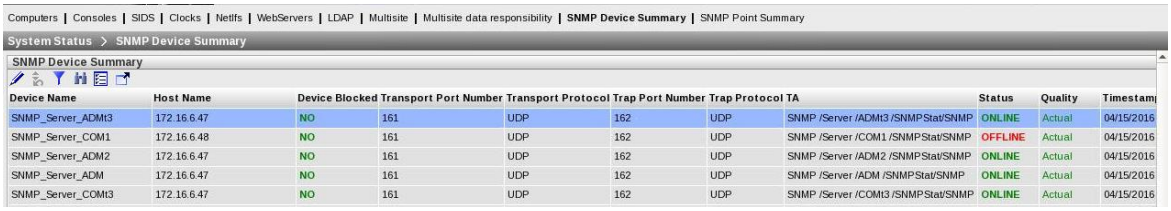


[sc_cnstat-snmpdevicesummary, 2, en_US]

Figure 13-10 SNMP Device Summary

13.2.14 SNMP Point Summary

This tab shows the configured point values for selected SNMP device:



[sc_cnstat-snmpdevicesummary, 2, en_US]

Figure 13-11 SNMP Point Summary

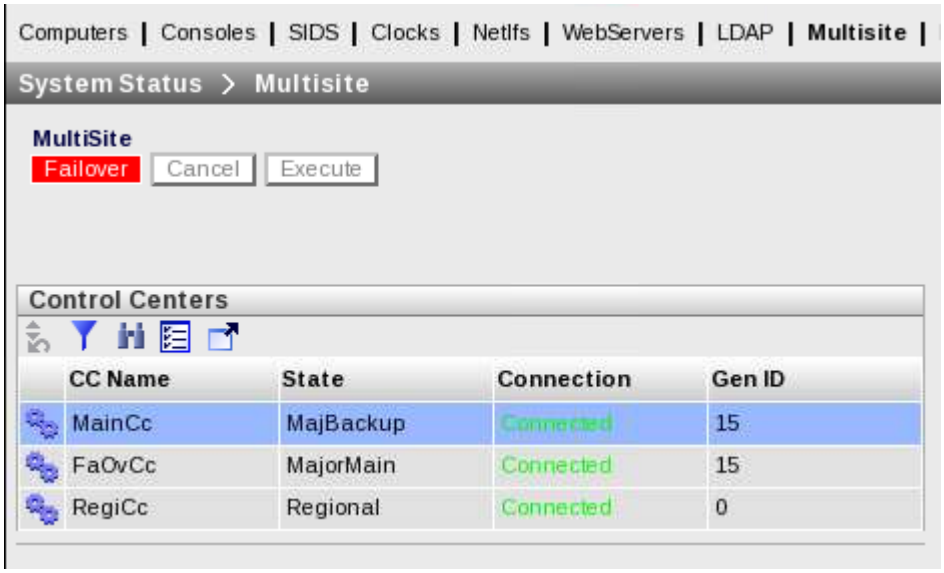
13.3 Operational Steps

13.3.1 Displaying the Connection Status of Control Centers in a Multisite Network

To display the connection status of control centers in a multisite network, perform the following:

- Select the **Multisite** tab.
- In the Control Centers table, click the line of a control center.

  The Connection field of all other control centers in the table shows whether they have a connection to the selected control center.



[sc_multisite, 3, en_US]

Figure 13-12 Connection Status of Control Centers in a Multisite Network

14 TA-Related Information Displays

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14.1 Introduction

The TA-related display provides the following features:

- Displaying information for a TA (Technological Address).
- A name (short name), text (long name) (as configured in IMM) or ID (numeric) can be displayed for each TA.
- Viewing the status of alarms and flags, activating, or deactivating the processing of individual alarms for a TA.
- Setting, displaying, editing, and deleting tags.
- Modifying the limits of a TA.
- Modifying the handling of redundant data source.
- Setting, displaying, editing, and deleting the tags Permit to Work and Permit to Test.

These displays can be opened by selecting the corresponding item of the element context menu in a network diagram.

14.2 User Interface

14.2.1 Title Bar

Title bar icons are used to navigate to the previous and next display, refresh the display, print, and so on.

Table 14-1 Title Bar Icons

Icon	Description
	Opens the previous window.
	Opens the next window.
	Stops loading of data.
	Updates the table in the display.
	Zoom out
	Zoom in
	Reset zoom
	Displays general information about the display.

14.2.2 TA Selection Area

This section shows the TA, for which the information is displayed. Some information can be directly changed for the selected TA directly in the display. For these displays, besides the TA fields, the following buttons are available:

- **Call Up**

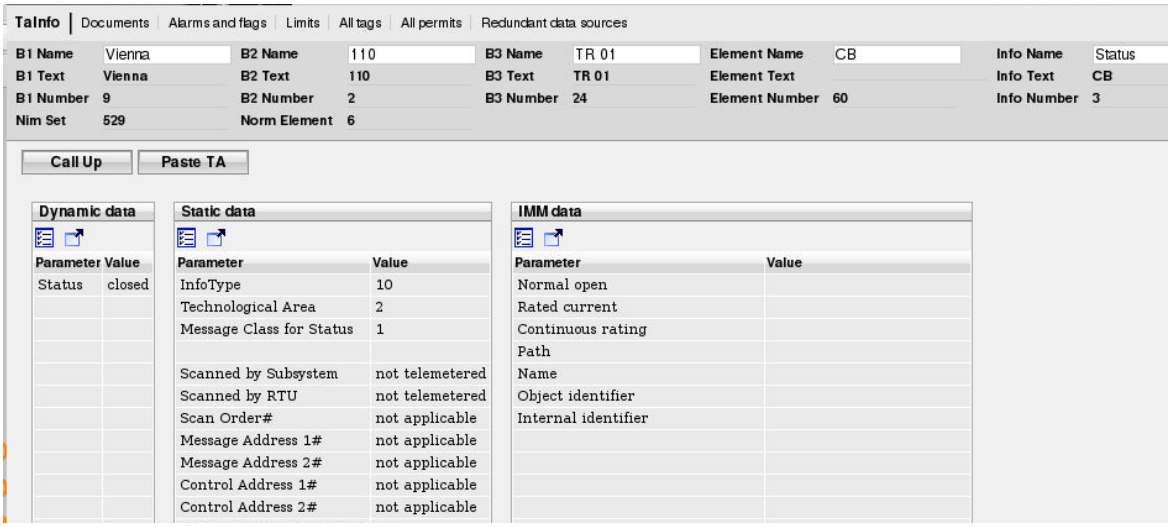
  This button opens the network diagram with the selected element.

- **Paste TA**

  Inserts the TA that is copied into the clipboard by using the **Copy TA** feature, for example, in the worldmap.

14.2.3 TAInfo Display

14.2.3.1 Overview



[sc_tainfodisplay, 4, en_US]

Figure 14-1 TAInfo

Dynamic Data

This section shows dynamic information for the selected TA. The data in this table are continuously updated while the display is open.

Static Data

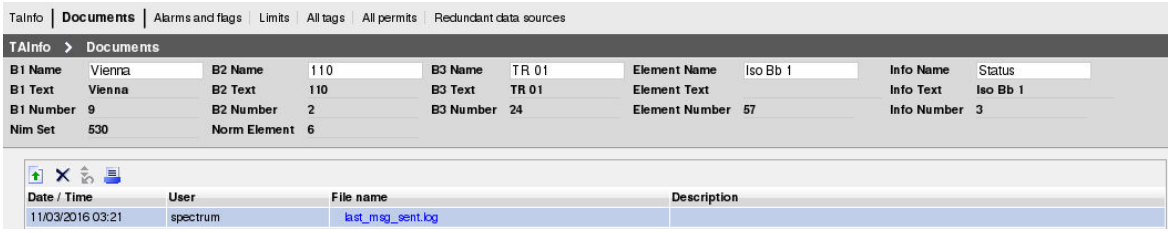
This section shows static information for the selected TA.

IMM Data

This section shows IMM data for the selected TA.

14.2.4 Documents

The Documents tab provides a list of uploaded documents. If a document is uploaded for a TA, it is listed for the substation to which the element or TA belongs. These documents can be opened using a mouse click, additional documents can also be uploaded.



[sc_tainfo-documents, 2, en_US]

Figure 14-2 Documents

- Date/Time

  Date and time when the document has been uploaded

- User

  User who uploaded the document

- File name

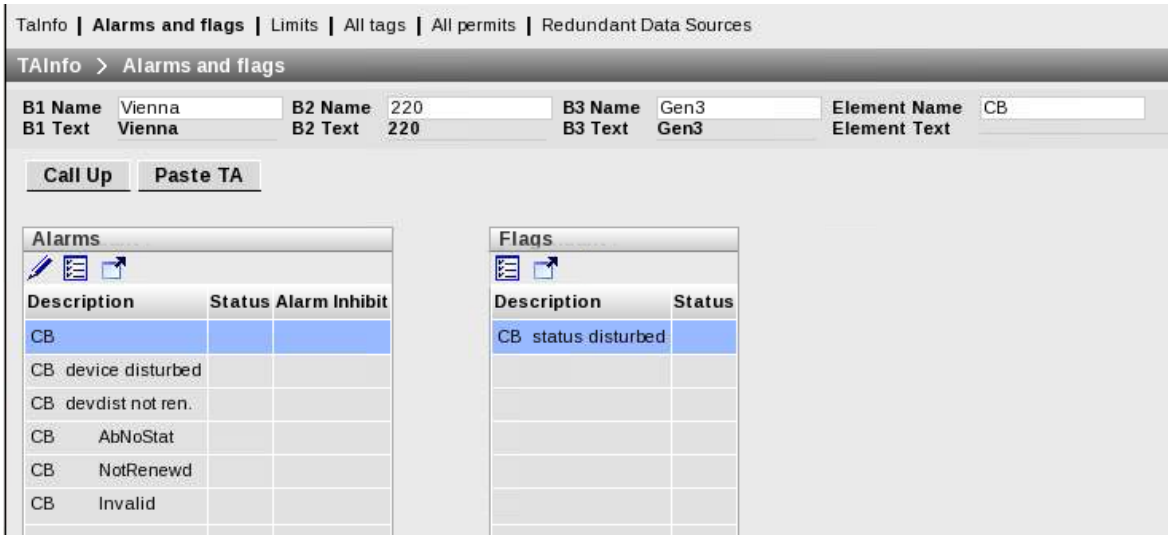
  File name of the uploaded document. Clicking the file name opens the respective document.

- Description

  Optional description of the document

14.2.5 Alarms and Flags Display

14.2.5.1 Overview



[sc_alarmsandflags, 3, en_US]

Figure 14-3 Alarms and Flags

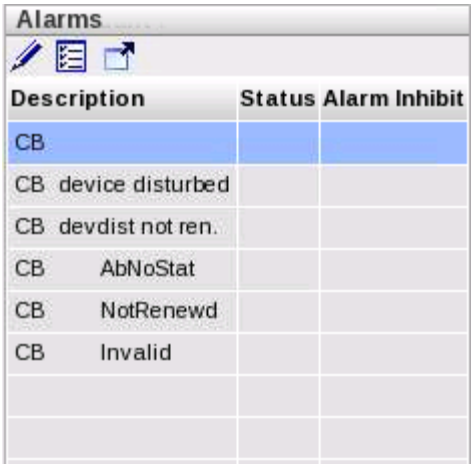
14.2.5.2 Alarms Table

Toolbar

Toolbar has the following options:

- 

  Open in new Window.



[sc_Alarms, 1, en_US]

Figure 14-4 TAInfo - Alarms and Flags Edit

- 

  Edit Alarm

- 

  Adjust Columns

Table

Table has the following options:

- Description

  Name of flag.

- Status

  State of alarm: To check whether limit is currently violated.

- Alarm inhibit

  Whether limit violations are processed.

14.2.5.3 Flags Table

Toolbar

Toolbar has the following options:

- 

  Prints the table

- 

  Adjust Columns

- 

  Opens the table in a new window.

Table

Table has the following information:

- Description

  Name of flag.

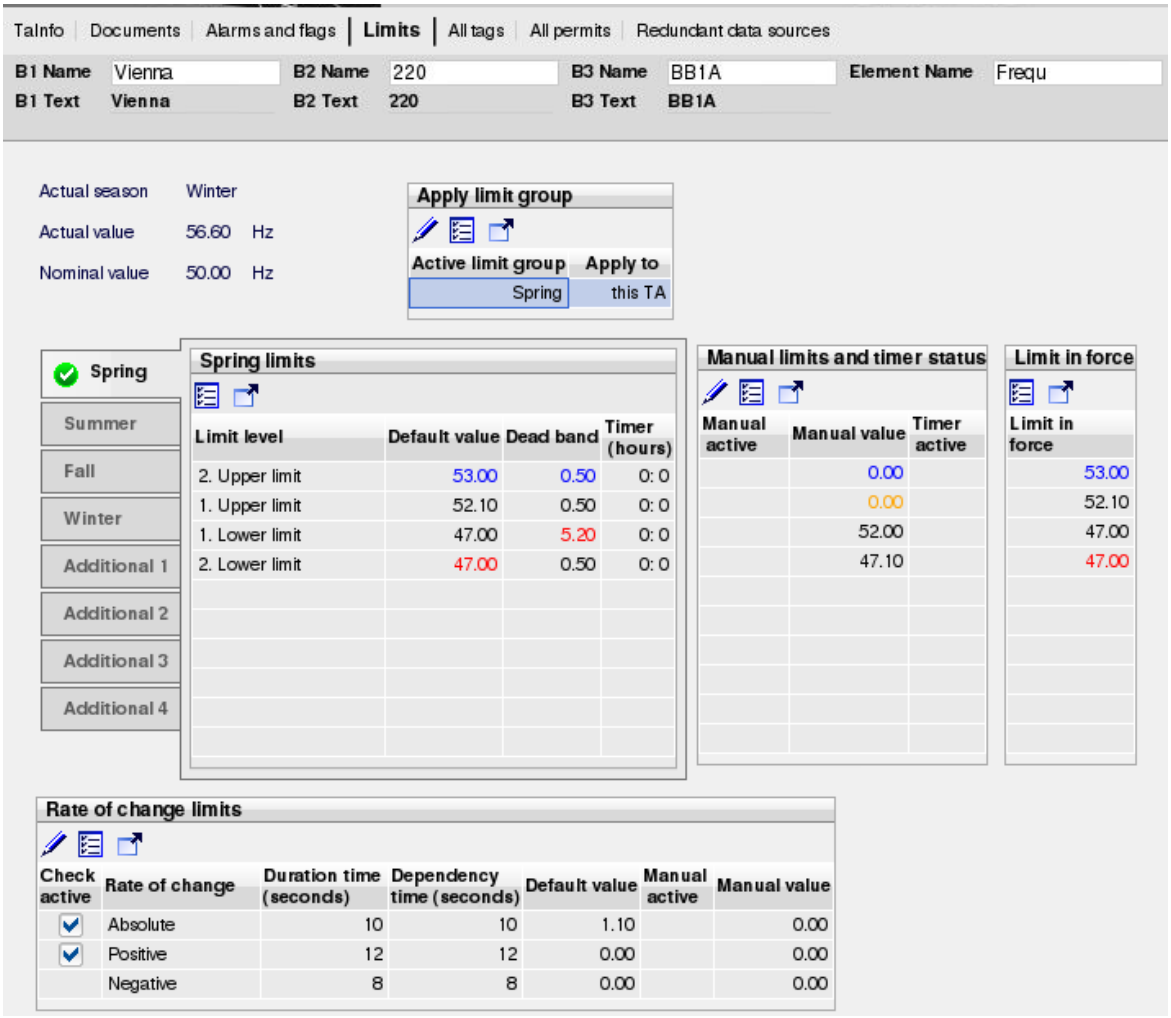
- Status

  When selected (hook is displayed), the flag is set, otherwise the field is empty

14.2.6 Limits Display

14.2.6.1 Overview

This display allows the operator to select the limit group to be used for the selected TA, to enter manual limits and to enable/disable alarm timers for limits.



[sc_limits, 6, en_US]

Figure 14-5 Limits

14.2.6.2 Limits Section

Current Settings

- Actual season

  Current season

- Actual value

  Current value of the TA

- Nominal value

  Nominal value of the TA

Apply Limit Group

- Active limit group

The user selects the limit group that is used.

- Apply to

  The user selects whether the settings shall be applied to the selected TA, all TAs of the same type, or all analog values.

Limit Group Tabs

For each limit group, a tab is provided. Selecting the corresponding tab shows a table with the default limit values and the alarm timer setting for a maximum of five upper and five lower limits. For each limit the following information is displayed:

- <Limits Group> limits:
  - Limit level

    Type of the limit
  - Default value

    Shows the default value of the limit.
  - Deadband

    Deadband of the limit
  - Timer

    Shows the timer value after which the alarm is issued.

Manual Limits and Timer Status

The table on the right side of the tabs allows to override the settings (limit values and enable/disable timer):

- Manual active

  Checkbox to define whether the default limit value shall be used, or whether a manually entered value shall be used
- Manual value

  In case the **Override** checkbox has been selected, a new value for the limit can be entered here which is then used instead of the default value.
- Timer active

  Checkbox to enable or disable the alarm timer of the selected limit

Limit in Force

This table shows the limit values that are currently used for the TA.



NOTE

Colors in the **Limit Group** tab, and the tables **Manual Limits and Timer Status** and **Limit in Force** have the following meaning:

- Black: The limit value is valid.
- Blue: The limit is currently not active due to manual deactivation.
- Red: The limit value is not valid (error in parameterization)
- Orange: The value is 0,0

Rate of Change Limit Check

This table shows the rate of change limits. The operator can activate/deactivate individual checks and to define manual values for the rate of change limits and the dependency time.

- Check active

  Checkbox to activate/deactivate a rate of change limit check

- Rate of change

  Type of the rate of change limit check

- Duration time

  This time indicates how long the limit violation message is delayed. The limit violation message is only issued when the violation persists for more than this time period.

- Dependency time

  Defines the time in history whose value shall be used for calculating the rate of change

  Note: Duration time and Dependency time are mutually exclusive. If values have been entered in both fields, only the dependency time is considered.

- Default value

  Value of the rate of change limit defined in IMM.

- Manual active

  Checkbox to determine whether the default value (checkbox not checked) or a manual value (checkbox checked) shall be used.

- Manual value

  In case the checkbox in column **Manual** is checked, the manual value to be used instead of the default value can be entered.

14.2.7 Tags Displays

14.2.7.1 Overview

There are different displays to edit, delete, and view the tags set on a TA. These displays can be selected using the corresponding item of the context menu.

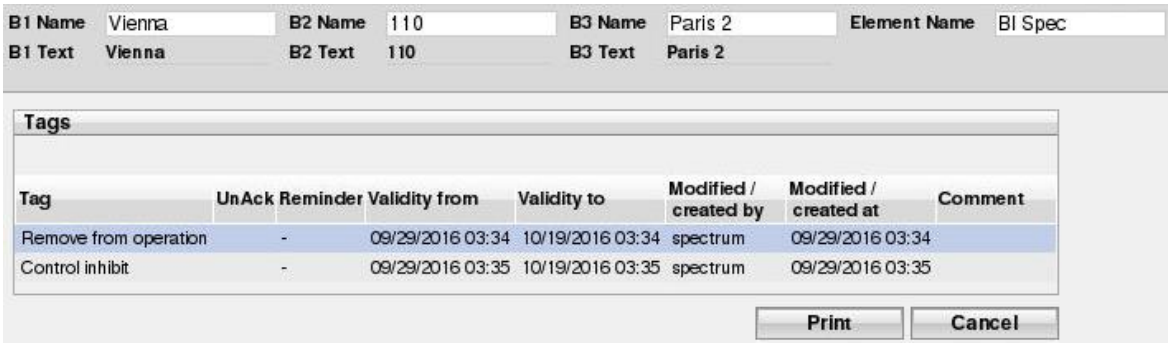
- View all tags
- Set <Tag name>
- View/Edit <Tag name>
- Delete <Tag name>

The following chapters provide a description of these displays.

14.2.7.2 Tags Overview

The Tags Overview display provides a list of all tags set on the selected TA. It can be opened via the menu item **View all tags** of the context menu in the network diagram.

The TA cannot be changed in this display.



[sc_view-all-tags, 6, en_US]

Figure 14-6 View All Tags

Table

The table shows a list of all tags set on the network element with the following information:

- Tag

  Name of the tag

- UnAck

  If there are any unacknowledged reminders for the tag.

- Reminder

  If a reminder has been defined, this field shows date/time of the reminder. If a reminder has expired, it is displayed in red and an alarm is inserted in the reminder summary.

- Validity from

  Date from which on the tag is valid

- Validity to

  Date until which the tag is valid

- Modified/created by

  User who created or last modified the permit to work tag

- Modified/created at

  Date/time when the permit to work tag was created or last modified

- Comment

  Comment for the tag

Buttons

- Print

  Prints the display

- Cancel

  Closes the display

Context Menu

- View/Edit

  Opens the View Tag display for editing the tag.

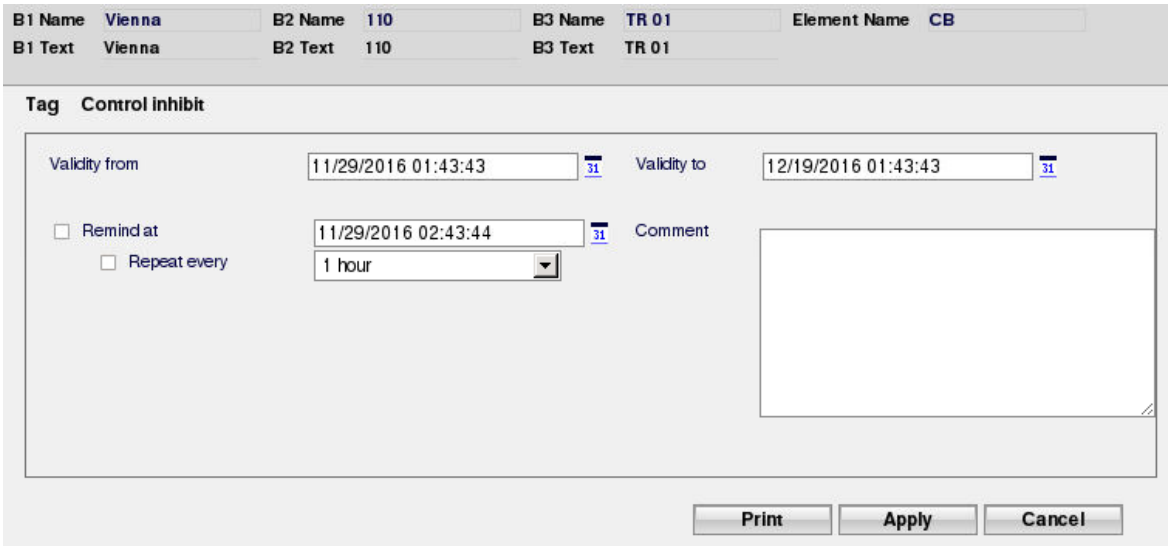
- Delete

  Deletes the selected tag after confirmation

14.2.7.3 Set Tag Display

The Set Tag display allows to set a tag on the selected TA. It can be opened using the menu item **Set<Tagname>** of the context menu in the network diagram.

The TA cannot be changed in this display.



[sc_Settag, 3, en_US]

Figure 14-7 Set Tag

- Tag

  Shows the type of tag to be set

- Validity from

  Input field for the date from which on the tag shall be valid

- Validity to

  Input field for the date until which the tag shall be valid

- Remind at

  Checking this checkbox and selecting a date/time defines a reminder.

- Repeat every

  Checking this checkbox and selecting an interval defines the interval at which the reminder shall be issued (indicated in the Basic Signaling Display, HUD, the Reminder Summary, and the display View all Tags.

- Comment

  Input field for a comment for the tag

Buttons

- Print

  Prints the display.

- Apply

  Saves the input made, closes the display, and sets the tag on the network element.

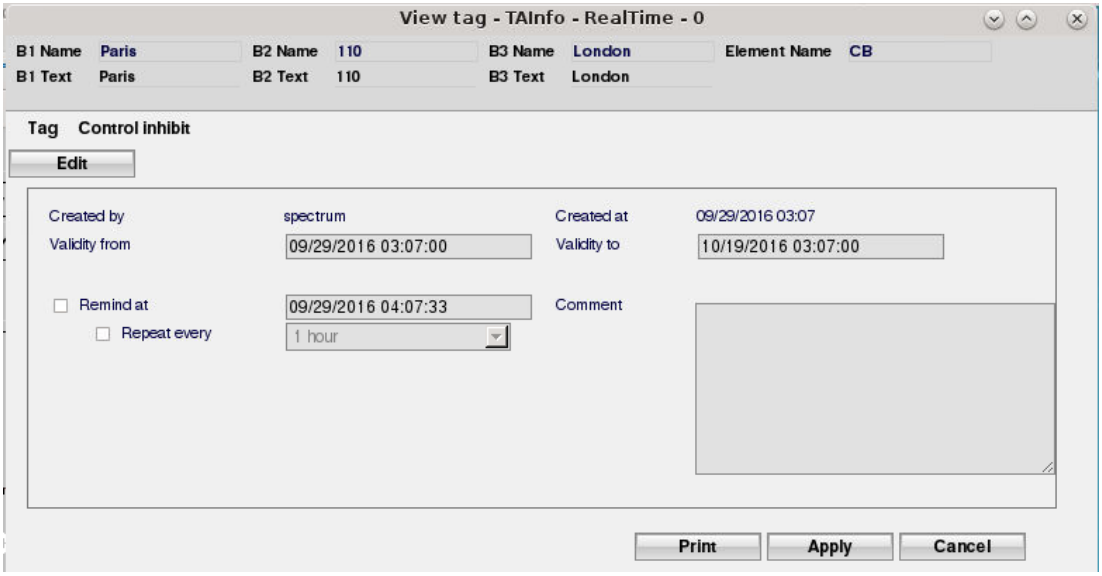
- Cancel

Closes the display without setting the tag.

14.2.7.4 View/Edit Tag Display

The View/Edit Tag display allows to edit the tag set on the selected TA. It can be opened via the menu item **View/Edit <Tag name>** of the context menu in the network diagram.

The TA cannot be changed in this display.



[sc_edit-tag, 3, en_US]

Figure 14-8 View Tag

- Tag

  Shows the type of tag to be edited

- Edit

  Clicking this button switches the display to edit mode and the content of the input fields can be modified

- Modified/created by

  User who created or last modified the permit to work tag

- Modified/created at

  Date/time when the permit to work tag was created or last modified.

- Validity from

  Input field for the date from which on the tag shall be valid

- Validity to

  Input field for the date until which the tag shall be valid

- Remind at

  Checking this checkbox and selecting a date/time defines a reminder.

- Repeat every

  Checking this checkbox and selecting an interval defines the interval at which the reminder shall be issued (indicated in the Basic Signaling Display, HUD, the Reminder Summary, and the display View all Tags.

- Comment

  Input field for a comment for the tag

Buttons

- Print

  Prints the display

- Apply

  Saves the changes and closes the display.

- Cancel

  Closes the display and discards the changes.

14.2.8 Permits Displays

14.2.8.1 Overview

There are different displays to set, edit, delete, and view the Permit to Work and Permit to Test tags set on a TA. These displays can be selected using the corresponding item of the context menu.

- Set Permit to work/Permit to test
- View/Edit Permit to work/Permit to test
- Delete Permit to work/Permit to test
- View all permits

The following chapter provides a description of these displays.



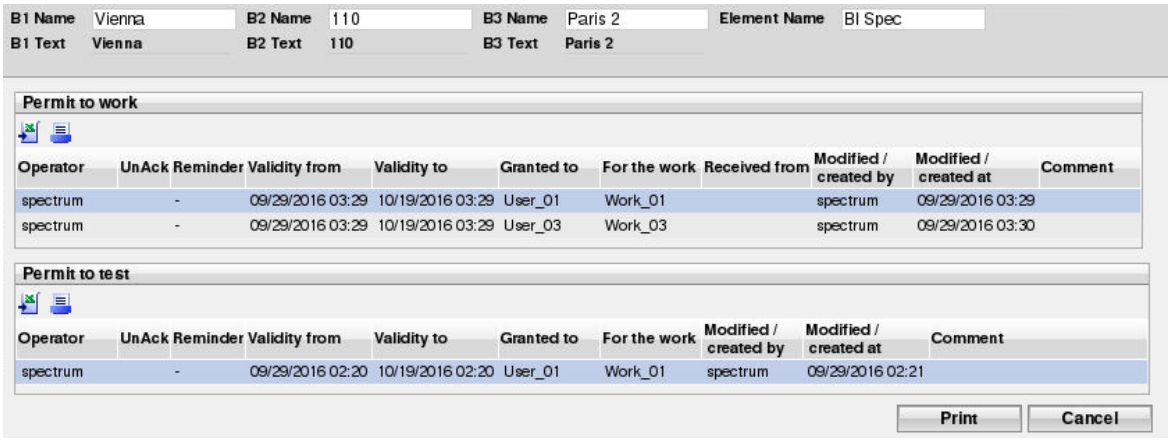
NOTE

Only the displays for setting, editing, and deleting a Permit to Work tag are described. The displays for a Permit to Test tag are identical, except for the External Permit section, which does not exist in the Permit to Test displays.

14.2.8.2 Permit Overview

The Permits Overview display provides a list of all permit to work and permit to test tags set on the selected TA. It can be opened via the menu item **View all permits** of the context menu in the network diagram.

The TA cannot be changed in this display.



[sc_view-all-permits, 5, en_US]

Figure 14-9 Permits Overview

It consists of 2 tables. The upper table provides a list of all Permit to work tags (maximum 6 Permit to work tags can be set on a TA) set on the selected network element. For each Permit to work tag, the following information is displayed.

- Operator

  Operator who is setting the permit to work tag.

- UnAck

  If there are any unacknowledged reminders.

- Reminder

  If a reminder has been defined, this field shows date/time of the reminder.

- Validity from

  The time from when the permit is valid.

- Validity to

  The time range for which the permit to work is granted

- Granted to

  User who is granted permit to work.

- For the work

  Work for which the permit to work is granted

- Received from
- When it is an external permit to work, this field shows the name of the user from whom the Permit to work has been received.
- Modified/created by

  User who created or last modified the permit to work tag.

- Modified/created at

  Date/time when the permit to work tag was created or last modified.

- Comment

  Comment for the permit to work tag

If a Permit to test tag has been set on the selected network element, the lower table provides the following information on this tag:

- Operator

  Operator who is setting the permit to work tag.

- UnAck

  If there are any unacknowledged reminders.

- Reminder

  If a reminder has been defined, this field shows date/time of the reminder.

- Validity from
- Validity to

  Time range for which the permit to work is granted

- Granted to

  User who is granted permit to test.

- For the work

  Work for which the permit to test is granted

- Modified/created by

  User who created or last modified the permit to test tag.

- Modified/created at

  Date/time when the permit to test tag was created or last modified.

- Comment

  Comment for the permit to test tag

Buttons

- Print

  Prints the display

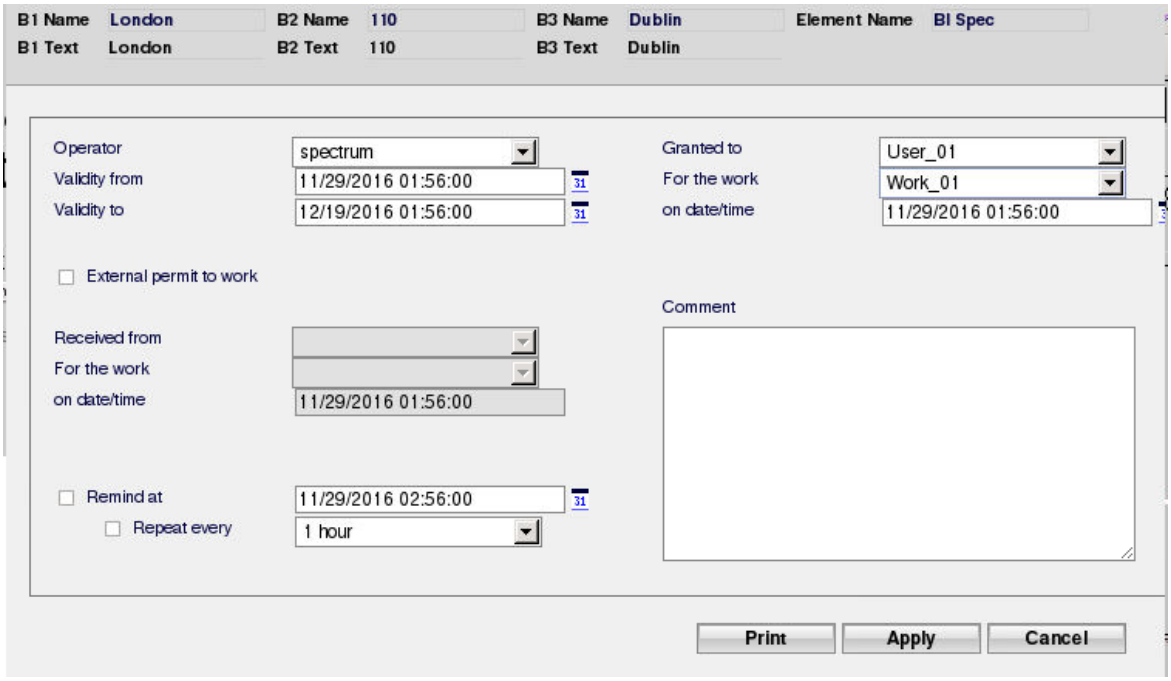
- Cancel

  Closes the display

14.2.8.3 Set Permit to Work/Permit to Test Display

The Set Permit to Work/Permit to Test display allows to set a Permit to Work/Permit to Test tag on the selected TA. It can be opened via the menu item **Set Permit to work/Set Permit to test** of the context menu in the network diagram.

The TA cannot be changed in this display.



[sc_set-permit, 5, en_US]

Figure 14-10 Set Permit to Work

- Operator

  Operator who set the permit to work tag.

- Validity from

  Input field for the date from which on the tag shall be valid

- Validity to

  Input field for the date until which the tag shall be valid

- Granted to

  User who is granted permit to work.

- For the work

  Work for which the permit to work is granted.

- On Date/Time

  Date and time of the work

- External Permit to Work

Select this checkbox in case of external permit to work. When selected, the following fields can be edited:

- Received from

  User from whom the permit to work is received.

- For the work

  Work for which the permit to work is received.

- On Date/Time

  Date and time of the work

- Remind at

  Checking this checkbox and selecting a date/time defines a reminder.

- Repeat every

  Checking this checkbox and selecting an interval defines the interval at which the reminder shall be issued (indicated in the Basic Signaling Display, HUD, the Reminder Summary, and the display View all Tags.

- Comment

  Input field for a comment for the tag

Buttons

- Print

  Prints the display

- Apply

  Saves the input made, closes the display, and sets the tag on the network element.

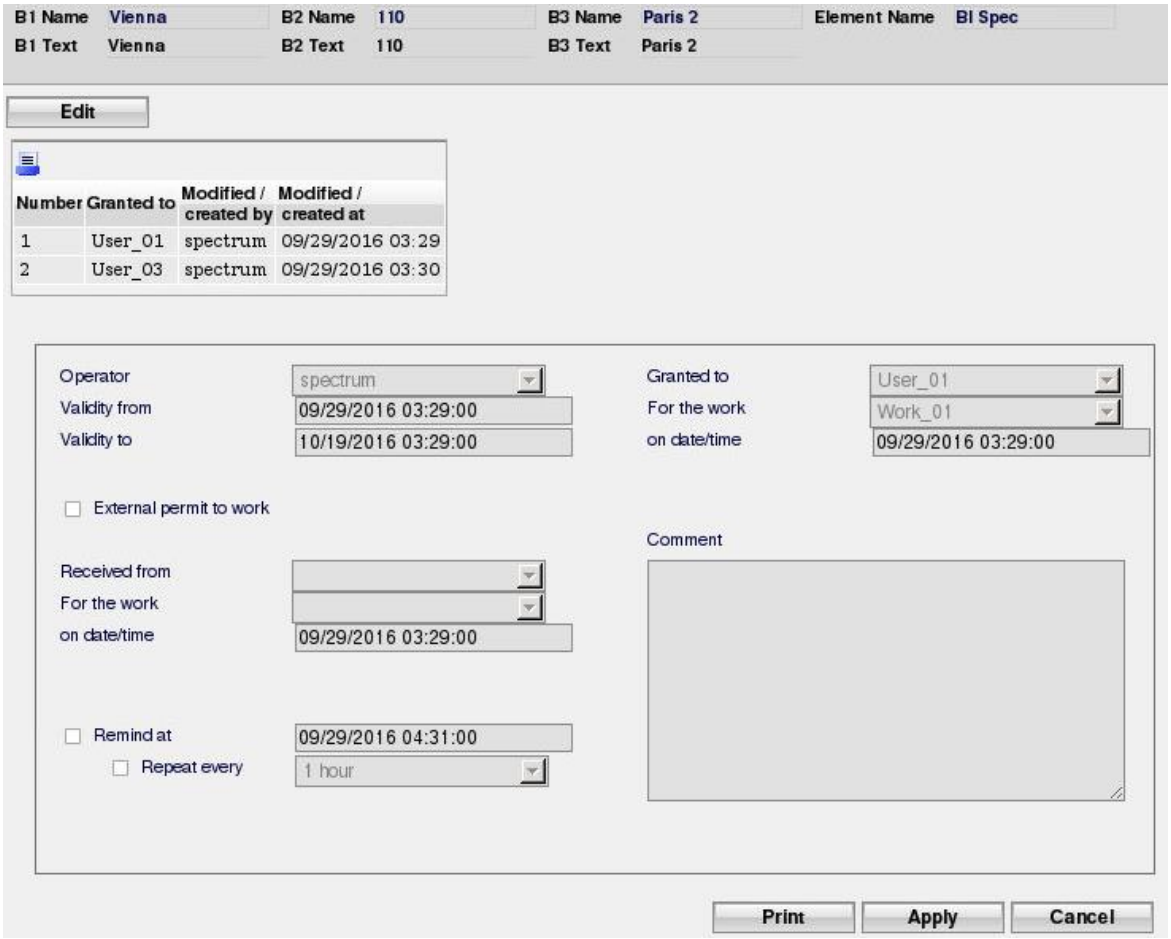
- Cancel

  Closes the display without setting the tag

14.2.8.4 View/Edit Permit to Work/Permit to Test Display

The Permit to Work/Permit to Test display is used to edit a Permit to Work or Permit to Test tag on the selected TA. It can be opened via the menu item **View/Edit Permit to work/Set Permit to test** of the context menu in the network diagram.

The TA cannot be changed in this display.



[sc_edit-permit, 4, en_US]

Figure 14-11 Edit Permit

- Edit

  Clicking this button switches the display to edit mode and the content of the input fields can be modified

- Table showing the Permit to work tags set on the TA with number, user to whom it has been granted, user who created or last modified the tags and the date and time when the tag was created or last modified.
- Operator

  Operator who set the permit to work tag.

- Validity from

  Input field for the date from which on the tag shall be valid

- Validity to

  Input field for the date until which the tag shall be valid

- Granted to

  User who is granted permit to work.

- For the work

  Work for which the permit to work is granted.

- On Date/Time

  Date and time of the work

- External Permit to Work

  Select this checkbox in case of external permit to work. When selected, the following fields can be edited:

  - Received from

    User from whom the permit to work is received.

  - For the work

    Work for which the permit to work is received.

  - On Date/Time

    Date and time of the work

- Remind at

  Checking this checkbox and selecting a date/time defines a reminder.

- Repeat every

  Checking this checkbox and selecting an interval defines the interval at which the reminder shall be issued (indicated in the Basic Signaling Display, the Reminder Summary, and the display View all Tags.

- Comment

  Input field for a comment for the permit

Buttons

- Print

  Prints the display

- Apply

  Saves the input made, closes the display, and sets the tag on the network element.

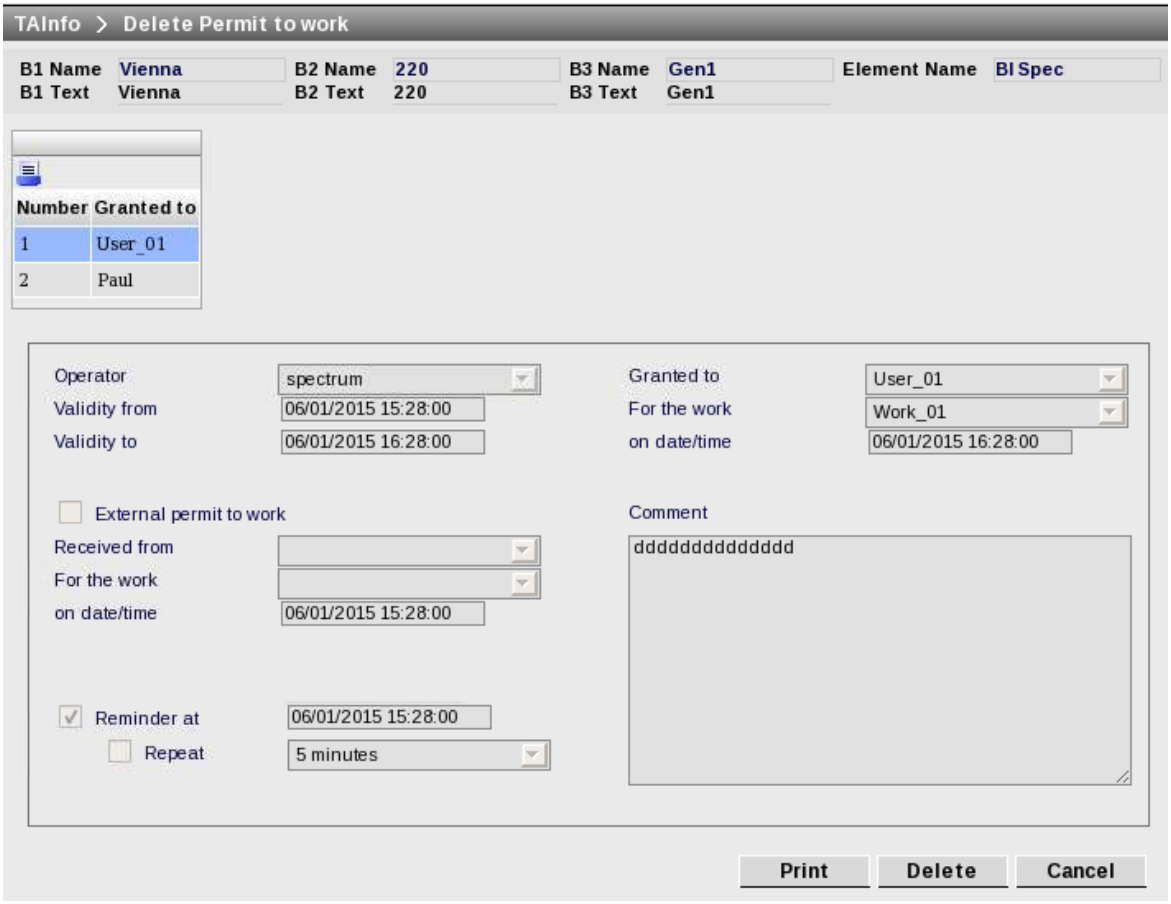
- Cancel

  Closes the display without setting the tag

14.2.8.5 Delete Permit to Work/Permit to Test Display

The Permit to Work/Permit to Test display allows to delete a Permit to Work/Permit to Test tag set on the selected TA. It can be opened via the menu item **Delete Permit to work/Set Permit to test** of the context menu in the network diagram

The TA cannot be changed in this display.



[sc_clear-permit, 3, en_US]

Figure 14-12 Delete Permit to Work

- Table showing the Permit to work tags set on the TA with number and user, to whom it has been granted, user who created or last modified the tags and date/time when the tag was created or last modified.
- Operator

  Operator who set the permit to work tag.
- Validity from

  Shows the date from which on the tag shall be valid
- Validity to

  Shows the date until which the tag shall be valid
- Granted to

  User who is granted permit to work.
- For the work

  Work for which the permit to work is granted.
- On Date/Time

  Date and time of the work
- External Permit to Work

When this checkbox is selected, the following fields show:

- Received from

  User from whom the permit to work is received.

- For the work

  Work for which the permit to work is received.

- On Date/Time

  Date and time of the work

- Reminder at

  Indicates whether a reminder is defined.

- Repeat

  Defines the interval at which the reminder is issued (indicated in the Basic Signaling Display, the Reminder Summary, and the display View all Permits).

- Comment

  Shows the comment for the tag

Buttons

- Print

  Prints the display

- Delete

  Deletes the tag and closes the display.

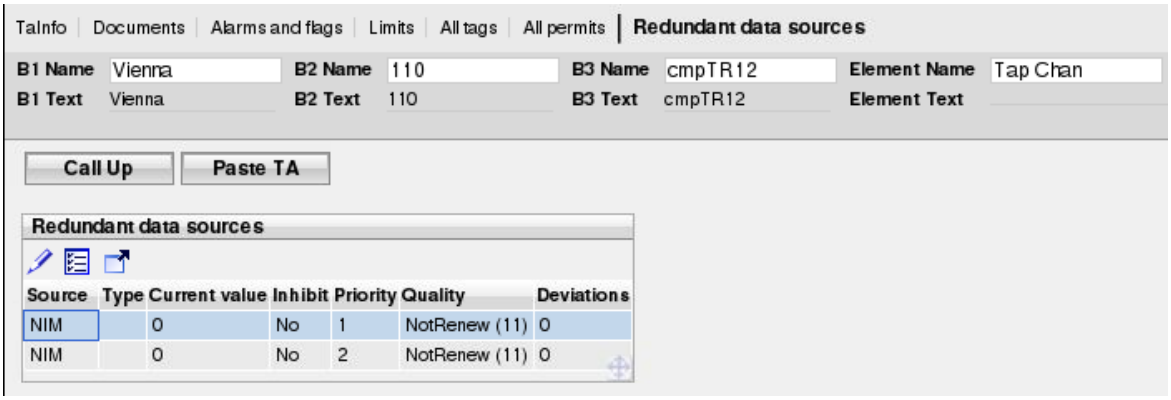
- Cancel

  Closes the display without deleting the tag.

14.2.9 Redundant Data Sources Display

14.2.9.1 Overview

This display provides a list of all data sources used for a TA.



[sc_redundantdatasources, 6, en_US]

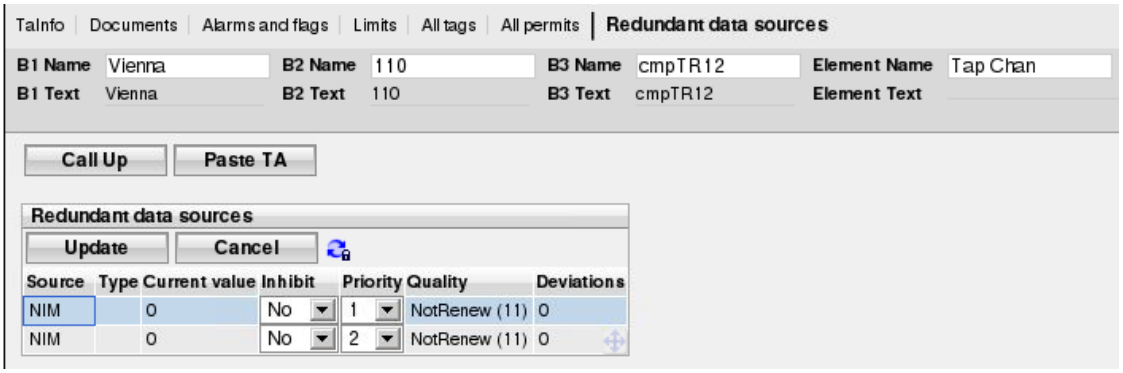
Figure 14-13 Redundant Data Sources

14.2.9.2 Toolbar

Toolbar has the following options:

- 

  Switches the table to edit mode.



[sc_TAInfo - Redundant Data Sources Edit, 1, en_US]

Figure 14-14 TAInfo - Redundant Data Sources Edit

  - Cancel

    Leaves edit mode and discards the changes.

  - Update

    Saves the changes and leaves edit mode.

  - 

    Indicates that the display is not updated with current values while it is in Edit mode. Type, Inhibit, and Priority can be modified.

- 

  Adjust columns

- 

  Opens the table in a new window.

14.2.9.3 Table

Table has the following options:

- Source Name

  Name of the data source.

- Type

  Auto: Source is selected automatically according to priority.

  Manual: The selected source is always used.

- Current value

  Current value of the source.

- Inhibit

  No: This source can be used.

  Yes: This source cannot be used.

- Priority

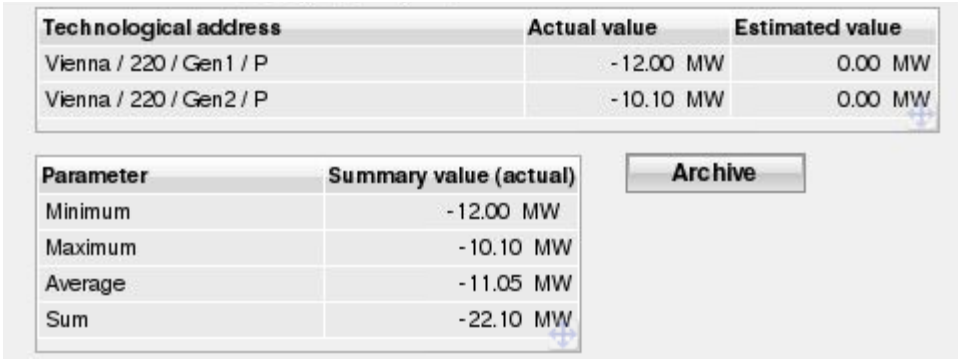
  Priority according to which the source is selected.

- Quality

  Quality of the source.

14.2.10 Summary Values

After selecting multiple elements in a network diagram, this display shows summary values (minimum, maximum, average, sum) for these elements. The values are continuously updated while the display is open.



[sc_SummaryMultipleTAs, 1, en_US]

Figure 14-15 Summary Multiple TAs

Title Bar

Table 14-2 Title Bar Icons

Icon	Description
	Opens the previous window.
	Opens the next window.
	Stops loading of data.
	Updates the table in the display.
	Zoom out
	Zoom in
	Reset zoom
	Displays general information about the display.

Table of Elements

This table displays a list of all selected elements with the following information:

- Technological address

  Contains the technological addresses of the elements

- Actual value

  Shows the current values of the elements

- Estimated value

  Shows the estimated values of the elements

Summary Values Table

- Parameter

  Type of parameter whose value is displayed:

  - Minimum
  - Maximum
  - Average
  - Sum

- Summary value (actual)

  Value of the parameter

Button

- Archive

  Opens the HIS display for the selected elements

14.3 Operational Steps

14.3.1 Opening the Display with TA-Related Information

To open a display with TA-related information, perform the following:

- On a network diagram, right-click the TA.

  A context menu opens.

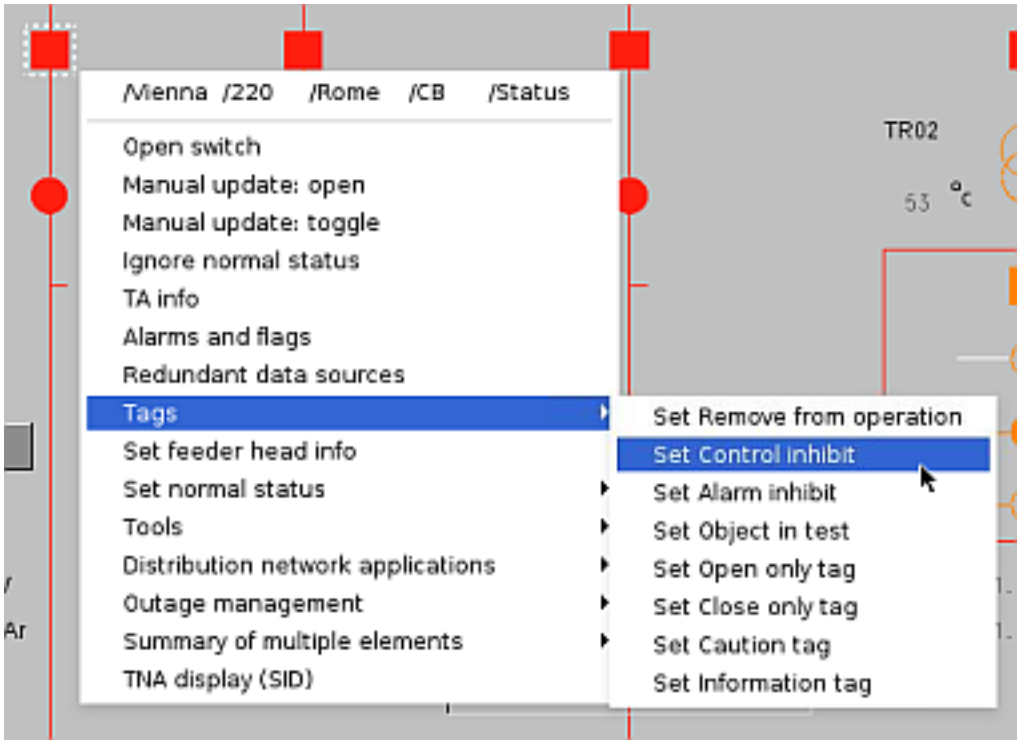
- Select one of the following items (depending on the type of the selected TA, not all items are available):
  - TA info
  - Alarms and flags
  - Tags
  - Permits
  - Limits
  - Redundant data sources
  - Summary of multiple elements

  The respective display opens. For a description of these displays, refer to chapter 14.2.3.1 Overview.

14.3.2 Setting a New Tag

To set a new tag (in the following example, Control Inhibit) on a TA, perform the following steps:

- Right-click the TA in a network diagram and from the context menu select **Tags -> Set Control Inhibit**.

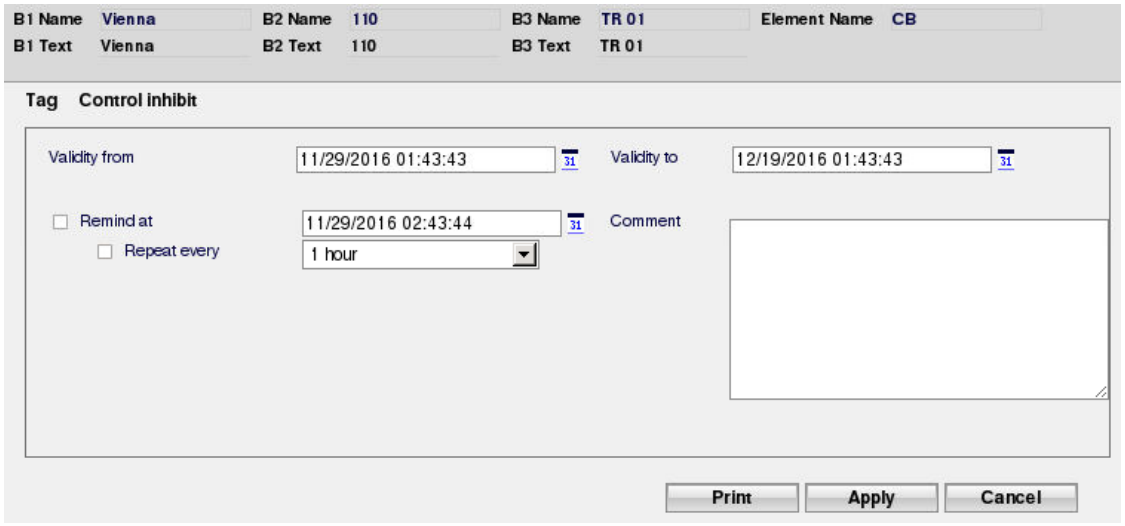


[sc_operation_settag1, 3, en_US]

Figure 14-16 Set Control Inhibit

The display for setting a new tag opens.

- Enter the dates when the tag shall be valid and, optionally, enter a comment.



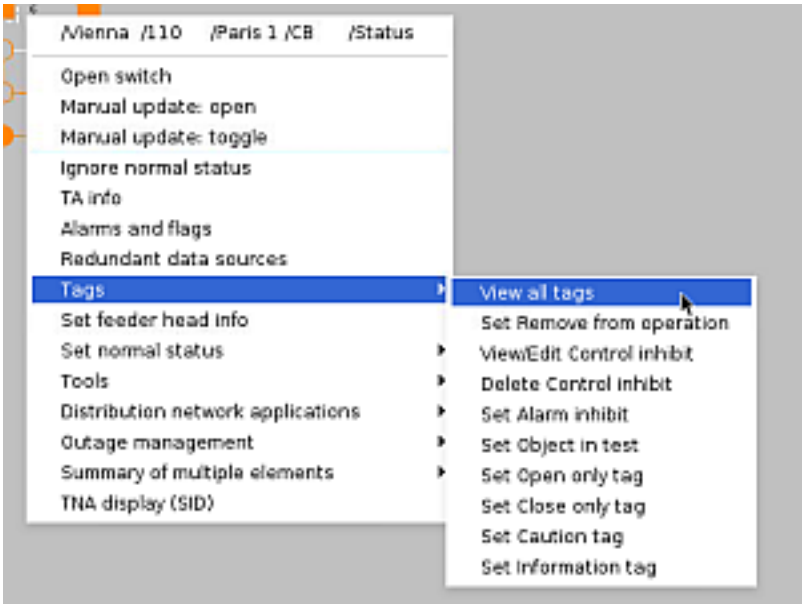
[sc_Settag, 3, en_US]

Figure 14-17 Set Tag

- Click **Apply**.

  The display is closed and the tag is set on the TA.

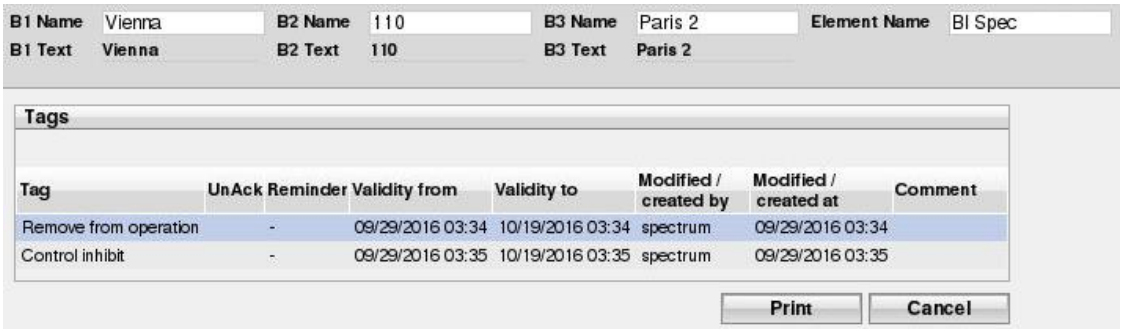
- To display the tags that have been set on the TA, select **Tags -> View all tags**.



[sc_tag-viewalltags-menu, 2, en_US]

Figure 14-18 View all Tags Menu

A display showing a list of all tags set on the selected TA opens.



[sc_tag-view-alltags, 4, en_US]

Figure 14-19 View all Tags

14.3.3 Deleting a Tag

To delete a tag, there are three options:

- Right-click the element on which the tag is set and click Delete
- Open the display View all Tags for the element, right-click the corresponding tag and select Delete tag from the context menu.
- Open the corresponding Tag summary, right-click the corresponding tag and select Delete tag from the context menu.

In all three cases, a confirmation dialog opens.



[sc_delete-tag, 3, en_US]

Figure 14-20 Delete Tag

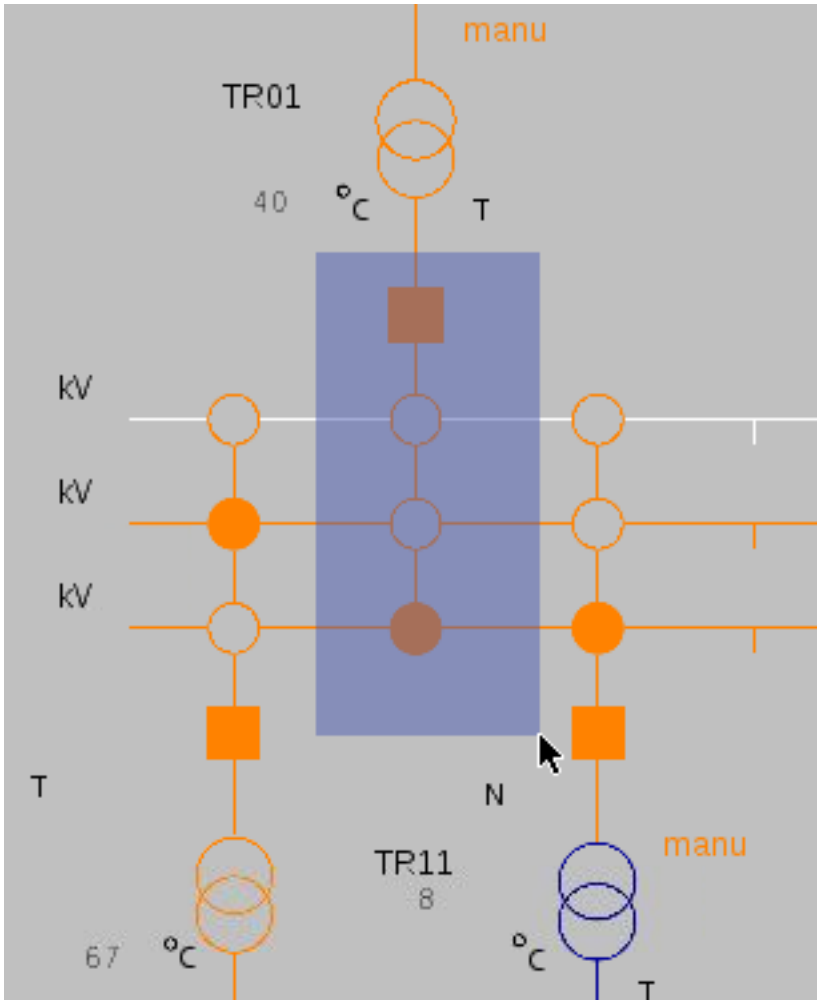
- Click **Delete**.

  The tag is removed from the selected element.

14.3.4 Setting and Deleting a Tag on Multiple Elements

It is possible to set and delete tags on multiple elements within one step. In order to set a tag on multiple elements at once, perform the following steps:

- Select the desired elements by clicking one element and selecting additional ones keeping the Ctrl-key or the Shift-key pressed, or by drawing a rubberband rectangle, for example:

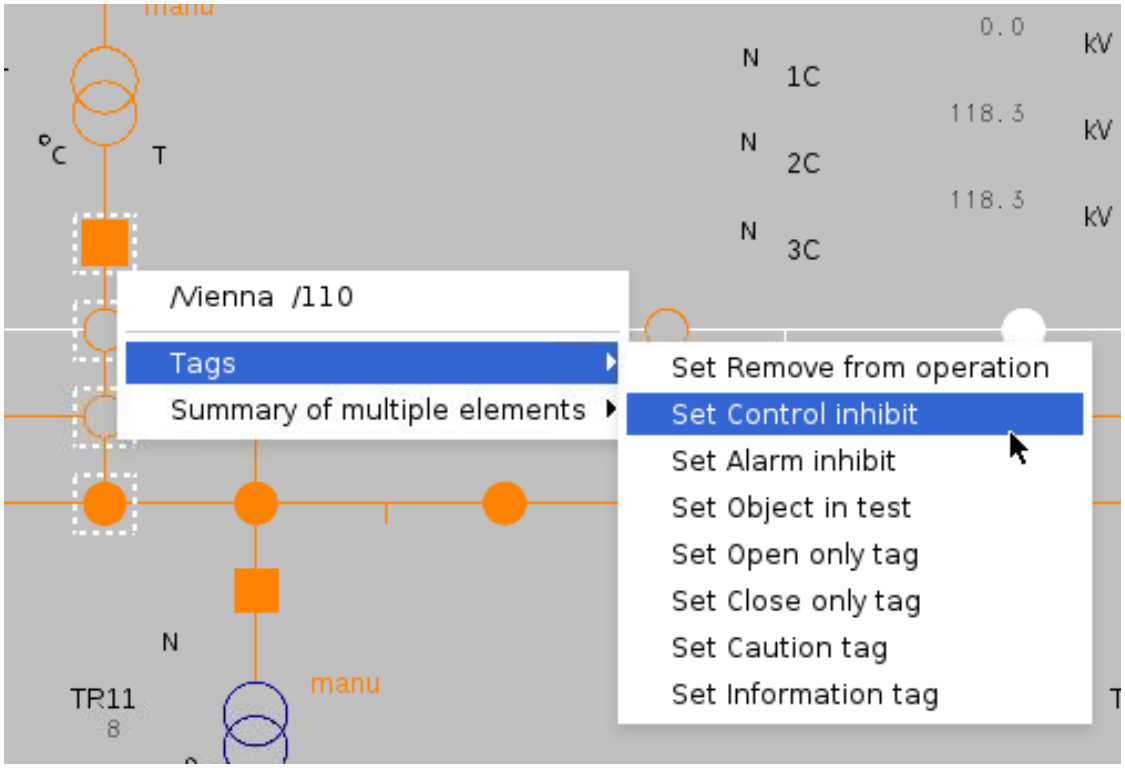


[sc_grouptagging1, 2, en_US]

Figure 14-21 Selecting Multiple Elements

The ctrl or shift key can be used to deselect elements as well.

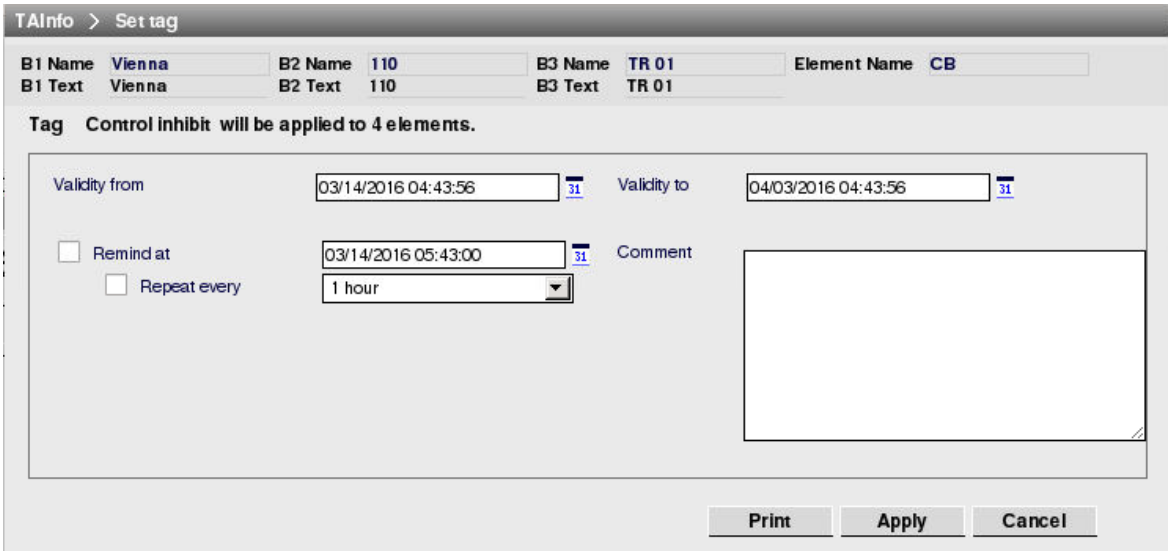
- Right-click one of the selected network elements to open a context menu and select the tag you want to apply to the selected elements (for example, Control Inhibit).



[sc_grouptagging2, 2, en_US]

Figure 14-22 Setting Tag

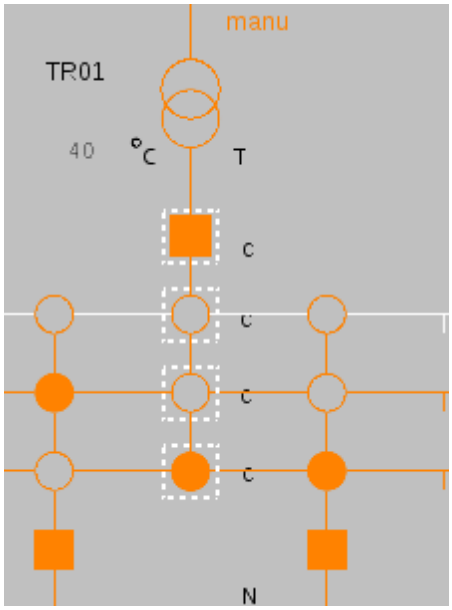
Similar to setting a tag on a single element, a display is opened where you can define additional parameters for the tag.



[sc_grouptagging3, 1, en_US]

Figure 14-23 Setting Tag Parameters

- Click **Apply**.



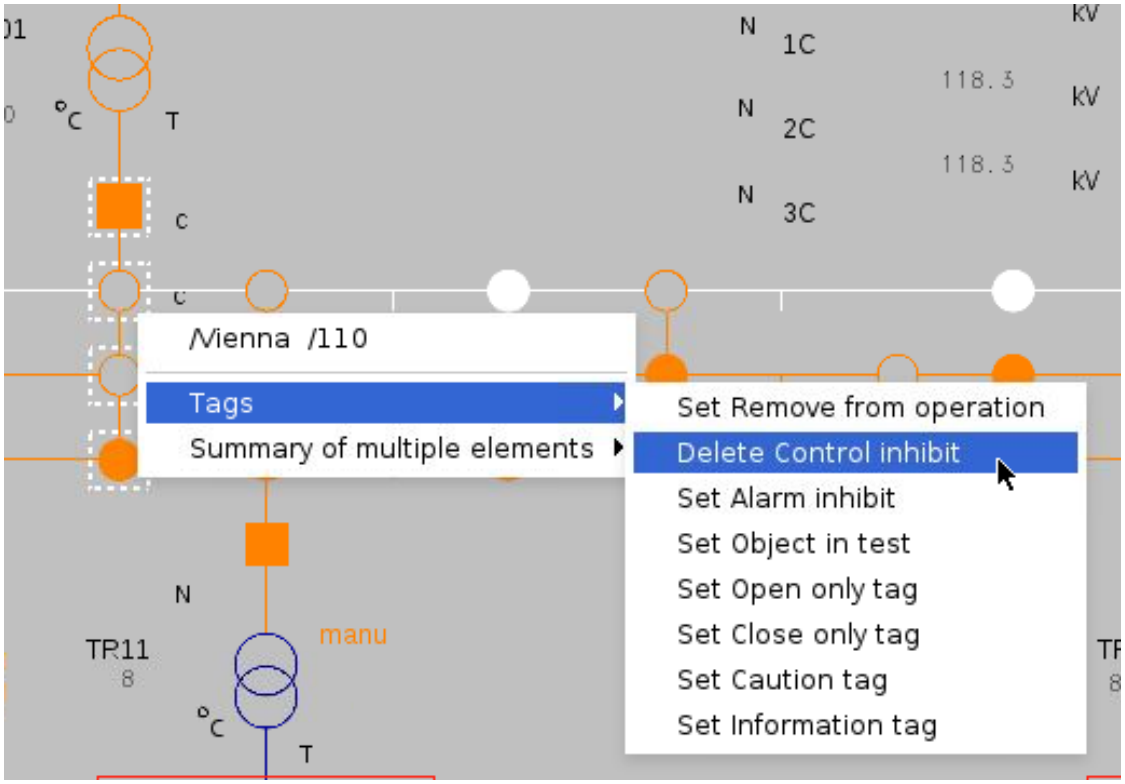
[sc_grouptagging4, 2, en_US]

Figure 14-24 Tag Set on Multiple Elements

The tag Control Inhibit has been set on all previously selected network elements.

To delete a tag from multiple elements, perform the following steps:

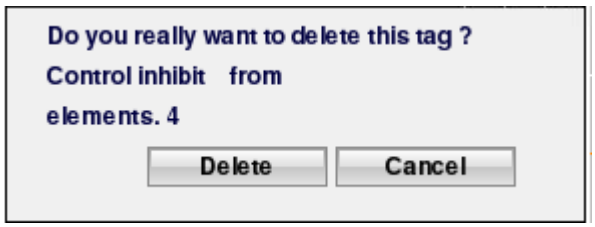
- As described for setting the tag, select the elements from which the tag shall be removed.
- Right-click one of the selected network elements to open a context menu and select the tag you want to delete.



[sc_grouptagging-delete1, 2, en_US]

Figure 14-25 Deleting Tags

A display opens where you have to confirm that you want to delete the tag from the selected elements.



[sc_grouptagging-delete2, 3, en_US]

Figure 14-26 Deleting Group Tag

- Click **OK**.

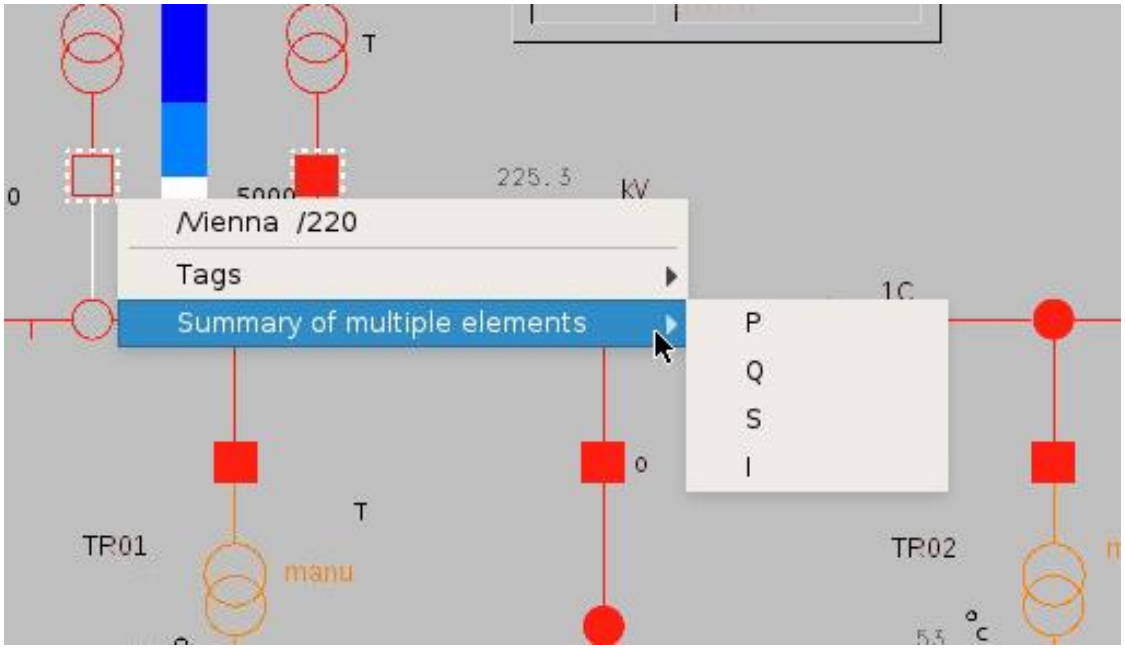
  The tag is removed from the elements.

14.3.5 Displaying Summary Values of Multiple TAs

To display the values of several TAs in one display, perform the following steps:

- Select the elements for which you want to display the values and right-click to open the context menu.
- From the context menu select the corresponding item, and select the value type to be displayed from the sub-menu:

The items provided in the context menu depend on the selected elements.



[sc_SelectSummaryMultipleTAs, 1, en_US]

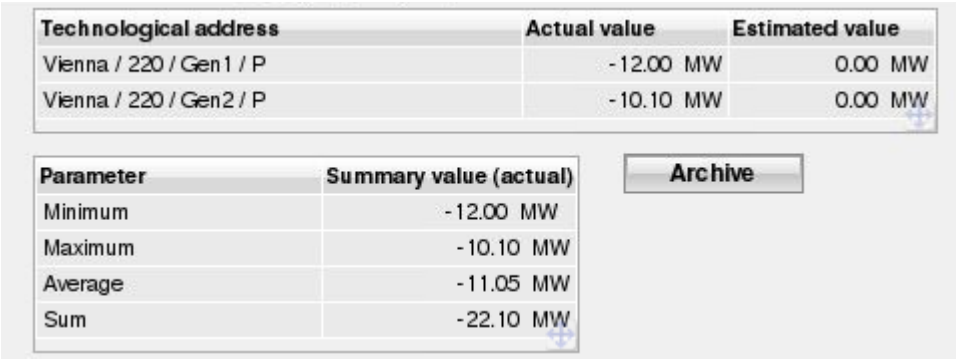
Figure 14-27 Summary of Multiple Elements

Depending on the selected context menu item, those elements providing this information start flashing.

The display with a list of all these TAs is opened.

The upper table shows actual and estimated values for each TA.

The lower table shows minimum, maximum, sum, and average of the actual values.



Technological address	Actual value	Estimated value
Vienna / 220 / Gen1 / P	-12.00 MW	0.00 MW
Vienna / 220 / Gen2 / P	-10.10 MW	0.00 MW

Parameter	Summary value (actual)
Minimum	-12.00 MW
Maximum	-10.10 MW
Average	-11.05 MW
Sum	-22.10 MW

Archive

[sc_SummaryMultipleTAs, 1, en_US]

Figure 14-28 Summary Multiple TAs

- Click the **Archive** button.

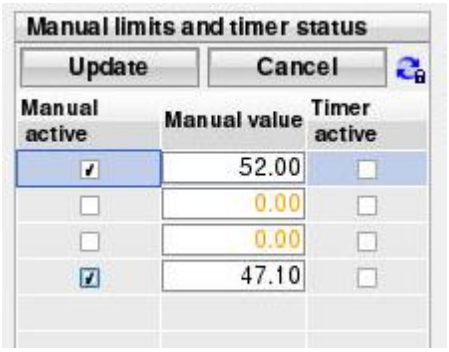
  The HIS display is opened showing data for the selected TAs.

14.3.6 Modifying the Limits of a TA

Replacing a Limit Value of a Limit Group by a Manual Limit

To change the limit group to be used for a TA, perform the following steps:

- Open the **Limits** display by selecting **Limits** of the context menu of an analog value.
- Click the icon in the toolbar of the **Manual limits and timer status** table.
- Click the checkbox of the **Manual active** column in the line of the limit to be changed and enter the new limit value in the column **Manual** value



[sc_limits-manuallimit, 2, en_US]

Figure 14-29 Limits - Manual Limits

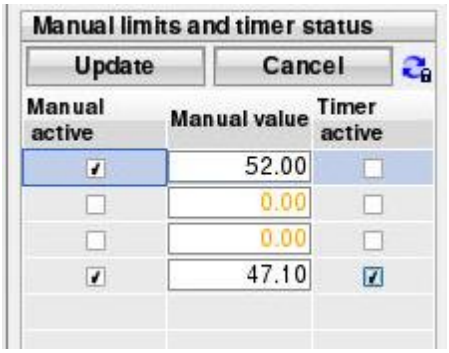
- Save the change by clicking **Update**.

  The manually entered limit value is displayed in the table **Limit in force**.

Enabling or Disabling the Alarm Timer for a Limit

To enable or disable the alarm timer for a limit of a TA, perform the following steps:

- Click the icon in the toolbar of the **Manual limits and timer status** table.
- To enable an alarm that is currently disabled, click the checkbox of the **Timer active** column in the line of the limit for which the alarm timer shall be enabled.



[sc_limits-enabletimer, 2, en_US]

Figure 14-30 Limits - Enable Timer

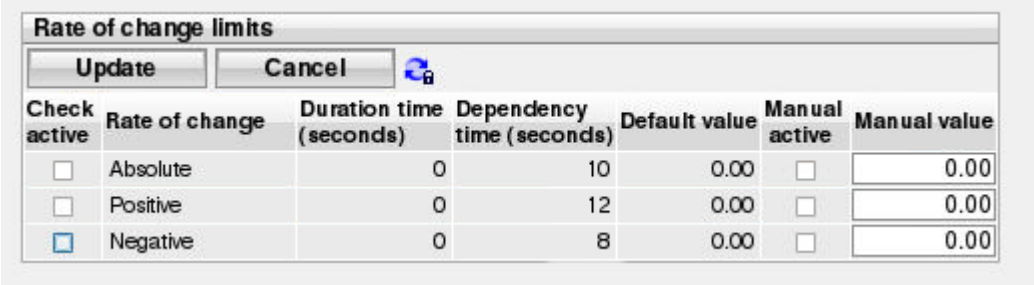
- Save the change by clicking **Update**.

The change is saved and the checkbox is checked.

14.3.7 Enabling or Disabling Rate of Change Limits

To enable or disable a rate of change limit check and to enter a manual rate of change limit value, perform the following steps:

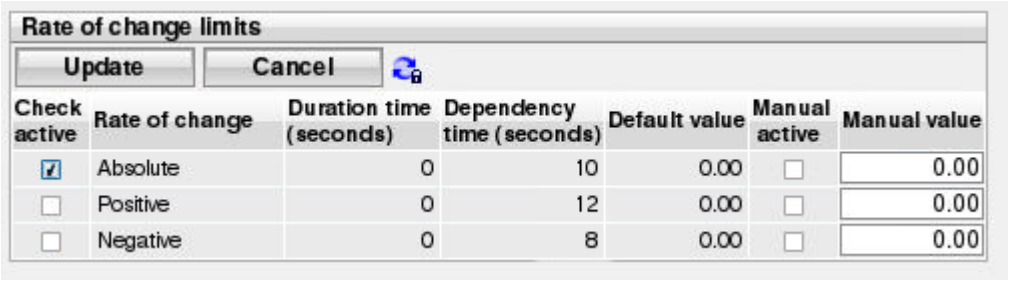
- Open the **Limits** display by selecting **Limits** of the context menu of an analog value.
- Click the icon in the toolbar of the **Rate of change limits** table.



[sc_editrateofchangelimit1, 1, en_US]

Figure 14-31 Edit Rate of Change Limit 1

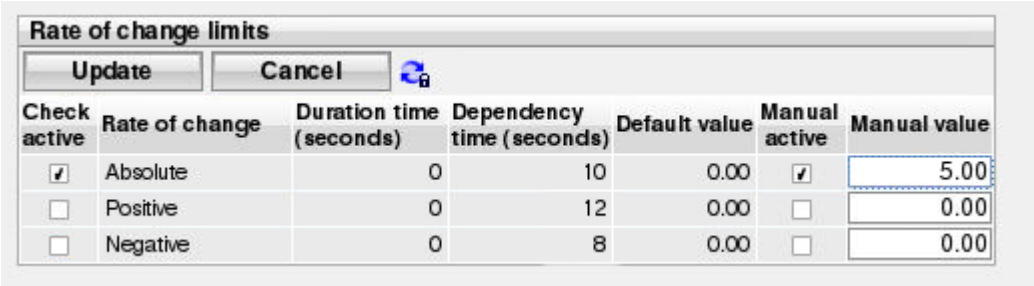
- Click the checkbox of the **Check active** column in the line of the rate of change limit to be activated.



[sc_editrateofchangelimit2, 1, en_US]

Figure 14-32 Edit Rate of Change Limit 2

- If you want to use a rate of change limit value different from the default value, click the checkbox of the **Manual active** column in the corresponding line of the rate of change limit and enter the new rate of change limit value in the column **Manual value**.

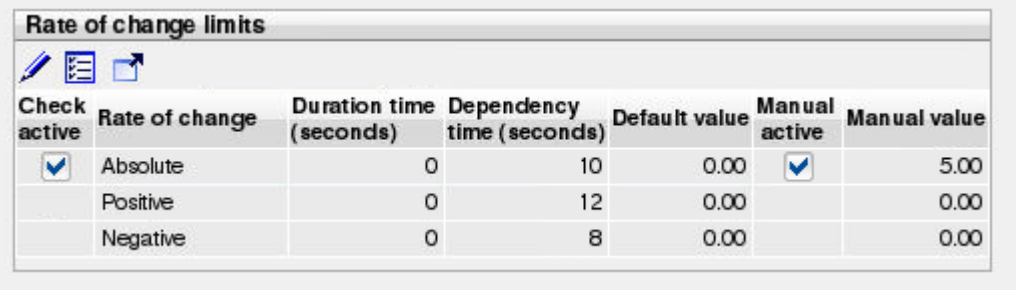


[sc_editrateofchangelimit3, 1, en_US]

Figure 14-33 Edit Rate of Change Limit 3

- Click Update

  The rate of change limit check is activated and the manual value entered before is used.



[sc_editrateofchangelimit4, 1, en_US]

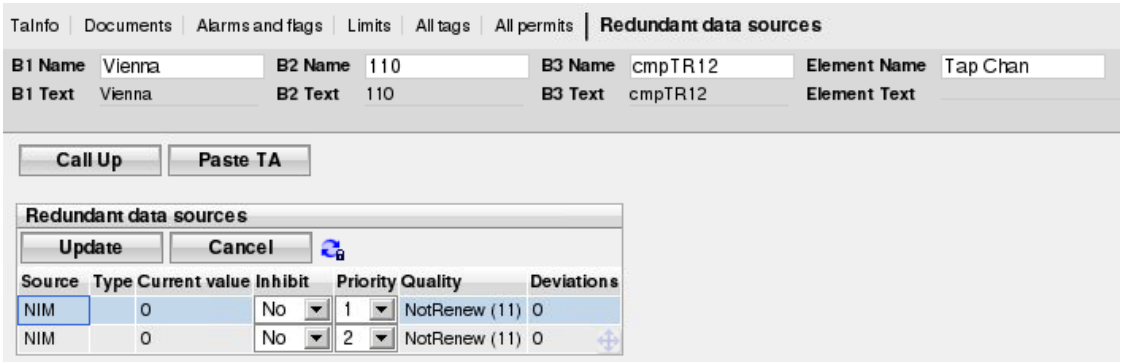
Figure 14-34 Edit Rate of Change Limit 4

14.3.8 Editing Redundant Data Sources

To edit the redundant data sources, perform the following:

- Open the **Redundant Data Sources** tab of the TAInfo display by selecting **Redundant Data Sources** of the context menu.
- Click icon in the toolbar of the Redundant Data Sources table.

  The table switches to Edit mode.



[sc_redundantedit, 1, en_US]

Figure 14-35 TAInfo - Redundant Data Sources

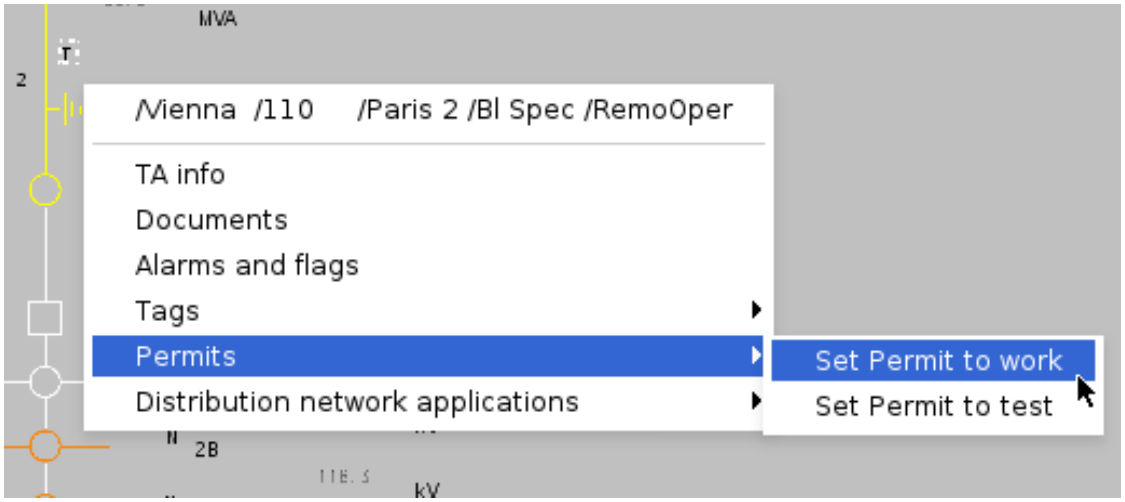
- Click **Update** to save the changes.

  The new data are displayed in the table.

14.3.9 Setting a Permit to Work Tag

To set a new Permit to Work tag on a TA, perform the following steps:

- Right-click the TA and select **Permits -> Set Permit to work** from the context menu.

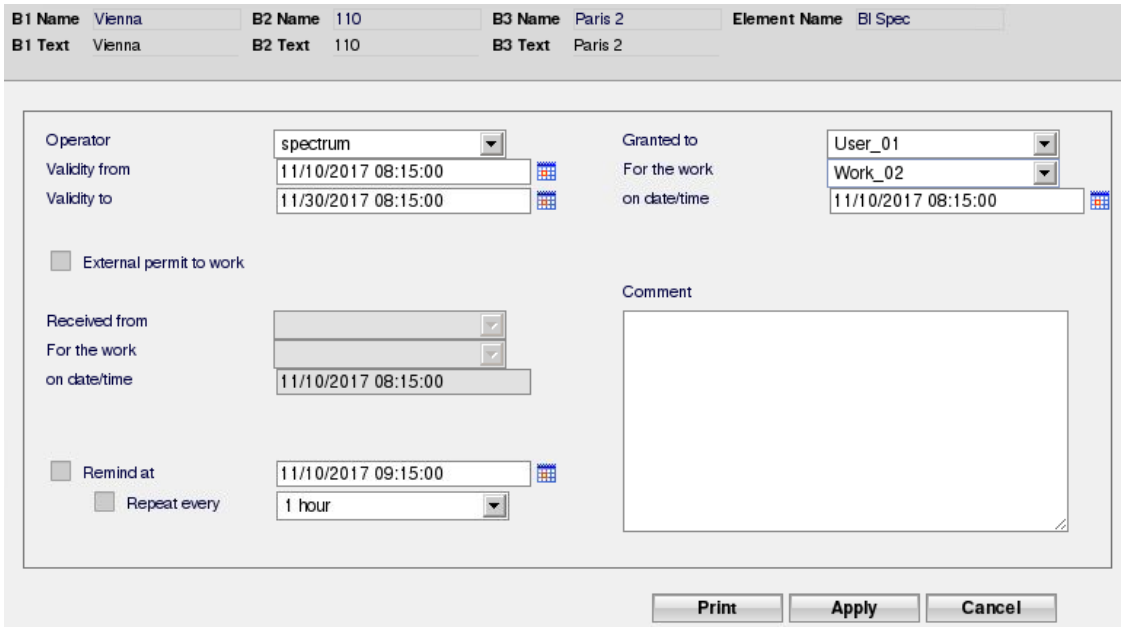


[sc_operation_setpermit1, 3, en_US]

Figure 14-36 Context Menu - Set Permit to Work

The display for setting a new Permit to Work tag opens.

- Enter the dates when the Permit to Work tag shall be valid, select the user and the work for which the permit to work is granted as well as the date for this work, a reminder, if required and, optionally, enter a comment.



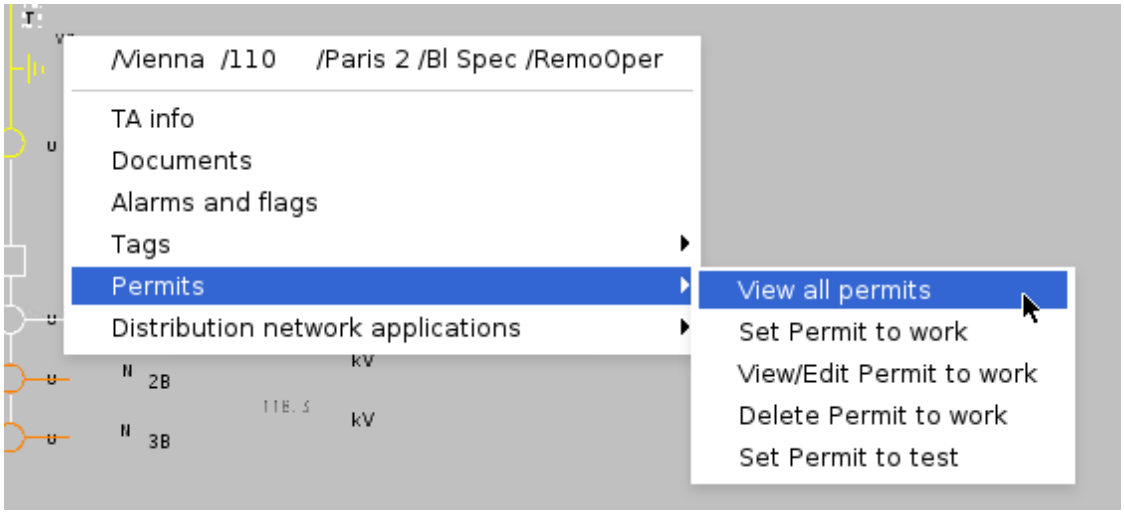
[sc_operation_setpermit2, 4, en_US]

Figure 14-37 Set Permit to Work

- Click **Apply**.

  The display is closed and the tag is set on the TA.

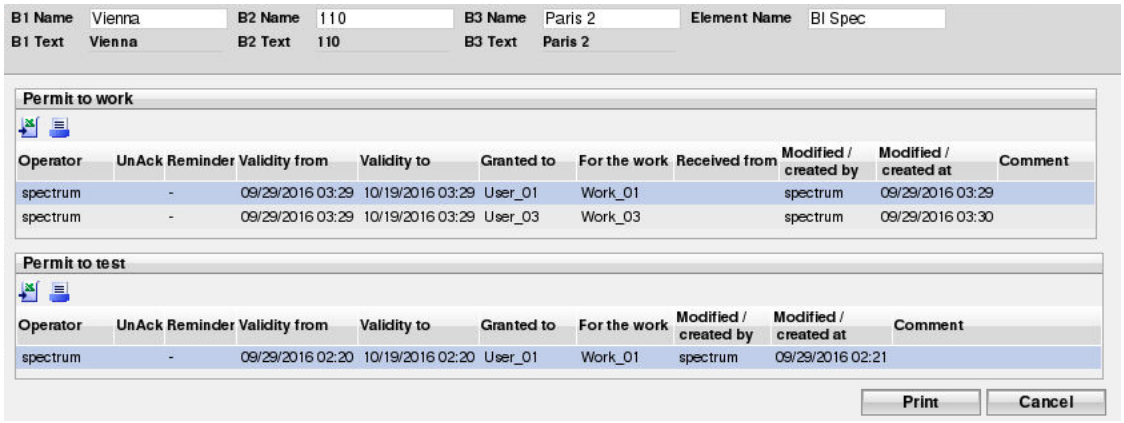
- To display the Permit to Work tags that have been set on the TA, select **Tags -> View all permits**.



[sc_operation_setpermit3, 3, en_US]

Figure 14-38 Context Menu - View all Permits

A display showing a list of all Permit to Work tags set on the selected TA opens.



[sc_view-all-permits, 5, en_US]

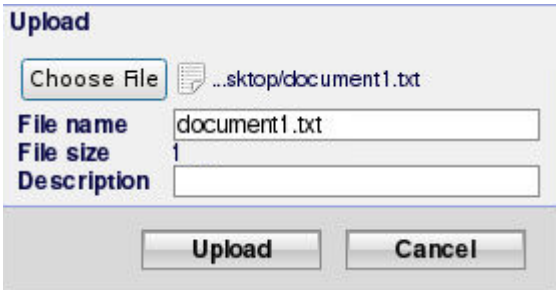
Figure 14-39 View all Work Permits

14.3.10 Upload a Document

To upload a document:

- Open the Documents tab of the TAInfo Display by selecting the corresponding tab.
- Click the  icon above the table.

  A dialog to select the document opens.

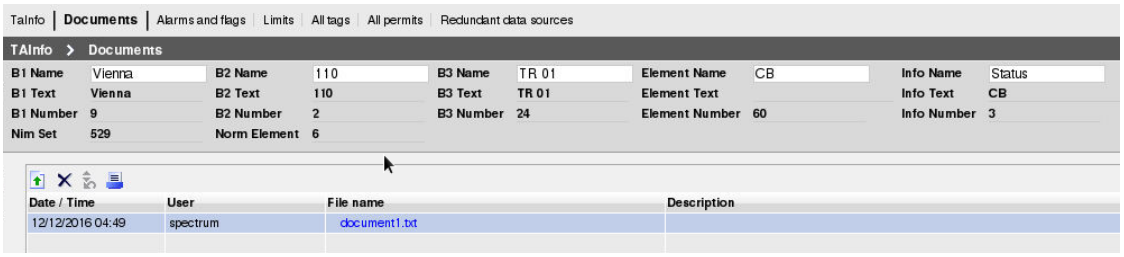


[sc_tainfodocumentupload1, 2, en_US]

Figure 14-40 Select Documents to Upload

- Select the file and click Upload.

  The uploaded file is displayed in the table of documents.



[sc_tainfodocumentupload2, 2, en_US]

Figure 14-41 Documents Uploaded

15 Reminder Summary

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15.1 Introduction

The reminder summary is used to display and acknowledge reminders set on a note, a tag, or a permit.

15.2 User Interface

15.2.1 Overview

The reminder summary contains a list of all active reminders for tags, permits, and notes.

Overview



[sc_reminder-summary, 3, en_US]

Figure 15-1 Reminder Summary

Toolbar

	Restore sort Restores the current sorting order to the default sorting.
	Filter Opens the Filter display for defining filter criteria.
	Reset filter Resets the current filter settings to the default filter.
	Export Opens a submenu for selecting the file type and scope (all rows or only displayed rows) for the export.
	Print Opens a display with options for what to print and where to print it.

List of Reminders

Shows all active reminders set for tags, permits, and free notes with the following information:

- A

  Indicates whether this reminder has to be acknowledged (Y) or not (N)

- Reminder Time

  Time when the reminder expires

- Repetition type

  Interval at which the reminder is repeated, or “one time”, if the reminder is only issued once.

- TA/Theme

  TA and tag/permit for which the reminder has been defined.

15.3 Operational Steps

15.3.1 Acknowledging a Reminder



NOTE

For information on setting a reminder for a tag or permit, refer to chapters 14.3.2 Setting a New Tag or 14.3.9 Setting a Permit to Work Tag.

When a reminder has expired, it is highlighted in the reminder summary and the button in the Basic Signaling Display is flashing. To acknowledge the reminder, do the following:

- Open the Reminder Summary by clicking the corresponding button in the Basic Signaling Display. The Reminder Summary opens.
- Right-click the reminder you want to acknowledge to open the context menu.
- In the context menu, select **Acknowledge**.

  If the reminder will not be repeated in the future, the line is removed from the Reminder Summary.

  Otherwise the letter in the first column changes to “N” and the reminder is not highlighted any more.

  The button in the Basic Signaling Display stops flashing.

16 Favorites

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16.1 Introduction

On web applications, the operator can create favorites for storing user specific adjustments (filter, sort, column size and order) for later usage. Additionally, user specific changes are auto-saved, so that the user finds the display unchanged when he returns to it. Using favorites saves clicks for user-specific adjustments.



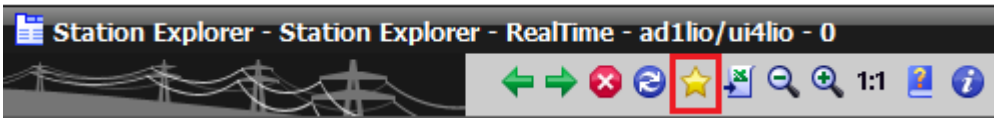
NOTE

Authority roles are defined in CUMAN. For additional information, refer to 6 Console and User Management.

16.2 User Interface

Application displays are enhanced with Favorite  , so that the operator can create and save favorites.

The operator who saves a favorite, can reuse it later privately for his/her own purposes or share it with all operators or only those having the same authority role.



[sc_Favorites Station Explorer, 1, en_US]

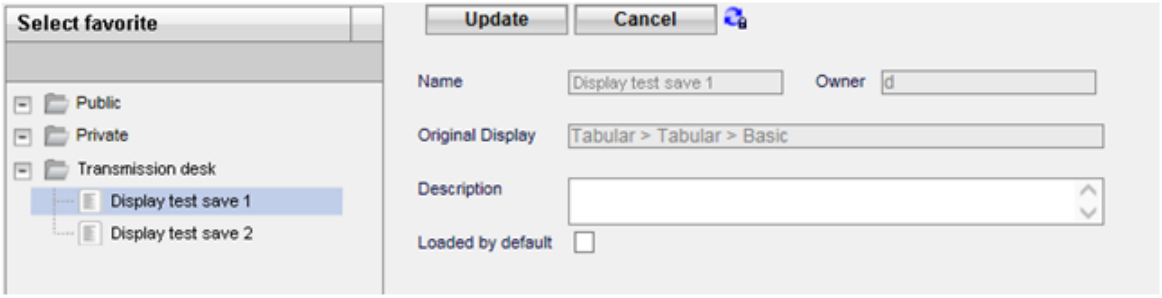
Figure 16-1 Favorite Icon

The favorite window shows a tree view of existing favorites on the left and other options on the right. The tree on left includes a 'Public' folder, 'Private' folder, and folder for current authority role that the operator is logged in as.

While creating a favorite, the operator can decide if he/she wants to share it with other users by selecting the corresponding folder at the left side.

When a favorite is saved in 'Public' folder, the favorite is visible to all other users in the Spectrum system. In case, the operator saves the favorite in 'Private' folder, only the operator can view and use the favorite later.

If the operator wants to share the saved favorite with other operators having the same authority role, then the favorite can be saved in the folder of the current authority role.



[sc_Favorite Window, 1, en_US]

Figure 16-2 Favorite Window

Fields on right side of the window:

- Name

  It shows the user defined name.

- Owner

  Name of the operator who created and saved the favorite.

- Original Display

  Displays path of the original display

- Description

  Shows description of the favorite, if the operator added any.

- Loaded by default

  When this is checked, favorite is opened every time when the operator navigates to that display.



NOTE

System administrator can view all public and role-specific favorites.

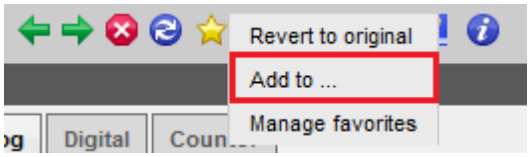
System administrator can save, copy/paste, and delete favorites for all authority roles.

16.3 Operational Steps

16.3.1 Creating a Favorite

To create a favorite:

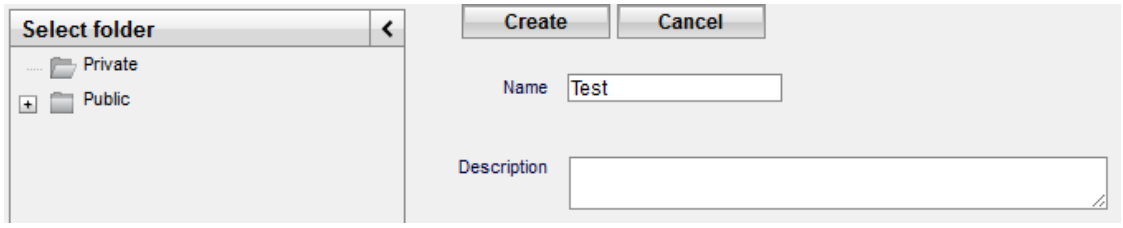
- From application toolbar, click on favorite icon  .
- Slect **Add to**.



[sc_Addto, 1, en_US]

Figure 16-3 Create Favorite

- Favorite window opens. Based on availability of favorite to other operators, select the corresponding folder accordingly from tree on left side of the favorite window.
- Update 'Name' and 'Description' fields in the favorite window.
- Click **Create**.



[sc_CreateFav, 1, en_US]

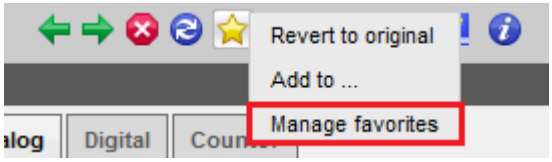
Figure 16-4 Favorites Window

Favorite is created and is saved in the corresponding authority role folder.

16.3.2 Modifying Favorite

To modify an already saved favorite:

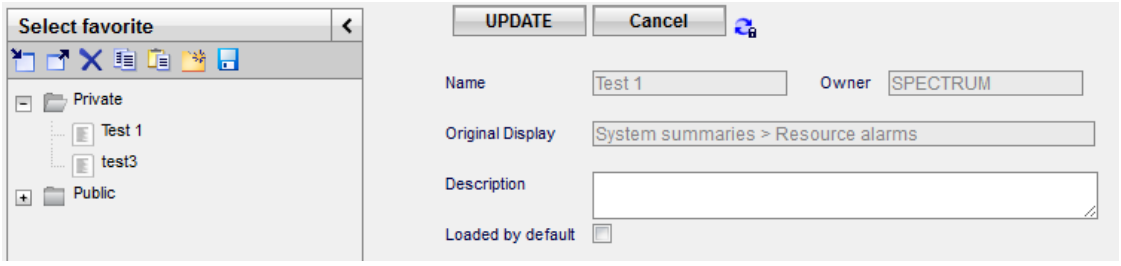
- Click on the favorite icon  and select **Manage favorites** to open the Manage favorites window.



[sc_ManageFav, 1, en_US]

Figure 16-5 Manage Favorites

- Select an already saved favorite under 'Select favorite' tab.



[sc_Modify favorite, 1, en_US]

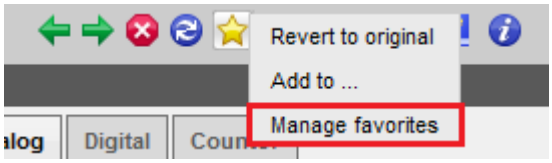
Figure 16-6 Modify Favorite

- Modify 'Description' and Loaded by default' as necessary'.
- Click **Update**.

16.3.3 Setting Favorite as Default

To open an already saved favorite every time when the operator navigates to the display:

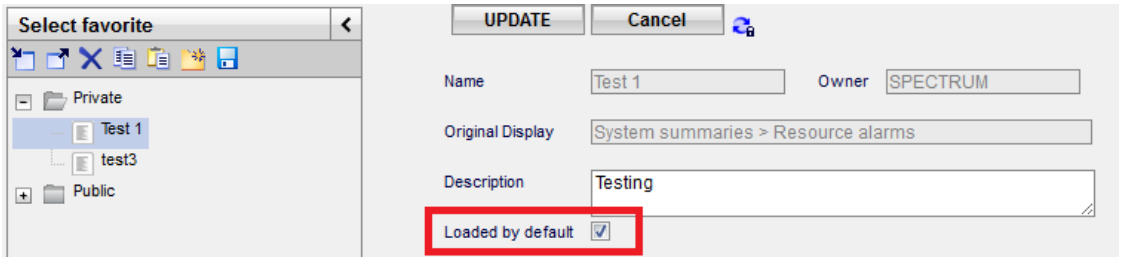
- Click on the favorite icon and select **Manage favorites** to open the Manage favorites window.



[sc_ManageFav, 1, en_US]

Figure 16-7 Manage Favorites

- Navigate and select the favorite in authority roles folder under 'Select favorites' tab.
- Click on 'Loaded by default' checkbox.
- Click **Update**.



[sc_Load by default, 1, en_US]

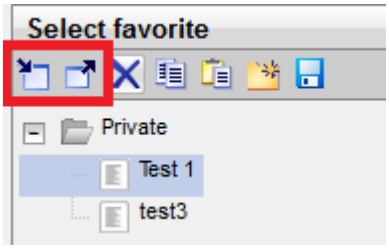
Setting Favorite as Default

16.3.4 Opening Favorites

To open an already saved favorite:

- Open the Manage favorites window.

- Select the desired favorite under 'Select favorite' tab.
- Click to open the favorite in the current window, or to open favorite in new window.



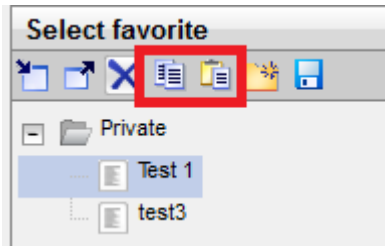
[sc_Open, 1, en_US]

Figure 16-8 Open Favorites

16.3.5 Copying and Pasting existing Favorite

To copy favorite from one folder and paste it in another folder:

- Open Manage favorites window.
- Select the desired favorite under 'Select favorite' tab.
- Click to copy the favorite.
- Select another folder where the copied favorite is to be pasted.
- Click to paste the favorite in the selected folder.



[ic_CopypastingFav, 1, en_US]

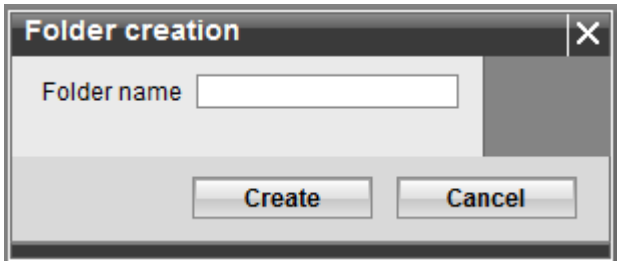
Figure 16-9 Copy and Paste favorite

16.3.6 Creating Folder for Favorite

To create sub-folder within authority role folder:

- Open Manage favorites window.
- Select the folder under 'Select favorite' tab in which a sub-folder is to be created.

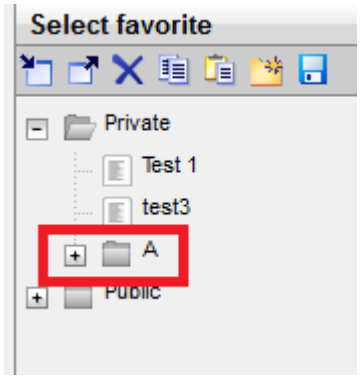
  Click to open folder creation window.



[sc_FolderCreation, 1, en_US]

Figure 16-10 Folder Creation Window

- Provide a name to the new folder.
- Click **Create**. New sub-folder is created.



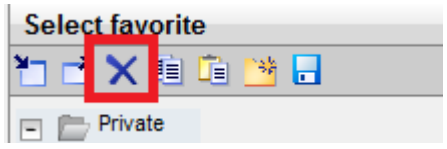
[sc_Subfolder, 1, en_US]

Figure 16-11 Example of Sub-folder for Favorite

16.3.7 Deleting Favorite

To delete an already saved favorite:

- Open Manage favorites window.
- Select the favorite from under 'Select favorites' tab.
- Click ✕ to delete favorite from the folder.



[sc_DeleteFav, 1, en_US]

Figure 16-12 Delete Favorite

NOTE

Delete favorites carefully as deleted favorites cannot be restored.

17 Curve Tool

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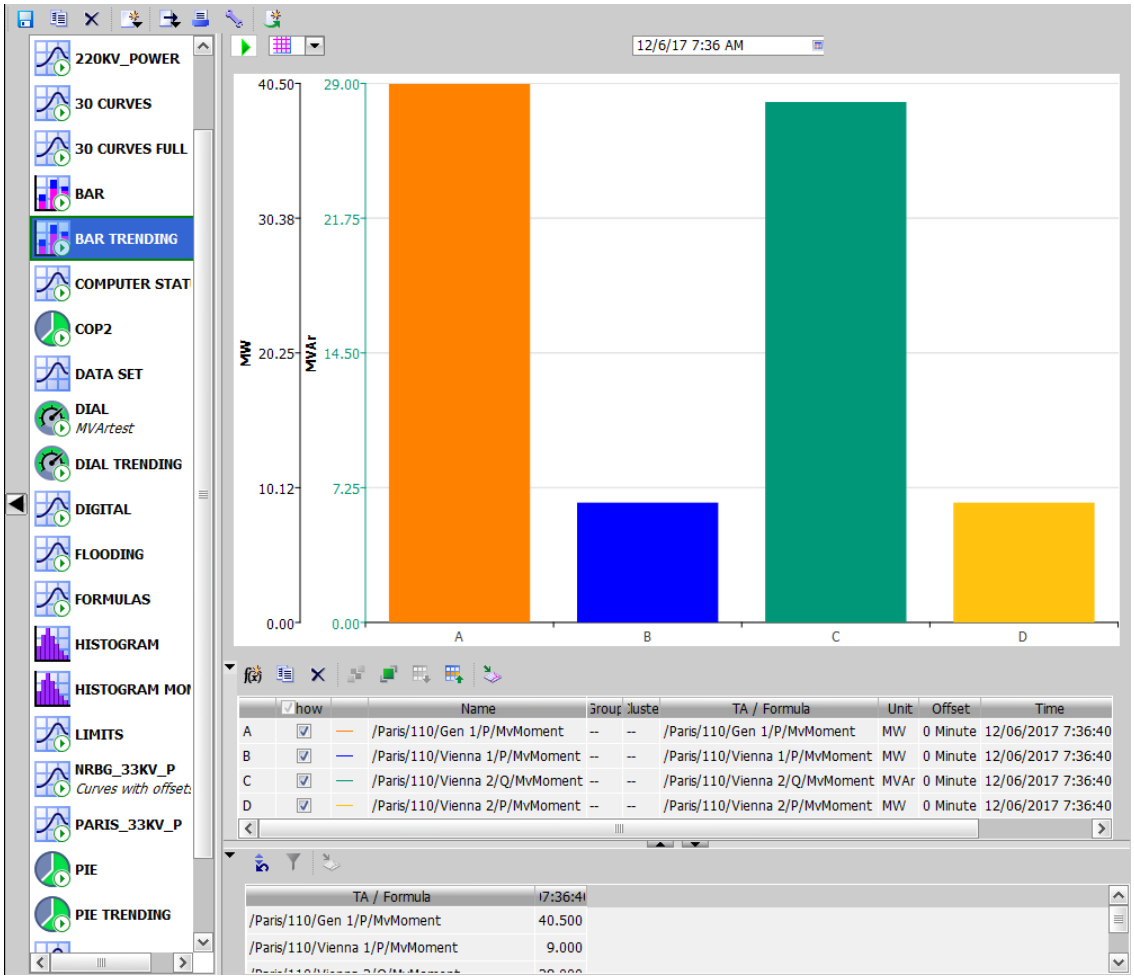
17.1 Introduction

The Curve Tool displays charts that represent analog or digital values. The values can be taken from the operational database, from an archive, or from a schedule.

17.2 User Interface

17.2.1 Overview of Curve Tool Display

Click the **Curves** button under the **Curves and Tables** tab in the Basic Signalling Display to open the Curve tool. A second possibility is to open it from the context menu of a TA in the network diagram. In this case, the Curve Tool is opened showing a new set containing the curve for the selected TA. Tooltips appear upon mouse over on the legend or the curve.



[sc_CurveToolwithValuetable, 1, en_US]

Figure 17-1 Curve Tool

The basic display structure consists of the following:

- Toolbar
- Chart selection area
- Chart view area
- Legend
- Value Table

The Curve tool supports line charts, bar charts, pie charts, dials, and histograms.

17.2.2 Toolbar

The toolbar consists of a row of buttons providing the following functionality:

- 
  New set. Opens a dropdown, where user can select type of new data set: Curve, Bar chart, Pie chart, Histogram, or Dial chart.

  A new (empty) data set of this type is created.
- 
  Save current loaded set. Saves the changes made to the currently loaded set.
- 
  Copy set. Creates a copy of the currently selected set.

  After creating a copy, the duplicate set is automatically loaded. This means that if there is unsaved data on the initial set, curve tool provides a chance to cancel the copy so that changes on the original set can be saved first.
- 
  Remove set. Deletes the selected set.
- 
  Export current visible time window. Exports data points that are currently displayed in the curve display area to a CSV, Microsoft® Excel, or image file.
- 
  Print current loaded set. Prints the currently loaded set to a printer.
- 
  Open chart settings display. The chart settings display is used to modify data points and to configure appearance of chart.
- 
  Adds the set to a new Operational dashboard.

The following icons are displayed in the toolbar in chart display area when a data set is loaded:

- 
  Trend set.

  Initially when this icon is selected the curve display section shows the past values defined by a time range and the current values of the TAs, shifting forward as time elapses.

  Clicking this icon to stop trending mode. A bar on top of the curve display section allows to scroll forward and backward in time.
- 
  Grid lines option. Add grid lines to the set. The display of this icon and the number of options depend on chart type:

  – Curves: Horizontal and vertical, Horizontal, Vertical, or No gridlines

  - Bar chart/Histogram: Horizontal or No gridlines
  - Pie chart/Dial: Icon is not displayed

- 
  Recalculate chart.

  Available only for Histogram. Calculates and plots the whole histogram chart based on values of data points and configured time range.

The legend toolbar consists of a row of buttons providing the following functionality:

- 
  New calculated data point. Adds a new calculated data point to the set
- 
  Duplicate data point within a set.
- 
  Remove data point. Deletes the selected data point from the set.
- 
  Send data point to back
- 
  Send data point to front
- 
  Move data point up in the legend.
- 
  Move data point down in the legend.
- 
  Displays TA for the data point on a network diagram.

17.2.3 Chart Selection Area

This area contains all defined sets. Loading one of them displays the data points stored in this set in the curve display area.

17.2.4 Chart View Area

This area displays the chart as defined in the set that is currently selected in the set area.

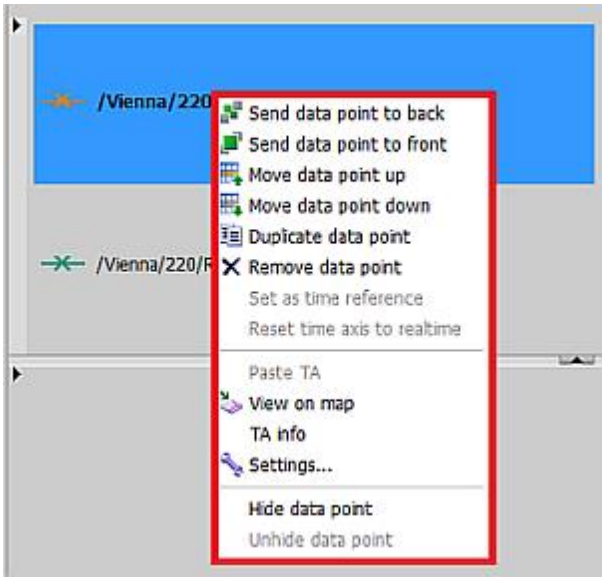
17.2.5 Legend

This area displays information about the data points in the set. Operations on the data points, such as calculations, view on map, and so on can be done when a data point is selected from the legend.

Click the arrow next to the legend to minimize or maximize it.

The operator can move the legend to another position on screen (left/right/top/bottom) to optimize the display layout. The operator can save the legend position and size for future use. When the operator changes order of TAs in legend, the value table also gets updated accordingly.

Right-click on a data point displays the context menu for the selected data point.



[sc_Context menu, 1, en_US]

Figure 17-2 Context Menu

The entries of context menu in line chart are:

- 

  Send data point to back

  Selected curve is sent to the background in chart view (lowest z-order).

- 

  Send data points to front

  Selected curve is brought to the foreground in chart view (highest z-order).

- 

  Move data point up

  Selected data point is moved one row up in table. The order of the table represents for

  – Line charts: the z-order for plotting the curves/flooding
  – Bar charts: the order of bars or stacking order of segments
  – Pie charts: the order of slices

  In dials, only the uppermost data point of legend table is displayed. Therefore, user can use this menu item to move a data point step by step to the uppermost position.

  Not visible in Histogram.

- 

Move data point down

Selected data point is moved one row down in table. The order of the table represents for

- Line charts the z-order for plotting the curves/flooding
- Bar charts the order of bars or stacking order of segments
- Pie charts the order of slices

Not visible in Histogram.

- 
  Remove data point

  Removes current selected data point from chart

- 
  View on map

  ODV is opened and TA of selected data point gets displayed on a network diagram.

  Alternatively, a TA can be dragged on and dropped from the legend onto an open ODV window to display information about the TA.

- 
  Settings

  Open chart settings display. The chart settings display is used to modify data points.

Depending on chart type, the context menu entries differ:

- Bar chart (not stacked/grouped): Send data point to back, Send data point to front, Move data point up, Move data point down, Duplicate data point, Remove data point, Paste TA, View on map, TA info, Settings, Hide data point, Unhide data point
- Bar chart (stacked/grouped): Edit group, Remove data point, Paste TA, View on map, TA info, Settings, Hide data point, Unhide data point
- Histogram (context menu appears only when legend is maximized): Remove data point, Paste TA, View on map, TA info, Settings
- Pie chart: Move data point up, Move data point down, Duplicate data point, Remove data point, Paste TA, View on map, TA info, Settings, Hide data point, Unhide data point
- Dial: Move data point up, Move data point down, Duplicate data point, Remove data point, Display on dial, Paste TA, View on map, TA info, Settings

Click **Hide data point** to temporarily hide a data point.

Right-click on a data point and select **TA info** to open the TA info window for the data point.

17.2.6 Value Table

This area displays the detailed values of the data points. The value table supports sorting, filtering, and changing of column width.

The value table is constructed differently depending on the current chart type.

17.2.7 Chart Settings Display

17.2.7.1 Introduction

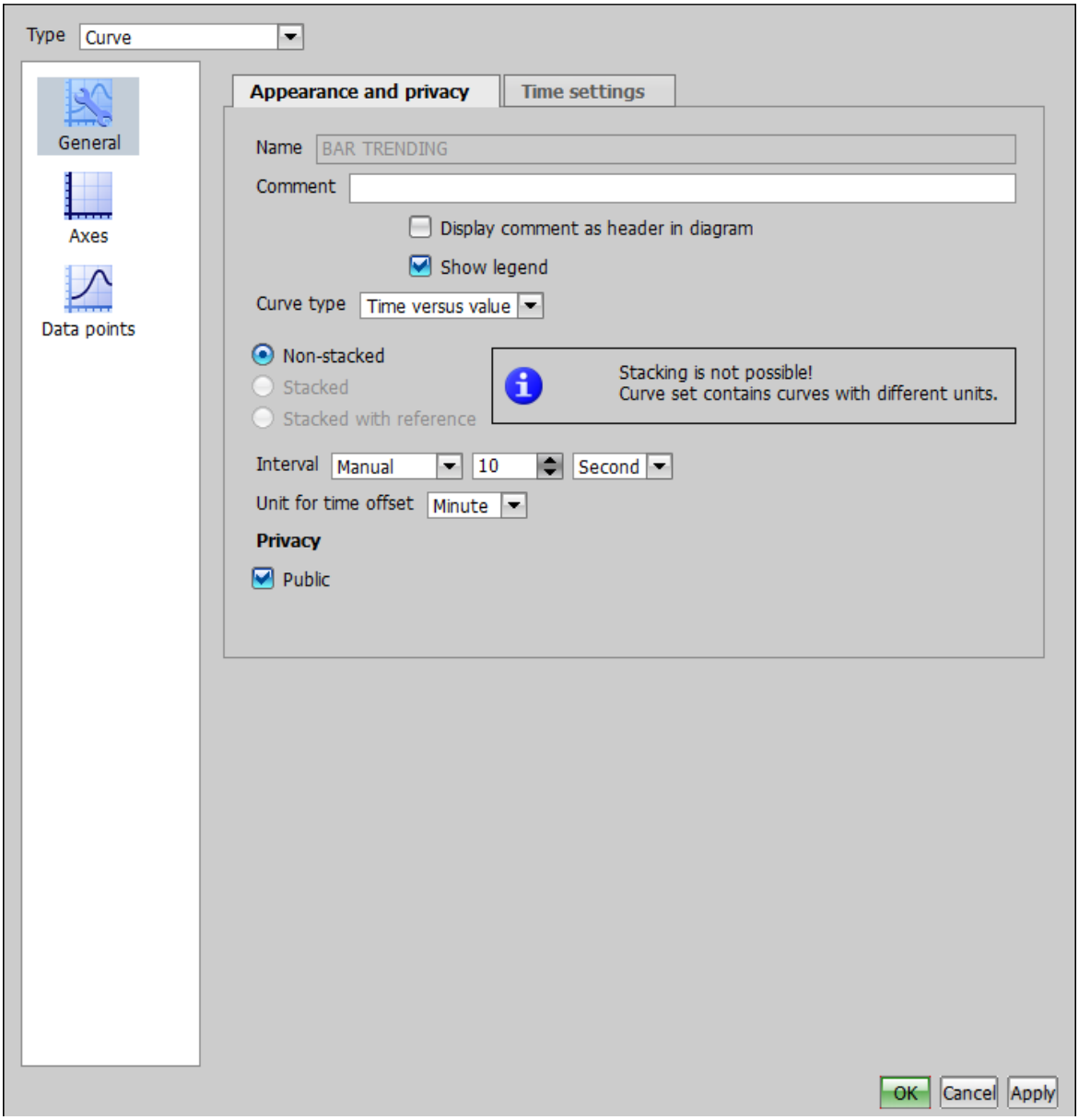
The chart settings display is used to define the chart type and make definitions for the currently loaded set and its data points.



NOTE

The chart settings display changes depending on the type of chart.

17.2.7.2 General



[sc_PreferenceDialog, 4, en_US]

Figure 17-3 Chart Settings - General Trending

Appearance and Privacy

- Name

  Shows the name of the set (is shown in the set area for easier user orientation).

- Comment

  Comment for the set area.

  – Display comment as header in diagram

    When selected, comment is shown as background text on the curve display

– Show legend

When selected, the legend table is displayed.

- Chart Type and definition

The values and selections used to define the chart, for example, horizontal or vertical for bar charts. The fields change depending on the type of selected chart.

Time Settings

NOTE

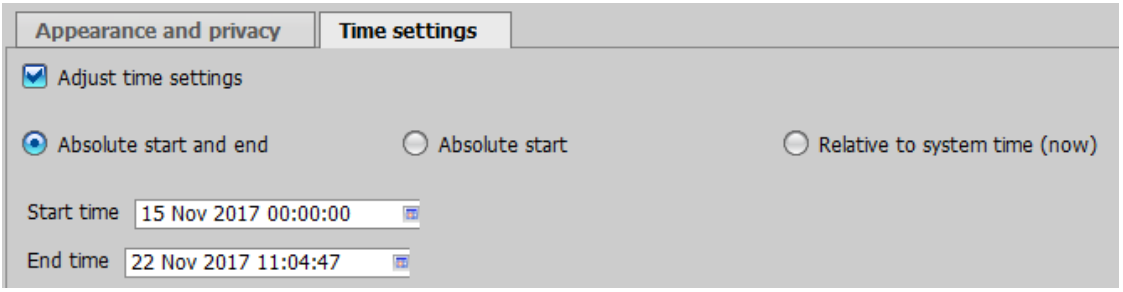
This tab appears for line charts and histograms only.

The start time, end time and the update cycle for the curve is defined on the **Time settings** tab.

To make it clear for the user whether time parameters should be taken over and to avoid accidentally over-writing of time settings done by time navigation bar, a checkbox **Adjust time setting** is added. This parameter is unchecked by default and not stored with chart. When it is unchecked, all time parameters are disabled. When it is checked, all time setting values are applied to chart (independent of whether the operator had changed it or not) when operator confirms it. After successful application of changes the checkbox is un-checked again.

The tab **Time settings** has three different modes of setting time for line chart:

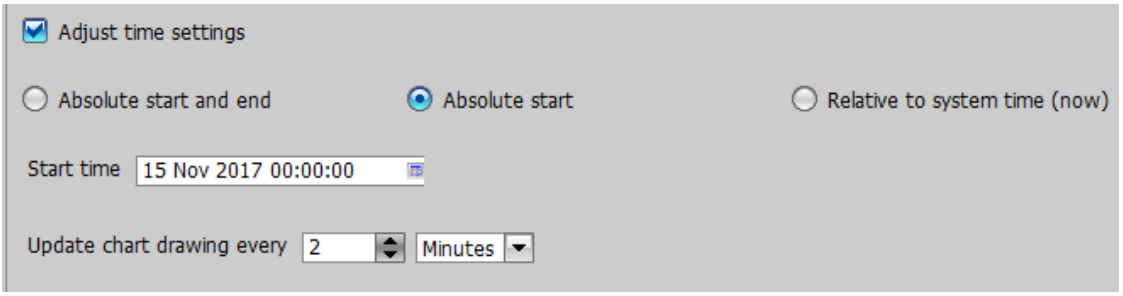
- Absolute start and end time



[sc_AbsoluteStartEnd, 1, en_US]

Figure 17-4 Absolute Start and End Time Settings

- Absolute start

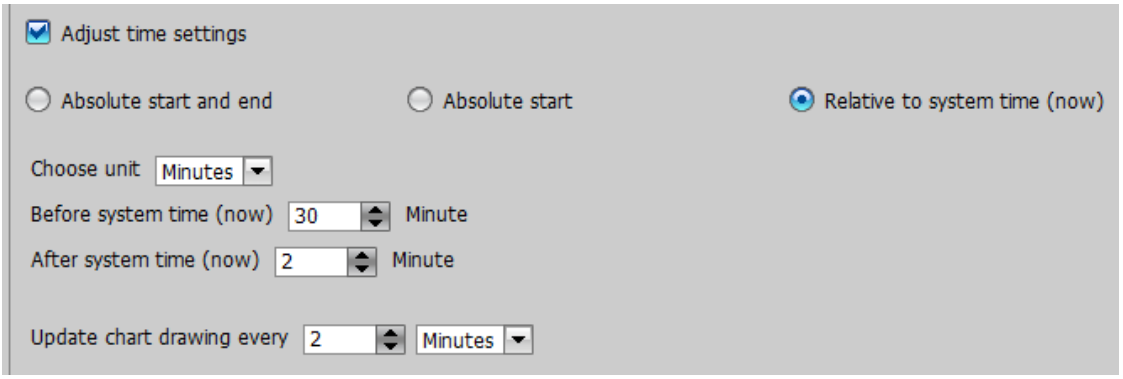


[sc_Absolute start, 1, en_US]

Figure 17-5 Absolute Start Time Settings

Default value for **Update chart drawing every** is 2 minutes.

- Relative to system time (now)



[sc_relative to systemtime, 1, en_US]

Figure 17-6 Relative to System Time (now)

This option includes real-time trending.

When time setting is **Relative to system time (now)** the parameters have following default values:

- Before system time (now) - 30 minutes
- After system time (now) - 2 minutes
- Update chart drawing every - 2 Minutes

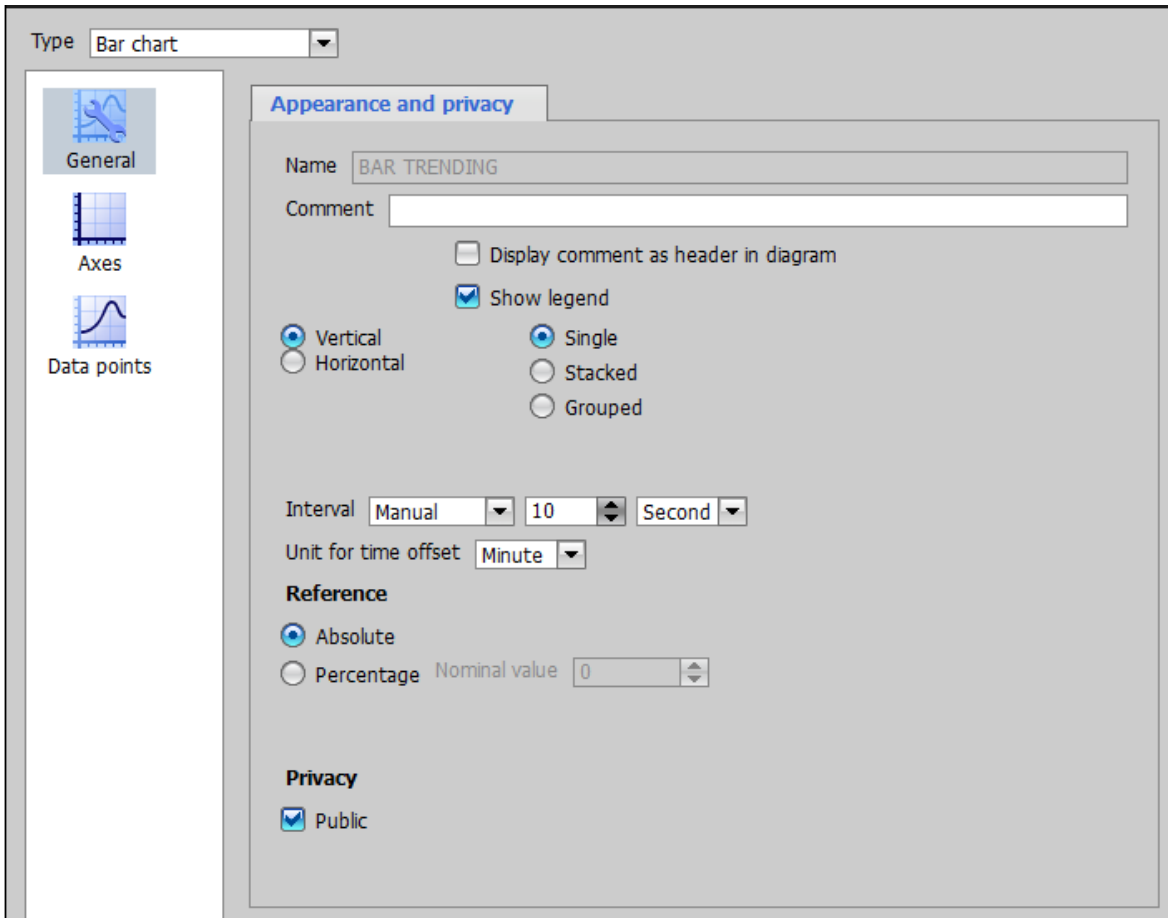
Assignments

The **Assignments** tab is used to create clusters in bar charts. Refer 17.3.12 Creating and Modifying Bar Charts for more information.

Privacy

Defines whether the set is private for the user or available for all users.

General Settings for Bar Charts



[sc_chartsettingsGeneralbarcharts, 2, en_US]

Figure 17-7 Chart Settings for Bar Charts - General

Appearance and Privacy

The appearance and privacy fields of the bar charts contain similar fields to the line chart apart from:

- Vertical or Horizontal:

  Bar charts can either be displayed horizontally or vertically. Select the corresponding radio button to choose the orientation of the bars.

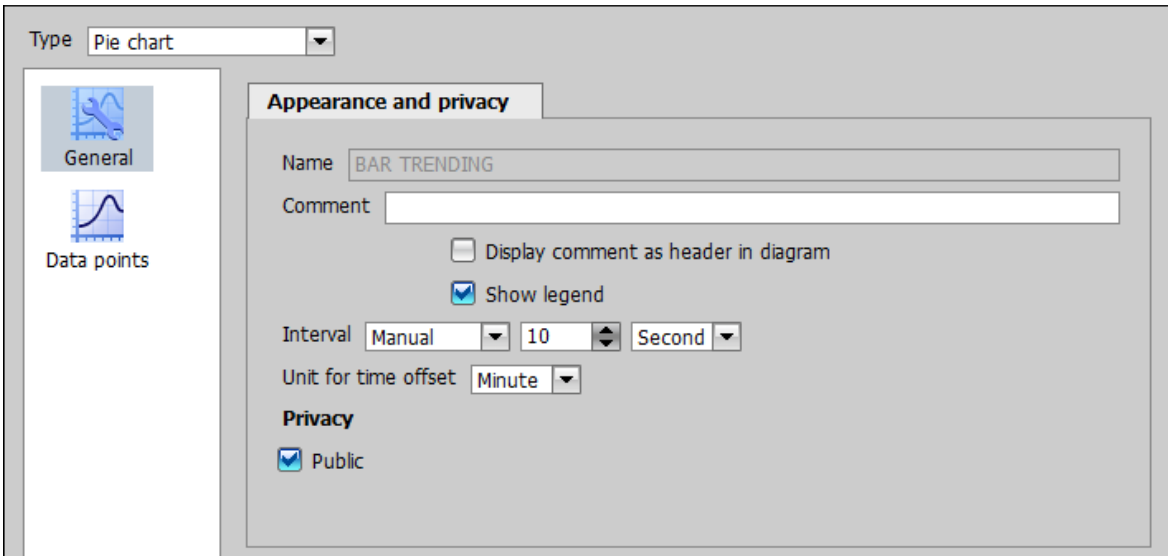
- Single, Stacked, and Grouped

  By default data points are displayed as single bars. Data points can be stacked or grouped using the **Assignments** tab. Grouped and stacked bars are organized using clusters and groups. For further information, refer to 17.3.12 Creating and Modifying Bar Charts

- Reference

  The data points can be displayed in absolute or percentage values. To display the chart in percent, a nominal value has to be defined. The percentage is calculated with reference to the nominal value.

General Settings for Pie Charts

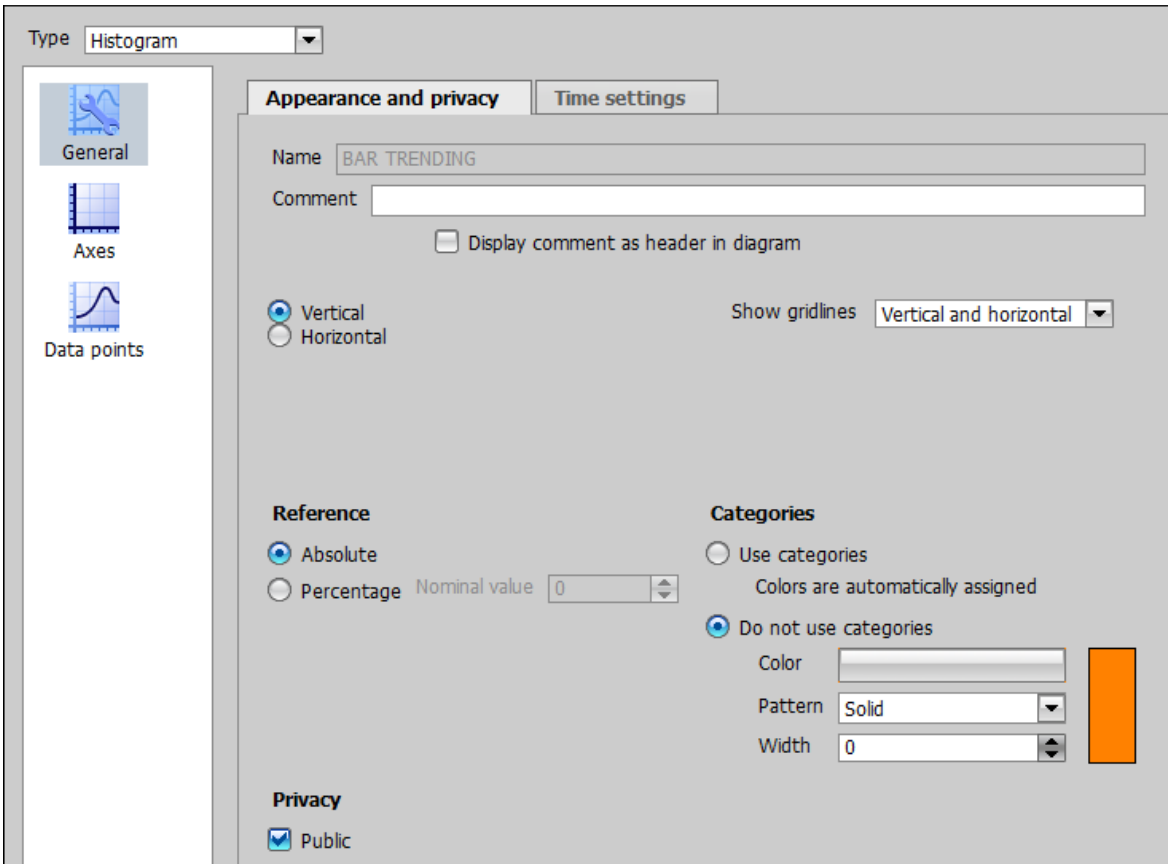


[sc_Generalsettingspiecharts, 2, en_US]

Figure 17-8 Chart Settings - General - Pie Charts

The **General** tab for pie charts contains only the appearance and privacy fields. The fields are similar to the line chart.

General Settings for Histogram



[sc_Generalsettingshistogram, 2, en_US]

Figure 17-9 Chart Settings - General - Histogram

Appearance and Time Settings

The **General** tab for histograms contain only the appearance and privacy fields similar to bar charts apart from:

- Show gridlines

  The histogram can be displayed with horizontal, vertical, both horizontal and vertical, or no gridlines.

- Vertical or Horizontal:

  Histograms can either be displayed horizontally or vertically. Select the corresponding radio button to define the orientation of the bars.

- Reference

  The data points can be displayed in absolute or percentage values. To display the chart in percent, a nominal value has to be defined. The percentage is calculated with reference to the nominal value.

- Categories

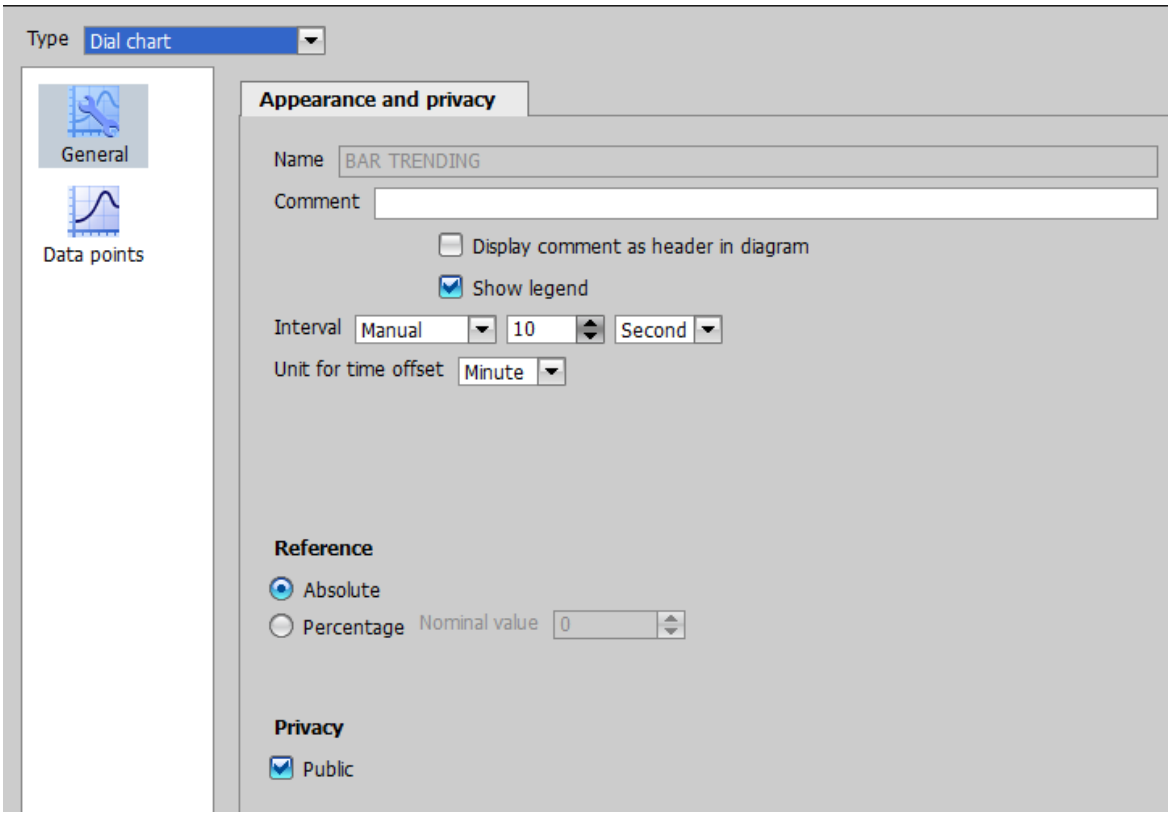
  Categories divide the selected time range into equal time slices which are displayed using a stacked bar chart.

The color is predefined in the curve tool and depends on the time interval (for example, hour, week, and month)

Time Settings

A start time and the time duration (day, week, month, or year) is defined to calculate and display the histogram.

General Settings for Dials



[sc_General Settings for Dials, 1, en_US]

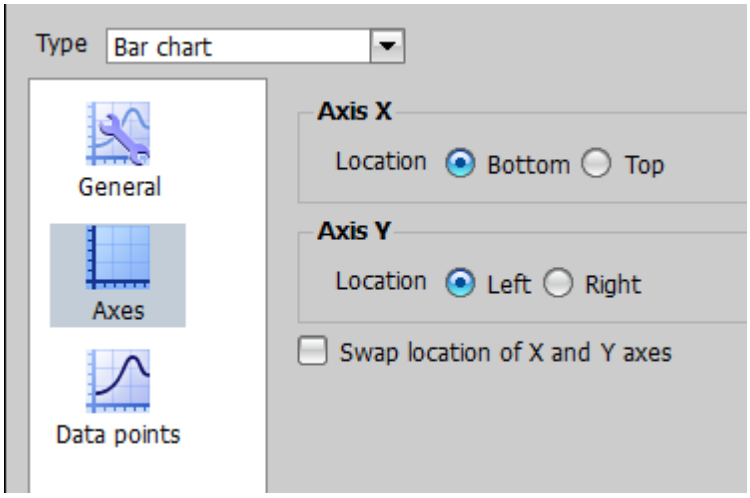
Figure 17-10 Chart Settings - General - Dials

The **General** tab for dials contains only the Appearance and privacy fields. The fields are similar to the line chart.

Reference

The dial shows the difference between an actual value and a nominal value. The scale of the dial can be an absolute (default) or percentage value.

17.2.7.3 Axes



[sc_curve-preferences-axes, 3, en_US]

Figure 17-11 Chart Settings - Axes

Used to define the axis settings and swap the x-axis and y-axis.

NOTE

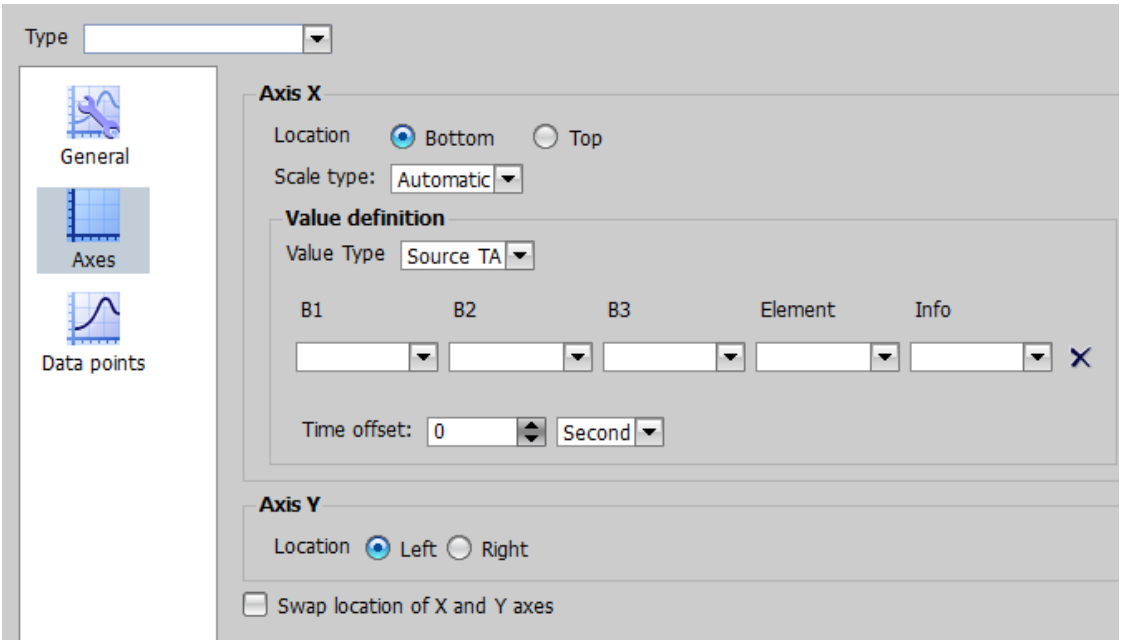
Axes are not present for Pie charts and Dials.

Axis X

For line charts (time versus value or value versus value), the definition of the axis X differs:

- Time vs. value
  - Scale direction

    Left to right, right to left, top to bottom, or bottom to top.
- **Value Vs. value - Autoscale Option**



[sc_Axes dialog, 2, en_US]

Figure 17-12 Chart Settings - Autoscale

- Scale location

  Bottom or top.

- Scale type

  Automatic or manual can be selected. In case of automatic scale type, the following fields are provided for value definition:

- Value Definition
  - ◊ Source TA

    TA whose values are displayed. A new TA can be entered manually or using drag and drop from a network diagram, or a formula can be used.

  - ◊ Formula

    Formula for the new data point.

  - ◊ Time offset

    Defines the starting point of the curve. It can be used to compare the values of a TA at two different points in time.

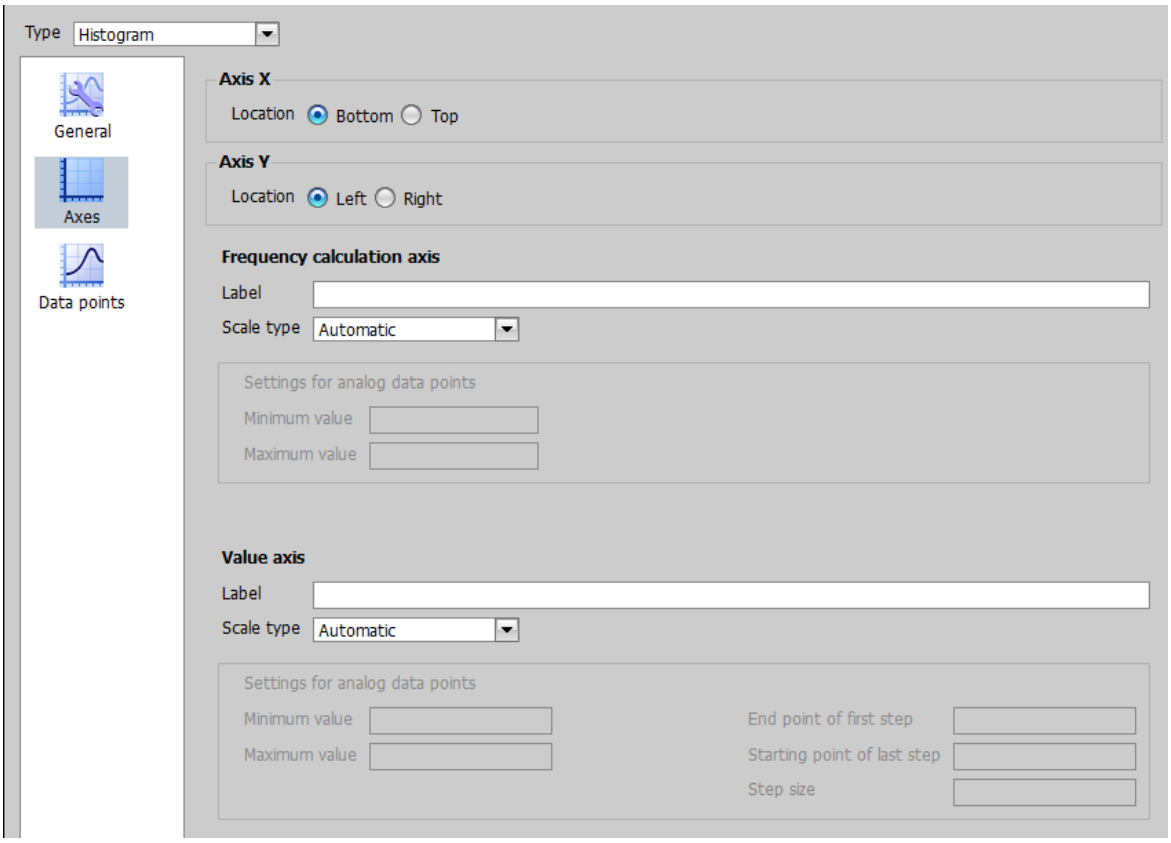
    For example, offset of 0 sec means that the curve starts at the current date and time, offset of -1 day means that the curve starts at the same time of the previous day.

Axis Y

The Y axis can be positioned left or right.

The location of X and Y axes can be swapped.

Axes for Histograms



[sc_Axishistogram, 2, en_US]

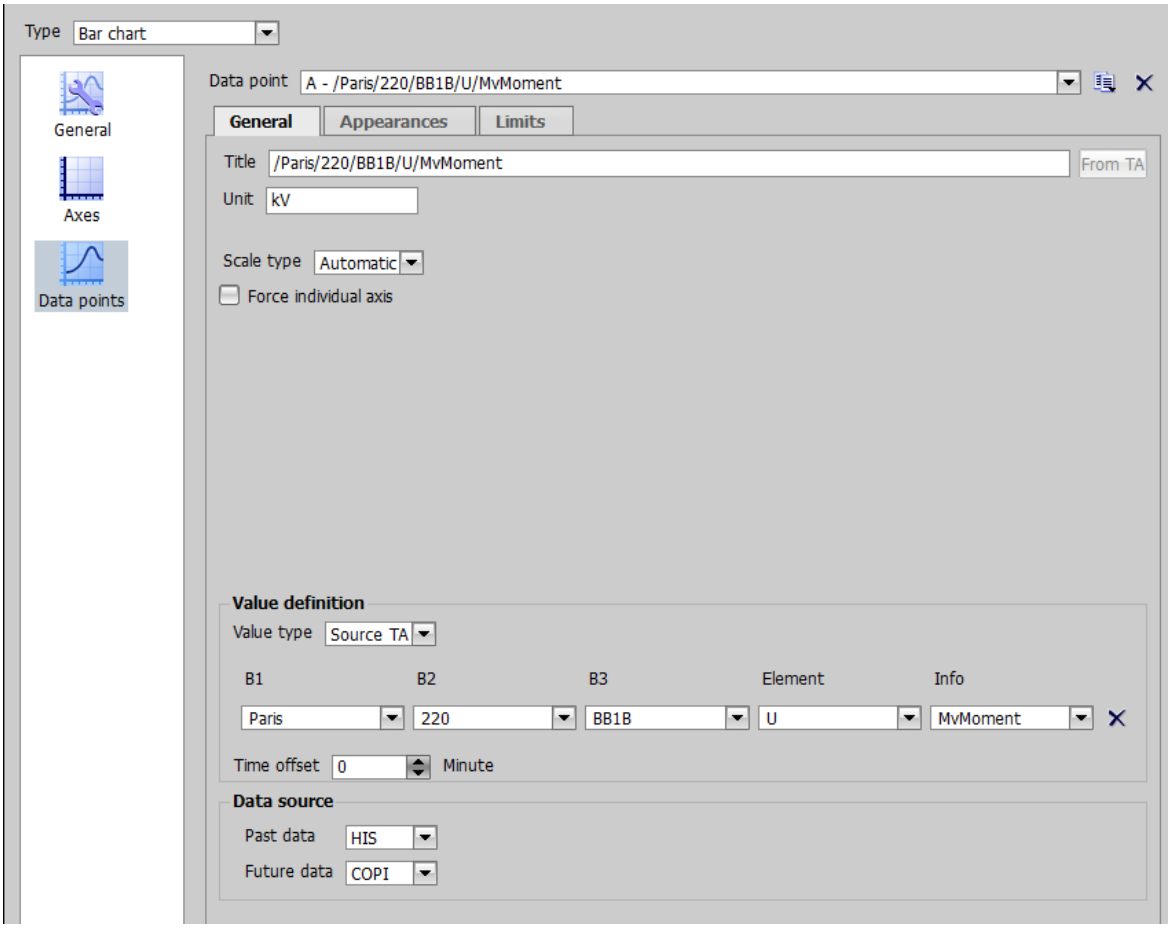
Figure 17-13 Axes for Histograms

The frequency and the value axis can be labelled for histograms. By default, the scale type is automatic. The scale type can also be manually defined with a minimum and maximum value for frequency calculation and the minimum and maximum value, end point of first step, starting point of last step, and step size for the value axis.

17.2.7.4 Data Points

Data point is used to set parameters for each individual data point. The fields change depending on the chart type. The following example considers a line chart type.

At the top of the section, there is a drop down menu box to select the data point.



[sc_Data-Series, 1, en_US]

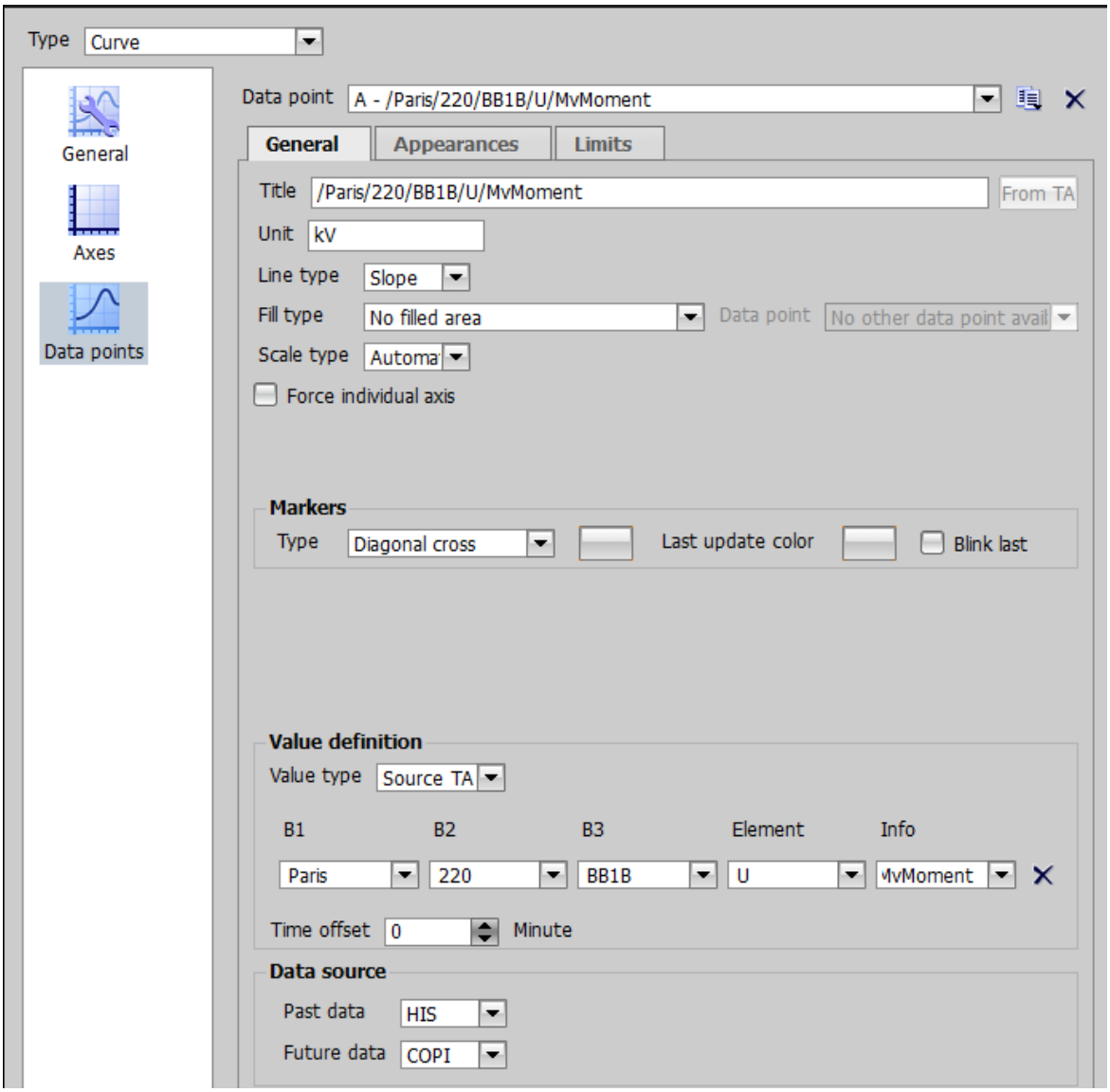
Figure 17-14 Chart Settings - Data Points

Following icons copy the data point setting and delete the data point:

- 
  Opens a display to choose the data point to which the settings of the current data point will be copied.
- 
  Deletes the selected data point from the set.

The lower part of the section has the following three tabs.

General



[sc_curve-preferences-dataseries-general, 3, en_US]

Figure 17-15 Chart Settings - General Settings

General Settings

General Settings has the following options:

- Title

  A title for the data point can be entered. By default, TA is entered. When entering a new title, the From TA button is enabled to reset the title to default.

- Unit

  Unit of the data point.

- Line type

  Slope, step, and scatter

- Fill type

  Used for flooding curves.

- Scale type

  Possible values are automatic and manual. In case of manual scaling, fields for input of upper and lower limit are displayed.

- Force individual axis

  Specifies whether to display an individual axis for the curve.

  The curve tool combines TAs with the same measurement unit, for example, MW, on the same axis by default. When the value axis represents more than one data point, the combined axis is shown in a neutral color, for example, black.

Markers

Markers setting has the following options:

- Type

  Whether markers are displayed as well as the type of the marker. For example, square, cross, and so on.

- Last update color

  If a marker type has been selected, the color for the marker is displayed and can be modified.

- Blink last

  A check box to define if the last marker should always blink.

Value Definition

Value definition setting has the following option:

- Source TA

  TA whose values are displayed. A new TA can be entered manually or using drag and drop from a network diagram.

- Formula

  A formula can be used to define a data point.

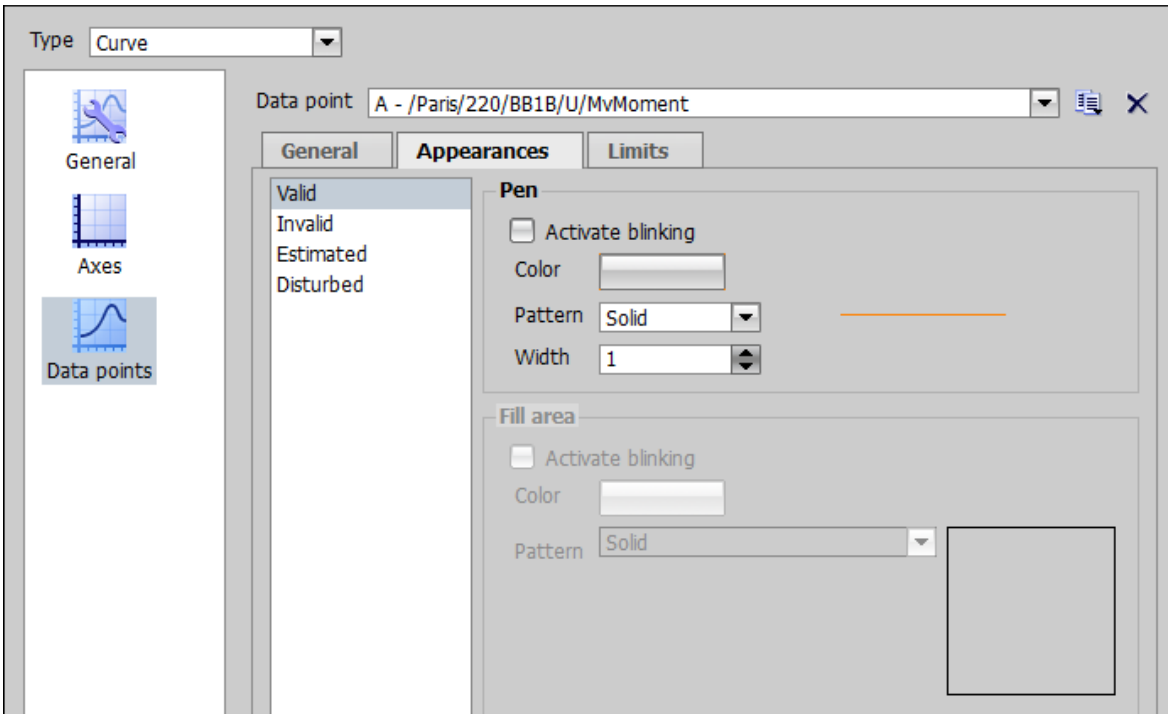
- Time offset

  Defines the starting point of the data point. It can be used to compare the values of a TA at two different points in time. For example, offset of 0 sec means that the curve starts at the current date and time, offset of -1 day means that the curve starts at the same time of the previous day.

- Data Source

  The data source for past data and future data of the curve.

Appearances



[sc_dataseriesappearances, 2, en_US]

Figure 17-16 Chart Settings - Data Points for Appearances Tab

Appearances tab has the following options:

- Qualities

  Value qualities are valid, invalid, estimated, and disturbed.

- Activate blinking

  Click Activate blinking checkbox to activate blinking in chart display area.

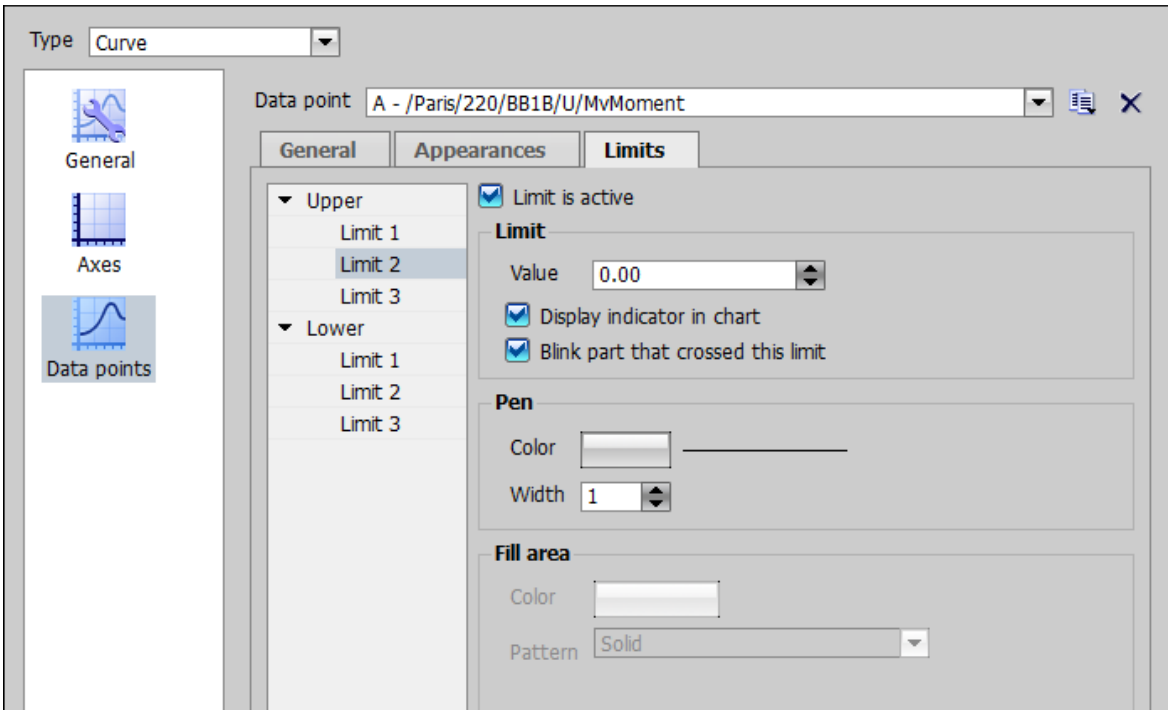
- Pen

  Color, pattern, and width of the line can be defined.

- Fill area

  If Filled Area is selected in the **General** tab, color and pattern of the fill area can be defined.

Limits



[sc_dataserieslimits, 2, en_US]

Figure 17-17 Chart Settings - Limits

Up to three upper limits and three lower limits can be defined. If a limit is set to active, a limit value as well as the representation of the limit in the curve display area can be defined.

Limits tab has the following options:

- Limit is active

  When this check box is selected, parameters for representation of the limit can be entered and the visual representation of the limit will be displayed in chart display area.

- Limit

  Limit has following options:

  – Value

    Input field for entering a value for the limit. In case a seasonal limit is defined for the TA, the field contains this value.

  – Show line indicator on plot area

- Limit Appearance

  Part of the curve that has crossed this limit blinks.

  Check box to determine whether the part of the curve that crossed the selected limit is blinking.

  In case filled has been selected on the **General** tab, this area is used to select the color of the fill area. The color of the line can always be defined independent of filling.

Data Points for bar charts, pie charts, and dials have fields similar to line charts. Histograms have no limits.

17.2.7.5 Stacked Curves

For information on stacked bar charts, refer to 17.3.12 Creating and Modifying Bar Charts.

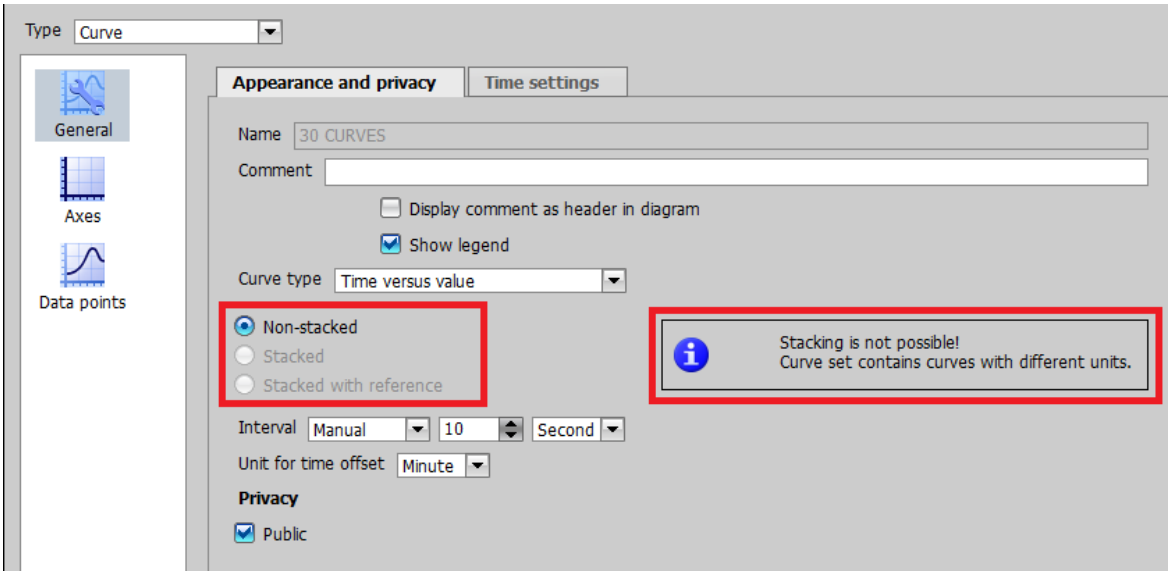
Line charts can be displayed as stacked or non-stacked.

In chart settings dialog, operator can select mode for line charts by selecting **Non-stacked**, **Stacked**, or **Stacked with reference**.

Stacked means that all curves are stacked over each other in reference to zero line. The stacking order is defined by line order in maximized legend.

Stacked with reference means that all curves are stacked in reference to another curve, which is configured in **Chart settings** > **Data points** > **General**. Minimum one data point must exist which is related to zero line.

Changing a non-stacked set to a stacked mode is only possible when all curves have same unit. If a set contains curves with different units the additional modes are disabled and a hint for the operator is displayed.



[sc_StackedCurvehint, 1, en_US]

Figure 17-18 Chart Settings Dialog

In **Non-stacked** and **Stacked** mode the parameters **Reference for stacking** and **Stacking** in Chart settings > **Data points** > **Appearance** are not displayed.

In both stacking modes chart displays only one y-axis. Therefore, the parameter **Force individual axis** is disabled. The features for definition of location of x and y axis and swapping their location are the same as for a non-stacked line chart.

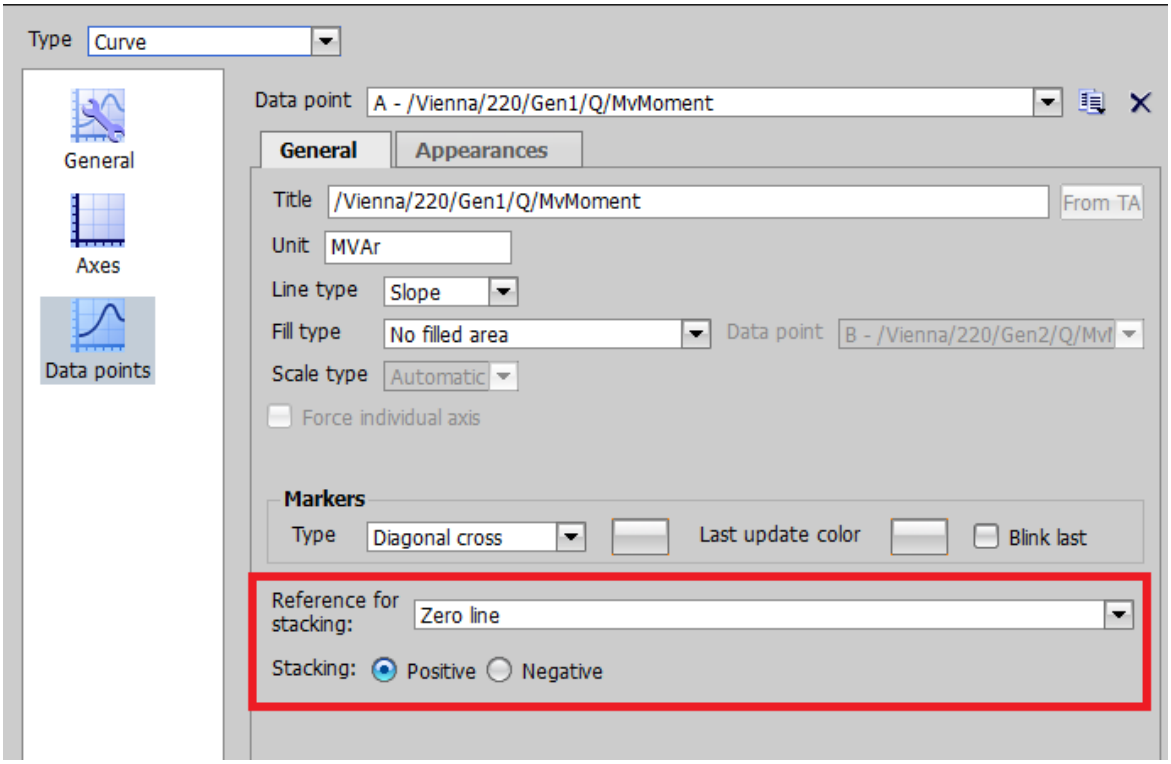
In **Stacked with reference** mode, for every data point, a reference where stacking is related to can be configured in **Chart settings** > **Data points** > **General**. The **Reference for stacking** is set by default to **Zero line**.

Zero line means that curve of current data point is plotted based on zero line (no stacking). Digital TAs cannot be used as reference.

The operator can change **Reference for stacking** from **Zero line** to one of the data points from the drop-down list. This means the curve of current data point is plotted based on the data point that is used as a reference for stacking.

Using **Stacking**, the operator can define how stacking is done. **Positive** means that the values of the data point are added to values of reference data point. **Negative** means that the values of the data point are subtracted from values of reference data point.

Default setting is Zero line and positive stacking. If a new TA is dropped, the new data point will get the same default settings.

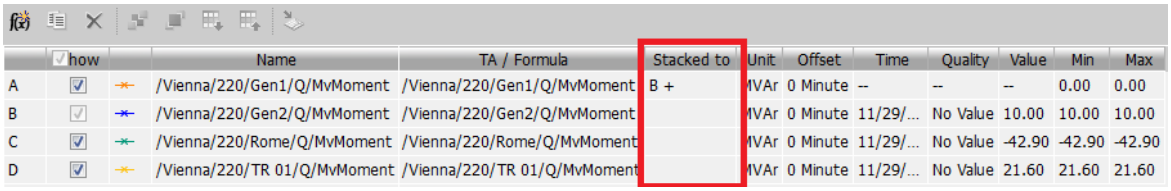


[sc_ChartSettingsstackedreference, 1, en_US]

Figure 17-19 Chart Settings Dialog in Stacked with Reference mode

In **Non-stacked** and **Stacked** mode the maximized legend is unchanged.

In **Stacked with reference** mode, the maximized legend displays the reference to which every data point is related regarding stacking in an additional column. The column **Stacked to** displays the identifier of the related reference data point which is used as base for stacking. If it is empty, stacking is related to zero line. The type of stacking is displayed using a plus or minus sign as suffix.



	how	Name	TA / Formula	Stacked to	Unit	Offset	Time	Quality	Value	Min	Max
A	☑	/Vienna/220/Gen1/Q/MvMoment	/Vienna/220/Gen1/Q/MvMoment	B +	MVAr	0 Minute	--	--	--	0.00	0.00
B	☑	/Vienna/220/Gen2/Q/MvMoment	/Vienna/220/Gen2/Q/MvMoment		MVAr	0 Minute	11/29/...	No Value	10.00	10.00	10.00
C	☑	/Vienna/220/Rome/Q/MvMoment	/Vienna/220/Rome/Q/MvMoment		MVAr	0 Minute	11/29/...	No Value	-42.90	-42.90	-42.90
D	☑	/Vienna/220/TR 01/Q/MvMoment	/Vienna/220/TR 01/Q/MvMoment		MVAr	0 Minute	11/29/...	No Value	21.60	21.60	21.60

[sc_Maxstackedreference, 1, en_US]

Figure 17-20 Legend in Stacked with Reference mode

In case of invalid values in stacked curves , the operator can configure to hide invalid values, and curve section of the related data point is not plotted.

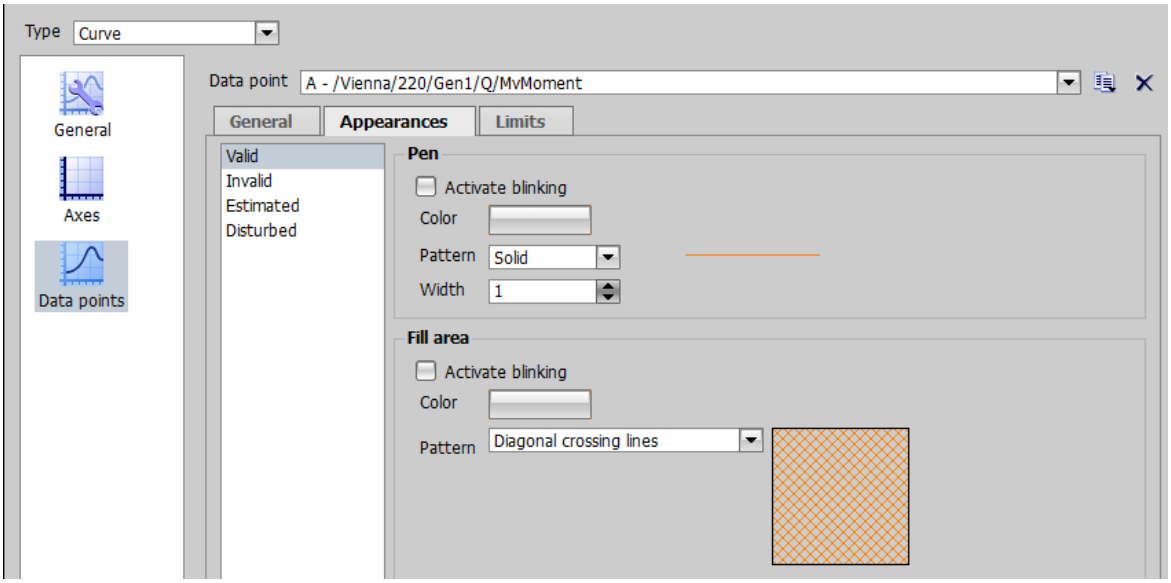
17.2.7.6 Blinking of Curves Based on Quality Codes and Limit Violations.

Curve tool supports displaying of value states. Curves and flooded areas can be configured to blink depending on their state, for example, in case of limit violation or disturbed values.

Blinking of curves is available for line charts, bar charts, and pie charts. Only the part of the curve or chart that is affected blinks. Blinking is only visible when set is open and a new event occurs while trending.

Blinking is configured between actual color and neutral color which is different from background, so that the data point is visible when it blinks.

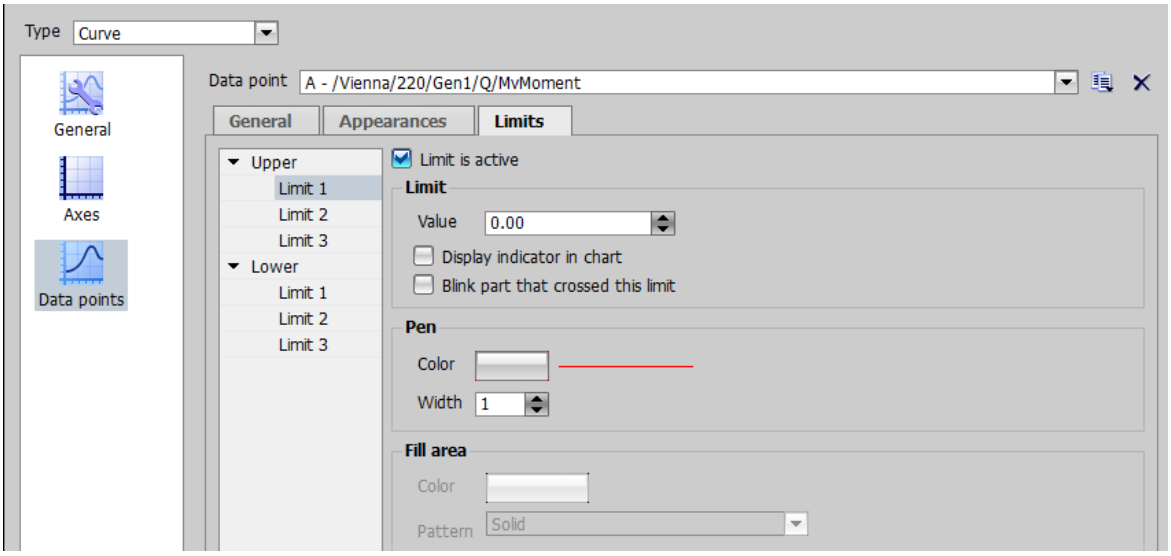
Configuring blinking based on quality codes is available in **Chart settings** > **Data points** > **Appearance**.



[sc_Appearancetab, 1, --_--]

Figure 17-21 Appearance Tab

Configuring blinking based on limit violation is available in **Chart settings** > **Data points** > **Limits**.



[sc_limitstab, 1, en_US]

Figure 17-22 Limits Tab

The operator can turn off blinking for open sets by:

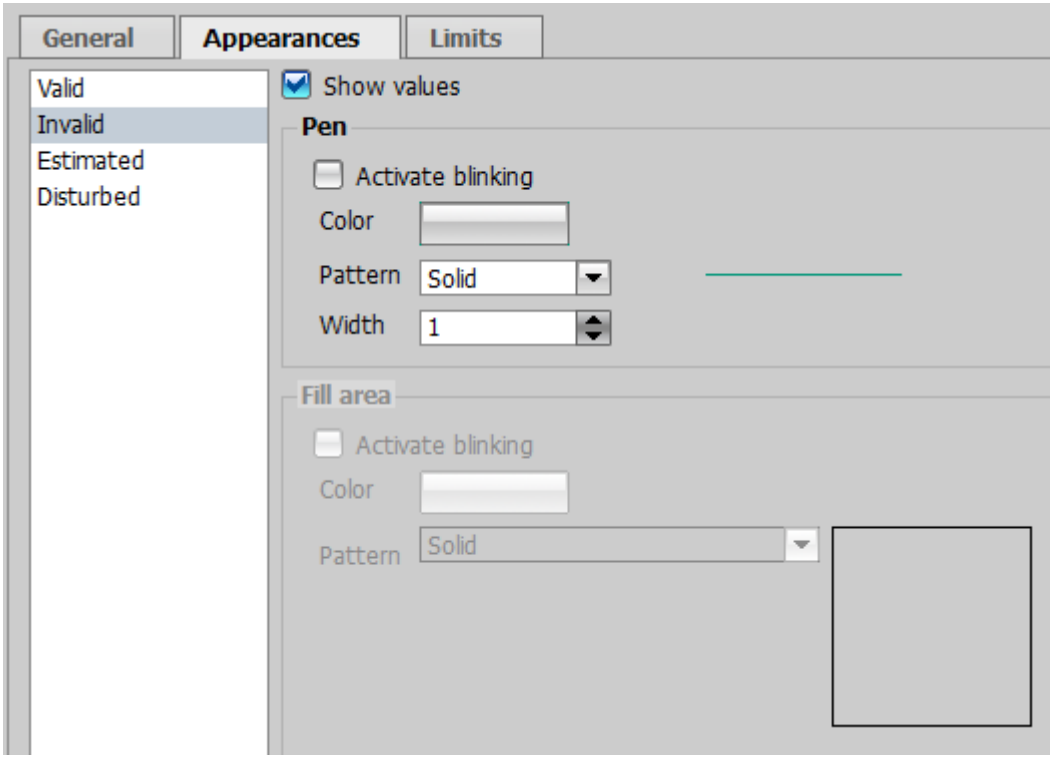
- Right-click on blinking area and open context menu in chart view.
- Select **Stop blinking**. This will disable all blinking sections/area in chart.

When a new limit violation or change of quality codes appear (after blinking has been stopped), the related section/area starts blinking again.

17.2.7.7 Hide Invalid Values

Invalid data is hidden by default, so that the operator is not confused by wrong information. There will be an empty space between the two valid values to represent the invalid value. If needed, the operator can manually change the appearance for invalid values. The options are the same as for other quality codes (valid, estimated, and disturbed).

Hiding invalid values is available only for Line charts, bar charts and pie charts.



[sc_Invalid values, 1, en_US]

Figure 17-23 Chart Settings - Invalid Value

17.3 Operational Steps

17.3.1 Opening the Curve Tool

Following are the two possibilities to open the Curve Tool:

- **Opening using BaSiDi or Heads-Up Display**

  BaSiDi:

  – Click on **Curves** in **Curves and Tables** tab.

  HUD:

  – Using Navigation menu: Click on **Tools > Curves**.
  – Using Cards: Double-click on Trending Card.

  The Curve Tool opens, showing the already defined sets, if available.

- **Opening using Analog Value Context Menu of Network Diagram**
  – In a network diagram, right-click the TA you want to display as a curve.

    The context menu opens.

  – In the context menu, select **Tools** and then select **Show curve**.

    The Curve Tool opens, showing a new set with the curve of the selected TA.

- **Calling-up Curve Tool from Grid/Pivot Table on WebUI**
  – Select one or more rows of a Web applications. Show Curve icon gets enabled.
  – Click on .

  A new instance of curve tool will open with the selected TAs in a new set.

17.3.2 Adding a Data Point

To add a data point, perform the following:

- Open the Curve Tool.
- In a network diagram, click the TA you want to display as a data point and drag it to the set area of the Curve Tool.

  A new set containing the data point of the selected TA is created and shown in the curve display area.

- To add an additional data point to this set, click the desired TA in the network diagram and drag it to the chart display or legend area.

  The data point is added to the currently selected set and displayed in addition to the already displayed curves in the curve display area.

17.3.3 Changing Position of Legend

Legend can be moved to another position on the screen to optimize the window layout.

To move a legend:

- On the legend header, hold down the left mouse to start dragging.
- Drag the legend to one of the allowed areas: right, left, above, or below the chart area and leave the left mouse button.

  The legend will be placed where the user drops it.

Legend does not overlay any chart content and the position of legend can be saved with set.

17.3.4 Saving Legend Properties

To save the legend state for future use:

- Change the legend position or visibility. The modification indicator ***** will be visible on the application title bar.
- On the application toolbar, click the Save icon.

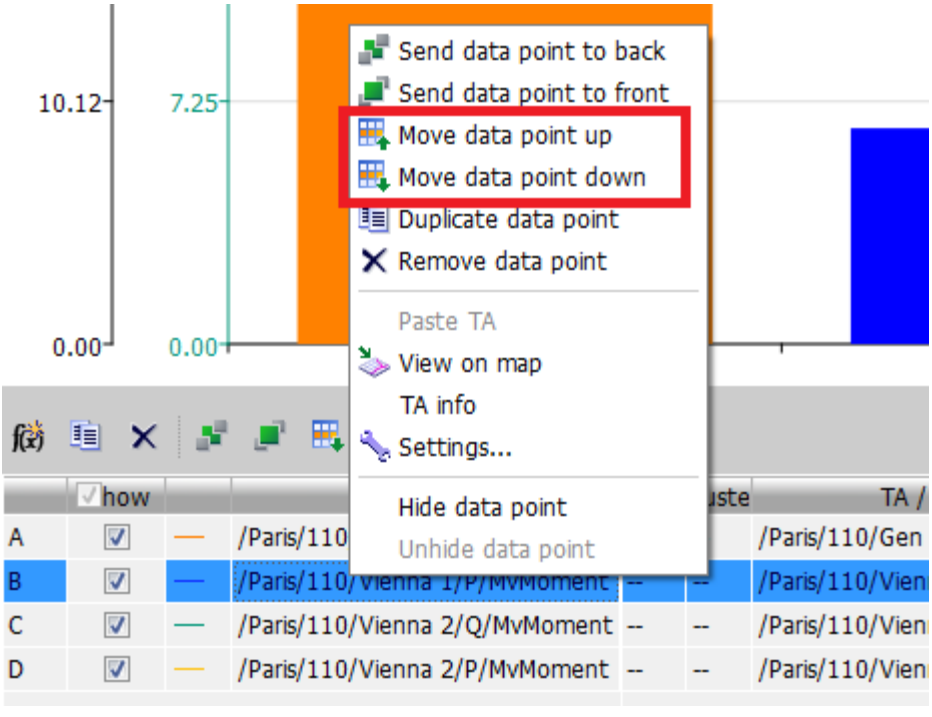
  The set is saved with the legend as it is.

Legend properties, for example, position (up, down, left or right), visibility, and size can be saved.

17.3.5 Adjusting Order of Data Points Using Legend

To adjust the order in which data points are displayed in line chart, bar chart, pie chart, and dials:

- Expand the legend.
- Right click on a data point and select or from the context menu to move data point up or down.



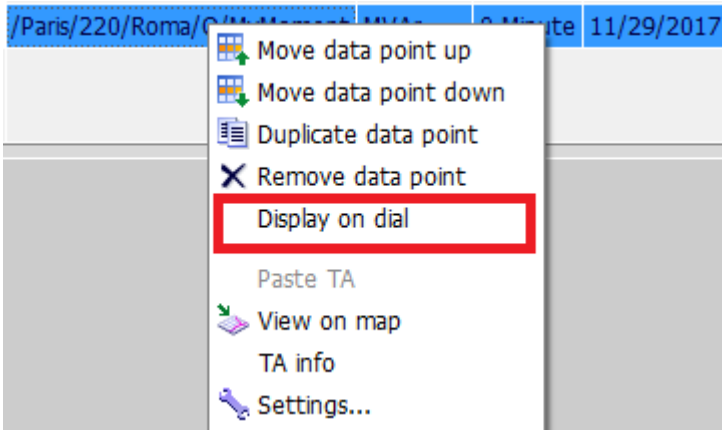
[sc_AdjustOrder, 1, en_US]

Figure 17-24 Adjust Order of Data Points

Legend and chart are updated accordingly.

When the order of data points is changed, z-order and value table are also adjusted accordingly. For example, when a selected data point is moved up to top of the legend, it has the highest priority and is plotted over all other curves, and the corresponding z-order and the value table is updated accordingly.

To move selected data point to legend top position and display it in the dial, right-click on data point and select **Display on dial** from the context menu. **Display on dial** is available only for dials.



[sc_Displayondial, 1, en_US]

Figure 17-25 Display on Dial

17.3.6 Defining the Chart Type

Line charts, bar charts, pie charts, dials, and histograms can be created using the curve tool. To create a particular chart type:

- Open the Curve Tool.
- In a network diagram, click the TA you want to display as a curve and drag it to the set area of the Curve Tool.

  A new set containing the curve of the selected TA is created and the data point is displayed in the curve display area as a line chart.
- Click icon in the toolbar.

  The chart settings dialog opens.
- Select the chart type from the drop down and click **Apply**.

  The corresponding chart type is created.

17.3.7 Modifying the Appearance of a Line Chart

To modify the parameters of a set, perform the following:

- Select the desired set by clicking it in the set area.
- Click  icon in the toolbar.

  The chart settings display is opened.

Example for Modifying General Parameters

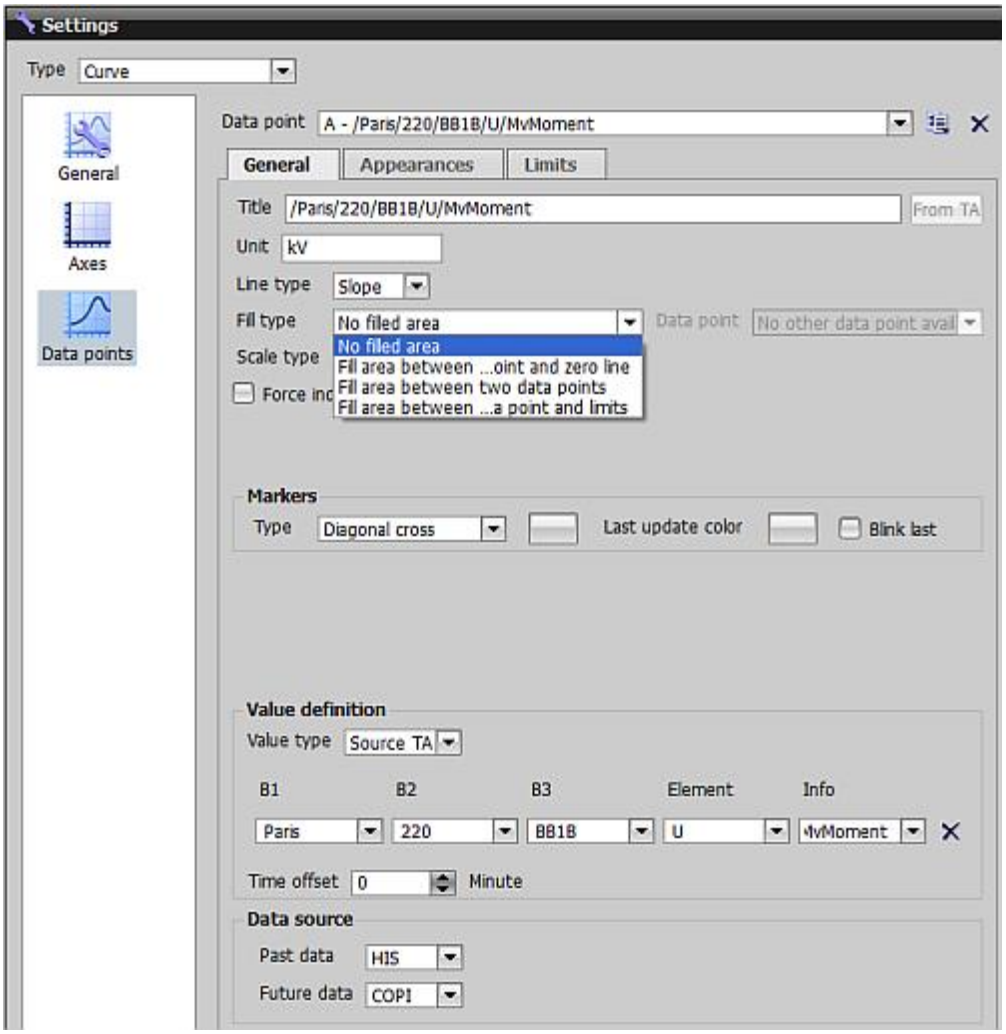
In the General section, enter a comment in the corresponding field and select **Display comment as header in diagram**.

The comment is displayed as background text in the set area.

Flooding of Curves

To flood a curve, perform the following:

- Open chart settings dialog, go to data points and select related data point.
- In the **General** tab, select **Line type**, and **Fill type** (area to fill) for example, Fill area between curve and zero line.



[sc_Flooding settings, 3, en_US]

Figure 17-26 Chart Settings - Fill Type

- Select **Appearances** tab, select quality, for example, Valid, and set the color on right side (Fill area).
- Click **OK** or **Apply**.

17.3.8 Switching Trending On/Off

Initially, trending is switched on. That means that the curve display area is continuously updated with the latest values. By default, the curve shows some values of the recent past (for example, last 30min) together with the real-time trend.

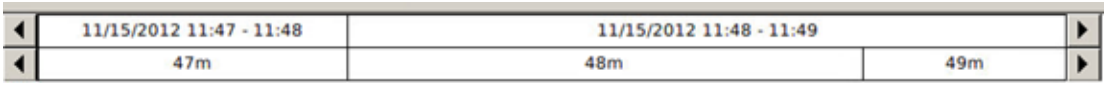
To view historical or future data, perform the following:

- Select the desired set by clicking it in the set area.

  The curves stored in this set are displayed in the curve display area and trending is switched on.

- To switch trending off, click icon in the toolbar.

  Trending is switched off and the time navigation bar can be used to scroll forward and backward in time.



[ugsca047-310113-01.tif, 1, en_US]

Figure 17-27 Curve Tool - Tracking

- When trending is switched off, clicking the trending icon in the toolbar switches trending on again.

NOTE

The trending feature is only available for line, bar charts, dials and pie charts. The time navigation bar is only available for line charts.

17.3.9 Creating, Modifying, and Deleting Formulas

To create curve tool formulas, perform the following:

- Select the desired set by clicking it in the set area or create a new set by clicking .

  The curves in this set are displayed in the curve display area.

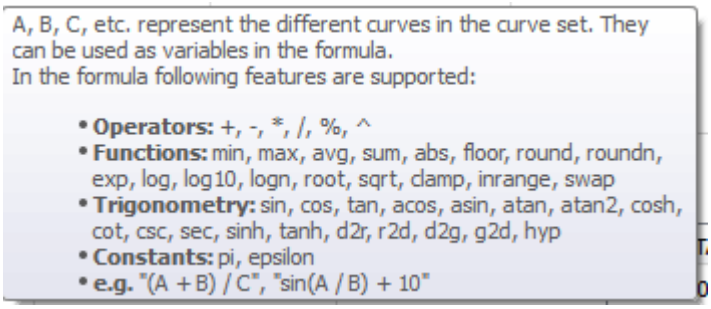
  Note: At least one data point from a TA should already be there in the chart view area to reference in the formula.

- Click in the legend toolbar.

  An entry for a new data point based on the formula is made in the legend at the bottom of the display.

- Enter the formula for the new data point in the legend, for example, if curve A is in the set, A+1 can be another curve as shown in the figure.

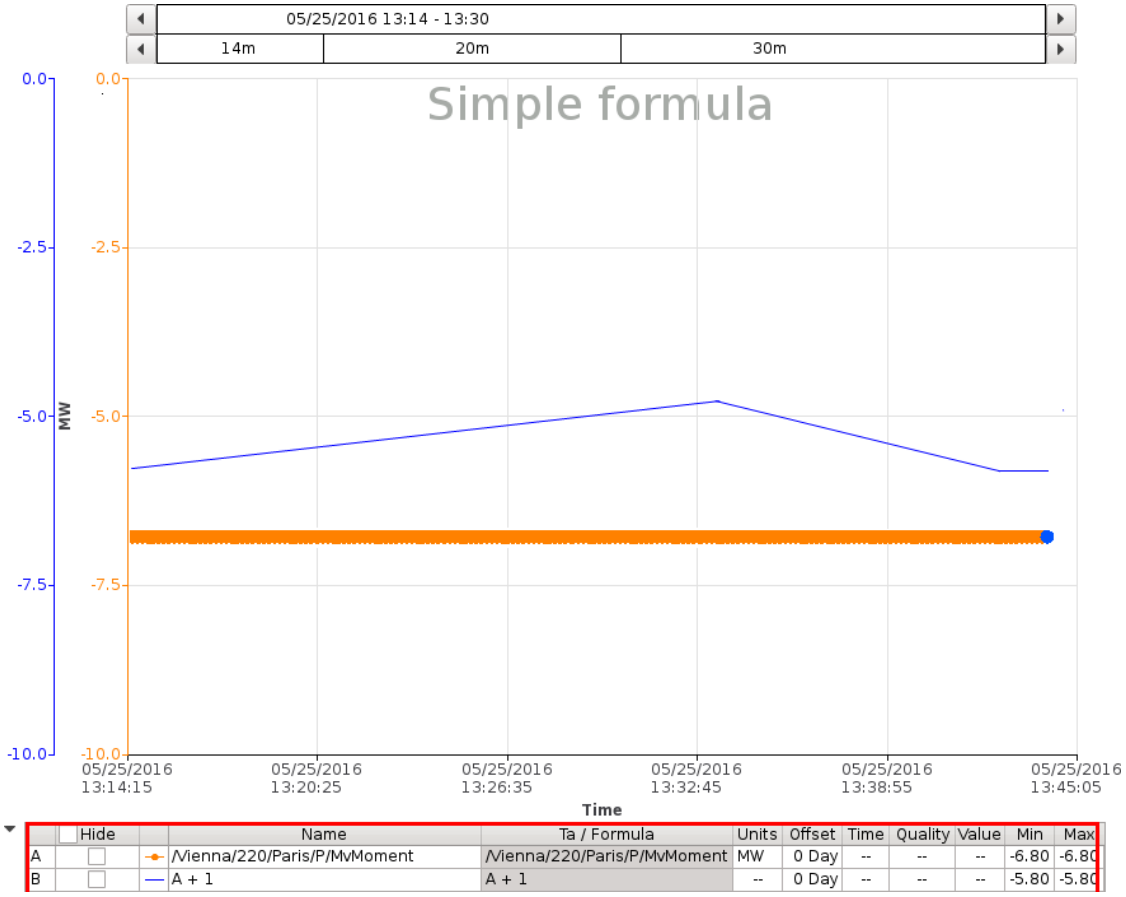
  A list of supported operators and tips on using formulas is displayed when you enter the formula.



[sc_operators, 2, --_--]

Figure 17-28 Operators and Formula Tips

The calculated data point is displayed in the set area.



[sc_CurveTool_Formula, 2, en_US]

Figure 17-29 Curve Tool Formulas

- Formulas can be edited by double-clicking them in the legend.
- To delete a formula and the corresponding calculated data point, select the formula in the legend and click ✕ in legend toolbar.

Alternatively, formulas can be directly entered in the chart settings display of the set.



[sc_formula_preferences, 3, en_US]

Figure 17-30 Entering Formulas in the Chart Settings

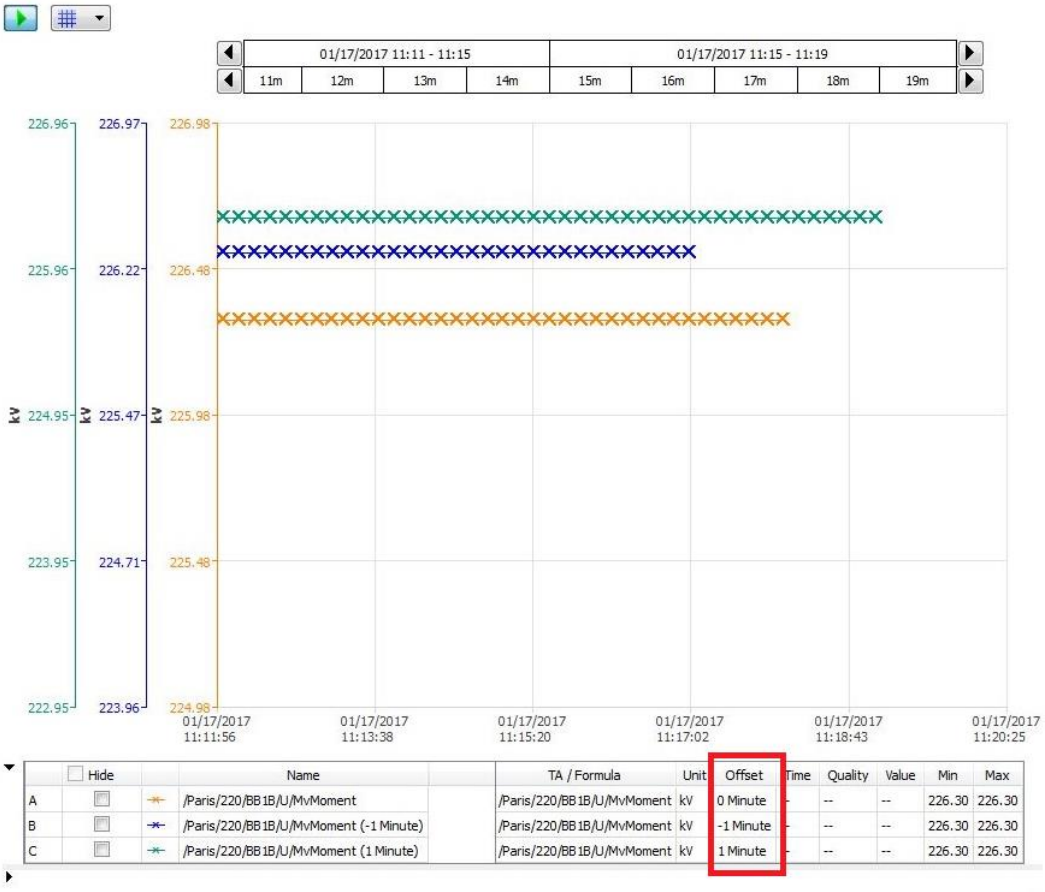
17.3.10 Displaying Data Points with Time Offsets

To create offsets for data points:

- Create a copy of a data point by clicking  .

  The duplicate data point is created.

- Open the chart setting dialog by clicking  .
- Define the time offset in the chart settings dialog.
  - Define offsets in the future: positive values, for example, 1.
  - Define offsets in the past: negative values, for example, -1.



[sc_curveoffset, 2, en_US]

Figure 17-31 Offset Curves

17.3.11 Exporting Data Points from the Curve Tool

Data points can be exported to CSV, image, and Microsoft Excel formats. To export all data points of a set:

- Create a chart of any type using the curve tool.
- Click the  icon.

A drop-down list with the export options opens.

- Select the required format.
- Enter the destination and file name for the exported file and click **OK**

  The data points are exported and saved.

17.3.12 Creating and Modifying Bar Charts

Bar charts are used to compare multiple TAs for one point in time or values of one TA in many points of time. The curve tool supports up to 30 bars per set.

To create a bar chart:

- Open the Chart settings dialog for a set.
- Select bar chart type and click **OK**

  A bar chart is created.

- The bar chart can be changed between vertical and horizontal from the general settings dialog of the chart settings display.
- The reference values of the data points can be set to absolute or percentage (with a nominal value) from the general settings dialog of the chart settings display
- Data points from a bar chart can be stacked and grouped.
- Changing the position of the data points in the legend creates a change in the bar chart as well.

Defining Stacked and Grouped Bar Chart

Stacked and grouped bar charts can be created in the Curve tool using clusters. To create stacked and group bar charts:

- Select the **Stacked** or **Grouped** option in the general tab of the chart settings display.
- Select the **Assignment** tab.
- Click + to define group or stack and enter a name for the group.

  Repeat the process to create multiple groups or stacks.

- Click + to define a cluster. Define the color and name for the cluster.

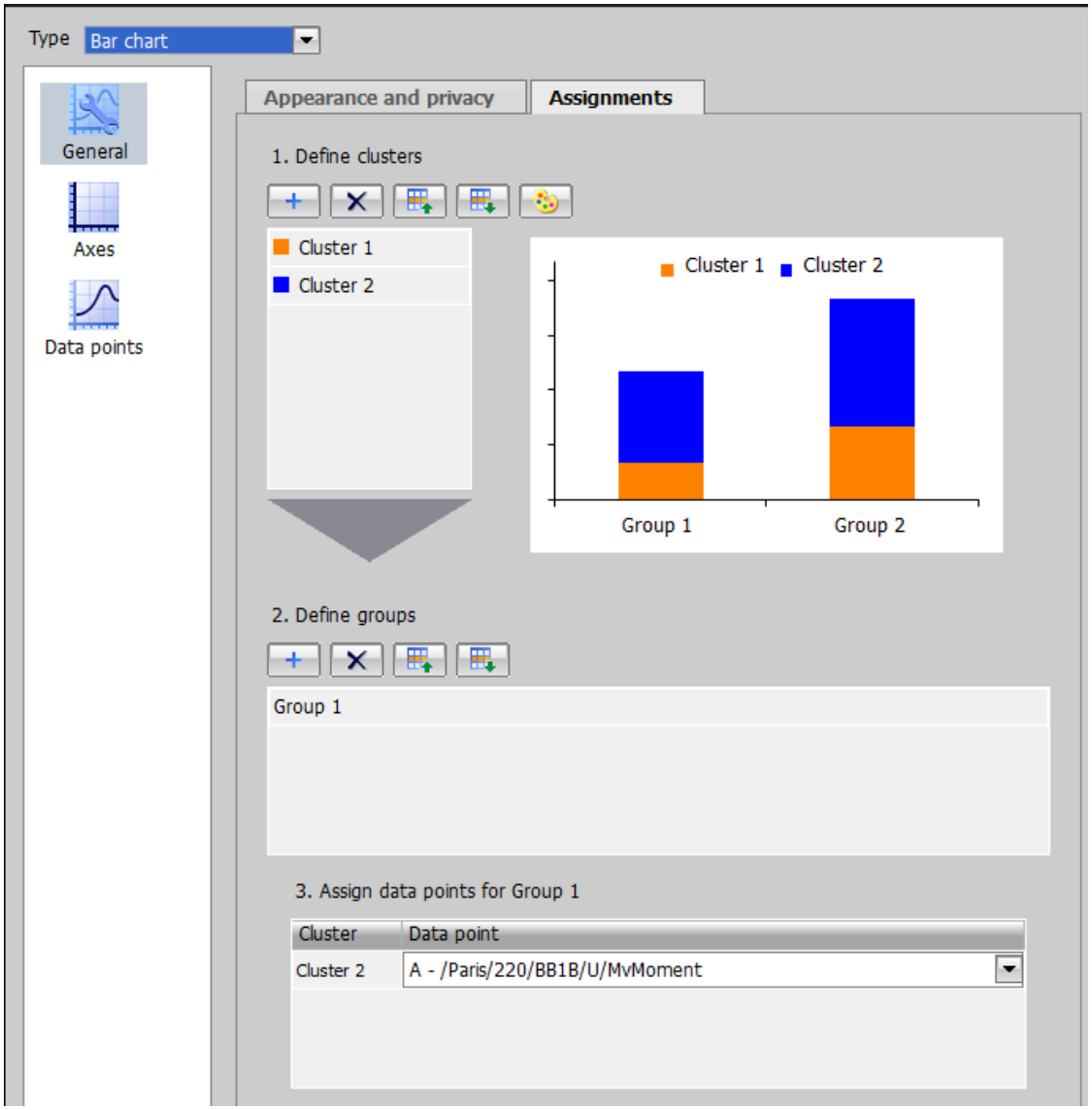
  Repeat the process to create multiple clusters.

- Assign data points to the clusters.

  The drop-down box shows all data points that are already part of the set.

- Click **Apply**

  The data points are grouped or stacked and the chart is created.



[sc_clusters, 2, en_US]

Figure 17-32 Clusters

17.3.13 Creating and Modifying Pie Charts

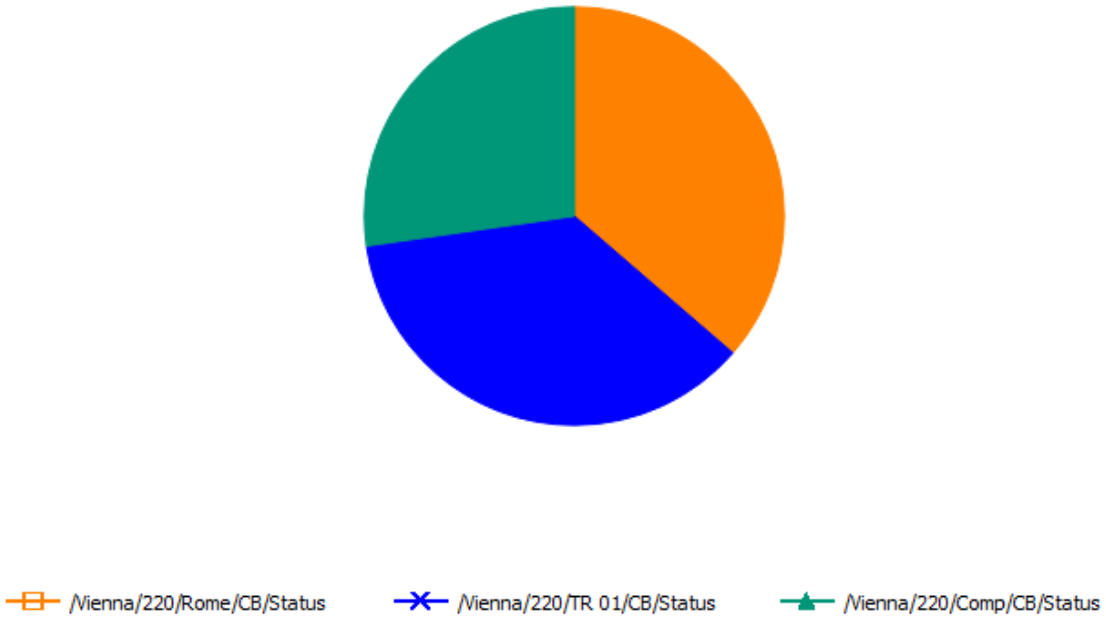
Pie charts are used to compare multiple TAs for one point in time or values of one TA in many points of time. The curve tool supports upto 30 segments. The angles of the chart segments are determined automatically based on the values

To create a pie chart:

- Open the Chart settings display for a set.
- Select pie chart type and click **OK**

  A pie chart is created.

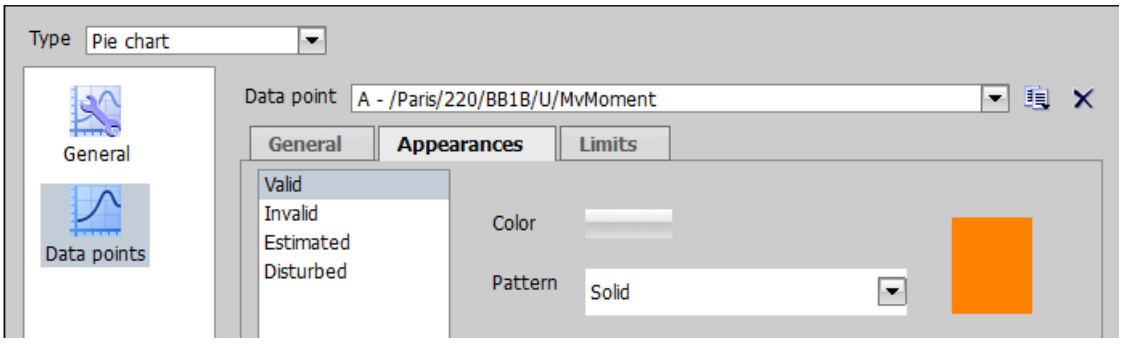
- The pie chart is always a full circle.



[sc_piechart, 1, en_US]

Figure 17-33 Pie Charts

- The colors for the segments can be defined from the appearance tab of the chart settings menu.



[sc_piechartcolor, 2, en_US]

Figure 17-34 Segment Color definition

- TAs can be dragged and dropped into the pie chart.

17.3.14 Creating and Modifying Histograms

Histograms display frequency distribution of one or more data points.

Calculating and displaying the bars of a histogram can be a very time consuming task depending on time range and number of TAs.

The histogram is only updated on demand. Click the icon to update a histogram. The curve tool supports upto 30 data points.

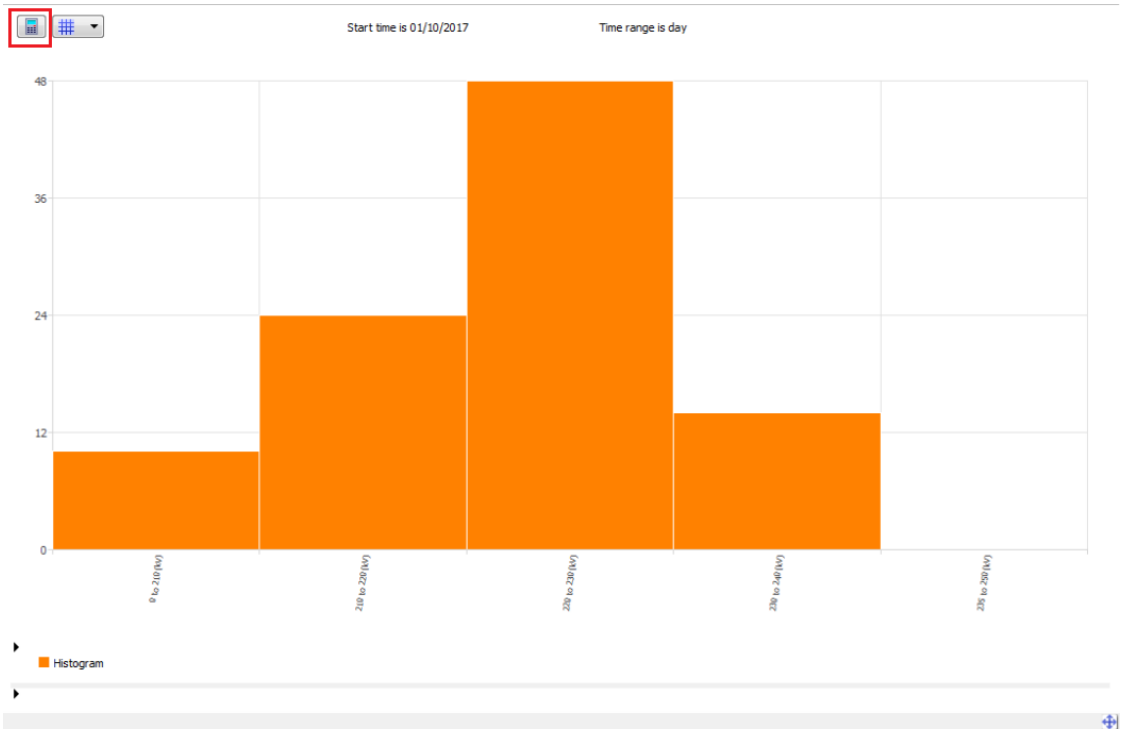
To create a histogram chart:

- Click on the icon in toolbar and select the icon in dropdown list to create an empty histogram.
- Drag and drop TAs into the legend area.
- Open chart settings dialog and define start time and calculation time range and press **OK**.
- Click the icon.

A histogram chart without categories is plotted (depending on available data of assigned TAs)

The values for frequency calculation axis and the value axis are automatically calculated, by default.

The operator can change it to manual and change the appearance of the histogram. The minimum and maximum value for frequency calculation, and minimum, maximum value, end point of first step, starting point of last step, and step size for value axis can be defined.

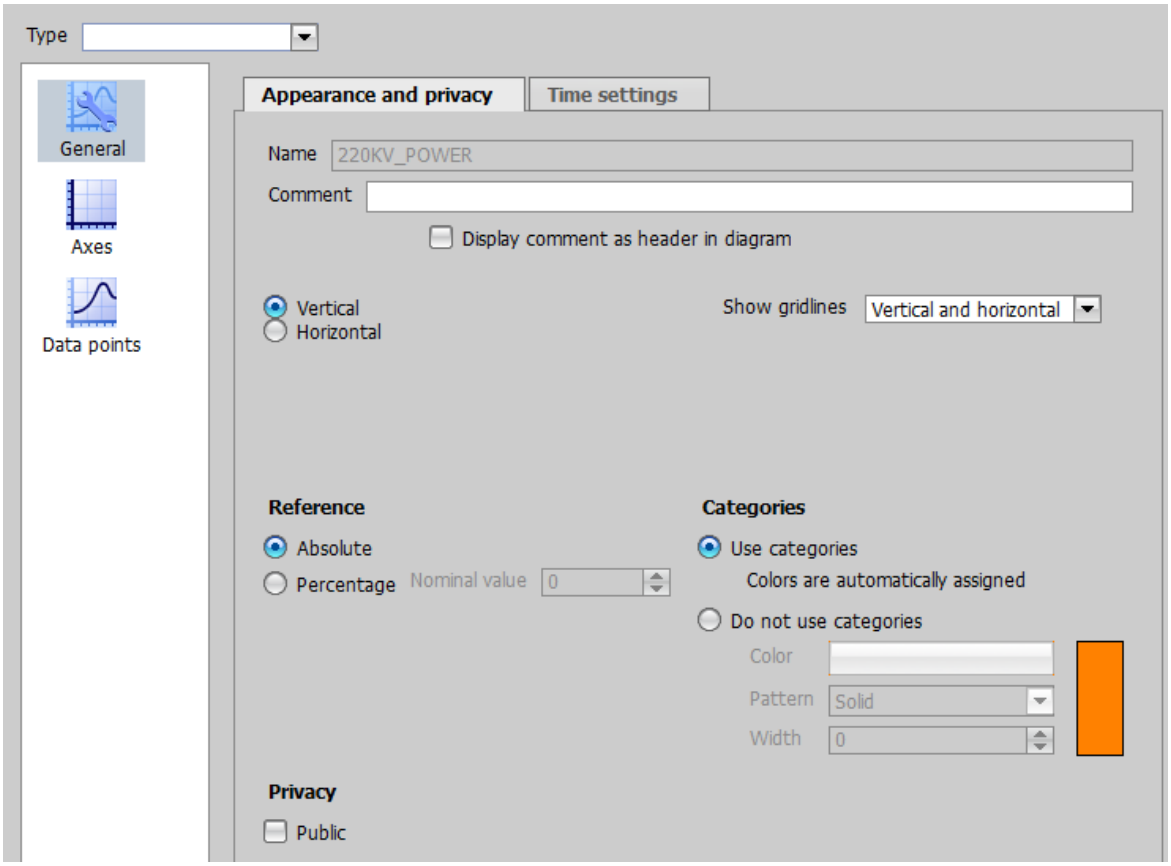


[sc_histogram, 1, en_US]

Figure 17-35 Histogram Without Categories

Histogram with Categories

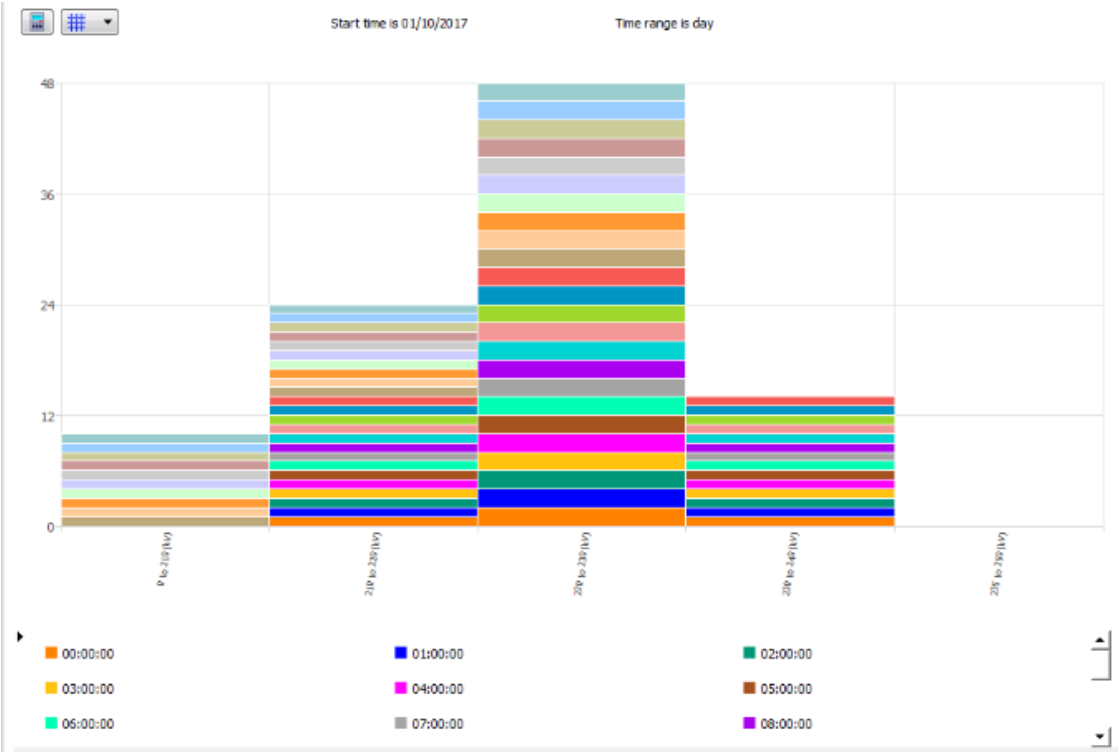
A histogram can be displayed with or without categories. A histogram with categories is displayed for a selected time range and is divided into equal time slices. The slices are displayed using a stacked bar chart.



[sc_Catagories, 2, en_US]

Figure 17-36 Histogram With Categories

- Select **Use Categories** in the chart settings display.
- Define a selected time setting in the chart settings display to create a histogram with categories. The selected time range is divided up into equal time slices which are displayed using a stacked bar chart.



[sc_Categorieshistogram, 1, en_US]

Figure 17-37 Histogram With Categories

The number, type, and colors of the categories are fixed depending on the calculation time range and cannot be configured.

Table 17-1 Histogram Calculation Time Range

Calculation Time Range	Number of Categories	Category	Example (Start time = 02/01/2016 09:00)
Day	24	hours	01/31/2016 10:00, 01/31/2016 11:00-12:00, …. 02/01/2016 08:00
Week	7	day	Monday 01/25/2016, Tuesday 01/26/2016, … Sunday 01/31/2016
Month	28, 29, 30, 31	days	Friday 01/01/2016, Saturday 01/02/2016, … Sunday 01/31/2016
Year	12	month	February 2015, March 2015, … January 2016

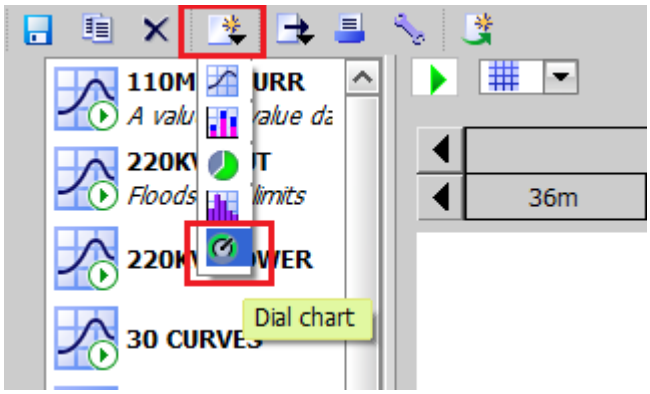
17.3.15 Creating and Modifying Dials

A dial shows the actual value of a data point. To show the difference between an actual value and a nominal value, the operator has to configure a data point as a formula which calculates this difference. The scale of the dial can be absolute (default) or percentage value.

Creating a New Dial

The operator can create a new chart type Dial and view the value of a data point in it.

- Click on new data set icon and select Dial from drop-down list.

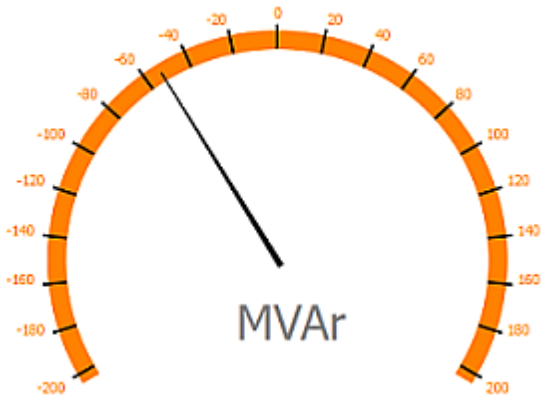


[sc_NewDial, 1, en_US]

Figure 17-38 Create New Dial

- Drag and drop a TA.

A dial chart is created. A dial can only display one TA at a time. When the set contains more TAs, only the topmost TA in the legend will be displayed. Note that digital TAs cannot be displayed on a dial.



[sc_Dial, 1, en_US]

Figure 17-39 Example of Dial

In case of any modification in dial, the modification indicator * is visible on the application title bar. The operator can save any modification by clicking on icon.

Defining Title for Dial

The operator can change the title for a dial.

- Click to open chart settings.
- Click on **General**. In **Appearance and Privacy** tab, configure comment (text and visibility)
- Click **Apply**.

Changing Scale from Absolute to Percentage

- Click to open chart settings.
- Click on **General**. In **Appearance and Privacy** tab, select **Percentage**.
- Define reference value by entering a positive value into the **Nominal value** field.
- Click **Apply**.

Comparing Two TAs in Dial

The operator can view the difference between two TAs in the dial by adding the two TAs to the set and then creating a formula for comparing the TAs.

- Add two TAs to set. For additional reference, refer to 17.3.2 Adding a Data Point.
- Add formula. For additional information, refer to 17.3.9 Creating, Modifying, and Deleting Formulas.
- Move formula to top on legend. Data point representing formula is shown on dial.
- Click to save the chart.

Configuring Display of Limits

The operator can set different limits for dials.

- Click to open chart setting.
- Click on **Data Points**. In the **Limits** tab, configure limits (visibility, value, color, and pattern).
- Click **Apply**.
- Click to save the chart.

Changing Data Points Shown in Dial Using Context Menu Option

The data point in the dial that is currently not shown on top of the legend can be displayed in dial:

- Right-click on a legend item that is not currently shown in dial to open the context menu.
- Select **Display on dial**.

The selected data point will be moved to the legend top position, and is shown in the dial. Dial can only display one data point at a time. When the set contains more data points, only topmost data point in legend will be displayed.

17.3.16 Stacking Curves

Stacking Line Chart (Without Reference)

The operator can view line charts as stacked chart in order to get a better understanding on the situation.

When a set with line chart containing at least 3 TAs is called up:

- Set line chart as **Stacked** to show stacked line chart.
- Turn on flooding between curves to display stacked line chart with flooded areas (optional).

Digital TAs are not displayed in a stacked line chart.

NOTE

Value table displays the data point values and not the calculated (stacked) values for stacked line charts.

Dropping TA to Stacked Line Chart (Without reference)

The operator can drop a TA into an existing stacked line chart without reference.

- Drop a TA in the chart area.

  TA is added in the next available position and the chart is recalculated to include values of dropped TA.

Moving Curve on Stacked Line Chart (With or Without Reference)

The operator can change the curves order in stacked line chart.

- Right-click on the corresponding data point to open context menu.
- Select **Move data point up**, **Move data point down**, **Send data point to front**, **Send data point to back**.

Legend is updated with new curves order, and the chart is recalculated.

Filling Area on Stacked Line Chart (With or Without Reference)

The operator can add fill areas to curves in stacked line charts.

- In settings dialog box, change **Fill area** of curve A from **No filled area** to **Stacked Fill Area**.

  The chart shows a filled area between the zero line and the Curve A. The filled area color and pattern are defined in **Appearance** tab.

- Click **Apply**.

Chart shows a filled area between Zero line and curve A.

Defining Stacked Line Chart Based on Reference

The operator can view line charts as stacked chart with a reference in order to get an overview of a generator schedule and its maximum and minimum operating ranges.

- Create or load a line chart.
- Convert line chart to **Stacked with reference**.

  Click **Apply**.

  All curves have **Reference for stacking** set to **Zero line** and curve values are recalculated related to zero line.

- In setting dialog of curve B, change **Reference for stacking** to curve A and **Stacking** to positive.

  Click **Apply**.

  Curve B is updated and value table shows curve values (not the calculated value).

Dropping TA to Stacked Line Chart (With Reference)

- Drop TA in the chart area.

  Dragged TA is added to set. The chart is recalculated, and TA values are added to the value table.

Hiding Curve on Stacked Line Chart (With or Without Reference)

The operator can temporarily hide a curve on a stacked line chart.

- Right-click on a data point to get the context menu.
- Click on **Hide data point**.

  Chart does not show the hidden curve, and any filled area related to the hidden curve is also hidden.

Data points which are used as reference, cannot be hidden.

17.3.17 Defining Time Settings for Line Charts

Defining Absolute Start and End Time

To define absolute start and end time:

- Open chart settings dialog and navigate to **General** > **Time settings**.
- Check **Adjust time settings** checkbox.
- Select **Absolute start and end**.
- Select a valid **Start time** from the calendar.
- Select a valid **End time** from the calendar.
- Click **Apply**.

Defining Absolute Start Time

- Open chart settings dialog and navigate to **General** > **Time settings**.
- Check **Adjust time settings** checkbox.
- Select **Absolute start**.
- Select **Start time** from the calendar.
- Define the **Update chart drawing every** field.
- Click **Apply**.

Defining Position of 'Now' relative to System Time

- Open chart settings dialog and navigate to **General** > **Time settings**.
- Check **Adjust time settings** checkbox.
- Select **Relative to system time (now)**.
- Set **Before system time (now)**.
- Set **After system time (now)**.
- Define the **Update chart drawing every** field and click **Apply**.

17.3.18 Hiding Invalid values

Defining Display of Invalid Values for Line Chart, Bar Chart and Pie Chart

The operator can specify how invalid values are shown.

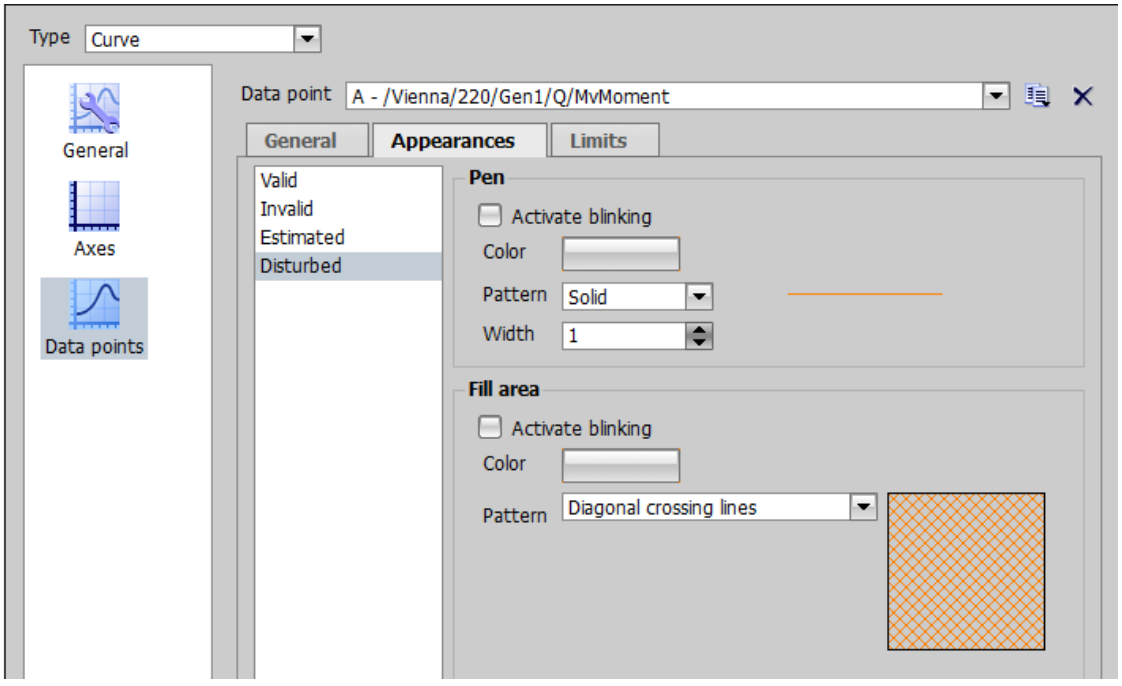
- Open Curve settings.
- Navigate to **Data points** > **Appearance** tab
- Select **Invalid** and configure appearance (visibility and color) of invalid values.
- Click **Apply**.

17.3.19 Defining Appearance for Quality Codes

To show the appearance of disturbed values on a set, display for data quality has to be defined.

The operator can specify how data qualities are shown.

- Open Chart settings.
- Navigate to **Data points** > **Appearance** tab, and select one of the data qualities (valid, invalid, estimated, or disturbed).
- Select **Activate blinking** checkbox to activate blinking of disturbed value.
- Configure appearance (width, color, pattern) of data qualities.



[sc_Define Data QualitiesDisturbed, 1, en_US]

Figure 17-40 Define Data Qualities

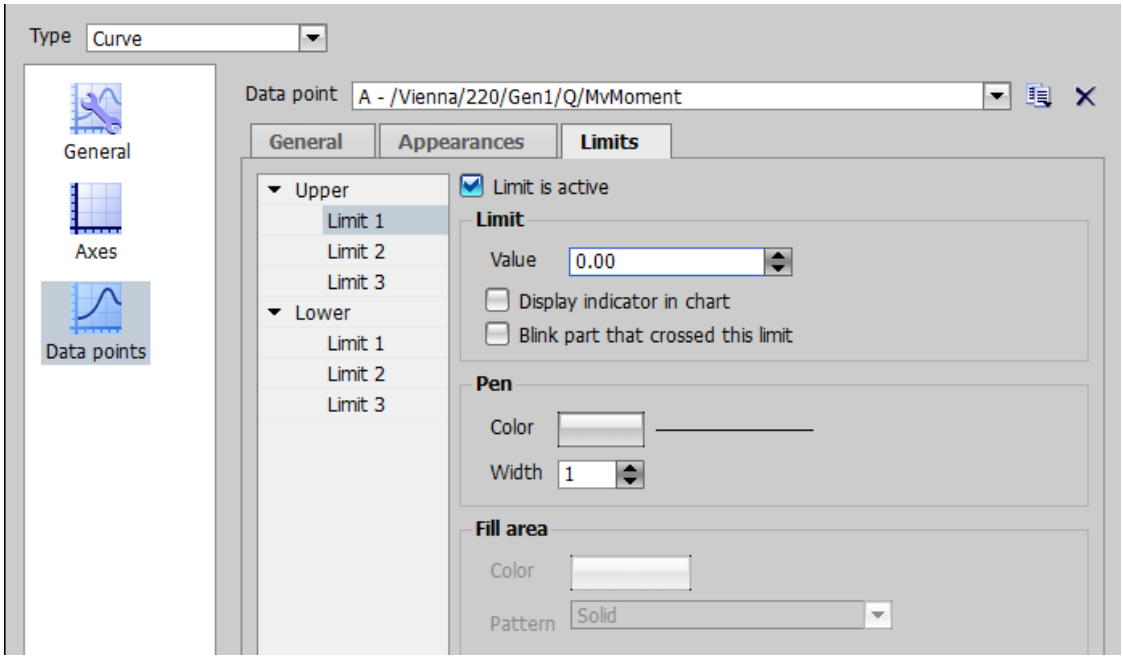
- Click **Apply**.

17.3.20 Defining Appearance for Limit Violations

To show the appearance for limit violations on a set, display for limits have to be defined.

The operator can specify limit values as follows:

- Open Chart settings.
- Navigate to **Data points** > **Limits** tab, and select one of the limit types (Limit 1, Limit 2, Limit 3).
- Click on **Limit is active** checkbox and configure blinking for limits.
- Configure appearance (width and color) of limits.



[sc_Definelimit, 1, en_US]

Figure 17-41 Define Limits

- Click **Apply**.

18 Operational Dashboard

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18.1 Introduction

The Operational dashboard allows operators to combine different displays (network diagrams, summaries, curves) into one window by simple dragging and dropping selected display (parts) onto the dashboard. The dashboard is used as a quick reference for operators.

The dashboard provides easy means for arranging contents like docked windows and tabs. The user can save dashboards for later usage and share existing dashboards with other users.

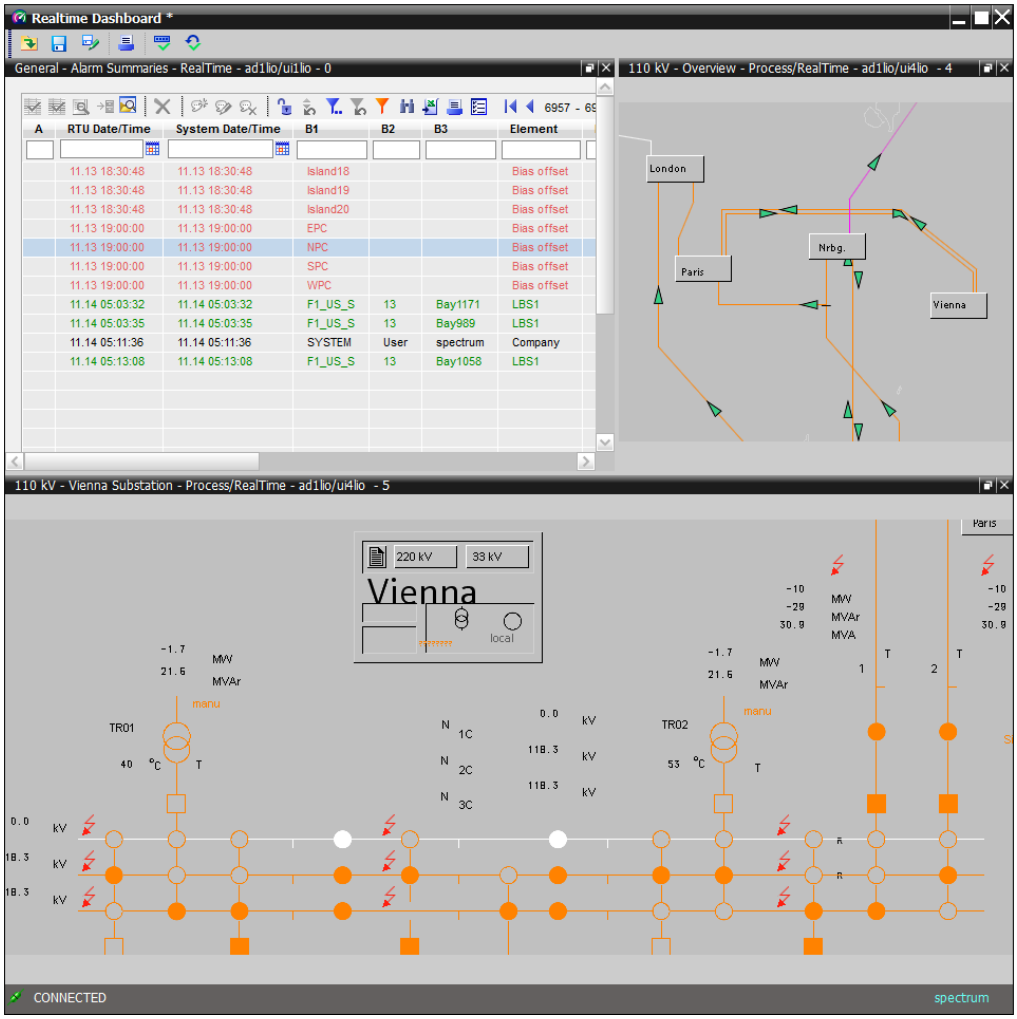


NOTE

Operational dashboard supports only real-time context.

18.2 User Interface

18.2.1 Overview



[sc_Realtime dashboard, 3, en_US]

Figure 18-1 Example of Operational Dashboard

18.2.2 Toolbar

The Operational dashboard has the following icons:

- Open dashboard. Opens the list of all available dashboard
- Save current dashboard. Select if the dashboard is public (visible to other users), give the name and a short description of the dashboard, and save dashboard.
-

Save current dashboard as. Save current dashboard with a new name and description.

- 

  Print current dashboard. For further reference, see 18.3.4 Printing an Operational Dashboard.

- 

  Display toolbars for dashboard displays. By default the toolbars are hidden in the Operational dashboard for saving screen space, click icon to display toolbars.

- 

  Turn synchronization on and off. When synchronization is on, all sets in the dashboard are synchronized regarding slider position, visibility of legend, visibility of data points and time axes.

  Note: Applies only to charts of the same type.

  Click the icon to turn synchronization off.

18.3 Operational Steps

18.3.1 Creating an Operational Dashboard

To create an Operational dashboard, perform the following:

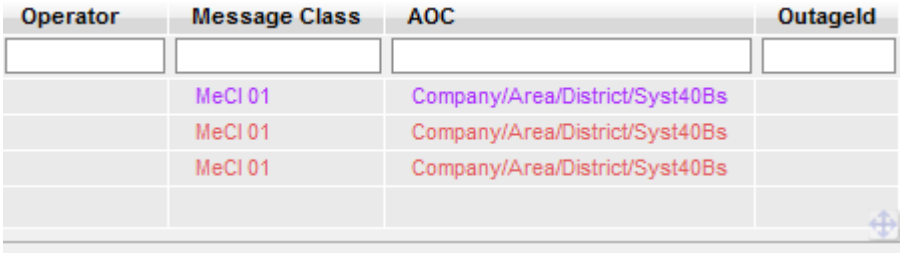
- Open the dashboard application from the **Curves and tables** tab in the BaSiDi. In HUD, the Dashboard can be opened through **New dashboard** option in **Tools** menu.
- Create a new empty dashboard.
- Drag and drop an ODV, Web application or curve tool display that you want to add to the Operational dashboard by pressing the icon in the right edge of the status bar and dragging it onto the dashboard window..

  The display gets added to the Operational dashboard.

  Web applications additionally provide drag icons for each table.

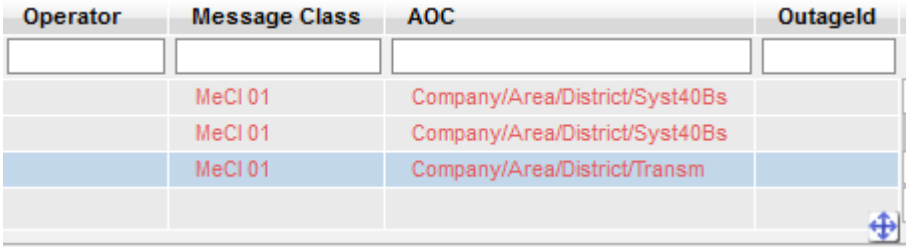
  The dashboard drag icon is rendered at the right bottom of the grid.

  – A grid with dashboard drag icon almost invisible.



[sc_Invisibledraganddrop, 1, en_US]

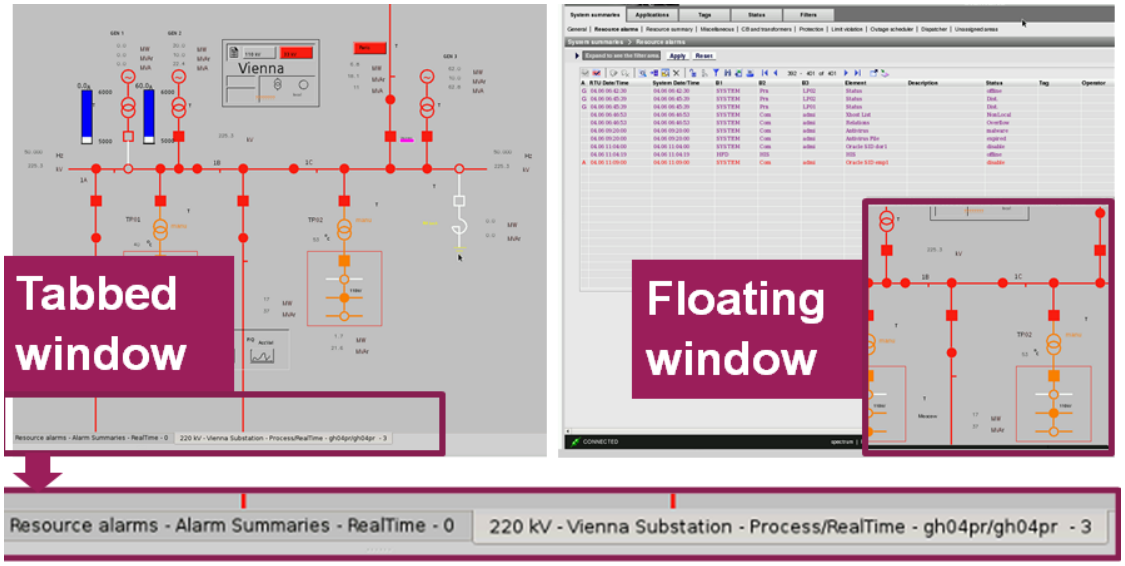
  – While hovering over the grid, the icon becomes visible.



[sc_Visibledraganddrop, 1, en_US]

  Alternatively, you can click the icon in the toolbar of an ODV or curve tool display to add it to a new Operational dashboard.

- You can dock a window onto the dashboard (tabbed window) by dragging the display and dropping it onto the dashboard.
- Double-click the title bar to restore the window to the original position.



[sc_tabbed vs Floating, 2, en_US]

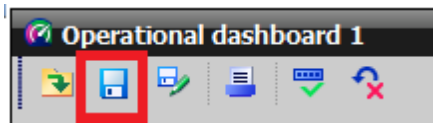
Figure 18-2 Tabbed Window and Floating Window

18.3.2 Saving an Operational Dashboard

To save an Operational dashboard, perform the following:

- Create the dashboard that you want to save.
- Click the Icon.

  The **Save dashboard as** display opens only for saving the dashboard for first time. The dialog **Save dashboard as** will not appear if the operator has already saved the dashboard before.



[sc-SaveDashboard, 1, en_US]

Figure 18-3 Save Dashboard

- Enter the name and a brief description of the Operational dashboard.
- Select the **Public** checkbox if you want to share the dashboard with other users.
- Click **Save**.

Alternatively you can open a dashboard, edit the dashboard, and click the (save as) icon to save the dashboard with a different name and description.

18.3.3 Opening an Operational Dashboard

To open a saved Operational dashboard, perform the following:

- Click on **Dashboard** from the **Curves and tables** menu in the BaSiDi.

The Operational dashboard display opens.

- Click the  icon.

  The **Open dashboard** Menu opens.

- Select the checkbox of the saved dashboard that you want to open.
- Click the  icon.

  The Operational dashboard opens.

18.3.4 Printing an Operational Dashboard

To print an Operational dashboard, perform the following:

- Open an Operational dashboard.
- Click the  icon.

  The print options menu opens with the following options

  - Print As Is
  - Print Invert
  - Print Greyscale (White Invert)
  - Print Greyscale (Black)

- Select the required option.

  The print menu opens.

- Select the printer and page range.
- Click the **Print** button.

  The Operational dashboard is printed.

18.3.5 Showing or Hiding Toolbars in an Operational Dashboard

To show or hide toolbars in an Operational dashboard, perform the following:

- Open or create an Operational dashboard
- Click the  icon.

  The toolbars for all displays appear.

- Click the  again.

  The toolbars for all displays in the dashboard are hidden.

18.3.6 Deleting an Operational Dashboard

To delete a dashboard, perform the following:

- Click the  icon.

  The **Open dashboard** Menu opens.

- Select the checkbox of the dashboard that you want to delete.
- Click the  icon.

  The confirmation dialog appears.

- Click the **Yes** button to confirm.

  The Operational dashboard is deleted.

19 Corporate Viewer

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19.1 Introduction

Spectrum Power Corporate Viewer is the access point for back office users to information from the Spectrum Power system. The corporate viewer consists of two main parts:

- Corporate network diagrams
- Corporate dashboards (based on Jasper)

Secure Access for Named Users

The Corporate authentication is based on user and password access control using HIS user administration. System administrators are responsible for defining user logins and associating **Areas of Responsibility (AORs)** to each user in order to restrict user access to the dashboards and network diagrams. AORs are enforced on corporate displays

19.2 User Interface

19.2.1 Corporate Viewer

19.2.1.1 Overview

The Corporate viewer can be accessed from the URL or through the BaSiDi. Login to access the Corporate Viewer

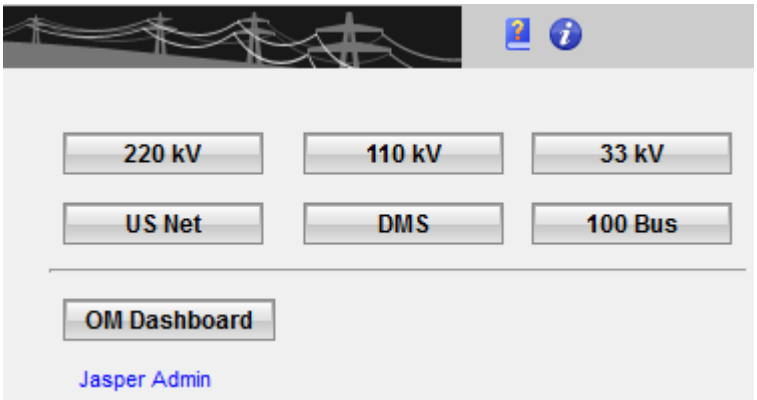


[sc_External_Corporate_viewer, 2, en_US]

Figure 19-1 Corporate Viewer Login Screen

The Corporate Viewer is the access point for back office users to information from the Spectrum Power system. All displays in the corporate viewer are shown in read only mode. The corporate viewer consists of two main parts:

- Corporate network diagrams
- Corporate dashboards (based on Jasper)



[sc_overview_display_16012015, 3, en_US]

Figure 19-2 Example of Corporate Viewer

19.2.1.2 Title Bar

Title bar icons are used to navigate to the previous and next display, refresh the display, print, and so on.

Table 19-1 Title Bar Icons

Icon	Description
	Set user preference
	Opens online help
	Application button
	Closes the window.

19.2.1.3 Buttons

The entry page of the corporate viewer contains a number of display call-up buttons. The display call-up buttons allow you to invoke Corporate Network diagrams in read only mode.

Besides network diagrams, you can access pre-defined dashboards (customer specific configuration).

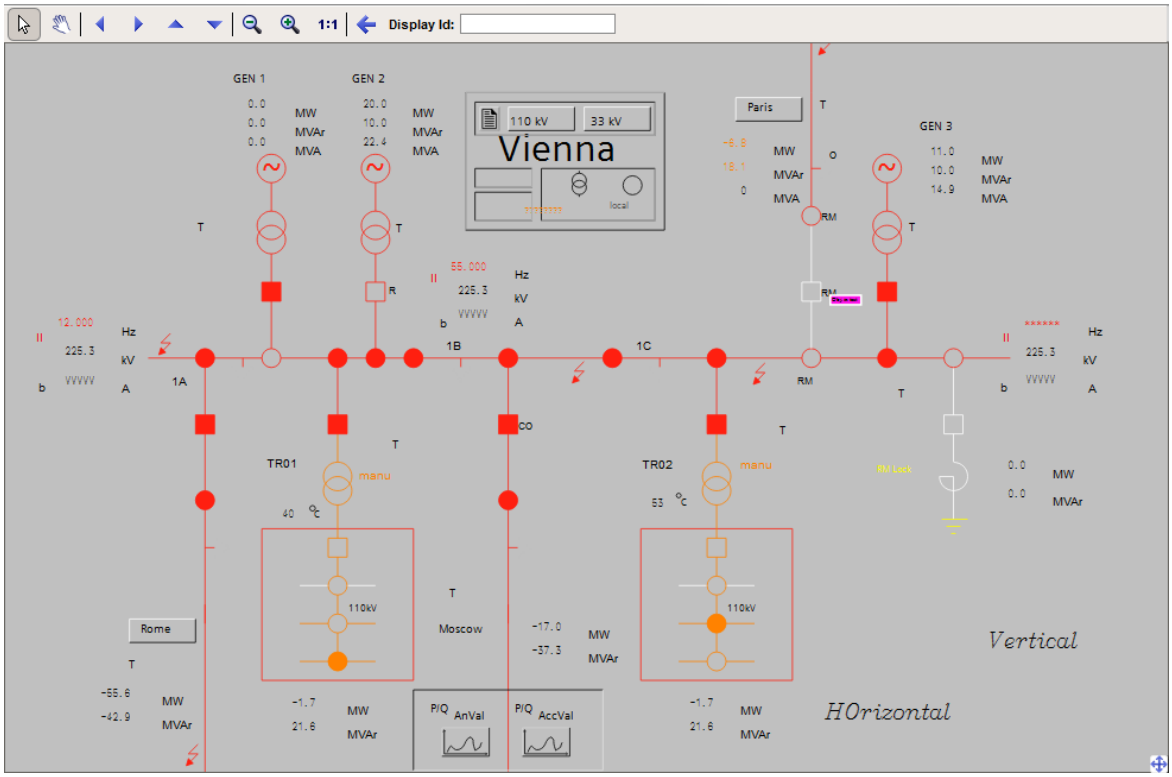
You can also open the Jasper soft admin tools from the corporate viewer. The Jasper report administrator tool allows you to define new dashboards and modify existing dashboards

19.2.2 Corporate Network Diagram

19.2.2.1 Overview

Corporate Network Diagrams are read-only diagrams that have near real-time updating capabilities. The data is displayed with a maximum delay of 30 seconds. Data updates include the following information:

- Analog data
- Switch positions
- Quality flags
- Tags or flag markers
- Topology coloring



[sc_corporate_nw_dis_20102014, 3, en_US]

Figure 19-3 Corporate Network Diagram Display

19.2.2.2 Toolbar

Table 19-2 Toolbar Icons

Icon	Description
	Pointer mode
	Pan mode
	Zoom out
	Zoom in
	Show default view
	Previous Display
	Allows the user to directly navigate to a Display

19.2.3 Corporate Dashboard

The Spectrum Power Corporate Viewer uses Dashboards from Jaspersoft for providing comprehensive views on information such as outage overviews. It displays the summaries of specific use-cases and has the ability to drill down to the lists and details from the displayed summary.

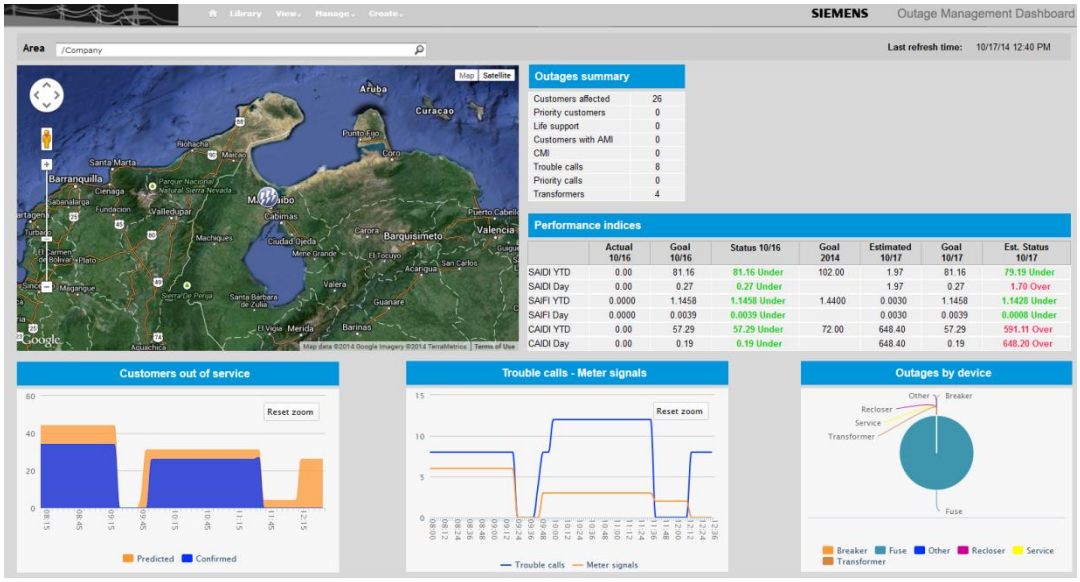
Jasper Dashboards provide powerful and flexible means for defining dashboards such as pie charts, line charts, and dials.

By accessing the JasperAdmin tools, you can perform the following operations on a corporate dashboard:

- Create a new dashboard.
- Modify an existing dashboard.
- Define or modify the access rights.

For further information on the JasperAdmin tools, refer to the Jasper Documentation.

Outage Management Dashboard and Crew Dashboard are examples of Corporate Dashboards.



[sc_om dashboard_27102014, 2, en_US]

Figure 19-4 Outage Management Dashboard